Clinical Trial Design Decision Framework

From Scientific Question to Defensible Design

J Kou

2026-02-21

Table of contents

Clinical Trial Design Decision Framework 12

What This Book Is—and Is Not 13
The Problem This Book Addresses . . . 13
Who This Book Is For . . . 14
How This Book Is Organized . . . 14
Four Questions That Will Not Go Away . . . 15
A Note on Tone . . . 15

Chapter 1: What Are We Trying to Show? 17
The Decision This Chapter Is About . . . 18
What This Chapter Covers . . . 19
The Risk of Starting Downstream . . . 19
What Follows from Getting This Right . . . 20

1.1 Purpose and Estimand 21
The question that precedes the hypothesis . . . 21
What the estimand framework requires you to specify . . . 21
Treatment policy versus hypothetical: the choice that changes everything . . . 22
Who specifies the estimand, and when . . . 23
The regulatory interface . . . 24
When the estimand is wrong . . . 25

1.2 Endpoint Choice 26
The question underneath the endpoint . . . 26
What an endpoint actually commits you to . . . 27
The surrogate problem . . . 27
Composite endpoints: combination as a decision . . . 28
Patient-reported outcomes: the ownership problem . . . 29
Who owns the endpoint decision . . . 30
The pressure to compromise . . . 31
What this section demands before proceeding . . . 31

1.3 Intercurrent Events and Estimand Strategy 33
Where the experimental logic breaks down . . . 33
Treatment policy: the question a prescriber asks . . . 33
Hypothetical: the question a mechanism scientist asks . . . 34
Composite: incorporating the event into the outcome . . . 35

While on treatment: limiting the question . . . 36
Principal stratum: asking about a defined subgroup . . . 36
The strategy is the estimand . . . 37
Multiple estimands and the primary/sensitivity relationship . . . 38
What must be settled before section 1.4 . . . 38

1.4 Closing: What Has Been Decided, and What It Costs **40**
What this chapter asked . . . 40
The decision this chapter made . . . 40
The most common failure mode . . . 41
The characteristic mistakes of this chapter . . . 42
What cannot be fixed downstream . . . 43
The professional risk . . . 43
What Chapter 2 requires from Chapter 1 . . . 44
The decision this chapter demands . . . 44
Chapter 1 risk summary . . . 45

Chapter 2: How Will We Measure the Difference? **46**
The question Chapter 1 leaves open . . . 47
The problem this chapter addresses . . . 47
What this chapter covers . . . 48
The decision structure of this chapter . . . 49
What this chapter is not about . . . 49
The connection to what follows . . . 50
The question Chapter 2 must answer . . . 50

2.1 Effect Measures **51**
The measure is not the outcome . . . 51
Binary outcomes: three measures, three commitments . . . 51
Time-to-event outcomes: when the hazard ratio assumption fails . . . 52
Continuous outcomes: the difference between average and distribution . . . 54
The scale dependency of effect measures . . . 54
What this section demands before proceeding . . . 55

2.2 Interpretability and Efficiency **56**
The tension that does not resolve . . . 56
When efficiency comes at interpretive cost . . . 56
When interpretability comes at efficiency cost . . . 57
The restricted mean survival time as a case study . . . 58
Responder analyses: when the distribution matters more than the mean . . . 59
The audience determines the measure . . . 59
What this section demands before proceeding . . . 60

2.3 The Non-Inferiority Margin **62**
The question only a non-inferiority trial can answer . . . 62

The two-step logic of margin justification . . . 62
Who owns the margin . . . 63
The constancy assumption and why it matters . . . 64
The margin and the effect measure . . . 65
The margin as a risk allocation . . . 65
What this section demands before proceeding . . . 66

2.4 Closing: The Measure Is a Commitment **68**
What this chapter asked . . . 68
What this chapter decided . . . 68
The characteristic mistakes of this chapter . . . 69
What cannot be recovered . . . 70
The connection to what follows . . . 70
Chapter 2 risk summary . . . 71

Chapter 3: What Are We Committing To? **72**
The question Chapter 2 leaves open . . . 73
The decision structure of this chapter . . . 73
What unites the chapter . . . 74
The connection to what surrounds this chapter . . . 75
What this chapter is not about . . . 75
The question this chapter must answer . . . 76

3.1 Effect Size Ownership **77**
The assumption no one wants to own . . . 77
Where effect size assumptions come from . . . 77
The optimism problem . . . 78
Who should own the effect size assumption . . . 79
The clinically meaningful effect versus the expected effect . . . 80
Sensitivity of power to effect size assumption . . . 81
What this section demands before proceeding . . . 82

3.2 Variance and Dropout **83**
The assumptions that compound quietly . . . 83
Variance: the number that hides heterogeneity . . . 83
The dropout problem . . . 84
Estimating control arm event rates . . . 85
Interaction between dropout and event rate assumptions . . . 86
The compounding error . . . 87
What this section demands before proceeding . . . 87

3.3 Event Rate Uncertainty **89**
Why event rates deserve a section of their own . . . 89
The declining control arm event rate . . . 89
Competing risks and cause-specific event rates . . . 90

Enrollment pace and the calendar time problem . . . 91
The information fraction and what it actually means . . . 92
When the event rate assumption proves wrong: the options . . . 92
What this section demands before proceeding . . . 93

3.4 Power as Risk Budget **95**
What power actually is . . . 95
The asymmetry of error consequences . . . 95
Power as a function of effect size: reading the curve . . . 96
The 80% convention and when it is wrong . . . 97
Allocating the risk budget . . . 98
The power report as a design document . . . 99

3.5 Closing: The Commitment Behind the Number **100**
What this chapter asked . . . 100
What this chapter decided . . . 101
The characteristic mistakes of this chapter . . . 101
What cannot be recovered . . . 102
The connection to Chapter 4 . . . 103
Chapter 3 risk summary . . . 103

Chapter 4: When Might We Stop Early? **105**
The question Chapter 3 leaves open . . . 106
Why this chapter exists separately from the sample size . . . 106
What this chapter covers . . . 107
The decision this chapter must make . . . 108
What this chapter is not about . . . 109

4.1 Why Sponsors Want Interim Analyses **110**
The honest starting point . . . 110
What early stopping for efficacy actually does . . . 110
Stopping for futility: a different incentive structure . . . 111
Regulatory interest in interim analyses . . . 112
The DSMB's interest: protection, not optimization . . . 113
Aligning the interests without conflating them . . . 113

4.2 Alpha Spending **115**
What is being spent . . . 115
Spending functions and what they commit to . . . 115
One-sided versus two-sided boundaries . . . 117
The information fraction problem . . . 117
What the spending function does not control . . . 118
Reporting the spending plan . . . 119

4.3 Operating Characteristics **120**
What the plan is actually committing to . . . 120

Under the null hypothesis . . . 120
Under the alternative hypothesis . . . 121
Under intermediate hypotheses . . . 122
Expected sample size and trial duration . . . 122
The consistency requirement . . . 123
Reporting the operating characteristics . . . 123

4.4 Futility, Efficacy, and Safety **125**
Three decisions that are not the same . . . 125
Efficacy stopping: what is being claimed . . . 125
Efficacy stopping: the information fraction question . . . 126
Futility stopping: binding versus conditional power . . . 127
The non-binding futility problem . . . 128
Safety stopping: a different authority structure . . . 128
When the three categories overlap . . . 129

4.5 Governance **131**
The gap between the plan and the decision . . . 131
The DSMB: composition and independence . . . 131
The charter: what it must specify . . . 132
Information flow: who sees what, when . . . 133
The open session, closed session, and executive session . . . 134
Documentation: the governance record . . . 134
When governance fails . . . 135

4.6 Closing: The Governed Decision **137**
What this chapter asked . . . 137
What this chapter decided . . . 138
The characteristic mistakes of this chapter . . . 138
What cannot be recovered . . . 139
The connection to what follows . . . 140
Chapter 4 risk summary . . . 141

Chapter 5: How Do We Protect Against Bias? **142**
The question Chapter 4 leaves open . . . 143
What bias is, and what it is not . . . 143
What this chapter covers . . . 144
The system, not the components . . . 145
What this chapter is not about . . . 145
The question this chapter must answer . . . 146

5.1 Randomization Logic **147**
What randomization guarantees . . . 147
Simple, blocked, and stratified randomization . . . 147
What randomization does not protect against . . . 148

Covariate-adaptive randomization . . . 149
Cluster randomization: when the unit of assignment is not the individual patient . 150
What this section demands before proceeding . . . 151

5.2 Stratification **152**
What stratification is for . . . 152
Choosing stratification factors . . . 152
Ascertainment at randomization . . . 153
Reflecting stratification in the analysis . . . 154
Stratification versus subgroup analysis . . . 155
Stratification in multi-regional trials . . . 155
What this section demands before proceeding . . . 156

5.3 Allocation Concealment **157**
The gap between randomization and assignment . . . 157
What concealment requires in practice . . . 157
Inadequate concealment in common trial designs . . . 158
The relationship between blinding and concealment . . . 159
Concealment in platform and adaptive trials . . . 160
What this section demands before proceeding . . . 160

5.4 Outcome Assessment and Validation **162**
The assessor who knows too much . . . 162
What must be blinded, and why . . . 162
Blinding in practice: maintaining it under pressure . . . 163
Independent adjudication committees . . . 164
Outcome validation: the adjudication charter as evidence . . . 165
Blinding of the statistical analysis . . . 165
What this section demands before proceeding . . . 166

5.5 Open-Label Designs **167**
When blinding is not available . . . 167
What open-label designs are at risk for . . . 167
Compensating designs for open-label trials . . . 168
The estimand for open-label trials . . . 169
Open-label extensions . . . 170
What this section demands before proceeding . . . 170

5.6 Closing: The System That Holds the Comparison Valid **172**
What this chapter asked . . . 172
What this chapter decided . . . 173
The characteristic mistakes of this chapter . . . 173
What cannot be recovered . . . 174
The connection to Chapter 6 . . . 175
Chapter 5 risk summary . . . 176

Chapter 6: What Are We Allowed to Claim? 177
The question Chapter 5 leaves open . . . 178
Why this chapter exists . . . 178
What this chapter covers . . . 179
The decision structure of this chapter . . . 180
What this chapter is not about . . . 181
The connection to what surrounds this chapter . . . 181

6.1 Claims, Not P-values 182
What a p-value is . . . 182
What a claim requires . . . 182
The gap between the result and the claim . . . 183
The p-value as a decision tool, not a truth statement . . . 184
When the primary fails: the secondary claim problem . . . 185
What this section demands before proceeding . . . 185

6.2 Co-Primary and Multiple Primary Endpoints 187
When two primary endpoints appear . . . 187
The conjunctive and disjunctive decision rules . . . 188
The type I error challenge . . . 189
Two primaries as a symptom of unresolved design . . . 189
Multiple primary endpoints in regulatory context . . . 190
What this section demands before proceeding . . . 191

6.3 Hierarchical Testing 192
The problem hierarchical testing solves . . . 192
Designing the hierarchy . . . 192
Fixed sequence testing versus graphical procedures . . . 193
The fallback and the fallback's failure modes . . . 194
What counts as pre-specified . . . 195
Interaction between the hierarchy and interim analyses . . . 195
What this section demands before proceeding . . . 196

6.4 Subgroup Discipline 197
The subgroup problem . . . 197
Confirmatory versus exploratory subgroup analyses . . . 197
What makes a subgroup analysis confirmatory . . . 198
The biomarker subgroup: a special case . . . 199
Interaction tests and what they cannot show . . . 200
The forest plot and its misreading . . . 201
What this section demands before proceeding . . . 202

6.5 Closing: The Boundary of What Can Be Said 203
What this chapter asked . . . 203
What this chapter decided . . . 204
The characteristic mistakes of this chapter . . . 204

What cannot be recovered . . . 205
The connection to what follows . . . 206
Chapter 6 risk summary . . . 207

Chapter 7: What If the Design Itself Adapts? 208
The question Chapter 6 leaves open . . . 209
Why adaptation is a risk amplifier . . . 209
What this chapter covers . . . 210
The regulatory context . . . 211
What this chapter is not about . . . 211
The question this chapter must answer . . . 212

7.1 Sample Size Re-estimation 213
The problem SSR solves . . . 213
Nuisance-parameter SSR versus effect-size SSR . . . 213
What the re-estimation rule must specify . . . 214
Operating characteristics of the adapted design . . . 215
The governance requirements . . . 216
SSR in event-driven trials . . . 217
What this section demands before proceeding . . . 217

7.2 Adaptive Enrichment 219
What adaptive enrichment does . . . 219
The estimand shift at enrichment . . . 219
Pre-specifying the enrichment rule . . . 220
The claim structure after enrichment . . . 221
The biomarker: validated before the enrichment? . . . 222
Adaptive enrichment in non-inferiority designs . . . 223
What this section demands before proceeding . . . 223

7.3 Estimand Shift 225
The adaptation that changes the question . . . 225
Explicit estimand shifts . . . 225
Implicit estimand shifts . . . 226
Seamless phase II/III designs . . . 227
The estimand for the adapted trial . . . 228
Documentation and the audit trail . . . 228
What this section demands before proceeding . . . 229

7.4 Non-Inferiority Failure Modes in Adaptive Designs 230
Why NI and adaptive design compound each other . . . 230
The constancy assumption under population shift . . . 230
The constancy assumption under duration extension . . . 231
The assay sensitivity problem under adaptation . . . 232
The switching strategy: from NI to superiority . . . 233

Pre-specifying the constancy re-evaluation . . . 233
What this section demands before proceeding . . . 234

7.5 Closing: The Pre-Specified Adaptation 236
What this chapter asked . . . 236
What this chapter decided . . . 237
The characteristic mistakes of this chapter . . . 237
What cannot be recovered . . . 238
The connection to Chapter 8 . . . 239
Chapter 7 risk summary . . . 240

Chapter 8: What Must Be Locked, and When? 241
The question the entire book leaves open . . . 242
Why governance of the design process matters . . . 242
What this chapter covers . . . 243
The four lock events . . . 244
What this chapter is not about . . . 244
The book's final question . . . 245

8.1 What Must Be Pre-specified 246
The pre-specification principle, made operational . . . 246
The protocol: what it must contain . . . 246
The SAP: what it must contain and when it must be locked . . . 248
The timing of SAP finalization: what compromises it . . . 249
When the deadline is missed . . . 250
What this section demands before proceeding . . . 250

8.2 Adaptive Rules and Their Lock Requirements 252
Why adaptive rules require earlier locks . . . 252
The SSR charter: content and lock timing . . . 252
The enrichment charter: content and lock timing . . . 253
Protocol amendments during the trial . . . 254
The verification requirement . . . 255
What this section demands before proceeding . . . 256

8.3 The Decision Log 257
What the decision log is for . . . 257
What the decision log must contain . . . 257
When entries must be made . . . 258
Who maintains the decision log . . . 259
The decision log and the clinical study report . . . 259
The decision log as a governance discipline . . . 260

8.4 Design Review and the Independent Challenge 262
Why independent review is a governance requirement . . . 262
What independent review must examine . . . 263

Who should conduct independent review . . . 263
The design review as a discipline, not an event . . . 264
After the design review: the design summary . . . 265

8.5 Closing: The Defensible Design **267**
What this chapter asked . . . 267
The characteristic mistakes of this chapter . . . 268
What cannot be recovered . . . 269
The four questions, answered . . . 269
The defensible design . . . 270
Chapter 8 risk summary . . . 271

Appendix **272**

Design Review Hearing **273**
Questions That Must Be Answered Before Enrollment . . . 273
Final Condition . . . 276

Design Scenarios **277**
Where the Principles Meet the Room . . . 277
Scenario 1: “We Want an Interim Analysis” . . . 277
Scenario 2: “The Stopping Rate Is Too High” . . . 279
Scenario 3: Continuous, Binary, or Time-to-Event? . . . 281
Scenario 4: “We Need Five Co-Primary Endpoints” . . . 282
Scenario 5: Dropout — Sample Size Problem or Estimand Problem? . . . 284
Scenario 6: The Statistician’s Boundary . . . 286
Scenario 7: Randomization and Double Programming . . . 288
Scenario 8: “The Primary Endpoint Failed, But Look at This Subgroup” . . . 289
Scenario 9: The NI Margin Discussion . . . 291
Scenario 10: “Can We Add an Endpoint?” . . . 293
Scenario 11: The Alpha Allocation Meeting . . . 294
Scenario 12: “We Need to Show Results in All Regions” . . . 296
Scenario 13: Surrogate Endpoints . . . 298
Scenario 14: Sample Size in Rare Disease and Small Populations . . . 299
Scenario 15: When the Statistician Disagrees With the Design . . . 301
On These Scenarios . . . 303

Clinical Trial Design Decision Framework

From Scientific Question to Defensible Design

What This Book Is—and Is Not

This is not a statistics textbook.

There are excellent textbooks on survival analysis, multiplicity correction, adaptive design methodology, and sample size derivation. This is not one of them. If you want formula derivations, this book will disappoint you. If you want to know why a margin exists, who is responsible for it, and what happens when it is wrong, keep reading.

This book exists because of a pattern that repeats itself in almost every trial design meeting: someone asks *"what's the sample size?"* before anyone has agreed on what the trial is actually trying to show. The statistician produces a number. The sponsor accepts it. The protocol gets written. And then, two years into the trial, someone asks a question that should have been asked first—and there is no good answer.

That pattern is not a failure of statistical competence. It is a failure of decision discipline.

The Problem This Book Addresses

Clinical trial design involves a sequence of commitments, each one constraining what comes after, each one allocating risk to someone.

When you define an estimand, you are committing to a version of the treatment effect that the trial can actually estimate. When you choose a non-inferiority margin, you are asserting that a certain magnitude of inferiority is acceptable, and you are accepting responsibility for that assertion. When you plan an interim analysis, you are deciding who gets to see what, and when, and what they are allowed to do with it.

These are not statistical decisions. They are design decisions with statistical consequences. The distinction matters because statistical decisions can often be revised; design decisions, once made, tend to stick, through the protocol, through the regulatory submission, through the label negotiation, and sometimes all the way into the courtroom.

The question this book asks, at every turn, is not *"how do we do this?"* but *"what are we committing to, and can we defend it?"*

Who This Book Is For

This book is written for people who sit in trial design meetings and need to speak clearly, not just technically correctly, about what the design is doing.

That includes statisticians who want to explain their choices to clinical teams without retreating into jargon. It includes clinical scientists who want to ask better questions of their statistical colleagues. It includes regulatory strategists who need to anticipate how design choices will be received by reviewers. And it includes anyone who has ever left a design review meeting unsure whether the right decision was made, or whether any decision was made at all.

You do not need to follow every mathematical detail to benefit from this book. You do need to be willing to sit with uncomfortable questions about who bears the risk when a design assumption turns out to be wrong.

How This Book Is Organized

The book follows the sequence of decisions that every trial design forces you to make, roughly in the order you have to make them.

Chapter 1 begins where every design must begin: with the question. Not the hypothesis, not the endpoint, but the question. What are we actually trying to show? For whom? Under what conditions? The estimand framework is introduced not as a regulatory compliance exercise, but as the structure that forces you to answer this question precisely.

Chapter 2 turns to measurement. Once you know what you are trying to show, you must decide how to measure it. Different effect measures carry different interpretive commitments. The choice of a ratio versus a difference is not cosmetic, it reflects a claim about how treatment effects scale across populations.

Chapter 3 is about commitment. Sample size calculation is usually presented as a technical exercise. This chapter presents it as a risk-allocation exercise. Every assumption you build into the calculation is an assumption someone is taking responsibility for.

Chapter 4 addresses interim analyses. The desire to stop trials early, for efficacy, for futility, for safety, is understandable. This chapter asks what early stopping actually protects, and what it costs.

Chapter 5 examines bias. Randomization and blinding are the trial's structural defenses against systematic error. This chapter explains not just how they work, but what fails when they are compromised, and why compromise is more common than the literature suggests.

Chapter 6 is about claims. What, exactly, are you allowed to assert when the trial is over? The answer depends entirely on what was pre-specified, in what order, and with what

protections for multiplicity. A p-value is not a claim. A defensible claim requires discipline applied before the data are seen.

Chapter 7 examines adaptive designs, not as innovations, but as risk amplifiers. Every adaptation changes the estimand, whether or not the protocol acknowledges it. This chapter traces those shifts.

Chapter 8 closes with what must be locked. Flexibility in design is valuable; flexibility in interpretation is dangerous. The line between them is pre-specification, and this chapter is about where that line is drawn and how to defend it.

Four Questions That Will Not Go Away

Across all eight chapters, four questions recur. They are not answered once and then set aside. They return in every context, slightly reframed, with different stakes.

What evidence are we defining? The trial can only generate evidence about what it was designed to measure. The definition of that evidence happens before enrollment, not after.

What risk are we committing to carry? Every design assumption is a risk. The question is not whether to accept risk, you must, but whether you have accepted it consciously and assigned it clearly.

How do we prevent the design from failing on its own terms? Bias, drift, protocol deviations, and unplanned adaptations all erode the design's integrity. Control is not automatic.

What are we finally allowed to say? The trial ends with a dataset and a set of pre-specified questions. The intersection of those two things determines what you can assert. Everything outside that intersection is speculation.

These four questions are the skeleton of this book. The chapters are the muscle.

A Note on Tone

This book tries to be sharp rather than comprehensive. It makes arguments rather than surveys. Where there are genuine disagreements, about the appropriate use of non-inferiority trials, about the value of response-adaptive randomization, about how aggressively to pursue subgroup analyses, the book takes positions. Not because those positions are certainly correct, but because clarity is more useful than false balance.

Where the book takes a position, it tries to show the cost of the alternative. The goal is not to be right. The goal is to make the cost of different choices visible, so that whoever is responsible for the decision can make it with open eyes.

Design decisions made under ambiguity tend to be made by the loudest voice in the room. This book is an attempt to give quieter voices something to say.

Disclaimer

These are my personal learning notes. I wrote them to organize my own understanding — not as professional advice of any kind. If you find them useful, great, but please validate anything here against your own study requirements and consult a qualified statistician before applying these methods in any regulated or clinical setting. I take no responsibility for how this material is used.

Chapter 1: What Are We Trying to Show?

Most trial designs fail before they begin.

Not in the statistical sense—the power calculation may be perfectly correct, the randomization scheme impeccable, the blinding procedures airtight. They fail in a prior sense: the team has not agreed on what the trial is actually trying to show. They have agreed on an endpoint. They have agreed on a hypothesis. But the question underneath those agreements—*what treatment effect, in what population, under what conditions of use, are we claiming to have demonstrated?*—has been left unresolved.

That unresolved question does not stay quiet. It surfaces during the interim analysis, when someone asks whether discontinuations should be counted as failures. It surfaces during the regulatory review, when an agency asks whether the primary analysis reflects the treatment policy or the biological mechanism. It surfaces at the label negotiation, when the sponsor and the agency discover they have been answering different questions with the same dataset.

The estimand framework—formally introduced in ICH E9(R1) but carrying a logic that predates the guidance—exists to force this question into the open before the trial starts. That is its purpose. Not to add regulatory paperwork, not to create new statistical objects, but to require that the question be answered clearly and publicly, by the people who are responsible for it, at the moment when the answer can still shape the design.

The Decision This Chapter Is About

Before any trial can be designed, someone must answer a question that sounds deceptively simple: *what are we trying to show?*

The deception is in the simplicity. Every word of that question is doing work.

What requires specifying an estimand—the precise quantity the trial is designed to estimate. Not a general direction ("we expect improvement") but a defined treatment effect, in a defined population, on a defined outcome, under a defined set of conditions.

We requires assigning ownership. Someone must stand behind the estimand. Someone must be able to say, in a regulatory meeting or a dispute with a payer, *this is what we intended to measure, this is why, and these are the conditions under which the estimate is valid.* That person cannot be the statistician alone. The estimand is a scientific and clinical commitment, not a statistical artifact.

Trying to show requires acknowledging that trials are designed to generate evidence for a specific claim, not to explore data until something appears. The claim must exist before the data. Its contours must be drawn before enrollment begins. Everything that happens after—endpoint selection, intercurrent event handling, analysis strategy—is in service of that prior claim.

When these three elements are not resolved before design begins, the trial does not become undefined. It becomes defined by default: by the path of least resistance, by the assumptions

embedded in the statistical software, by the endpoint that was easiest to measure, by the handling of discontinuations that someone decided informally during data cleaning. Default designs answer default questions. Those default questions are rarely the ones the sponsor, the clinician, or the regulator actually wanted answered.

What This Chapter Covers

This chapter has three sections and a closing.

Section 1.1 — Purpose and Estimand takes the estimand framework seriously as a decision tool, not a documentation requirement. The four estimand attributes—population, variable, intercurrent event strategy, population-level summary—are presented as four distinct decision points, each requiring an owner. The section asks what it means to specify each one, and what is lost when specification is deferred.

Section 1.2 — Endpoint Choice examines how endpoints are selected in practice—and why the selection is more consequential than it appears. A primary endpoint is not simply a measurement. It is a bet on what will move, on what regulators will accept, on what clinicians will find meaningful, and on what the trial's sample size can actually detect. These four considerations frequently point in different directions. The section traces those conflicts and asks how they should be resolved—and by whom.

Section 1.3 — Intercurrent Events is where the estimand becomes hardest. Intercurrent events—discontinuations, rescue medication use, treatment switches, deaths—are the points at which real clinical experience diverges from the clean experimental logic of the protocol. Every decision about how to handle them is a decision about what question the trial is answering. The section identifies the five ICH-recognized strategies and examines not what they are, but what they commit you to.

Section 1.4 — Closing returns to the chapter's core tension: the pressure to finalize the design before the question is fully resolved, and the consequences of yielding to that pressure.

The Risk of Starting Downstream

There is a standard sequence for trial design: endpoint, then sample size, then randomization scheme, then interim analysis plan. This sequence is not wrong, but it is incomplete. It presupposes that the question has already been answered, that what precedes "endpoint" has been resolved.

In practice, it often has not. The endpoint gets chosen because it was used in the previous trial, or because it is accepted by the relevant agency, or because it is the most sensitive measure available. These are reasonable starting points. They are not substitutes for asking what the trial is for.

Starting downstream—from the endpoint rather than from the question—creates a characteristic failure mode. The design is internally consistent but externally ambiguous. The protocol specifies the primary endpoint, the analysis population, and the handling of missing data. What it does not specify, because the question was never asked, is what the primary analysis is supposed to represent. Is it an estimate of what happens when patients are randomized and followed to the end of the protocol, regardless of what they do in the meantime? Or is it an estimate of what would happen if patients took the treatment as intended, without rescue or discontinuation? Or something else?

These are different questions. They have different answers. They require different trial designs. And they cannot all be answered by the same dataset with the same analysis, regardless of how many sensitivity analyses are appended.

The estimand framework, at its core, is a requirement to ask the question before it is too late to design around the answer.

What Follows from Getting This Right

A precisely specified estimand does not guarantee a successful trial. The treatment may not work. The outcome may not move. The population may be wrong. Getting the question right does not make the answer good.

What it does is protect the integrity of whatever answer the trial generates. A trial that answers a clearly specified question—even if the answer is disappointing—has produced real evidence. A trial that produces a technically valid p-value for a question no one has agreed on has produced noise with a confidence interval.

More concretely: getting the estimand right at the design stage determines what analyses are primary and what analyses are secondary, what intercurrent event handling is pre-specified and what is post-hoc, what subgroup hypotheses are confirmatory and what are exploratory. These distinctions have regulatory and commercial consequences that no statistical correction applied after the fact can fully repair.

The chapters that follow—on effect measures, sample size, interim analyses, bias control, multiplicity, and adaptive design—all assume that this chapter's work has been done. They are decisions about how to pursue a defined question. They cannot substitute for the decision about what the question is.

That decision is made here, at the beginning, or it is made by default later. The cost of making it by default is the subject of this chapter.

1.1 Purpose and Estimand

The question that precedes the hypothesis

A clinical trial has a hypothesis. It also has a purpose. These are not the same thing, and confusing them is one of the most consequential errors in trial design.

The hypothesis is statistical: the treatment effect is not zero, or the treatment is not inferior by more than a specified margin. The purpose is scientific and clinical: we want to know whether this treatment, given to these patients, in this way, produces a benefit that matters. The hypothesis is testable. The purpose is what makes the test worth running.

The estimand is the bridge between the two. It is the precise specification of the treatment effect that the trial is designed to estimate—the quantity whose value, once known, would answer the clinical question. Defined well, it makes the hypothesis a meaningful test of the purpose. Defined poorly, it allows the hypothesis to be tested rigorously while the purpose goes unanswered.

This is not a theoretical concern. It is the routine outcome of trial design processes that jump to the hypothesis before the purpose has been settled.

What the estimand framework requires you to specify

ICH E9(R1) defines the estimand through four attributes: the population, the variable, the handling of intercurrent events, and the population-level summary. Each attribute corresponds to a decision. Each decision has an owner. Each owner takes on responsibility for the consequences if the decision turns out to be wrong.

Population. Who are the patients in whom the treatment effect is being estimated? This is not identical to the eligibility criteria, though it is informed by them. The eligibility criteria define who can enroll. The population attribute of the estimand defines the patients in whom the estimated effect is intended to be interpreted. In a trial that enrolls a broad population but expects the treatment to work primarily in a biomarker-defined subgroup, the estimand population and the enrolled population may differ—and that difference matters for what the trial can actually claim.

Variable. What outcome is being measured, and how? The variable is the measurement at the patient level—the raw material from which the treatment effect will be estimated. It

must be defined precisely enough that two independent assessors, given the same patient at the same time, would record the same value. Ambiguity in the variable propagates through to ambiguity in the estimand: if the measurement is not clearly defined, the treatment effect it generates cannot be clearly interpreted.

Intercurrent events. What happens when the clinical course does not follow the clean experimental logic of the protocol? Patients discontinue. They switch to rescue medication. They die before the primary endpoint can be measured. They receive treatments outside the protocol. Each of these intercurrent events disrupts the simple experimental comparison the trial was designed to make—and the strategy for handling each one determines, more than almost any other design choice, what question the trial is actually answering. This is addressed in detail in Section 1.3. The point here is that the intercurrent event strategy is an estimand attribute, not an analysis decision. It must be settled at design, not resolved post-hoc.

Population-level summary. How is the patient-level variable aggregated into a single estimate of the treatment effect? A mean difference, a median difference, a proportion achieving response, a hazard ratio, a win ratio—each summarizes the individual-level distribution differently and implies a different claim about how the treatment effect distributes across the population. The choice of population-level summary is not purely statistical. It reflects a claim about whether the treatment effect is expected to be homogeneous or heterogeneous, concentrated in responders or distributed across the population, durable or transient.

These four attributes are not independent. The population shapes which intercurrent events are likely and in what frequency. The variable constrains the available population-level summaries. The intercurrent event strategy affects the interpretation of the summary. Specifying one attribute without examining its implications for the others produces an estimand that is formally complete but substantively incoherent.

Treatment policy versus hypothetical: the choice that changes everything

Of the decisions embedded in the estimand framework, the most consequential is the choice of intercurrent event strategy for the primary analysis. And within the available strategies, the choice between treatment policy and hypothetical estimands is the one that most sharply divides what the trial is claiming to show.

Under a **treatment policy** strategy, the outcome is measured regardless of whether the patient followed the protocol, took the assigned treatment, used rescue medication, or discontinued. The estimand reflects the effect of being assigned to treatment in the context in which the treatment would actually be used—including all the messiness of real-world treatment behavior. This is the question a payer or a clinician asks when deciding whether to prescribe: if I give this treatment to patients like these, what happens, accounting for

the fact that some will discontinue, some will need rescue, and some will deviate from the regimen?

Under a **hypothetical** strategy, the outcome is estimated under a counterfactual condition—typically, what would have happened if the intercurrent event had not occurred. What would the patient's outcome have been if they had not discontinued? What would the blood pressure have been if they had not needed rescue medication? This is the question a pharmacologist or a mechanism-focused clinical scientist asks: what does this treatment do, under conditions where its effect is not diluted by non-adherence or rescue?

These are both legitimate questions. They are not the same question. A trial designed to answer the treatment policy question will produce an estimate that is directly relevant to clinical practice but may underestimate the biological effect of the treatment. A trial designed to answer the hypothetical question will produce an estimate that is scientifically cleaner but may overestimate the benefit patients will actually experience. Neither estimate is wrong. Each answers a different question, and the choice between them must be made before the trial begins—because the choice determines what data need to be collected, how the analysis must be specified, and what the result will actually mean.

The difficulty is that this choice is often deferred. The protocol specifies an intention-to-treat analysis without asking which estimand that analysis serves. An ITT analysis is consistent with a treatment policy estimand—but only if the outcome is measured regardless of discontinuation or rescue. If discontinuations are handled by last-observation-carried-forward, or if rescued patients are censored, the ITT label is being applied to an analysis that does not serve the treatment policy estimand. The result is an analysis with an ITT label that answers neither the treatment policy question nor the hypothetical question clearly.

This is the specific failure mode the estimand framework was designed to prevent: not technical error, but conceptual mismatch between the analysis performed and the question being claimed.

Who specifies the estimand, and when

The estimand is a scientific document. Its four attributes—population, variable, intercurrent event strategy, population-level summary—are scientific and clinical choices, not statistical ones. Statisticians can evaluate the feasibility of estimating a proposed estimand, can flag inconsistencies between attributes, and can specify the analysis that will estimate it. They cannot substitute for the clinical judgment about which estimand is the right one.

This creates an accountability structure that is often poorly managed in practice.

The clinical team—physicians, clinical scientists, disease area experts—must own the question. What does it mean for this treatment to work, for these patients, in this clinical context? The answer to that question determines the estimand. The clinical team must be present for the estimand discussion, must understand what each attribute commits the trial to, and

must be able to defend the choices to a regulator, to an ethics committee, and eventually to the clinical community that will use the trial's results.

The statistical team must own the technical specification. Given the estimand the clinical team has defined, what analysis will produce an unbiased estimate of it? What data must be collected to support that analysis? What sensitivity analyses will evaluate the robustness of the primary result? The statistician who cannot answer these questions for a given estimand has not yet finished their work.

The regulatory team must own the claim. Given the estimand the trial is designed to estimate, what will the regulatory submission be able to assert? What label language will be defensible? What conditions will be attached to the approval, and are those conditions acceptable given the development strategy?

When these three ownership structures are in place, the estimand becomes a shared document that the entire team has contributed to and is accountable for. When ownership is missing—most commonly, when the estimand is treated as a statistical artifact to be finalized by the biostatistics team after the protocol is already drafted—it becomes a technical formality that no one fully understands and no one is responsible for defending.

The regulatory interface

ICH E9(R1) was finalized in 2019 and has been progressively incorporated into regulatory expectations at FDA, EMA, and PMDA. The practical consequence is that regulators now expect the estimand to be specified in the protocol and the statistical analysis plan, with the analysis explicitly connected to the specified estimand.

What this means for trial design is that the estimand conversation can no longer be deferred to the analysis stage. A protocol that specifies an ITT primary analysis without connecting it to an estimand attribute—or that specifies a per-protocol sensitivity analysis without explaining what estimand it targets—will receive questions during the review that could have been answered during the design.

More substantively, the regulatory expectation creates an opportunity. Pre-submission meetings—Type B meetings at FDA, scientific advice at EMA—are increasingly used to align on estimands before the trial begins. This alignment is valuable not because it guarantees approval, but because it forces the question into the open at the moment when the design can still accommodate the answer. A regulator who disagrees with the chosen estimand during a pre-submission meeting is generating information that the sponsor needs. A regulator who raises the same disagreement during the review of a completed trial is generating a problem that cannot be fixed.

The estimand framework, in this sense, is not primarily a regulatory compliance tool. It is a forcing function for a conversation that should happen anyway—among the clinical,

statistical, and regulatory members of the design team—about what question the trial is for.

When the estimand is wrong

An estimand can be wrong in two ways. It can be internally inconsistent—the attributes are incompatible with each other, the analysis cannot produce an unbiased estimate of the specified quantity, or the population-level summary does not correspond to the patient-level variable. This kind of error is detectable by careful specification and review. It is the statistician's domain.

The second kind of error is harder to detect and more damaging: the estimand is internally consistent but answers the wrong question. The treatment effect is estimated precisely, the analysis is technically valid, the result is statistically significant—and the clinical community receives the result with indifference because it does not tell them what they needed to know.

This failure is not a statistical failure. It is a design failure, and it is attributable to the people who defined the question. Which brings the discussion back to the point that opened this section.

The hypothesis is testable. The purpose is what makes the test worth running. The estimand is what connects them. If that connection is not made explicitly, by the right people, at the right moment in the design process, the trial may produce a valid answer to a question no one was asking.

That outcome is not a neutral result. It consumes resources, exposes patients to intervention, and advances nothing. The estimand framework asks: what question are we trying to answer, precisely? The discipline it demands is not bureaucratic. It is the minimum required to make the trial worth running.

References: ICH E9(R1) Addendum on Estimands and Sensitivity Analysis in Clinical Trials (2019); Lawrance et al., "What Is an Estimand, and How Does It Relate to Quantifying the Effect of Treatment on Patient-Reported Outcomes?" J Patient-Rep Outcomes 2020; Permutt, "A Taxonomy of Estimands for Regulatory Clinical Trials with Discontinuation," Stat Med 2016.

1.2 Endpoint Choice

The question underneath the endpoint

Every trial has a primary endpoint. What is less often acknowledged is that every primary endpoint is an answer to a prior question—a question that is usually not asked explicitly.

The prior question is: *what does it mean for this treatment to work?*

That question has at least four different interpretations, and they do not always converge on the same endpoint.

It could mean: the treatment produces a measurable biological change. This leads toward biomarkers, surrogate endpoints, mechanistic measures—things that move reliably and can be detected with a small sample.

It could mean: the treatment changes how patients feel or function. This leads toward patient-reported outcomes, quality of life instruments, functional scales—things that reflect lived experience rather than laboratory values.

It could mean: the treatment prevents the outcomes that matter most clinically—death, hospitalization, irreversible organ damage. This leads toward hard clinical endpoints, time-to-event outcomes, event rates—things that are unambiguous but slow to accumulate and expensive to detect.

It could mean: the treatment produces effects that regulators will accept as evidence of benefit. This leads toward endpoints with regulatory precedent—endpoints that have been accepted before, in related indications, by the relevant agency—regardless of whether they optimally capture the treatment's mechanism.

These four interpretations—biological, experiential, clinical, regulatory—are not always in conflict. Sometimes they converge. But often they point in different directions, and the choice of primary endpoint is implicitly a choice about which interpretation takes priority. That choice is rarely made explicitly. It is more often made by imitation: we use what was used before, in a similar trial, in a similar indication, without asking whether the prior choice was well-reasoned or merely precedent.

The cost of not asking is that the endpoint can be simultaneously technically valid and scientifically inappropriate. A trial can be powered, executed, and analyzed correctly—and still answer the wrong question—because the endpoint was chosen without examining what question it was supposed to answer.

What an endpoint actually commits you to

Selecting a primary endpoint is not selecting a measurement. It is making three simultaneous commitments.

The first is a commitment about what counts as evidence. A primary endpoint defines the evidentiary standard. When the trial is over, the evidence that the treatment works—or does not—will be expressed in terms of that endpoint. If the endpoint does not capture what matters, the evidence will not answer what matters, and no amount of secondary analysis will repair that gap.

The second is a commitment about the population. Different endpoints are sensitive in different populations. A six-minute walk test is meaningful in patients who are ambulatory but impaired; it is less meaningful at disease extremes. A composite endpoint including hospitalization is driven by the subgroup with the highest event rate. An endpoint that seems right for the overall trial population may be insensitive in the patients who most need the treatment, and vice versa. Choosing an endpoint without specifying the population in which it is expected to be sensitive is choosing a question that may not be answerable in the enrolled patients.

The third is a commitment about timing. Endpoints have natural timescales. Biomarkers change in weeks. Functional outcomes change over months. Hard clinical events accumulate over years. The endpoint must be matched to the duration of follow-up the trial can sustain—which depends on enrollment, retention, and resources that are often not fully characterized when the endpoint is chosen. An endpoint that requires three years of follow-up to show a signal is the wrong primary endpoint for a trial designed to run for eighteen months. This sounds obvious. It is routinely violated.

Each of these three commitments interacts with the others. A composite endpoint that includes both a biomarker component and a hard clinical event component is trying to be sensitive to biological change and clinically meaningful at the same time. It is also making a population claim—that both components will be relevant for the enrolled patients—and a timing claim—that the composite will accumulate meaningfully within the trial's follow-up window. If any one of these commitments is wrong, the endpoint is wrong, even if the measurement itself is technically sound.

The surrogate problem

A surrogate endpoint is one that is expected to predict a clinical outcome without measuring it directly. Tumor shrinkage as a surrogate for survival. Blood pressure reduction as a surrogate for cardiovascular events. CD4 count as a surrogate for AIDS-defining illness.

Surrogates are attractive because they are faster, cheaper, and more sensitive than the clinical outcomes they represent. A trial powered on HbA1c can enroll far fewer patients,

and follow them for far less time, than a trial powered on cardiovascular mortality. The efficiency gain is real.

The cost is a claim. When you choose a surrogate primary endpoint, you are claiming that improvement in the surrogate constitutes evidence of clinical benefit. That claim requires two things to be true: the surrogate must be reliably associated with the clinical outcome in this disease, and the treatment's effect on the surrogate must reliably predict its effect on the clinical outcome. Both conditions must hold. The first is sometimes well-established. The second is often assumed without examination.

The history of surrogate endpoint failure is long. Antiarrhythmic drugs that suppressed ventricular ectopy and increased mortality. Bone density improvements that did not translate to fracture prevention. Cholesterol changes that did not predict cardiovascular outcomes uniformly across drug classes. In each case, the mechanism connecting surrogate to clinical outcome was more fragile than the trial designers assumed—or the treatment affected the surrogate and the outcome through different pathways.

This does not mean surrogates should not be used. In indications with long clinical timelines, in populations with high unmet need, in early development stages where resource constraints are real, surrogates may be the right primary endpoint. The decision to use a surrogate is defensible. What is not defensible is using a surrogate without being explicit about the claim it requires, without identifying the evidence that supports the surrogate-to-outcome relationship, and without acknowledging the conditions under which that relationship could fail.

Regulatory agencies are increasingly attentive to this. The FDA's Project Optimus, the EMA's qualification process for novel endpoints, and the increasing prevalence of accelerated approval conditioned on confirmatory outcomes trials all reflect the same recognition: a surrogate endpoint is a provisional claim, not a settled one. The trial designed around a surrogate should acknowledge that provisionality and plan for what happens if the confirmatory evidence does not come.

Composite endpoints: combination as a decision

Composite endpoints combine multiple individual outcomes into a single primary measure—typically taking the first occurrence of any component event as the endpoint. Major adverse cardiovascular events (MACE), combining cardiovascular death, myocardial infarction, and stroke, is the canonical example. Composite endpoints are used when no single component occurs frequently enough to power a trial, or when the clinical question spans multiple manifestations of the same disease process.

The logic is straightforward: if any of these outcomes would represent treatment failure, combining them gives the trial more events and therefore more power. The efficiency gain is real.

The interpretive cost is also real, and it is proportional to how heterogeneous the components are.

A composite endpoint's result is driven by whichever component occurs most frequently. If cardiovascular death occurs in 3% of patients and hospitalization for heart failure occurs in 18%, a composite of the two will behave almost entirely like a hospitalization endpoint. A hazard ratio of 0.85 on the composite may reflect a large effect on hospitalization and no effect on mortality—or it may reflect a modest effect on both. The composite does not distinguish between these possibilities. The composite says: something happened less often. It does not say what, or to whom, or whether it was the thing that mattered most.

This creates a specific interpretive obligation. When a composite endpoint is chosen, the trial should be able to specify, in advance, which components are expected to be affected and in what direction. If the treatment is not expected to affect one of the components, that component should not be in the composite—including it dilutes the signal and generates an uninterpretable result. If the components are ordered by clinical importance, the analysis should reflect that ordering, not treat a hospitalization and a death as equivalent events.

The FDA and EMA have both issued guidance on composite endpoint specification that reflects these concerns. The practical implication is that choosing a composite endpoint is not an escape from hard choices about what the trial is measuring. It is a different set of hard choices, with different interpretive obligations. Making those choices explicitly, before the trial begins, is what separates a defensible composite from a convenient one.

Patient-reported outcomes: the ownership problem

Patient-reported outcomes—PROs—measure what patients experience directly: symptoms, function, health-related quality of life. They are, in many indications, the most clinically meaningful endpoints available. A treatment that reduces pain, improves sleep, restores mobility, or reduces anxiety is producing benefit that no biomarker or physician assessment fully captures. For conditions where patients live with chronic symptoms, PROs may be the only endpoints that matter.

The challenges with PROs as primary endpoints are well-known—measurement variability, the definition of clinically meaningful change, the difficulty of blinding in open-label settings, the burden of assessment in patients who are already symptomatic. These are real. They are also solvable, given sufficient rigor in instrument selection, definition of the minimally important difference, and pre-specification of the analysis.

What is less often discussed is the ownership problem.

A PRO endpoint requires someone to specify, before the trial, what magnitude of change on the instrument constitutes a clinically meaningful difference. This is not a statistical question. It is a clinical and patient-centered question: how much better does a patient have to feel for the improvement to matter? The answer depends on the disease, the instrument,

the population, and the treatment context. It cannot be derived from statistical convention. It must be owned by the clinical team, validated against patient and clinician perspectives, and documented before the trial is powered.

When this ownership is missing—when the minimally important difference is selected to match the expected treatment effect, or adopted from a different population without examination, or left ambiguous until after unblinding—the PRO endpoint becomes uninterpretable. A statistically significant improvement on a PRO scale is evidence of benefit only if the magnitude of improvement was pre-specified as clinically meaningful. If the threshold was set after the effect size was known, the evidence is circular.

Regulatory agencies have become sophisticated about this. The FDA's guidance on patient-focused drug development reflects a sustained effort to require that the patient's perspective be captured rigorously and prospectively, not retrofitted to whatever the trial happened to show. The implication for design is that PRO endpoints require as much pre-specification rigor as any other primary endpoint—arguably more, because the connection between the measurement and the clinical meaning is less self-evident.

Who owns the endpoint decision

In most trial design processes, the primary endpoint is proposed by the clinical team, reviewed by the statistician, and negotiated with the regulatory agency—either in a pre-submission meeting or during the review process. This sequence is reasonable. But it obscures an accountability question that surfaces later, when the trial is over and someone asks why the endpoint was chosen.

The endpoint is a scientific claim about what constitutes evidence of benefit. That claim should be owned by the people who understand the clinical context: what the disease does, how patients experience it, what physicians use to assess it, and what magnitude of change would alter clinical practice. Statisticians can evaluate the feasibility of measuring an endpoint, the sample size implications of different effect sizes, and the regulatory history of similar endpoints. They cannot substitute for clinical ownership of the underlying claim.

This matters because endpoint decisions made without clear ownership tend to drift toward what is measurable rather than what is meaningful. The path of least resistance is to use the endpoint that has been accepted before, by this agency, in this indication—regardless of whether it is the right question for this treatment in this population at this stage of development. Regulatory precedent is a reasonable input. It is not a substitute for clinical reasoning.

The cleaner structure is for the clinical team to own the question—what does it mean for this treatment to work?—and for that answer to drive the endpoint selection, which the statistician then evaluates for feasibility. Endpoint choice that starts with statistical feasibility and works backward to clinical meaning produces endpoints that are powered correctly and answer the wrong question.

The pressure to compromise

There is a characteristic moment in many trial design processes where the endpoint choice gets compromised. The clinical team wants a hard clinical endpoint—mortality, irreversible organ damage—that will take years to accrue. The development timeline will not support that. The surrogate endpoint that can be measured earlier does not have regulatory acceptance in this indication. The PRO endpoint the patients care most about has never been used as a primary endpoint before.

The resolution is typically a composite that combines elements of all three—hard enough to seem clinically relevant, accruing fast enough to fit the timeline, containing a PRO component that can be highlighted in the label. The composite is presented as capturing the full range of clinical benefit. What it actually captures is the range of compromises that were available.

This is not always wrong. Sometimes a composite genuinely reflects the clinical question, and the compromise is between competing goods rather than between rigor and convenience. But the distinction matters. A composite chosen to capture clinical complexity is a defensible design choice. A composite chosen to make the timeline work is a design risk—one that the trial will carry through to the regulatory review, where the agency will ask the same questions that should have been asked at design.

The pressure to compromise is real, and it will not be eliminated by recognizing it. What can be managed is the transparency of the compromise—being explicit about which consideration drove the endpoint choice, what was sacrificed, and under what conditions the endpoint may fail to answer the question it was supposed to answer.

Transparency before the trial does not protect against a negative result. It does protect against an uninterpretable one.

What this section demands before proceeding

Before moving to intercurrent events—which are, in many ways, the place where endpoint decisions get stress-tested—the following must be resolved.

The primary endpoint must be explicitly connected to the question the trial is answering. Not "we are using MACE because it is standard in this indication," but "we are using MACE because our treatment is expected to affect cardiovascular death and non-fatal MI through the mechanism we have documented, and stroke is included because the pathophysiology overlaps sufficiently that a stroke-reducing effect is plausible and clinically meaningful to include."

The population in which the endpoint is sensitive must be specified. If the endpoint is expected to be sensitive only in patients with a certain baseline severity, that should be reflected in the eligibility criteria, not discovered during subgroup analysis after the trial.

The timing must be matched to the follow-up. If the endpoint requires two years to show a signal and the trial is designed for eighteen months, that mismatch must be resolved before enrollment begins—not by extending the trial after the fact.

And ownership must be assigned. Someone must be able to say, after the trial is over: *this was our endpoint, this is why we chose it, and this is what the result means.* If no one can say that before the trial starts, the endpoint has not been chosen. It has been inherited.

References: ICH E9(R1) Addendum on Estimands and Sensitivity Analysis; FDA Guidance on Patient-Focused Drug Development (2022); FDA Guidance on Surrogate Endpoint Qualification; Pocock et al., "The Primary Outcome Fails — What Next?" N Engl J Med 2016.

1.3 Intercurrent Events and Estimand Strategy

Where the experimental logic breaks down

A randomized controlled trial is built on a simple premise: assign patients to treatment or control, follow them under identical conditions, and attribute any difference in outcomes to the treatment. The premise is powerful. It is also fragile, because real clinical trials do not follow identical conditions.

Patients discontinue the assigned treatment because of side effects, because they feel better, because they feel worse, because their insurance changes, or because they decide they would rather not continue. They receive rescue medication when their condition deteriorates beyond what the protocol permits to go untreated. They switch to the comparator, or to a third treatment not specified in the protocol. They die before the primary endpoint can be measured. They become pregnant and are withdrawn. They experience an event that, by protocol definition, ends their participation.

Each of these occurrences is an intercurrent event: something that happens after randomization and before the primary outcome measurement that complicates the interpretation of whatever outcome is eventually observed. They are not edge cases. In most trials of meaningful duration, they are the norm. The question is not whether intercurrent events will occur. The question is what they mean—and what the trial's answer to that question says about what the trial is trying to show.

The ICH E9(R1) framework names five strategies for handling intercurrent events in the primary estimand: treatment policy, hypothetical, composite, while on treatment, and principal stratum. Each strategy produces a different estimand. Each estimand answers a different question. And the choice among them is not a statistical decision. It is a scientific and clinical decision about the nature of the treatment effect that the trial is claiming to estimate.

Treatment policy: the question a prescriber asks

Under the treatment policy strategy, the outcome is measured and used regardless of whether the intercurrent event occurred. A patient who discontinues the assigned treatment at week four and takes nothing thereafter is followed to the primary endpoint time point. A patient who requires rescue medication at week eight is still analyzed at week twelve. The

intercurrent event is acknowledged—it happened—but it does not change what is measured or how the measurement enters the analysis.

The resulting estimand answers the question: what happens to patients who are assigned to this treatment, in a clinical environment where some will discontinue, some will need rescue, and some will deviate from the regimen? This is the question a prescribing physician asks when deciding whether to use a treatment in practice. It is the question a payer asks when deciding whether a treatment produces value in a real patient population. It captures the treatment effect in context—messy, imperfect, realistic context.

The cost of this realism is dilution. If a substantial proportion of patients in the active arm discontinue and derive no further benefit, the treatment policy estimand will reflect that. If rescue medication is available and effective, its use by patients in the active arm who are not responding will compress the observed difference between arms. The treatment policy estimate is typically smaller than the biological effect of the treatment, because it includes the patients for whom the treatment did not work as intended in clinical practice.

This is not a defect. It is the point. The treatment policy estimand is honest about what the treatment produces in practice. A treatment that shows a meaningful effect under a treatment policy estimand is a treatment that produces real-world benefit, not just biological activity in perfect adherers.

What the treatment policy strategy demands from the trial is rigorous outcome collection after intercurrent events. If patients who discontinue are lost to follow-up, or if missing outcomes are handled by imputation that assumes discontinuers behave like continuers, the analysis is not serving the treatment policy estimand—it is producing a number with a treatment policy label that reflects something else. The estimand requires the data. If the data cannot be collected, the estimand cannot be estimated, and the strategy should not be chosen.

Hypothetical: the question a mechanism scientist asks

Under the hypothetical strategy, the estimand is the outcome that would have been observed if the intercurrent event had not occurred. What would the patient's blood pressure have been at week twelve if they had not required rescue medication at week eight? What would the six-minute walk distance have been if the patient had not discontinued? The intercurrent event is treated as a deviation from the experimental ideal, and the analysis attempts to estimate what would have happened in its absence.

This estimand answers the question: what is the biological effect of this treatment, under conditions where the protocol is followed? It is the question a pharmacologist asks when trying to understand the mechanism. It is sometimes the question a regulatory agency asks when the treatment policy estimate is diluted by high rates of rescue or discontinuation and the agency wants to understand the treatment's underlying activity.

The hypothetical estimand is scientifically coherent, but it requires a causal assumption that is not directly verifiable: that the counterfactual outcome—what would have happened if the intercurrent event had not occurred—can be estimated from the observed data. This estimation requires statistical modeling, and the model encodes assumptions about how the outcome trajectory would have continued in the absence of the intercurrent event. These assumptions cannot be tested from the data; they must be argued from clinical knowledge about the disease and the treatment.

The practical implication is that a hypothetical estimand produces a primary result that depends on modeling assumptions in a way that a treatment policy result does not. Sensitivity analyses are not optional; they are essential, because the primary result is only as credible as the assumptions underlying the counterfactual model. A sponsor who chooses a hypothetical estimand must be prepared to defend not just the analysis but the assumptions that make the analysis interpretable.

There is also a regulatory risk. An estimand that answers the hypothetical question—what would have happened if patients had followed the protocol?—may be difficult to translate into a label claim that describes what will happen when the treatment is used in practice. The gap between the hypothetical estimand and the treatment policy question is precisely the gap that regulators and payers are asking about when they ask whether a treatment works in the real world. Choosing the hypothetical estimand is a decision to defer that question, and deferral has consequences.

Composite: incorporating the event into the outcome

Under the composite strategy, the intercurrent event is incorporated into the outcome definition rather than handled separately. Death before the primary endpoint becomes an outcome—typically the worst possible outcome on the scale being measured—rather than a complication that interrupts measurement of the primary endpoint. Discontinuation due to treatment failure is coded as failure. Rescue medication use is itself a component of a composite endpoint.

The composite strategy is most natural when the intercurrent event is clinically meaningful in its own right. If a patient dies before the primary endpoint can be measured, their death is not a missing data problem. It is an outcome—arguably the most important outcome—and treating it as a complication that requires imputation produces a result that is scientifically distorted. Incorporating death into a composite, or treating it as the worst rank in a win ratio analysis, acknowledges its clinical significance rather than modeling it away.

The composite strategy is also transparent. Unlike the hypothetical strategy, it does not rely on modeling assumptions about counterfactual outcomes. The outcome is what it is—the composite of the primary measure and the intercurrent event—and the analysis reflects that composite directly.

The cost is interpretability. A composite that combines a continuous outcome with a binary intercurrent event is no longer a continuous outcome in the usual sense. The scale has been extended, the distribution has changed, and the clinical meaning of a unit change in the composite measure may be less intuitive than the clinical meaning of the original continuous outcome. The composite strategy requires that the extended or transformed outcome be interpretable on its own terms, not just on the terms of its components. When it is, it is often the most honest available strategy. When it is not, it produces a primary result that is statistically clean and clinically opaque.

While on treatment: limiting the question

Under the while-on-treatment strategy, the estimand is restricted to the period during which the patient is receiving the assigned treatment. Outcomes after discontinuation are not included in the primary estimand. The question is not what happens to assigned patients overall, but what happens to patients while they are on treatment.

This strategy is appropriate when the scientific question is genuinely about the on-treatment period—when a short-term biological effect is the target, when the treatment is intended to be used for a defined duration and the off-treatment period is not clinically relevant, or when the mechanism of action is such that effects are expected only during active treatment. In oncology, where post-progression treatment switches are nearly universal and post-progression outcomes are governed more by subsequent therapy than by the index treatment, a while-on-treatment or related strategy may be the most scientifically honest framing of the question.

The risk of this strategy is that it can be used to avoid the treatment policy question rather than to answer a genuinely different one. If patients who discontinue due to tolerability are excluded from the estimand, and tolerability is a meaningful dimension of the treatment's clinical profile, the while-on-treatment estimand may produce an efficacy estimate that does not reflect the treatment's net benefit. The question of what happens while patients are on treatment is legitimate. The answer is incomplete if the reason patients stop being on treatment is itself clinically relevant.

Principal stratum: asking about a defined subgroup

Under the principal stratum strategy, the estimand is defined for a subgroup of patients identified by their potential intercurrent event status—patients who would not have experienced the intercurrent event regardless of treatment assignment. If the intercurrent event is death before the primary endpoint, the principal stratum might be patients who would

survive to the measurement time regardless of which treatment they received. The estimand is the treatment effect in that subgroup.

This strategy addresses a specific problem: when the intercurrent event itself may be affected by treatment, comparing outcomes across patients who experienced different intercurrent events may conflate the effect of the intercurrent event with the effect of the treatment. The principal stratum approach attempts to define a comparison that is not confounded by differential intercurrent event rates.

The difficulty is that the principal stratum is defined by potential outcomes that are not directly observable. Patients cannot be classified into the stratum from the data alone; the classification must be estimated, which requires strong and often untestable assumptions. Principal stratum estimands are scientifically important and methodologically demanding. They are most defensible in settings where the scientific question requires them and where the assumptions can be partially evaluated from clinical knowledge—and least defensible as a default choice made because the other strategies seemed complicated.

The strategy is the estimand

The practical consequence of the five-strategy framework is that the choice of intercurrent event strategy is not separable from the choice of estimand. They are the same decision. A trial that specifies a primary endpoint without specifying an intercurrent event strategy has not specified an estimand. It has specified a measurement. What that measurement means—what question it answers—remains open until the strategy is chosen.

This distinction has immediate design implications. Different intercurrent event strategies require different data collection procedures. A treatment policy strategy requires outcome data from patients after they discontinue—data that many trial designs do not collect because the assumption, often unstated, is that post-discontinuation data are irrelevant or unavailable. A hypothetical strategy requires the data needed to support the counterfactual model—which may include covariates, time-varying measurements, and auxiliary outcomes that were not originally planned for. A composite strategy requires the intercurrent event itself to be measured and adjudicated with the same rigor as the primary outcome—which it often is not.

If the intercurrent event strategy is chosen after the protocol is finalized, some of the data needed to implement it may not have been collected. The strategy then becomes constrained not by scientific reasoning but by what the database happens to contain. This is the design equivalent of letting the available data determine the question, rather than letting the question determine what data to collect.

Multiple estimands and the primary/sensitivity relationship

ICH E9(R1) explicitly endorses the use of multiple estimands—one primary and several secondary or sensitivity estimands—as a way of addressing the trial's question from multiple angles. This is not multiplicity in the statistical sense. It is scientific completeness.

A primary estimand of treatment policy, for example, answers the prescriber's question. A hypothetical sensitivity estimand, estimated under the assumption that discontinued patients would have continued on the same trajectory as completers with similar baseline characteristics, answers the mechanism scientist's question. Together, they provide a more complete picture than either alone.

The relationship between the primary and sensitivity estimands must be pre-specified, and the logic of the relationship must be transparent. Sensitivity estimands exist to stress-test the primary result—to ask what the result would look like if the assumptions underlying the primary strategy were modified. If the treatment policy and hypothetical estimands produce similar results, confidence in the primary result is strengthened. If they diverge substantially, the divergence itself is informative: it means the intercurrent events matter clinically, that the treatment's effect depends heavily on whether patients stay on treatment, and that prescribers and payers will need to account for that dependence.

The sensitivity estimand is not a fallback. It is not the analysis to run if the primary fails. It is a pre-specified scientific question that the trial is simultaneously designed to answer, with the relationship to the primary result specified in advance and the divergence, if it occurs, treated as a finding rather than a problem.

What must be settled before section 1.4

By the time a trial design moves from estimand specification to the questions of closing—what has been committed to, what remains uncertain, what can go wrong—the intercurrent event strategy must be resolved. Not tentatively, not with a note that it will be finalized in the SAP, but resolved: the strategy named, the data collection implications identified, the sensitivity approach pre-specified, and the owner identified.

The owner is the person who can stand up in a regulatory meeting and answer the question: why this strategy, for this trial, for these patients? The answer requires clinical reasoning about the disease, about the treatment mechanism, about the conditions under which patients will be prescribed this treatment in practice. It requires regulatory judgment about what the agency will accept and what they will challenge. And it requires statistical judgment about what the chosen strategy can actually produce given the trial's data collection plan.

If these three forms of judgment have not been brought together before the design is finalized, the intercurrent event strategy will be settled by default—by the software, by the previous protocol, by whoever drafted the SAP without instruction. Default strategies produce

defensible-looking results. They do not produce defensible science, because the defense requires knowing why the strategy was chosen, not just what it was.

The estimand is the question. The intercurrent event strategy is the part of the question that most people would rather not answer. Answering it is where trial design becomes honest.

References: ICH E9(R1) Addendum on Estimands and Sensitivity Analysis in Clinical Trials (2019); Lipkovich et al., "Causal Inference and Estimands in Clinical Trials," Stat Biopharm Res 2020; Qu et al., "Defining the Estimand in Clinical Trials with Treatment Switching," Pharm Stat 2021; Rubin, "Causal Inference Using Potential Outcomes," J Am Stat Assoc 2005.

1.4 Closing: What Has Been Decided, and What It Costs

What this chapter asked

Chapter 1 asked one question: what are we trying to show?

It asked that question three times, in three registers.

In Section 1.1, it asked what the estimand framework requires—the four attributes that must be specified before a trial has a question rather than an aspiration, and the three ownership structures that must be in place before those specifications are defensible.

In Section 1.2, it asked what endpoint choice commits a trial to—not just a measurement, but a claim about what constitutes benefit, for which patients, over what time horizon, and at what interpretive cost.

In Section 1.3, it asked what the intercurrent event strategy decides—which of five fundamentally different scientific questions the trial is claiming to answer, and what data collection and analytical obligations follow from that answer.

Together, these three sections constitute a single decision: the definition of the treatment effect the trial is designed to estimate. That definition is the estimand. Everything else in trial design—sample size, randomization, interim analysis, multiplicity control—is in service of it. If the estimand is not settled, nothing downstream is stable.

The decision this chapter made

By the end of this chapter, the following must be true.

The estimand is specified. All four attributes—population, variable, intercurrent event strategy, population-level summary—are stated clearly enough that two independent statisticians, reading the protocol, would construct the same primary analysis. If that condition is not met, the estimand has not been specified. A sentence in the protocol that names a primary endpoint and references an intention-to-treat analysis is not an estimand. It is a placeholder.

The endpoint is connected to the question. The primary endpoint is not present in the design because it was used before, or because the agency accepted it in a related indication, or because the software handles it conveniently. It is present because someone has articulated why this measurement, in this population, at this time point, captures the treatment effect the estimand defines. That articulation is documented and owned.

The intercurrent event strategy is explicit. The strategy is named—treatment policy, hypothetical, composite, while on treatment, or principal stratum—and the name is accompanied by the reasoning: why this strategy, for this trial, for these patients? The data collection implications of the strategy have been identified and reflected in the protocol. The sensitivity estimands are pre-specified, and their relationship to the primary estimand is stated.

Ownership is assigned. The clinical team can defend the question. The statistical team can specify the analysis that estimates it. The regulatory team can articulate what the result will be able to claim. These three responsibilities are not interchangeable, and they are not all held by the same person.

If any of these conditions is not met, the chapter's work is unfinished. Proceeding to Chapter 2—to the question of how the treatment effect will be measured and summarized—without settling Chapter 1 is proceeding without a destination. The measurement can be made. The summary can be calculated. The result will not be interpretable, because no one has agreed on what it was supposed to show.

The most common failure mode

The most common way Chapter 1's work goes unfinished is not through negligence. It is through deferral.

The protocol is written on a timeline. The estimand discussion is scheduled for the next design meeting. The intercurrent event strategy will be finalized in the SAP. The endpoint was chosen because it is what this team always uses, and the question of whether it is the right endpoint for this trial is deferred to the regulatory feedback after submission.

Deferral feels like flexibility. It is not. It is the accumulation of unresolved decisions that will have to be resolved later, under worse conditions, with less time, and without the ability to change the protocol that has already been running for two years.

The specific cost of each deferral is predictable.

Deferring the estimand specification to the SAP means the protocol may have been written with a data collection plan that cannot support the estimand eventually chosen. The SAP is not the place to discover that post-discontinuation outcome data were not planned for because the treatment policy strategy was not considered at design.

Deferring the endpoint justification to the regulatory submission means the agency's questions about why this endpoint arrive after the trial is complete, when the answer "we would

have chosen differently" has no practical consequence. The agency's question during a pre-submission meeting is an opportunity. The same question during a complete response letter is a problem.

Deferring the intercurrent event strategy to the analysis means the strategy will be chosen to fit the data rather than to answer the question. This is the definition of post-hoc analysis, and it is what the estimand framework was specifically designed to prevent.

Each deferral is a transfer of risk—from the design stage, where it can be managed, to the analysis stage, where it can only be disclosed.

The characteristic mistakes of this chapter

Three mistakes recur in the work this chapter covers. They are not rare. They are the normal product of design processes that move faster than the underlying questions.

The estimand that cannot be estimated. The four attributes are specified, the language looks precise, and the primary analysis is named. But the analysis cannot produce an unbiased estimate of the specified estimand given the data that will be collected. Post-discontinuation outcomes are needed but not planned for. The counterfactual model requires time-varying covariates that are not in the CRF. The composite requires adjudication of the intercurrent event as rigorously as the primary outcome, but no adjudication committee has been constituted for it. The estimand is formally complete and analytically empty.

The endpoint that answers the wrong question. The endpoint is clinically accepted, statistically feasible, and regulatory precedent supports its use. It also does not capture what the treatment does. The treatment's mechanism produces a biological change that the chosen endpoint is not sensitive to in the enrolled population at the planned follow-up duration. The trial will complete, the endpoint will be measured, and the result will be uninterpretable—not because the analysis failed but because the endpoint was chosen without examining what it would detect.

The strategy selected by default. No one discussed the intercurrent event strategy. The SAP analyst used the approach from the previous trial. The ITT label was applied to a completers-only analysis because that is what the software produced. The intercurrent event strategy is whatever fell out of the process—and the process did not include the question of what question the strategy is answering. When the agency asks, the answer is "that is what we did," which is not a defense.

These three mistakes have a common origin: the estimand conversation did not happen, or it happened without the people who needed to be in it, or it happened and produced language that was not precise enough to constrain the analysis. The framework is only as useful as the discipline with which it is applied.

What cannot be fixed downstream

Some design errors are recoverable. A sample size that turns out to be insufficient can sometimes be addressed through a pre-specified sample size re-estimation. A randomization imbalance can be controlled for in the analysis. A stratification factor that turns out to be prognostic but was not used in the randomization can be included as a covariate.

The errors of Chapter 1 are not recoverable in this sense.

If the estimand was never specified, no sensitivity analysis can specify it retroactively. A sensitivity analysis explores the robustness of a pre-specified primary result. If there is no pre-specified primary result—only an analysis that was run and a question about what it means—the sensitivity analysis has no anchor.

If the endpoint was wrong—not insensitive, but wrong, measuring something other than what the treatment does—no analytical adjustment can make the result answer the question the trial was supposed to answer. The data contain what they contain. If the endpoint does not capture the treatment effect, the data do not contain it.

If the intercurrent event strategy was never chosen, the primary result reflects whatever the software default produced. Regulatory reviewers are experienced at identifying this. The absence of documented reasoning for the strategy is itself a finding—it signals that the strategy was chosen for analytical convenience rather than scientific necessity, which is precisely the problem the estimand framework was designed to prevent.

The design errors of this chapter are recoverable only if they are caught before enrollment. After enrollment, they are findings. After unblinding, they are liabilities.

The professional risk

There is a version of this chapter's failure mode that is not just scientific but professional.

A statistician who signs off on a primary analysis strategy without a documented estimand has not made a statistical error. The analysis may be technically correct. But they have produced a result without a clear statement of what that result estimates—and when the question is asked, as it will be, there is no answer that does not involve admitting that the question was never resolved.

A clinical scientist who accepts an endpoint because it was used in the previous trial, without asking whether it answers the current trial's question, has not made a clinical error. The endpoint may be perfectly reasonable. But they have committed the trial to a measurement without committing it to a purpose—and the difference between those two things is the difference between evidence and data.

A regulatory strategist who plans a submission around a result without knowing what estimand that result estimates cannot defend the label claim when the agency asks. The

answer to "what does this estimate represent?" is the beginning of every label negotiation. If the answer is "the primary analysis result," the negotiation will not go well.

The professional risk of Chapter 1's failure is not that the trial will fail. It is that the trial will succeed—produce a statistically significant result, reach regulatory submission, enter label negotiation—and at each of those stages, the same question will be asked that was not answered at design. And at each stage, the answer will be harder to give, the options will be fewer, and the cost of the original deferral will compound.

What Chapter 2 requires from Chapter 1

Chapter 2 asks: how will we measure the difference? It addresses effect measures—risk differences, ratios, hazard ratios, win ratios—and their interpretive and efficiency implications. It addresses the non-inferiority margin and what it takes to defend one.

None of this can proceed without a settled estimand.

The effect measure must be appropriate for the estimand. A hazard ratio summarizes a time-to-event estimand in a specific way, under specific assumptions about the proportionality of hazards. A risk difference at a fixed time point summarizes a different quantity. The choice between them is not primarily statistical—it is a choice about what kind of summary is scientifically appropriate for the estimand that has been specified. If the estimand has not been specified, the effect measure cannot be chosen on any basis other than convention.

The non-inferiority margin must be connected to the estimand. The margin asserts that a treatment effect smaller than a specified threshold is clinically acceptable—that the treatment's other attributes justify tolerating some inferiority on the primary endpoint. That assertion requires knowing what the primary endpoint is estimating. A margin defined without a clear estimand is a number without a referent.

Chapter 2's work is technical in ways that Chapter 1's work is not. But the technical work of Chapter 2 is only as rigorous as the conceptual work of Chapter 1. If Chapter 1's question is unsettled, Chapter 2's answers are built on sand.

The decision this chapter demands

The chapter ends with a demand, not a summary.

Before the trial moves forward, someone must be able to answer, without consulting the SAP, without deferring to the statistician, and without referencing a previous trial's protocol:

what effect, in what patients, under what conditions, measured how, summarized as what, is this trial designed to estimate?

If that question can be answered—clearly, completely, and by the right people—the chapter's work is done. If it cannot, the chapter's work must continue, regardless of what the timeline says.

The pressure to move on is real. It is also the most common source of the errors this chapter describes. Every week spent resolving the estimand before enrollment is a week not spent discovering the same problem after unblinding, when the cost of resolution is not a delayed start but a compromised result.

That trade is not close. The estimand is the question. The question must come first.

Chapter 1 risk summary

The decision this chapter owns: what treatment effect, in what population, under what conditions, is this trial designed to estimate?

The most common mistake: treating endpoint selection as estimand specification—naming a primary endpoint and an analysis method without resolving what the analysis is designed to estimate or why.

The professional-level risk: proceeding to sample size, effect measure, and interim analysis planning without a settled estimand. These downstream decisions will be internally consistent. They will not be anchored to a defensible scientific claim. The inconsistency will surface at regulatory review, at label negotiation, or at the point when a payer asks what the result actually shows—and none of those moments allow for the answer "we should have decided this earlier."

Chapter 2: How Will We Measure the Difference?

The question Chapter 1 leaves open

Chapter 1 settled the estimand. It named the population, the variable, the intercurrent event strategy, the population-level summary. It assigned ownership. It required that the question be stated precisely enough that a second statistician, reading the protocol, would construct the same primary analysis.

What it did not settle is how the difference between arms will be expressed.

This is not a minor technical detail. It is the decision that determines what the trial can claim, how efficiently it can claim it, and whether the claim will be legible to the people who need to act on it—prescribers, payers, regulators, and patients. Two trials with identical estimands, identical populations, identical data, and identical true treatment effects can produce results that feel completely different depending on the effect measure chosen. One result reads as a modest improvement. The other reads as a clinically compelling advance. The data are the same. The framing is different. And framing, in the context of regulatory submissions and label negotiations, is not cosmetic.

The choice of effect measure is a commitment. It encodes assumptions about how the treatment effect distributes across the population, what kind of comparison is scientifically appropriate, and what claim the result is intended to support. Those assumptions should be explicit—not because explicitness is a regulatory requirement, though it increasingly is, but because implicit assumptions are the ones that fail silently.

The problem this chapter addresses

There is a standard way effect measures are chosen in clinical trial design: the clinical team proposes an endpoint, the statistician proposes the natural summary measure for that endpoint type—rate difference for binary outcomes, hazard ratio for time-to-event, mean difference for continuous—and the design proceeds. The choice of effect measure is treated as a consequence of the endpoint type, not as an independent decision.

This is wrong in a subtle but important way.

The endpoint type constrains the available effect measures. It does not determine which one is scientifically appropriate. A binary outcome can be summarized as a risk difference, a risk ratio, or an odds ratio. Each implies a different model of how the treatment effect scales across patient subgroups with different baseline risks. A time-to-event outcome can be summarized as a hazard ratio, a restricted mean survival time difference, a milestone survival difference, or a win ratio. Each makes different assumptions about the shape of the treatment effect over time. A continuous outcome can be summarized as a mean difference, a median difference, a proportion achieving response, or a distribution-free rank-based statistic. Each answers a subtly different question about what the treatment does and for whom.

The choice among these is a scientific choice, not a statistical one. It should follow from the estimand—from the question that has been settled—rather than from the convention for this outcome type. When it follows from convention instead, the effect measure may be technically appropriate and scientifically misaligned: producing a valid estimate of something that is not quite what the trial was designed to show.

This chapter asks the question that the standard approach skips: given the estimand that has been specified, which effect measure best expresses the treatment effect it defines, can be estimated with the available data, and will support the claim the trial is intended to make?

What this chapter covers

Section 2.1 — Effect Measures examines the major families of effect measures and what each one commits a trial to. The focus is not on the statistical properties of the measures—those are well-documented—but on the scientific and interpretive commitments they carry. A hazard ratio is a proportional measure; it implies that the treatment's relative benefit is constant across the follow-up period and homogeneous across patients with different baseline risks. That implication is an assumption. When the assumption holds, the hazard ratio is efficient and interpretable. When it does not—when hazards are non-proportional, when the treatment effect diminishes over time, when the trial population is heterogeneous in ways that matter—the hazard ratio produces a number that is statistically valid and clinically misleading. The section examines when this happens and what the alternatives are.

Section 2.2 — Interpretability and Efficiency examines the tension between what is statistically efficient and what is clinically meaningful. These two objectives do not always align. The most powerful test of a treatment effect is not always the test whose result is easiest to interpret. The number needed to treat is interpretable but inefficient. The odds ratio is efficient but frequently misread. The restricted mean survival time is interpretable and robust to non-proportional hazards but requires a time horizon choice that is itself a commitment. The section asks how this tension should be navigated—not by prescribing a single answer, but by making the trade-offs visible.

Section 2.3 — The Non-Inferiority Margin addresses the specific and consequential decision that arises when the trial is designed to show that a new treatment is not unacceptably worse than an active comparator. The non-inferiority margin is the threshold of acceptable inferiority—the amount by which the new treatment may fall short of the comparator and still be considered acceptable. Defining this margin requires answering a question that is simultaneously clinical, historical, and ethical: how much efficacy can be traded for something else—safety, convenience, tolerability, cost—without betraying patients? The section examines who should own this question and what happens when the answer is borrowed instead of earned.

The decision structure of this chapter

The three sections of this chapter are not independent. They address a single decision sequence.

The effect measure determines what the primary result will say. It shapes the statistical power of the primary test, the interpretability of the estimate, and the kind of claim the result will support. It must be chosen before sample size can be calculated—because the sample size required depends on the variance structure of the chosen measure—and before the non-inferiority margin can be defined—because the margin is expressed in the units of the effect measure.

This sequence is often violated. Sample sizes are calculated before effect measures are examined. Non-inferiority margins are borrowed from historical trials that used different effect measures. The result is a design where the components are each defensible in isolation but incoherent in combination—where the sample size was calculated for a risk difference but the primary analysis reports a hazard ratio, or where the non-inferiority margin was defined in absolute risk terms but the trial is powered on a relative measure.

These inconsistencies do not always surface during the trial. They surface at the regulatory review, when the agency asks how the margin was justified given the effect measure, or during the label negotiation, when the sponsor discovers that the result cannot support the claim they intended. At that point, the inconsistency is not a design problem. It is a submission problem, and it is much harder to fix.

What this chapter is not about

This chapter does not provide a guide to selecting statistical tests. It does not compare the type I error properties of different test statistics, explain the computational mechanics of the restricted mean survival time, or evaluate the relative performance of different estimators under various missing data mechanisms.

Those questions matter, and they have answers, documented in statistical literature that is more current and more complete than this book can be. What this chapter provides is the prior question: before selecting a test or an estimator, what is the effect measure supposed to represent, who is it for, and what claim will it support?

A trial team that can answer that question will select appropriate statistical methods from the available literature. A trial team that selects statistical methods without answering that question will produce technically valid analyses of unclear scientific meaning—which is the specific failure mode this book exists to prevent.

The connection to what follows

Chapter 2's decisions constrain Chapter 3. The effect measure determines what effect size must be assumed when calculating sample size—and the assumptions embedded in that effect size are among the highest-stakes commitments the trial makes. A hazard ratio assumption implies a model of how events accumulate over time. A risk difference assumption implies a model of baseline event rates in the control arm. A mean difference assumption implies a model of within-patient variability. Each of these models can be wrong, and each one being wrong has a different consequence for whether the trial succeeds.

Chapter 2 also connects to Chapter 6. The effect measure shapes what can be claimed when the trial is over. A result expressed as a hazard ratio supports a different label claim than the same result expressed as a restricted mean survival time difference, even if the underlying data are identical. The claim discipline of Chapter 6 begins with the effect measure choice of Chapter 2—and a measure chosen for statistical efficiency without considering what claim it will support may produce a result that is statistically clean and commercially marginal.

These connections are not incidental. They are the reason the effect measure decision belongs at the beginning of the design process, immediately after the estimand is settled, rather than as a consequence of endpoint type that no one explicitly chose.

The question Chapter 2 must answer

By the end of this chapter, the following question must have a documented answer: given the estimand specified in Chapter 1, what effect measure will be used to express the treatment difference, why is that measure scientifically appropriate for the question being asked, what assumptions does it carry, and what claim will a statistically significant result support?

If the non-inferiority design is being used, there is an additional question: what is the non-inferiority margin, who derived it, and on what evidence does it rest?

These are not questions that can be deferred to the SAP. The effect measure and the non-inferiority margin—if applicable—must be in the protocol, because the sample size is calculated from them, the primary hypothesis is expressed in their terms, and the regulatory agency will review them as part of the design justification. They are design commitments, not analysis parameters.

Chapter 1 defined what is being estimated. Chapter 2 defines how the estimate will be expressed and what it will be compared against. Both are required before the trial can be sized.

2.1 Effect Measures

The measure is not the outcome

When a trial is over, the result is expressed as a number: a hazard ratio of 0.78, a risk difference of 4.2 percentage points, a mean difference of 1.8 units on a validated scale. This number is how the trial communicates its finding to regulators, prescribers, and payers. It is also the number that will appear in the label, in the meta-analyses, in the cost-effectiveness models, and in the clinical guidelines that govern whether this treatment gets used.

The number is not the outcome. It is a summary of the outcome—a compression of patient-level data into a single quantity that describes the difference between arms. Different summaries of the same data tell different stories. A treatment that reduces the annual event rate from 8% to 6% can be described as producing a relative risk reduction of 25%, an absolute risk reduction of 2 percentage points, a number needed to treat of 50, or an odds ratio of approximately 0.73. Each of these is arithmetically correct. Each emphasizes a different aspect of the same underlying effect. Each implies a different comparison to alternative treatments and a different framing for clinical decision-making.

The choice among them is not aesthetic. It is scientific and strategic, and it should be made deliberately—before the trial is designed, not after the data are in hand.

Binary outcomes: three measures, three commitments

For binary outcomes—event occurred or did not—three effect measures are in common use: the risk difference, the risk ratio, and the odds ratio.

The **risk difference** is the absolute change in event probability attributable to treatment. A risk difference of minus 4 percentage points means that in a population identical to the one enrolled, the treatment reduces the probability of the event by 4 percentage points relative to control. The risk difference is directly interpretable in clinical terms—it is the number needed to treat when inverted—and it is the measure that scales most naturally with absolute patient benefit. Its limitation is that it is not transportable: a risk difference of 4 percentage points estimated in a high-risk population may not apply to a lower-risk population in which the treatment is subsequently used. If baseline event rates differ between the trial population and the target clinical population, the absolute benefit will differ even if the relative effect is the same.

The **risk ratio** is the proportional reduction in event probability. A risk ratio of 0.75 means the event probability in the treatment arm is 75% of the event probability in the control arm. The risk ratio is more transportable than the risk difference—if the relative effect is stable across baseline risk, the risk ratio estimated in the trial applies to populations with different absolute risks—but less interpretable in terms of direct patient benefit. A 25% relative risk reduction means different things to a patient with a 2% baseline risk and a patient with a 20% baseline risk, and the risk ratio does not make that difference visible.

The **odds ratio** is the ratio of the event odds in the treatment arm to the event odds in the control arm. It is the natural output of logistic regression and is therefore the most common effect measure in observational research and in trials where covariate adjustment is central. It has one important interpretive property and one important interpretive problem. The property: odds ratios are collapsible only under specific conditions, which means that an odds ratio estimated from a heterogeneous population may not equal the weighted average of stratum-specific odds ratios. The problem: odds ratios are systematically misread as risk ratios, even by clinicians who know better. When the event rate is high, an odds ratio of 0.73 and a risk ratio of 0.75 are materially different; when the event rate is low, they approximate each other. The clinical literature is littered with effect claims that conflate the two, producing overestimates of relative benefit that survive into guidelines.

The choice among these three is a scientific commitment about which kind of comparison is appropriate for the estimand. A trial designed to estimate the absolute benefit of treatment in the enrolled population—the treatment policy question for a specific clinical context—should favor the risk difference. A trial designed to estimate a relative effect that will be applied to populations with varying baseline risks should favor the risk ratio. A trial in which covariate adjustment is central and the event rate is low enough that odds ratios and risk ratios approximate each other may reasonably use the odds ratio. These are not interchangeable defaults.

Time-to-event outcomes: when the hazard ratio assumption fails

For time-to-event outcomes, the hazard ratio is the dominant effect measure in regulatory submissions and clinical trial literature. It has legitimate advantages: it uses all available follow-up time, it can be estimated efficiently even when events are sparse, and it is the natural output of the Cox proportional hazards model, which accommodates covariate adjustment and stratified analyses in a well-understood framework.

It also carries an assumption that is routinely violated and rarely examined: proportional hazards.

The proportional hazards assumption requires that the ratio of the hazard rates in the two arms is constant over the entire follow-up period. In practice, this means the treatment's relative effect must be the same at month three as at month thirty-six. For a treatment that produces an early reduction in hazard and then loses its effect as resistance develops, or for

a treatment whose benefit accumulates slowly and strengthens over time, this assumption does not hold. The hazard ratio estimated from such data is a weighted average of the time-varying hazard ratios—a summary statistic that may not correspond to the treatment's effect at any specific time point and may not be the quantity the estimand was designed to capture.

Non-proportional hazards are not rare in clinical trial data. They are common in oncology, where treatments that produce durable responses in a subset of patients generate crossing survival curves. They are common in cardiovascular disease, where early procedural risk may offset long-term benefit. They are common in any indication where the treatment effect requires time to accumulate and then stabilizes. The question for any trial using a time-to-event primary endpoint is not whether the proportional hazards assumption is exactly true—it never is—but whether the departure from proportionality is large enough to make the hazard ratio a misleading summary.

When the assumption is questionable, two alternatives deserve consideration.

The **restricted mean survival time** (RMST) difference is the difference in mean survival time up to a pre-specified time horizon. It does not assume proportional hazards. It is directly interpretable as a difference in average time free from the event—a quantity that clinicians and patients find meaningful. It requires specifying the time horizon in advance, which is a commitment; the choice of time horizon determines which part of the survival curve the estimate reflects, and a poorly chosen horizon can make a modest treatment effect look large or a meaningful effect look small. But the horizon must be chosen on clinical grounds, not on statistical convenience.

The **win ratio** is a more recent alternative that generalizes naturally to hierarchical composite outcomes: patients in the treatment arm are compared to patients in the control arm on a hierarchy of outcomes, and the treatment "wins" if the treatment patient has a better outcome on the most important component. It handles tied and censored observations, accommodates clinical hierarchies, and produces an effect measure that is interpretable in terms of the probability that a randomly selected treatment patient has a better outcome than a randomly selected control patient. Its limitations include sensitivity to the choice of outcome hierarchy and the time horizon over which comparisons are made.

Neither alternative is universally superior. Each is more appropriate in some settings than others. The decision between them—and between both and the hazard ratio—should be made before the trial begins, based on the expected shape of the treatment effect and the clinical question the estimand is designed to answer.

Continuous outcomes: the difference between average and distribution

For continuous outcomes, the mean difference is the default. It is interpretable, familiar, and efficient when the outcome is approximately normally distributed. It is also the measure most sensitive to the distributional assumptions that are rarely examined in practice.

The mean difference describes the average treatment effect in the enrolled population. When the treatment effect is homogeneous—when the treatment produces approximately the same benefit in every patient—the mean difference is both scientifically appropriate and practically meaningful. When the treatment effect is heterogeneous—when some patients benefit substantially, some benefit minimally, and some are harmed—the mean difference may describe a quantity that no patient actually experiences. An average improvement of 3 units on a validated scale may reflect a 6-unit improvement in half the population and no improvement in the other half, or a 3-unit improvement in nearly everyone. The mean does not distinguish between these scenarios.

This matters for the estimand. The population-level summary attribute of the estimand should specify what kind of average is appropriate. If the question is whether the treatment produces a clinically meaningful improvement in the typical patient, the mean difference may be the right summary. If the question is whether the treatment produces a clinically meaningful improvement in a substantial proportion of patients, a responder analysis—the proportion achieving a pre-specified threshold of improvement—may be more appropriate. If the distribution of outcomes is expected to be skewed, or if the variance is expected to differ substantially between arms, a median difference or a rank-based statistic may be more honest than a mean difference estimated under normality assumptions.

The choice between mean-based and distribution-sensitive summaries is not primarily a statistical question. It is a question about what the estimand is claiming—whether the trial is designed to detect an average effect or a distribution shift—and it should follow from the clinical question, not from the availability of convenient software.

The scale dependency of effect measures

One source of confusion in effect measure selection is that different measures capture different aspects of the same underlying effect, and the relationship between them depends on the baseline event rate or outcome distribution. This means that a trial that reports a large relative effect and a small absolute effect is not being inconsistent—it is reporting two different summaries of the same data, each appropriate for different purposes.

The clinical relevance is that effect measures should be matched to the decision context. For a decision about whether to prescribe a treatment to an individual patient, the absolute effect measure—risk difference, number needed to treat, restricted mean survival time difference—is

most relevant, because it describes what the patient can expect in terms of personal benefit. For a decision about whether a treatment is more efficacious than alternatives across a range of patient populations, the relative effect measure may be more relevant, because it is more likely to be transportable. For a regulatory decision about whether a treatment should be approved, both are relevant: the relative measure demonstrates that the treatment has biological activity, the absolute measure demonstrates that the activity is clinically meaningful.

A trial designed without attention to which decision context the effect measure serves may produce a result that is compelling in one frame and unimpressive in another—a situation that creates problems not at the time of design but at the time of submission and negotiation.

What this section demands before proceeding

The effect measure must be specified before sample size can be calculated. The sample size depends on the assumed variance structure, which is a property of the effect measure—the variance of a risk difference, a risk ratio, and an odds ratio are not the same, and neither are the sample sizes required to detect a given effect with a given power. A trial that calculates sample size for one effect measure and reports another has not made a statistical error, but it has produced a sample size that may not correspond to what is actually being tested.

The effect measure must also be connected to the estimand. The population-level summary attribute of the estimand should name the effect measure—not just the statistical test that will be used, but the quantity the test is designed to detect. If the estimand specifies a hazard ratio but the clinical question is better answered by a restricted mean survival time difference, the estimand has not been correctly specified.

And the effect measure must be chosen before the non-inferiority margin is defined. The margin is expressed in the units of the effect measure. A margin defined as a hazard ratio is incommensurable with a margin defined as a risk difference, and borrowing a margin from a historical trial that used a different measure requires justification that is rarely provided. This is the subject of Section 2.3. Before the margin can be discussed, the measure must be settled.

References: Hernán, "The Hazard of Hazard Ratios," Epidemiology 2010; Royston and Parmar, "Restricted Mean Survival Time," BMC Med Res Methodol 2013; Pocock et al., "The Win Ratio," Eur Heart J 2012; Senn, "Disappointing Dichotomies," Pharm Stat 2003.

2.2 Interpretability and Efficiency

The tension that does not resolve

Every effect measure sits somewhere on a spectrum between two properties that pull in opposite directions: statistical efficiency and clinical interpretability.

Statistical efficiency is the property of extracting maximum information from the available data. An efficient test uses all the available follow-up time, accommodates the covariance structure of the outcome, and produces the narrowest confidence interval for a given sample size. Efficiency is what allows a trial to detect a real treatment effect without enrolling more patients than necessary.

Clinical interpretability is the property of producing a result that the people who must act on it can understand and apply. An interpretable result tells a prescribing physician what a typical patient can expect, tells a payer what population-level benefit the treatment produces, and tells a patient what the treatment is likely to do for them specifically. Interpretability is what makes a result actionable.

The tension between these two properties is real and does not have a general resolution. The statistically optimal measure is not always the most clinically interpretable, and the most interpretable measure is not always the most efficient. The design team must decide, explicitly, how to navigate this tension for their specific trial—based on the estimand, the decision context, and the audience for the result.

Treating this as a technical question with a technical answer is the specific mistake this section addresses.

When efficiency comes at interpretive cost

The hazard ratio is the canonical example of a measure that is efficient and frequently misinterpreted.

Its efficiency derives from the Cox proportional hazards model, which uses all follow-up time—including the time of patients who are censored before the primary endpoint—and produces a precisely estimated relative measure. In a trial where events are sparse, the hazard ratio may be the only effect measure that can be estimated with acceptable precision

from the available data. In this setting, choosing a less efficient measure in the name of interpretability may mean the trial cannot detect a real treatment effect at all.

Its interpretive problem is that a hazard ratio is not, to most clinicians, an intuitively accessible quantity. A hazard ratio of 0.78 does not immediately convey what a patient experiences. It is a ratio of rates, not a difference in probabilities. It does not specify a time horizon. It does not answer the question a patient asks—"will this treatment help me, and by how much?"—in terms a patient can evaluate.

The response to this is often to supplement the hazard ratio with milestone survival rates: "at three years, survival was 68% in the treatment arm versus 58% in the control arm." This is helpful, but it introduces a secondary effect measure that is not the primary basis for the hypothesis test. The milestone survival difference is interpretable; the hazard ratio is what the trial was powered to detect. The two may tell different stories—a hazard ratio that is statistically significant may correspond to a milestone survival difference that is clinically unimpressive, or a non-significant hazard ratio may correspond to a meaningful absolute difference at a specific time point. When the supplementary measure tells a different story than the primary measure, the trial result is not ambiguous—it is informative about the shape of the treatment effect. But only if the relationship between the two was pre-specified and the difference in their messages was anticipated.

When interpretability comes at efficiency cost

The number needed to treat—the inverse of the absolute risk reduction—is the most clinically interpretable effect measure in common use. A number needed to treat of 25 means that 25 patients must receive the treatment for one additional patient to avoid the outcome. This directly connects the treatment effect to clinical effort and clinical benefit in terms that any prescriber or patient can evaluate.

It is also inefficient by construction. Because it is derived from the absolute risk difference rather than a relative measure, it is sensitive to the baseline event rate in the control arm—which is often estimated with considerable uncertainty, especially in prevention trials where the overall event rate is low. The confidence interval around the number needed to treat is wide. The estimate is unstable. When the trial is underpowered, or when the baseline event rate is uncertain, the number needed to treat may not be estimable with useful precision even when the relative effect is statistically significant.

This asymmetry—significant relative effect, imprecise absolute effect—is common and uncomfortable. A relative risk reduction of 20% is statistically significant in a trial where the absolute risk difference cannot be estimated within a factor of three, because the baseline event rate in the control arm was uncertain. The trial demonstrates that the treatment works, in the sense that the relative effect is not zero. It does not demonstrate, with precision, how much work the treatment does in a given clinical population. These are different questions, and the trial answered the first but not the second.

The resolution is not to abandon absolute measures—they are essential for clinical and policy decisions—but to be explicit about what each measure is answering and why the trial was powered for one and not the other.

The restricted mean survival time as a case study

The restricted mean survival time difference has emerged over the past decade as an effect measure that attempts to combine reasonable efficiency with genuine interpretability for time-to-event outcomes. It is worth examining in detail, not because it is always the right choice, but because it illustrates the trade-offs involved in any effect measure decision.

The RMST at a pre-specified time horizon T is the expected survival time up to T, estimated from the area under the Kaplan-Meier survival curve. The RMST difference between arms is the difference in this expected survival time—the average number of additional days, months, or years that treatment patients survive event-free compared to control patients, up to the time horizon T.

Its interpretive advantage is direct: an RMST difference of 4.2 months means that, on average, patients in the treatment arm survive 4.2 months longer without the event, up to the specified time horizon. This is a statement about what a patient experiences that a hazard ratio cannot provide. It does not require the listener to know what a hazard is, or what proportional hazards means, or how to convert a ratio of rates into a patient-level expectation.

Its efficiency relative to the hazard ratio depends on the shape of the survival curves. When hazards are proportional, the hazard ratio test is more powerful—it uses more of the information in the data. When hazards are non-proportional, particularly when survival curves cross, the RMST test may be more powerful, because it is not distorted by the canceling effects of early and late hazard differences. In a world where non-proportional hazards are common—and they are, particularly in oncology and in trials of treatments that produce durable responses in a subset—the RMST may be more efficient in practice than the theoretical comparison under proportional hazards would suggest.

Its cost is the time horizon. The RMST must be evaluated at a specified T, and the choice of T influences the result. An RMST difference evaluated at two years may be small even if the treatment produces a large benefit at five years. An RMST difference evaluated at five years may be dominated by the late portion of the survival curve, where few patients remain under observation and the estimate is imprecise. The time horizon must be chosen on clinical grounds—the period over which the treatment's benefit is clinically relevant—and that choice must be pre-specified and defended. When the horizon is chosen after the data are seen, the RMST becomes a post-hoc measure that is no more interpretable than a hazard ratio evaluated at a cherry-picked time point.

Responder analyses: when the distribution matters more than the mean

A responder analysis converts a continuous outcome into a binary one: did the patient achieve at least a pre-specified magnitude of improvement? The proportion of responders in each arm is then the primary effect measure.

Responder analyses are advocated on interpretive grounds: they translate a continuous scale into a clinical decision—did this patient benefit?—that is more meaningful than a mean difference of 1.8 units on a validated scale. This argument has force when the minimally important difference is well-established and clinically meaningful, when the distribution of responses is bimodal rather than continuous, and when the clinical decision context is genuinely binary—the treatment is used or not used, not titrated to a continuous response.

The efficiency cost is real. Converting a continuous outcome to binary discards information and reduces statistical power. A trial powered on a mean difference may not be adequately powered for a responder analysis at the same sample size. If the responder analysis is primary, the sample size must reflect it—which typically means a larger trial.

There is also a hidden assumption: that the threshold defining response is clinically meaningful and was established independently of the trial data. A threshold chosen to maximize the observed responder rate is not a clinical determination. It is data dredging with a clinical label. The minimally important difference must be pre-specified on the basis of patient and clinical evidence, not selected from the distribution of trial outcomes. When this requirement is not met—and it frequently is not—the responder analysis produces an interpretable-looking number that reflects neither the treatment effect nor the clinical threshold it claims to represent.

The audience determines the measure

One organizing principle for navigating the efficiency-interpretability tension is to ask: who needs to act on this result, and what kind of information do they need to act well?

Regulatory agencies need to determine whether the treatment has demonstrated efficacy—that the treatment effect is real and not attributable to chance. For this purpose, statistical significance and the precision of the effect estimate are paramount. An efficient measure with a narrow confidence interval is more useful for regulatory determination than an interpretable measure with a wide one.

Prescribers need to determine whether to recommend the treatment to a specific patient, or to a class of patients with specific characteristics. For this purpose, absolute effect measures expressed in clinically meaningful units are most relevant. The prescriber's question is "what will this treatment do for my patient?"—a question the hazard ratio does not answer but the number needed to treat or the RMST difference approximates.

Payers need to determine whether the treatment produces value relative to its cost, often in comparison with existing alternatives. For this purpose, absolute measures in the trial population and estimates of transportability to other populations are most relevant. A risk ratio, if the treatment effect is transportable, or a risk difference specific to a defined population, are more useful than an odds ratio that may not translate clearly.

Patients need to understand what the treatment is likely to do for them personally. For this purpose, the most interpretable measure—in absolute terms, at a clinically relevant time horizon, expressed as a probability rather than a ratio—is most appropriate. The format in which clinical trial results are communicated to patients has received increasing attention in the patient-focused drug development literature, and the evidence suggests that relative risk reductions systematically overstate perceived benefit compared to absolute risk reductions.

The practical implication is that a trial may need more than one effect measure—not as sensitivity analyses, but as pre-specified estimates designed for different audiences and different decision contexts. The primary measure is chosen for the primary purpose—usually regulatory—and secondary measures are pre-specified for clinical and patient-facing communication. This is not multiplicity. It is epistemic completeness.

What this section demands before proceeding

Before the non-inferiority margin can be discussed, the efficiency-interpretability balance must be explicitly resolved for the primary effect measure. This means the design team must have answered: for whom is the primary result being produced, and what information does that audience need? If the answer is "the regulatory agency," the primary measure should be the most efficient available for the estimand. If the primary result must simultaneously serve regulatory and prescriber needs, the interpretability cost of the primary measure must be acceptable, and efficiency may need to be purchased through a larger sample size rather than through measure selection.

It also means that secondary effect measures—those intended for clinical communication rather than regulatory determination—must be pre-specified, with their relationship to the primary measure stated. A secondary measure that tells a different story than the primary measure is not a problem to be managed at the analysis stage. It is a finding to be anticipated at the design stage, with the discordance explained in advance rather than rationalized after the fact.

The effect measure for the primary analysis is settled. The secondary measures are identified. Their audience and purpose are documented. Only then can the non-inferiority question—Section 2.3—be addressed.

References: Sormani and Bruzzi, "Can We Measure Long-Term Treatment Effects via Short-Term Surrogates?" Mult Scler 2015; Uno et al., "On the Utility of RMST," Stat Med 2014; Sedgwick, "Relative Risks, Absolute Risks, and Numbers Needed to Treat," BMJ 2013; FDA Guidance on Patient-Focused Drug Development: Methods for Developing and Selecting Patient-Reported Outcomes (2022).

2.3 The Non-Inferiority Margin

The question only a non-inferiority trial can answer

A superiority trial asks: does the new treatment produce a better outcome than the comparator? A non-inferiority trial asks a different question: is the new treatment acceptably similar to an active comparator, in the sense that whatever inferiority exists does not outweigh the treatment's other attributes—its safety profile, its route of administration, its cost, its suitability for patients who cannot tolerate the comparator?

This is a legitimate scientific question, and it arises in legitimate clinical contexts. When an effective treatment exists and withholding it from a control arm would be unethical, a placebo-controlled superiority trial is impossible. When a new treatment offers meaningful advantages in tolerability or convenience, demonstrating that it is not inferior to the standard on the primary efficacy endpoint—combined with demonstrating the other advantages—constitutes a clinically meaningful result. Non-inferiority trials are not a regulatory convenience or a statistical workaround. They answer a real question.

But the non-inferiority question is harder to answer than the superiority question, because it requires specifying in advance what "acceptably similar" means. The non-inferiority margin—call it M—is that specification. It is the largest difference in the primary effect measure, in favor of the comparator, that the trial is willing to accept while still concluding non-inferiority. If the treatment arm falls short of the comparator by no more than M, the trial concludes non-inferiority. If it falls short by more than M, it does not.

The margin is not a statistical parameter. It is a clinical and ethical judgment, expressed as a number in the units of the effect measure, that the design team—and ultimately the regulatory agency—must accept before the trial begins. The number must be defensible. And it must be earned, not borrowed.

The two-step logic of margin justification

The regulatory standard for non-inferiority margin justification, articulated in FDA guidance and ICH E10, requires reasoning in two steps.

The first step is the **M1 step**: establishing the effect of the active comparator over placebo. This requires evidence—usually from historical randomized controlled trials—that the

comparator produces a meaningful effect. The historical estimate of this effect, at some conservative fraction of its confidence interval, defines the ceiling from which the margin is derived. The logic is: if the comparator produced an effect of size E over placebo, and the new treatment is non-inferior to the comparator within margin M, then the new treatment has preserved at least E minus M of the comparator's effect over what placebo would have produced. This preserved effect is the assurance of efficacy.

The second step is the **M2 step**: determining what fraction of M1 is acceptable to sacrifice for the other benefits the new treatment offers. M2 is always smaller than M1. The proportion preserved is a clinical judgment about how much of the comparator's established efficacy must be retained to justify the new treatment's use. For treatments with serious or irreversible outcomes—mortality, stroke, major organ failure—the preserved proportion is typically high: regulators expect 50% or more of M1 to be preserved, and may require higher fractions for outcomes where any loss of efficacy is clinically unacceptable. For outcomes with effective rescue options, or where the new treatment's advantages are large, the preserved proportion may be smaller.

This two-step structure has an important implication: the margin is not negotiable independently of the comparator's evidence base. A sponsor who sets M to be 20% of the comparator's effect is claiming that 80% of the comparator's effect will be preserved. That claim is only defensible if the estimate of the comparator's effect is itself credible—if the historical trials are recent, conducted in populations similar to the current trial, using endpoints that are still clinically relevant, and analyzed with methods that produce an honest estimate of the effect size.

When the historical evidence base is weak—few trials, small samples, effect estimates with wide confidence intervals—the margin derived from it will be correspondingly uncertain, and a conservatively chosen margin will be small. A sponsor who finds that a rigorous application of the two-step logic produces a margin that makes the trial infeasible—too small to detect a meaningful treatment difference—has learned something important: the historical evidence does not support a non-inferiority design. The correct response is to reconsider the design, not to relax the margin.

Who owns the margin

The non-inferiority margin is a clinical judgment expressed as a number. It is not a statistical parameter, and it cannot be determined by statistical analysis alone. It requires a synthesis of historical evidence about the comparator's effect, clinical reasoning about what magnitude of inferiority is acceptable, and regulatory judgment about what the agency will accept.

This creates an ownership problem that is almost universally mishandled.

In most trial design processes, the margin is assigned to the statistician. The statistician reviews the historical trials, estimates the comparator's effect, applies a conservative fraction,

and proposes a margin. The clinical team accepts the proposal because it is expressed as a number they cannot easily challenge. The regulatory team is consulted, if at all, after the protocol is already drafted.

This sequence is backward. The statistician can estimate M1 from the historical data; that is an appropriate statistical task. But the M2 step—how much of M1 is acceptable to sacrifice?—is a clinical and ethical judgment that belongs to the clinical team and the clinical community that will use the treatment. What level of inferiority, on this endpoint, in this disease, for this patient population, is the profession willing to accept in exchange for the new treatment's advantages? That question cannot be answered from historical trial data. It requires clinical reasoning, patient input where available, and professional consensus that the statistician is not positioned to substitute for.

When the clinical team has not owned the M2 judgment—when the margin was set by statistical convention or borrowed from a related indication without examination—the margin is indefensible under scrutiny. Not because the number is necessarily wrong, but because the reasoning behind it has not been developed. And when the reasoning has not been developed, it cannot be recovered. The agency's question—"how did you determine that this magnitude of inferiority was acceptable?"—cannot be answered with "the statistician proposed it."

The constancy assumption and why it matters

The two-step margin justification assumes that the comparator's effect in the new trial is the same as its effect in the historical trials from which M1 was estimated. This is the constancy assumption: the treatment effect of the active comparator over placebo would be the same in the current trial as it was in the historical trials, if a placebo arm were included.

The constancy assumption is never literally true, and in many trials it is under serious pressure. The historical trials that established the comparator's efficacy may have been conducted in different patient populations, with different definitions of the primary endpoint, with different standards of background therapy, or with different disease severities. The effect of the comparator may have changed because the patient population has evolved—better background therapy means that the comparator's marginal effect is smaller than it was when background therapy was minimal. The endpoint may have been redefined in ways that change the comparator's estimated effect. The historical control event rate may be substantially different from the current trial's expected event rate, which matters for absolute effect measures.

When the constancy assumption is not credible—when the historical trials and the current trial are so different that the comparator's historical effect cannot be expected to transfer—the entire margin justification collapses. The M1 estimate is not an estimate of the comparator's effect in the current trial; it is an artifact of a different clinical context. No M2 adjustment can repair this.

Evaluating the constancy assumption requires comparing the historical trials to the current trial design: same population, same definition of the endpoint, same background therapy, similar event rates. Where substantial differences exist, they should be documented and their implications for M1 assessed. If the differences are large enough that M1 cannot be credibly estimated, the non-inferiority design should be reconsidered.

This evaluation is not optional. Regulatory agencies now routinely ask for it. A non-inferiority protocol that cannot address the constancy assumption will receive questions that cannot be answered after the trial is complete.

The margin and the effect measure

The non-inferiority margin is expressed in the units of the primary effect measure. A margin of 0.10 on the risk difference scale means something entirely different from a margin of 0.10 on the log hazard ratio scale. A margin defined in the original units of a continuous outcome means something different from a margin defined as a proportion of the standard deviation.

This is why Section 2.1 must precede Section 2.3. The margin cannot be defined before the effect measure is settled, because the margin is an expression in the language of the effect measure. Borrowing a margin from a historical trial that used a different effect measure—common in practice, particularly when the historical evidence base is thin—requires converting the historical margin to the current scale, which requires assumptions about the relationship between the two scales that are themselves empirical claims.

When the effect measure changes between the historical trials and the current design, the margin must be recalculated, not imported. This is not a technicality. It is the difference between a margin that reflects a coherent clinical judgment and a margin that is a number without a referent.

The margin as a risk allocation

A non-inferiority margin allocates a specific risk: the risk that the new treatment is inferior to the comparator by an amount that is clinically meaningful, but that the trial—by design—will not detect.

If M is large, the trial is easy to run and easy to succeed in. It is also willing to accept substantial inferiority. If a treatment that is truly 15 percentage points worse than the comparator would still be declared non-inferior because M is 20 percentage points, the margin has allocated the risk of that 15-point inferiority to patients who will subsequently

receive the treatment—patients who are not in the trial and who did not consent to that risk.

This is the ethical dimension of the non-inferiority margin, and it is rarely discussed explicitly. The margin is a design decision that affects not just the enrolled patients but all future patients who will be treated based on the trial's result. A margin set generously to make the trial feasible is a margin set at the expense of those future patients' expected benefit.

The appropriate framing is not "how large can the margin be for this trial to be feasible?" It is "how large can the margin be for the trial's result to be trustworthy—for a non-inferiority conclusion to genuinely mean that the treatment preserves adequate efficacy?" If the answer to the first question exceeds the answer to the second, the non-inferiority design is not appropriate for this trial at this moment in the treatment's development. A trial that can be run but whose result cannot be trusted is not a trial worth running.

What this section demands before proceeding

Before Chapter 2's closing and before Chapter 3 addresses sample size, three things must be resolved about the non-inferiority margin if the trial is a non-inferiority design.

The historical evidence base for M1 must be identified and assessed for the constancy assumption. The relevant trials, their population, their endpoint definitions, their effect size estimates and confidence intervals, and their comparability to the current design must be documented. If the constancy assumption cannot be credibly made, the design must be reconsidered.

The M2 judgment must be owned by the clinical team, with documented reasoning about what magnitude of inferiority is acceptable for this indication, this patient population, and this trade-off. The reasoning must be expressible in clinical terms—not "we used 50% of M1" but "we judged that a treatment that preserves at least X of the comparator's established effect is acceptable because Y"—where Y is a clinical argument that a physician or patient could evaluate.

And the margin must be expressed in the units of the primary effect measure. If the effect measure was settled in Section 2.1, the margin can be expressed in those units. If it was not, it cannot be, and the effect measure must be settled first.

A margin that has not been through this process is a number. A margin that has is a commitment—one that can be defended, by name, by the person who owns it, in a regulatory meeting.

References: FDA Guidance for Industry, Non-Inferiority Clinical Trials to Establish Effectiveness (2016); ICH E10, Choice of Control Group and Related Issues in Clinical Trials (2000); D'Agostino et al., "Non-Inferiority Trials," Drug Inf J 2003; Snapinn and Jiang, "Preserving the Comparator's Treatment Effect in a Non-Inferiority Trial," Pharm Stat 2008.

2.4 Closing: The Measure Is a Commitment

What this chapter asked

Chapter 2 asked one question: how will we measure the difference?

It asked that question in three registers, each exposing a different layer of the decision.

Section 2.1 asked what each class of effect measure commits a trial to—not just statistically, but scientifically. A hazard ratio commits to a proportionality assumption that the data may not support. An odds ratio commits to a model that clinicians systematically misread as something it is not. A restricted mean survival time commits to a time horizon that must be chosen before the data are seen. A risk difference commits to an absolute claim that may not be transportable across clinical populations. These are not defects; they are properties. Every measure carries commitments. The decision is which commitments are appropriate for this trial, this estimand, this clinical question.

Section 2.2 asked how to navigate the tension between efficiency and interpretability—not by resolving it in the abstract, but by identifying who needs to act on the result and what kind of information that audience requires. A result optimized for regulatory efficiency may be clinically opaque. A result optimized for prescriber interpretability may be statistically underpowered. The choice is a design decision, not an afterthought, and it must be made before the sample size is calculated.

Section 2.3 asked what it takes to define a non-inferiority margin that is defensible—not just numerically reasonable but epistemically earned. The two-step logic of M1 and M2 is not a checklist; it is a chain of reasoning in which each link must hold. If the historical evidence for M1 is weak, the chain breaks. If the clinical judgment behind M2 is unowned, the chain breaks. If the constancy assumption is not credible, the chain breaks. A margin that looks like a number but is not supported by this chain of reasoning is not a non-inferiority margin. It is a wish.

What this chapter decided

By the end of this chapter, four things must be settled.

The primary effect measure is named, connected to the estimand, and its assumptions are stated. Not “we will use a hazard ratio because this is a time-to-event endpoint” but “we will

use a hazard ratio because the treatment effect is expected to be proportional throughout the follow-up period, and this expectation is supported by the following clinical reasoning." Or: "we will use the restricted mean survival time difference at thirty-six months because the treatment is expected to produce non-proportional hazards due to its mechanism of action, and the thirty-six-month time horizon corresponds to the duration of clinical relevance defined by the treatment strategy." The choice is documented and owned.

The secondary effect measures for clinical and patient communication are pre-specified. Their relationship to the primary measure is stated. Anticipated discordances—scenarios in which the primary measure is significant but a secondary measure is clinically unimpressive, or vice versa—are acknowledged and explained. Not resolved—they may be genuine findings—but acknowledged, so that the result is not surprised by its own messages.

For non-inferiority trials: the margin is defined in the units of the primary effect measure, the historical evidence for M1 is documented with an assessment of the constancy assumption, the M2 judgment is owned by the clinical team and expressed in clinical terms, and the margin is endorsed—in writing—by the person who will defend it at the regulatory meeting.

Sample size can now be calculated. Not before.

The characteristic mistakes of this chapter

Three failures recur across trial designs in the territory Chapter 2 covers.

The measure chosen for its familiarity, not its fit. The hazard ratio because this is oncology. The odds ratio because logistic regression is what the statistician knows. The mean difference because continuous outcomes always use mean differences. Familiarity is not a justification. If the measure's assumptions do not hold for this trial, the result will be valid by the standards of the method and misleading by the standards of the science.

The margin borrowed without reasoning. The historical trial used a margin of 1.25 on the hazard ratio scale, so this trial will too. The related indication used a 10% absolute risk difference margin, so this indication will use the same. The margin is a number without a referent—it cannot be derived from the current trial's estimand, cannot be connected to the historical comparator's evidence, and cannot be defended as the product of a clinical judgment about acceptable inferiority. It is precedent masquerading as reasoning.

The efficiency-interpretability choice deferred to the label negotiation. The primary measure was chosen for regulatory purposes; the clinical measure for prescriber communication was not pre-specified. At the label negotiation, the sponsor discovers that the result in the primary measure supports a modest label claim, but the absolute effect is either large (a favorable surprise) or small (an uncomfortable one) in ways that were not anticipated. The negotiation occurs at a disadvantage because the clinical communication strategy was not designed—it was improvised from the data.

These failures share an origin: the effect measure decision was treated as a consequence of endpoint type, not as a design choice requiring the same deliberate reasoning as the estimand or the randomization scheme.

What cannot be recovered

Some decisions made badly in Chapter 2 can be addressed at the analysis stage, with cost. Some cannot.

An effect measure whose assumptions are clearly violated—non-proportional hazards when a hazard ratio is the primary measure, evidence of substantial heterogeneity in the treatment effect when a mean difference is the primary measure—can sometimes be supplemented with alternative analyses. The supplement does not become the primary; it provides context for a primary result whose limitations are now documented. This is honest science. It is also a worse position than having chosen the right measure at design.

A non-inferiority margin that was not justified at design cannot be justified after the trial. The reasoning chain—M1 from historical data, constancy assumption evaluated, M2 owned by the clinical team—must exist before the trial begins. After the trial, the reasoning cannot be developed without the suspicion that it is being tailored to the result. The agency knows this. The question "why this margin?" cannot be answered after unblinding without raising questions about whether the answer is being chosen because it supports a favorable interpretation.

The secondary effect measures for clinical communication cannot be claimed as confirmatory evidence if they were not pre-specified as confirmatory. If the absolute risk reduction turns out to be impressive and it was not pre-specified as a secondary endpoint with a clear analysis plan, it is a descriptive statistic. It can be reported. It cannot be claimed.

Chapter 2's errors tend to look like analysis choices. They are design choices that happened to remain invisible until the analysis.

The connection to what follows

Chapter 3 asks what the trial is committing to through its sample size calculation. Every assumption in the sample size—the expected effect size, the expected variance, the expected event rate, the expected dropout rate—is a prediction about what the trial will find. Getting these predictions wrong does not mean the trial fails statistically; it means the trial may be underpowered to detect a real effect, or overpowered and wasteful, or powered for an effect measure that does not align with the primary analysis.

All of this depends on the effect measure settled in this chapter. The variance of a risk difference is different from the variance of a log odds ratio. The number of events required for a hazard ratio analysis differs from the person-time required for an RMST analysis. The effect size assumed for an NI margin-based power calculation is expressed in the units of the primary effect measure—and if the units change between Chapter 2 and Chapter 3, the calculation is incoherent.

Chapter 3 also inherits this chapter's ownership structure. The clinical team that owns the estimand and the effect measure also owns the assumed effect size in the sample size calculation. That assumption is not a statistical parameter; it is a clinical prediction about what the treatment will do, in this population, at this time point, under these trial conditions. If the statistician owns it alone, and it turns out to be wrong, the trial has failed without clear accountability. If the clinical team owns it alongside the statistician, the responsibility is distributed correctly—and the assumption is more likely to reflect genuine clinical expectation rather than the number that makes the sample size convenient.

Chapter 2 risk summary

The decision this chapter owns: which effect measure will express the treatment difference, and—for non-inferiority designs—what magnitude of inferiority is acceptable?

The most common mistake: treating effect measure selection as a downstream consequence of endpoint type, and borrowing non-inferiority margins from historical trials or related indications without deriving them from first principles for the current trial.

The professional-level risk: the non-inferiority margin that cannot be defended. Not because the number is wrong, but because the reasoning behind it was never developed—the M2 judgment was never made explicitly, the constancy assumption was never assessed, and the clinical ownership of "acceptable inferiority" was never established. At the regulatory meeting, the question is asked. The statistician has a number. No one has a reason.

Chapter 3: What Are We Committing To?

The question Chapter 2 leaves open

Chapter 2 settled the effect measure. It named the primary summary statistic, connected it to the estimand, stated its assumptions, and—for non-inferiority designs—derived the margin from first principles. What it did not answer is: how large does the trial need to be?

That question sounds technical. It is not. It is the question of what the trial is willing to commit to believing before the data are seen.

A sample size calculation is not a formula. It is a prediction—about the size of the treatment effect, the variability of the outcome, the rate at which events accumulate, the proportion of patients who will complete the trial. Every assumption in the calculation is a claim about what the trial expects to find. When the assumptions are right, the trial has adequate power. When they are wrong, the trial may be underpowered and unable to detect a real effect, or overpowered and wasteful of patients and resources, or misspecified in ways that invalidate the power claim entirely.

The commitments embedded in a sample size calculation are not symmetric in their consequences. Underestimating the treatment effect produces an underpowered trial—a trial that may miss a real benefit and expose patients to the intervention and the control condition without generating a conclusive answer. Overestimating the treatment effect produces a false precision—a trial designed to detect a magnitude of effect that was never plausible, powered for a question that was never the right one. Both failures are expensive. Both are common. And both are attributable to assumptions that were not examined carefully at design.

This chapter is about those assumptions—about what the trial is committing to when it commits to a sample size.

The decision structure of this chapter

The sample size calculation contains four distinct categories of assumption, each corresponding to a section of this chapter.

Section 3.1 — Effect Size Ownership addresses the assumption that is simultaneously the most consequential and the most contested: the expected treatment effect. This is the effect size the trial is designed to detect—the magnitude of benefit that, if real, the trial will have adequate power to confirm. The assumed effect size is a prediction. It may be informed by prior trials, biological reasoning, clinical experience, or dose-ranging studies. But it is always, ultimately, a prediction, and predictions are wrong at a rate that should be uncomfortable for anyone who has looked at the history of phase III clinical trials. This section asks who should make this prediction, on what basis, and what it means to own it.

Section 3.2 — Variance and Dropout addresses the assumptions about outcome variability and patient retention that interact with the effect size assumption to determine

the required sample size. For continuous outcomes, the variance of the primary measure must be estimated. For time-to-event outcomes, the hazard rate in the control arm must be estimated. For binary outcomes, the baseline event probability must be estimated. Each of these estimates carries uncertainty, and each uncertainty compounds with the others. The section examines how these uncertainties propagate through the sample size calculation and what happens when they are wrong.

Section 3.3 — Event Rate Uncertainty focuses specifically on the challenge of estimating event rates in time-to-event trials—a problem that deserves separate treatment because it combines baseline rate uncertainty with follow-up duration assumptions, recruitment pace assumptions, and censoring assumptions in ways that are multiplicatively, not additively, wrong when they fail together.

Section 3.4 — Power as Risk Budget reframes the power calculation. Power is conventionally described as the probability of detecting a real effect. This section reframes it as a budget—a statement of the risk the trial is willing to accept of missing a real effect. That budget is allocated across the design: more power here means fewer resources there, higher power means larger trial means longer enrollment means later evidence for patients who need it. Power is not a target to be maximized. It is a resource to be allocated.

What unites the chapter

The sections of this chapter are unified by a single theme: the sample size calculation is a public commitment, and its assumptions must be owned by the people who make them.

In current practice, sample size calculations are most commonly produced by statisticians and presented to clinical and regulatory teams as deliverables. The clinical team may review the assumed effect size—sometimes—and accept or challenge it. The assumptions about variance, event rates, and dropout are rarely examined by anyone outside the biostatistics function. The power level is set at 80% or 90% by convention, with minimal examination of whether those levels are appropriate for this trial, this indication, and this decision context.

This allocation of responsibility is backward. The effect size assumption is a clinical prediction about what the treatment will do. The event rate assumption is a clinical prediction about what will happen to control arm patients over the trial's follow-up. The dropout assumption is a clinical prediction about patient behavior in the trial context. These are not statistical inputs; they are clinical judgments that have been delegated to the statistician because they must be expressed as numbers, and the statistician is the person who works with numbers.

When these assumptions turn out to be wrong—and many of them will be wrong—the trial either fails or is extended or is redesigned. The accountability for those consequences belongs to the people who made the predictions, not just the people who translated the predictions into a power calculation. When those people are identified as "the statisticians," the accountability structure is wrong, and the incentives for careful prediction are weak.

This chapter asks for correct accountability. Not as an organizational reform, but as a design discipline: every assumption in the sample size calculation should be traceable to a person who made it, a rationale that supports it, and an acknowledgment of what happens if it is wrong.

The connection to what surrounds this chapter

Chapter 3 sits between the definition of what is being measured (Chapter 2) and the question of when the trial might stop early (Chapter 4). Its position is not incidental.

The connection to Chapter 2 is direct: the effect measure determines the form of the power calculation. The variance structure of a risk difference is not the same as that of a log hazard ratio. The number of events required for adequate power under a hazard ratio analysis is not the same as the sample size required for an RMST analysis at a specified time horizon. If the effect measure was not settled in Chapter 2, the sample size calculation in Chapter 3 cannot be correctly specified.

The connection to Chapter 4 is also direct: the sample size determines the context in which interim analyses are planned. An interim analysis at 50% of the target information fraction means something different in a trial with 200 events than in a trial with 800 events. The operating characteristics of an interim stopping rule—the probability of stopping early under the null, the probability of stopping early when the treatment effect is real—depend on the total information the trial is designed to accumulate. If that total is misspecified, the interim analysis plan is misspecified along with it.

These connections are sequential and irreversible. Getting Chapter 3 wrong does not just waste resources. It distorts Chapter 4. And a distorted Chapter 4 produces stopping rules that behave differently than planned—which is exactly the kind of design failure that is most difficult to explain after the fact.

What this chapter is not about

This chapter does not provide formulas for sample size calculation. Those are available in every statistical textbook and in every sample size software package. This chapter is not about how to calculate a sample size. It is about what the calculation is for, what it commits the trial to, and what happens when the commitments are not examined.

It is also not about adaptive sample size re-estimation, which is addressed in Chapter 7. The fixed sample size design—in which the total sample size is determined before enrollment begins and does not change in response to interim data—is the baseline against which adaptive

designs are evaluated. Understanding what the fixed design commits to is prerequisite to understanding why and when adaptation is warranted.

The question this chapter must answer

By the end of this chapter, the following must be documented and owned.

The assumed effect size—the magnitude of treatment benefit the trial is designed to detect—is specified in the units of the primary effect measure, derived from sources that are identified and examined for their applicability to this trial, and owned by the clinical team as a clinical prediction about what the treatment will do.

The nuisance parameters—variance for continuous outcomes, baseline event rates for time-to-event and binary outcomes, expected dropout rates—are specified, with their sources identified and their uncertainty quantified. Where the uncertainty is large, its implications for the sample size calculation are documented.

The power level is justified—not merely stated. The justification connects the chosen power to the decision context: why 80% and not 90%? Why 90% and not 80%? Who bears the consequences if the trial is underpowered?

These are not administrative requirements. They are the minimum needed to make the sample size calculation a commitment rather than a calculation—a statement of what the trial believes about its treatment and its patients, expressed as a number, owned by the people whose judgment produced it.

Chapter 2 defined how the difference will be expressed. Chapter 3 defines how much evidence will be generated to express it. Both are required before the trial can begin.

3.1 Effect Size Ownership

The assumption no one wants to own

Every sample size calculation begins with a number: the expected treatment effect. A hazard ratio of 0.75. A risk difference of 5 percentage points. A mean change from baseline of 2.1 units. This number is the effect size the trial is designed to detect—the magnitude of benefit that, if real, the trial will have adequate power to confirm.

This number is a prediction. It may be informed by prior trials, by dose-ranging studies, by meta-analyses of related treatments, by biological reasoning about mechanism, or by extrapolation from a different population or a different stage of disease. But it is always, in the end, a prediction about what a future trial will find—and predictions made before a trial are frequently wrong. In a 2016 analysis of phase III oncology trials, approximately 40% failed to confirm the primary hypothesis. In cardiovascular disease, where large trials have been run for decades and the prediction inputs are rich, underpowered trials are still common. The history of phase III development is substantially a history of effect size predictions that turned out to be optimistic.

This is not an indictment of clinical scientists or statisticians. Effect sizes are genuinely difficult to predict in advance. The patients enrolled in the definitive phase III trial are not the same as the patients in the earlier trials from which the effect size was borrowed. The control arm event rates differ. The background therapy differs. The diagnostic criteria differ. The effect size assumption absorbs all these differences and converts them into a single number that is then treated as though it were a known quantity in a formula.

The question this section addresses is not how to make better predictions—though that is worth pursuing. It is who should make the prediction, what basis it should rest on, and who should be accountable when it is wrong.

Where effect size assumptions come from

In practice, the assumed effect size comes from one of four sources, each with characteristic strengths and weaknesses.

Prior trials of the same treatment. Phase II results, dose-ranging studies, or earlier phase III trials in related populations provide the most direct evidence about what the treatment

does. The problem is that prior trials are subject to selection bias in both directions: positive results are more likely to proceed to phase III, inflating the apparent effect size, while negative results in closely related populations may not be weighted appropriately if the design team is committed to a specific development hypothesis. The phase II to phase III attrition rate reflects, in part, the systematic overestimation of effect size at earlier stages of development.

Historical data from the control arm. For trials comparing a new treatment to an active comparator, the expected control arm outcome—the baseline against which the treatment effect is measured—is often derived from historical trials of the comparator. This estimate is subject to the same concerns about population differences and treatment context evolution that affect the non-inferiority margin. Control arm event rates in historical trials may not reflect what will happen in the current trial, particularly if standard-of-care background therapy has changed substantially.

Expert clinical judgment. When quantitative data are insufficient, the expected effect size may be based on the clinical team's assessment of what the treatment should do, given its mechanism of action and the nature of the disease. Expert judgment is a legitimate input, but it is also the source most likely to reflect optimism bias—the systematic tendency for people developing a treatment to believe it will work better than the evidence strictly supports. Expert judgment should be documented, subjected to independent review, and connected to a specific biological or clinical rationale that can be evaluated by others.

The minimum clinically important difference. Rather than predicting what the treatment will do, the trial can be powered to detect the smallest treatment effect that would be clinically meaningful. This approach avoids the prediction problem by asking a different question: if the treatment produces at least this much benefit, is that enough to change practice? Powering for the minimum clinically important difference is conservative—it may require a larger trial than powering for the expected effect—but it produces a design that is coherent in a specific way: a positive trial result will always be clinically meaningful, because the minimum meaningful threshold was the design target.

Each of these sources has a place. In most trials, the effect size assumption draws on all four, and the design team must make a judgment about how to weight them. The judgment should be documented—not as a sensitivity analysis note in the SAP, but as a primary design rationale that can be reviewed and challenged before the trial begins.

The optimism problem

There is substantial empirical evidence that assumed effect sizes in sample size calculations are systematically optimistic. The mechanism is not hard to understand: the people designing the trial are also the people most committed to the hypothesis that the treatment works. They have invested in the development program. They have seen the early efficacy signals. They believe in the mechanism. These are not disqualifying biases; they are the natural

disposition of a team doing its job. But they are biases, and they push the assumed effect size upward.

The consequences of optimism are asymmetric. A trial powered for an effect size of 30% relative risk reduction that finds a 20% relative risk reduction is underpowered—it may not reach statistical significance even if the treatment genuinely works. This outcome is damaging in multiple ways: the treatment may be abandoned or delayed, patients who might have benefited from early approval do not receive it, and the sponsor must either run an additional trial or accept a negative result that does not reflect the treatment's actual benefit. An underpowered trial is not a failed trial; it is a trial that asked a question too boldly and could not answer it.

The correction for optimism is calibration—comparing assumed effect sizes in prior trials in similar indications to the effect sizes those trials actually found, and using that comparison to apply a systematic discount to the current assumption. This approach is rarely used in practice, because it requires a database of prior assumptions that are not routinely published, and because it produces a larger trial than the uncorrected assumption would require. But the evidence supports it: in indications where calibration data are available, the calibrated assumption has been closer to the realized effect than the uncalibrated one.

The more common correction is scenario analysis: the sample size is calculated for two or three effect size assumptions—an optimistic assumption, a central assumption, and a conservative assumption—and the trial size is chosen based on a judgment about which scenario the design team is willing to commit to. This approach at least makes the uncertainty explicit. It does not resolve the optimism problem, but it converts an unexamined assumption into an examined range with an explicitly chosen operating point.

Who should own the effect size assumption

The effect size assumption is a clinical prediction. It is a claim about what the treatment will do—how much better, on the primary endpoint, the treatment arm patients will do compared to control arm patients, in the population defined by the estimand. This prediction is not statistical. It is clinical and biological. The statistician translates it into a power calculation, but the translation is not the judgment.

The judgment belongs to the clinical team: the physicians, the clinical scientists, the disease-area experts who understand what this treatment does in this disease, based on the available evidence. They should be the ones to propose the assumed effect size, to defend it against scrutiny, and to acknowledge its uncertainty. When the statistician proposes the effect size because the clinical team did not engage with the question, the prediction is made without the judgment of the people most capable of making it—and the accountability for the prediction's accuracy is assigned to the wrong party.

This is not merely an organizational preference. It has practical consequences. A clinical team that owns the effect size assumption will examine it more carefully—will ask whether it

is consistent with the mechanism, whether it matches the phase II data, whether it assumes something about the patient population that is actually true. A clinical team that accepts the statistician's proposed assumption without engagement will not. And when the trial misses its power calculation—when the observed effect is smaller than assumed—the question of who made the wrong prediction will matter for whether the program continues and on what terms.

Ownership also matters for documentation. A regulatory agency reviewing a sample size justification wants to know: where did this assumed effect come from, and why is it credible? A justification that says "based on prior literature, a hazard ratio of 0.75 was assumed" is not a justification. A justification that says "based on the following three trials in comparable populations, with the following analysis of their effect size estimates and the following reasoning about the differences between those trials and the current design, the central expected hazard ratio is 0.75, with a plausible range of 0.70 to 0.82"—that is a justification, and it is the clinical team's document, not the statistician's.

The clinically meaningful effect versus the expected effect

There is a design choice embedded in the effect size assumption that is rarely made explicit: should the trial be powered for the expected treatment effect, or for the minimum clinically meaningful treatment effect?

These are different targets, and they produce different trial designs.

Powering for the expected effect produces the smallest trial that has adequate probability of detecting what the treatment is actually expected to do. It is efficient. It is also contingent: if the expected effect turns out to be optimistic—if the actual treatment effect is smaller than assumed—the trial will be underpowered, and the result may be inconclusive even if the treatment genuinely works.

Powering for the minimum clinically meaningful effect produces a larger trial that can detect any effect large enough to be worth using the treatment. It is conservative. It has the property that if the trial succeeds, the result is clinically meaningful by construction—the effect was at least as large as the minimum useful threshold. It has the cost that if the expected effect is substantially larger than the minimum clinically meaningful threshold, the trial is larger than it needs to be, enrolling more patients than required to answer the question.

The choice between these two approaches depends on the consequences of an underpowered trial versus the consequences of a larger trial. In indications where an underpowered trial would delay access to a beneficial treatment for patients with serious unmet need, the cost of underpower is high, and a more conservative sample size may be justified. In indications where the treatment is adjunctive and the clinical decision can be revisited with additional trials, the cost of underpower is lower. In indications where the minimum clinically

meaningful effect is not well-established—where the profession has not reached consensus on what magnitude of benefit would change practice—powering for the minimum meaningful effect requires resolving that prior question, which is often not done.

Both choices are defensible. Neither should be made by default. And whichever is made should be documented: we are powering for the expected effect because X, or we are powering for the minimum meaningful effect because Y. The distinction affects how the result is interpreted if the trial barely misses statistical significance, and how it is interpreted if it succeeds by a narrow margin.

Sensitivity of power to effect size assumption

Power is highly sensitive to the effect size assumption, and this sensitivity is systematically underappreciated in trial design.

For a typical superiority trial, halving the assumed effect size approximately quadruples the required sample size. This is a consequence of the squared relationship between effect size and required sample size in most power formulas—a relationship that means small errors in the assumed effect size translate into large errors in the required sample. An effect size assumption that is 20% too optimistic produces a trial that is approximately 40-60% underpowered for the true effect—a trial that will fail to reach statistical significance even when the treatment genuinely works at the true magnitude.

This sensitivity should prompt scenario analyses in every trial design. What is the power if the true effect is 20% smaller than assumed? 30% smaller? If the power at the pessimistic scenario is unacceptable—below 60%, say—the design team must either increase the assumed sample size (by assuming a smaller effect), accept the risk of underpower, or plan a pre-specified sample size re-estimation to address uncertainty prospectively. All three options have costs. The decision among them is a risk allocation decision that the design team must make explicitly.

What it should not be is a decision made by ignoring the sensitivity—by reporting power at the central assumed effect size and moving on to the next design decision without examining the power curve. The power at the central assumption is a single point on a distribution of possible outcomes. The full distribution is what the trial is actually committing to, and the full distribution should be examined before the commitment is made.

What this section demands before proceeding

Before the variance and dropout assumptions of Section 3.2 can be addressed, the effect size assumption must be locked—not as a final determination that will never be revisited, but as a documented commitment to a specific prediction with a specific rationale.

The lock requires: a specific numerical value in the units of the primary effect measure, the sources from which that value was derived with an assessment of each source's applicability to the current trial, an explicit acknowledgment of the optimism risk and any calibration or discount applied to address it, the name of the clinical team member who owns the prediction, and a power curve that shows power across a plausible range of true effects.

Without this, the variance and dropout assumptions of Section 3.2 will be calculated to a sample size whose effect size foundation has not been examined. The numbers will compound. The result will look precise. The precision will be false.

References: Ioannidis, "Why Most Published Research Findings Are False," PLoS Med 2005; Button et al., "Power Failure: Why Small Sample Size Undermines the Reliability of Neuroscience," Nat Rev Neurosci 2013; Altman, "Statistics and Ethics in Medical Research: III. How Large a Sample?" BMJ 1980; Schulz and Grimes, "Sample Size Calculations in Randomised Trials: Mandatory and Mystical," Lancet 2005.

3.2 Variance and Dropout

The assumptions that compound quietly

The effect size assumption is the one assumption in the sample size calculation that the clinical team usually notices. It is discussed, challenged, sometimes revised. It is the number whose magnitude the sponsor cares about and whose plausibility the regulator will scrutinize.

The variance and dropout assumptions are different. They are the assumptions that compound quietly—accepted without examination, translated directly from historical estimates or rule-of-thumb conventions into the sample size formula, and then forgotten. They are also the assumptions that are most often wrong in ways that are predictable and preventable.

For continuous outcomes, the critical nuisance parameter is the within-group variance of the primary measure—the variability of the outcome across patients receiving the same treatment. For time-to-event outcomes, the critical nuisance parameters are the baseline event rate in the control arm and the follow-up duration. For binary outcomes, it is the baseline event probability. In all designs, the dropout rate—the proportion of enrolled patients who will not complete the primary endpoint assessment—acts as a multiplier on the required sample size.

Each of these parameters is estimated before the trial. Each estimate can be wrong. And when multiple estimates are simultaneously wrong in the same direction—variance higher than assumed, dropout greater than assumed, control arm event rate lower than assumed—the compounding error can reduce the effective power of the trial from 90% to below 50% without anyone noticing until the interim analysis or the final database lock.

Variance: the number that hides heterogeneity

For continuous outcomes, the variance of the primary endpoint determines how much of the observed difference between arms is signal and how much is noise. A large variance means that individual patient outcomes vary widely, even within a treatment arm, making it harder to detect the difference between arms. A small variance means that patient outcomes cluster tightly around the arm mean, making differences easier to detect.

The variance assumption is typically borrowed from prior studies—the same endpoint, similar populations, comparable treatment contexts. This borrowing is reasonable as far as it goes.

Its limitation is that variance is not a stable property of an endpoint; it is a property of an endpoint in a specific patient population at a specific stage of disease with specific measurement conditions. When the current trial's population differs from the prior studies in ways that affect outcome variability—younger or older, more or less heterogeneous in disease severity, measured more or less frequently—the borrowed variance may be wrong.

A more subtle problem is that variance estimates from prior studies reflect the realized variability under those studies' conditions, which may have included informative censoring, differential dropout, or endpoint measurement protocols that differ from the current design. These differences are invisible in the aggregate variance estimate but can substantially affect the variance in the new trial.

The practical consequence is that the variance assumption in the sample size calculation should be accompanied by a range—a plausible lower and upper bound—and the sample size should be evaluated at both extremes. If the power at the upper bound of the variance range is unacceptable, the design is fragile with respect to this assumption, and either the sample size should be increased or a sample size re-estimation should be planned prospectively.

What the variance assumption should not be is the point estimate from the closest available historical study, accepted without examination and translated directly into the formula. Point estimates from historical studies have confidence intervals. Those confidence intervals reflect uncertainty that will be realized, one way or another, in the current trial.

The dropout problem

Dropout—the failure of enrolled patients to complete the primary endpoint assessment—is the assumption in the sample size calculation that is most reliably underestimated and most consequentially wrong.

The mathematics of dropout is straightforward. If 20% of enrolled patients are expected to drop out before the primary endpoint, the effective sample size is 80% of the enrolled sample. To achieve the planned power with a 20% dropout rate, the trial must enroll 25% more patients than the power calculation without dropout adjustment would require. This adjustment is standard and well-understood.

What is less well-understood is that dropout assumptions are routinely optimistic, for the same reasons that effect size assumptions are routinely optimistic—the people making the assumption are the people most committed to the trial's success, and a higher assumed dropout rate means a larger, more expensive, more time-consuming trial.

The available data on dropout rates in clinical trials suggests that actual dropout rates consistently exceed assumed rates, often by substantial margins. In chronic disease trials, where the treatment period is long and the patient burden is high, the discrepancy is particularly marked. A trial that assumes 10% dropout based on the experience of a shorter phase II study in a more motivated patient population may find 25% dropout in the

longer, more burdensome phase III context. This is not a random error; it is a predictable consequence of borrowing dropout assumptions from contexts that differ from the current trial in ways that matter.

Three sources of dropout deserve explicit examination at design. **Administrative dropout** is dropout that occurs because the patient moves, loses insurance, or is withdrawn at the site's discretion for reasons unrelated to treatment response. **Dropout due to adverse events** is withdrawal from study treatment because of tolerability problems, which may be related to the treatment and therefore informative about the treatment's effect. **Dropout due to lack of efficacy** is withdrawal because the patient is not improving and seeks alternative treatment, which is highly informative about the treatment's effect under a treatment policy estimand and confounding under a hypothetical estimand.

The distinction among these types of dropout matters for both the sample size calculation and the estimand. Administrative dropout is approximately uninformative; its frequency should be estimated from the trial logistics and patient population, and the sample size should be inflated accordingly. Dropout due to adverse events and lack of efficacy is potentially informative; under a treatment policy estimand, outcome data from these patients after dropout should be collected, which changes both the data collection plan and the effective sample size. If the dropout assumption treats all dropout as uninformative administrative loss, and a substantial portion of actual dropout is informative, the trial is both underpowered and using the wrong analysis strategy—two problems that compound each other.

Estimating control arm event rates

For time-to-event outcomes, the required number of events—not the required number of patients—is the primary driver of sample size. The number of events depends on the total follow-up time accumulated by enrolled patients, which depends on the control arm event rate, the recruitment pace, and the follow-up duration. These three quantities interact, and errors in any one of them affect the trial's power.

The control arm event rate is the most consequential of these three. If the event rate is lower than assumed, events accumulate more slowly, the trial takes longer to reach the target number of events, and—if the trial is terminated at a pre-specified calendar time rather than at a target event count—the trial will be underpowered. If the event rate is higher than assumed, events accumulate faster; the trial may reach the target events sooner, but the faster accumulation may reflect a different patient population than intended, with different prognostic characteristics.

Control arm event rates are estimated from historical data—prior trials, registries, natural history studies. The fundamental challenge is that event rates in historical controls are often not transferable to the current trial because standard-of-care background therapy has improved. A cardiovascular prevention trial designed in 2010 using control arm event rates from 2000-era trials will enroll patients who are receiving more intensive statin therapy,

better blood pressure management, and more aggressive lifestyle intervention than the historical controls received. Their event rates will be lower. The trial will take longer to accumulate the target events, will enroll longer than planned, and may find that the anticipated absolute benefit of the new treatment is smaller than assumed because the competing risk reduction from background therapy has compressed the opportunity for the new treatment to demonstrate incremental benefit.

This is not a hypothetical concern. It has been observed repeatedly in cardiovascular outcome trials conducted over the past two decades, where declining control arm event rates have extended trial timelines and reduced the absolute effect sizes observed, even when the relative effects were as expected. The design team must examine the trend in control arm event rates over time in the relevant patient population, not just the point estimate from the most recent available trial.

Interaction between dropout and event rate assumptions

In time-to-event trials, dropout and event rate assumptions interact in a way that is not always appreciated in the sample size calculation.

Informative censoring—dropout that is associated with the likelihood of having an event—reduces the information content of the censored patients and changes the effective hazard ratio the trial is estimating. If patients who are not responding to treatment are more likely to drop out, the observed event rate in the treatment arm will be lower than it would have been if all patients had remained on treatment, because the non-responders who would have had events are no longer being followed. This produces an apparent treatment effect that overestimates the true effect—and it is not corrected by the standard sample size formula, which assumes non-informative censoring.

The sample size calculation for a time-to-event trial that expects substantial informative dropout should include a sensitivity analysis that accounts for the informative censoring mechanism. This is not a common component of standard sample size packages, and it requires explicit modeling assumptions about the relationship between dropout propensity and event risk. But in trials where informative dropout is expected—oncology trials where disease progression leads to treatment switches, chronic disease trials where lack of efficacy leads to withdrawal—the standard calculation may be materially misleading.

The compounding error

The central concern of this section is not any single misspecification of variance, dropout, or event rate. It is the compound effect of multiple simultaneous misspecifications in the same direction.

In a trial where variance is 20% higher than assumed, dropout is 10 percentage points higher than assumed, and the control arm event rate is 15% lower than assumed, the effective power may fall from 90% to below 60%—a catastrophic loss that was not visible in any single assumption but is the product of three independently plausible errors. This kind of compound misspecification is not rare. It is the normal outcome of sample size calculations that borrow assumptions from multiple sources, each reasonable in isolation, without examining their joint behavior.

The guard against compound misspecification is scenario analysis that jointly varies the key assumptions. Not three separate univariate sensitivity analyses—each informative but each showing only the effect of one wrong assumption—but a joint scenario analysis that asks: if variance is at the upper end of the plausible range, dropout is at the upper end, and event rate is at the lower end, what is the power? If the answer to that question is unacceptable, the design is fragile in a way that needs to be addressed—by a larger sample size, by a pre-specified re-estimation rule, or by acknowledging the fragility explicitly and accepting the associated risk.

Joint scenario analysis is more work than univariate sensitivity analysis. It is also the only analysis that is honest about the actual distribution of outcomes the trial design is committing to. A design that looks robust in isolation but collapses under joint pessimism is not a robust design. It is a design that has not been examined carefully enough.

What this section demands before proceeding

Before Section 3.3 addresses event rate uncertainty in detail, the variance and dropout assumptions must be documented with the same care as the effect size assumption.

The variance estimate requires a source, an applicability assessment, a plausible range, and a power evaluation at both ends of the range. The dropout estimate requires a breakdown by dropout type—administrative, adverse event, lack of efficacy—with separate estimates and sources for each type. The control arm nuisance parameter—event rate, baseline probability, or variance, depending on the endpoint type—requires a trend analysis over time in the relevant patient population, not just a point estimate from the nearest available study.

And the joint scenario analysis—varying all three categories of nuisance parameter simultaneously to their pessimistic values—must be conducted and its result documented. If the power at the joint pessimistic scenario is acceptable, the design is robust. If it is not,

the design is fragile, and the fragility is a design decision that must be acknowledged and addressed before enrollment begins.

References: Friedman et al., Fundamentals of Clinical Trials, 5th ed. (2015); Lachin, "Introduction to Sample Size Determination and Power Analysis," Control Clin Trials 1981; Proschan, "Two-Stage Sample Size Re-Estimation Based on a Nuisance Parameter," J Biopharm Stat 2005; Hernán, "The Hazard of Hazard Ratios," Epidemiology 2010.

3.3 Event Rate Uncertainty

Why event rates deserve a section of their own

Section 3.2 introduced event rate uncertainty as one of several nuisance parameters that compound quietly in the sample size calculation. This section treats it separately, because time-to-event designs are the dominant paradigm in the indications—cardiovascular disease, oncology, infectious disease—where the largest and most consequential trials are run, and because event rate uncertainty in these designs has a distinctive structure that is not captured by the general discussion of nuisance parameters.

In a continuous outcome trial, a wrong variance assumption produces an incorrect sample size. In a time-to-event trial, a wrong event rate assumption produces incorrect sample size and incorrect trial duration simultaneously. These two errors interact. A trial that is too small and too short cannot be rescued by running it longer; a trial that is the right size but based on an overly optimistic event rate will run longer than planned, recruiting beyond the originally projected enrollment completion, consuming resources and delaying the result. The cascade from a single wrong assumption propagates through the entire trial conduct timeline.

The second reason for separate treatment is that event rate assumptions embed multiple independent sub-assumptions, each of which can be wrong independently. The total number of events the trial needs depends on: the baseline event rate in the control arm, the proportion of events attributable to the cause under study versus competing causes, the follow-up duration per patient, the enrollment pace, and the censoring pattern. Getting the baseline rate right but the enrollment pace wrong produces the same kind of under-event problem as getting the event rate wrong. The trial ends up at the planned calendar date with fewer events than planned, facing either an extension or an underpowered primary analysis.

The declining control arm event rate

The most consequential and most predictable form of event rate misspecification is the failure to account for secular trends in the background event rate.

In almost every major indication where randomized controlled trials have been conducted over multiple decades, the event rate in control arm patients has declined over time. In

cardiovascular prevention, the event rates observed in control arms of major trials conducted in the 1980s and 1990s were substantially higher than those observed in trials conducted in the 2010s, even in nominally similar patient populations. The reasons are well-understood: better background pharmacotherapy, more intensive management of risk factors, higher rates of revascularization, and improved secondary prevention practices have reduced the absolute risk that placebo-treated or standard-care patients face over a given follow-up period.

For trial design, the consequence is specific: a sample size calculation that uses event rates from trials conducted five or ten years earlier will systematically overestimate the event rate in the current control arm, leading to a trial that is adequately sized on paper but under-powered in practice because events accumulate more slowly than assumed.

This is not a theoretical concern. Several landmark cardiovascular outcome trials conducted in the 2010s—including trials mandated as post-approval commitments following accelerated approvals based on surrogate endpoints—enrolled substantially and unexpectedly lower control arm event rates than the power calculations assumed, leading to longer trial durations, larger enrolled sample sizes than originally planned, and in some cases primary results that were statistically underpowered despite correct recruitment of the target number of patients.

The correction is systematic analysis of secular trends. The design team should identify all available trials in the relevant patient population over a span of at least ten years, plot the control arm event rates over calendar time, and use the trend to project the expected event rate at the time of the current trial. If the trend is clearly downward, the sample size calculation should use a conservative (lower) event rate assumption rather than the most recent observed rate, because the trend is likely to continue during the years the new trial is being conducted.

Competing risks and cause-specific event rates

In many time-to-event trials, the primary endpoint is a specific type of event—cardiovascular death, cancer-specific survival, time to hospitalization—rather than all-cause mortality. When the primary endpoint is a specific event type, competing events—events that prevent the primary endpoint from occurring—affect the event rate in a way that is not always accounted for in the sample size calculation.

If the primary endpoint is cardiovascular death and patients are also at risk of non-cardiovascular death, patients who die from non-cardiovascular causes before experiencing a cardiovascular death are censored from the cardiovascular-specific analysis. In a trial with a high competing risk of non-cardiovascular mortality—for example, a trial in elderly patients with multiple comorbidities—a substantial fraction of enrolled patients will be censored due to competing events before reaching the cardiovascular endpoint. This reduces the effective event rate for the primary endpoint, requiring a longer follow-up or a larger sample to accumulate the target number of primary events.

Competing risk misspecification is most common in elderly populations, in indications where non-primary mortality is high, and in trials that focus on disease-specific rather than all-cause outcomes. The sample size calculation for such trials should include an explicit accounting for the competing risk—using cause-specific hazard models that separate the primary event rate from the competing event rate—rather than treating the primary endpoint as if it were the only event the patient could experience.

When competing risks are ignored, the sample size calculation overestimates the primary event rate (because it implicitly assumes that censored patients would have had the primary event if they had survived longer), leading to a trial that is undersized for the number of primary events it will actually observe.

Enrollment pace and the calendar time problem

A time-to-event trial accumulates events over calendar time as a function of the number of patients enrolled, the follow-up duration per patient, and the underlying event rate. These three quantities interact, and errors in any one of them affect when the trial will have accumulated the target number of events.

The enrollment pace assumption is systematically optimistic in clinical trials. In a 2015 analysis of industry-sponsored trials, actual enrollment rates were on average 30-40% below projected rates. The reasons include slower-than-anticipated site activation, competing trials competing for the same patient population, more restrictive eligibility criteria as applied in practice versus as written in the protocol, and seasonal variation in patient presentation. These are predictable sources of enrollment delay that should be reflected in the design's enrollment assumptions.

The interaction between enrollment pace and event accumulation is important and not always intuitive. In a trial with a target of 500 events and a planned enrollment of 2,000 patients over two years with three additional years of follow-up, an enrollment pace that takes three years instead of two does not simply extend the trial by one year. It changes the follow-up distribution across enrolled patients—later-enrolling patients have shorter follow-up, fewer of them accumulate the long-term follow-up that produces most events in a chronic disease trial, and the event rate per patient-year of observation may be lower because the later-enrolled patients have shorter observation time. The relationship between enrollment delay and event accumulation delay is superlinear, not linear.

Sample size simulation—rather than analytical power formulas—is the appropriate tool for quantifying this interaction. A simulation that models enrollment pace, event accumulation, censoring, and calendar time can produce realistic estimates of when the trial will reach its event target under a range of assumptions, including pessimistic ones. This kind of simulation is more informative than an analytical power calculation and should be the primary tool for event-driven time-to-event trial design.

The information fraction and what it actually means

In event-driven time-to-event trials, the "information fraction"—the proportion of the target events that have been observed at any given interim analysis—is the operational currency of the monitoring plan. An interim analysis at 50% information fraction is an analysis conducted when half the target events have been observed. An alpha-spending function that allocates significance boundaries as a function of information fraction assumes that this fraction is measured correctly.

This assumption fails when the total target information—the planned number of events—is wrong.

If the trial was designed for 500 events based on an event rate assumption that turns out to be too optimistic, and the trial will actually accumulate only 400 events over the planned follow-up period, the 50% information fraction interim analysis occurs when 200 events have been observed—but 200 events represent 50% of 400 (the actual target the trial can achieve), not 50% of 500 (the planned target). The alpha-spending function that was calibrated to the 500-event plan is now being applied to a 400-event reality. The resulting operating characteristics—the probability of stopping early under the null, the probability of stopping early when the effect is real—are not what was planned.

This is not a minor technicality. It means that a trial with a miscalibrated event rate assumption will have interim analysis operating characteristics that differ from what the design team planned and what the regulatory agency approved. The DSMB will make stopping recommendations based on a monitoring plan that no longer corresponds to the trial's actual informational structure.

The guard against this is prospective sample size re-estimation based on the observed event rate, conducted at an interim time point before the interim efficacy analysis, by an independent statistician who does not see the treatment arm-specific data. Section 3.4 discusses the broader context of adaptive sample size adjustments; the event rate-specific application is that the planned number of events should be revisited as the trial accumulates data on the control arm event rate, and adjusted before the primary interim analysis is conducted.

When the event rate assumption proves wrong: the options

When it becomes clear during a trial that the event rate is lower than assumed and the trial will not reach the target events within the planned follow-up period, the design team faces a constrained set of options, none of them costless.

Extend the follow-up. If the event rate is lower than assumed but the treatment effect is real and time-stable, extending the follow-up will eventually accumulate the needed events. This option is feasible when the trial infrastructure can be maintained and funding extended, but it delays the primary result, increases total cost, and may strain site and patient participation. It also requires regulatory notification and often a protocol amendment, which introduces additional scrutiny.

Enroll additional patients. If follow-up extension alone is insufficient—because the enrollment window is closing and later-enrolling patients will not have adequate follow-up—additional enrollment can add patient-time that accelerates event accumulation. This requires reopening enrollment, which requires regulatory and ethics approvals, and may require sites that have already been closed.

Accept a reduced target. The trial may be completed with a target event count that is lower than originally planned, accepting correspondingly lower power. This option is defensible when the observed event rate, while lower than assumed, still provides adequate information to detect a clinically meaningful effect, but it requires acknowledging that the trial was designed for a different operating characteristic than it will actually achieve.

Expand the primary endpoint. In some trials, the primary endpoint can be broadened—adding an additional component to a composite, or expanding the follow-up window for event ascertainment—to increase the event count without additional enrollment or extended calendar time. This option requires careful examination for its implications for the estimand: broadening the endpoint changes what the trial is estimating, and the change must be scientifically defensible and pre-specified (or at minimum, transparently post-hoc and not data-driven).

None of these options is attractive, and all of them are more expensive than getting the event rate assumption right at design. The practical lesson is that event rate assumptions deserve more scrutiny at design than they typically receive—specifically, that secular trend analysis, competing risk analysis, and enrollment pace simulation should be standard components of sample size planning for every event-driven time-to-event trial.

What this section demands before proceeding

Before Section 3.4 reframes power as a risk budget, the event rate assumption must have been examined with the following components:

A secular trend analysis of control arm event rates in comparable trials over the past ten or more years, with a projected event rate at the time of the current trial based on the observed trend. A competing risk analysis identifying the major competing events in the patient population, with estimates of their frequency and their effect on the primary event rate. An enrollment pace simulation that models the distribution of follow-up time across enrolled

patients under a range of enrollment rate assumptions, and translates that distribution into an expected event accumulation curve over calendar time.

The event accumulation curve should be examined at both the central and pessimistic scenarios. If the pessimistic curve does not reach the target event count within an acceptable calendar time, the trial design is fragile—either the sample size must increase, the follow-up must extend, or the target event count must be reduced, with each option's consequences explicitly acknowledged.

This is the work that prevents the options described in the previous section from becoming necessary. Not all of them—event rate uncertainty cannot be eliminated—but the ones that arise from assumptions that were not examined.

References: Pocock and Stone, "The Primary Outcome Is Positive — Is That Good Enough?" N Engl J Med 2016; Wittes and Lachin, "Monitoring a Clinical Trial with Adaptive Interim Analyses," Stat Med 2001; Lachin and Foulkes, "Evaluation of Sample Size and Power for Analyses of Survival with Allowance for Nonuniform Patient Entry, Losses to Follow-up, Noncompliance, and Stratification," Biometrics 1986; Fine and Gray, "A Proportional Hazards Model for the Subdistribution of a Competing Risk," JASA 1999.

3.4 Power as Risk Budget

What power actually is

Power is defined as the probability of rejecting the null hypothesis when the null hypothesis is false—the probability that the trial will detect a real treatment effect when one exists. In a trial with 80% power at the assumed effect size, there is an 80% probability that the trial will produce a statistically significant result, and a 20% probability that it will not, if the true treatment effect is exactly as assumed.

This definition is correct and familiar. It is also misleading in a specific way: it frames power as a property of the trial, when it is better understood as a choice the design team makes about risk.

The 20% complement of power is the type II error rate—the probability of a false negative, of failing to detect a real treatment effect. That 20% is not a law of nature. It is a design decision. The design team chose 80% power, which means they chose to accept a 20% probability of missing a real effect. They could have chosen 85%, or 90%, or 95%. Each choice has a cost—in sample size, in trial duration, in resources, in the time that passes before patients who might benefit from the treatment can receive it. The choice among these options is not statistical. It is a judgment about risk: how much risk of missing a real benefit is acceptable, and who bears the consequence of that risk?

Reframing power as a risk budget makes the design decision visible. Instead of "we need 80% power," the question becomes: "we are willing to accept a 20% risk of missing a real treatment effect, and we are assigning that risk to the patients who would benefit from early access to this treatment if it is real. Is that allocation acceptable?"

The asymmetry of error consequences

The two error types in hypothesis testing—type I (false positive) and type II (false negative)—have asymmetric consequences in clinical trial design, and the asymmetry is not always reflected in the conventional choices of alpha and power.

A type I error is approving or recommending a treatment that does not work. The consequences are resource misallocation—patients receive a treatment that provides no benefit, paying in cost and burden and opportunity cost—and erosion of evidence quality, if the

false positive enters the literature and influences clinical practice. In regulatory terms, a type I error is an unsafe approval, and it is treated as a serious failure. This is why alpha is conventionally set at a stringent 5% (one-sided) or 2.5%: the consequences of falsely claiming efficacy are severe, and the threshold for rejection of the null is correspondingly high.

A type II error is failing to detect or recommend a treatment that works. The consequences are delayed access—patients who would benefit from the treatment do not receive it—and lost investment in a program that was real but failed to demonstrate its reality. In regulatory terms, a type II error is a missed approval. It is treated as unfortunate but not as a failure of the regulatory system.

The conventional asymmetry—alpha of 5% (one-sided), power of 80%—implies that the consequences of a false positive are four times more serious than the consequences of a false negative. For many treatments in many indications, this asymmetry is defensible: a false approval exposes many patients to an ineffective treatment, while a false negative delays access for a definable patient population that can be served by subsequent trials or approved alternatives.

But the asymmetry is not always defensible. For treatments in serious diseases with no effective alternatives, the consequence of a false negative is not merely delay—it is denial of access to a beneficial treatment to a population that has no other option and may not survive long enough for a subsequent trial to be completed and approved. In this context, treating a type II error as four times less serious than a type I error is a value judgment that should be explicit, not embedded in a conventional formula.

The FDA's accelerated approval pathway and the EMA's conditional marketing authorization both reflect, in part, a recognition that in serious conditions with high unmet need, the acceptable type I/type II error asymmetry may be different from the conventional 5%/20% ratio. Designs in these contexts should consider whether the conventional power level reflects the actual consequences of each error type in the specific indication.

Power as a function of effect size: reading the curve

Power is commonly reported as a single number—"the trial has 80% power"—at the assumed effect size. This single number is the least informative representation of the trial's operating characteristics.

The power curve—the relationship between power and the true treatment effect, plotted across a range of possible true effects—is the informative representation. It shows not just what happens if the true effect equals the assumed effect, but what happens if the true effect is smaller, and what happens if it is larger.

A typical power curve for a superiority trial has the following shape: power is near the type I error rate (alpha) when the true effect is zero; power rises monotonically as the true effect increases; power approaches 100% when the true effect is large relative to the assumed effect.

The slope of the rise determines how sensitive the trial is to departures from the assumed effect.

For a design decision, the most informative parts of the curve are not the peak (power at the assumed effect, where the trial was designed to be adequate) but the flanks. Specifically:

The power at an effect 20-30% smaller than assumed. This is the region where a plausibly wrong effect size assumption would place the trial. If the power at a 25% downward departure from the assumed effect is below 50%, the trial is highly sensitive to an assumption that may be too optimistic, and the design should be reconsidered.

The power at the minimum clinically important difference. If the power at the minimum meaningful effect is below an acceptable threshold—below 60%, say—the trial risks being inconclusive even when the treatment produces a real and clinically meaningful benefit. This is the specific failure mode that powering for the minimum clinically important difference is designed to prevent.

These two readings of the power curve should be reported alongside the power at the central assumed effect for every sample size report. A design team that has not examined the power curve—that knows only the power at the central assumption—has not understood what the trial is committing to.

The 80% convention and when it is wrong

The 80% power convention has no statistical basis. It is a rule of thumb that has persisted because it represents a plausible balance between the costs of insufficient power and the costs of excessive sample size, given the conventional alpha of 5% (two-sided) or 2.5% (one-sided). In a world where sample sizes are constrained and most clinical trials are in indications where subsequent trials are feasible if the first one fails, 80% is a reasonable operating point.

In several specific contexts, 80% is the wrong choice.

Confirmatory trials with no alternative path. When the trial is the definitive evidence base for an indication—when the treatment will be approved or not based on this trial, and no subsequent trial is planned—the cost of a false negative is borne entirely by the patients who would have benefited from the treatment. In this context, the 20% type II error rate is a 20% probability of failing the patient population, and 90% or higher power may be appropriate.

Trials with high assumed effect sizes. When the assumed effect size is generous—based on optimistic extrapolation from early trials or biological reasoning—the power at a more conservative effect size may be inadequate even if the power at the central assumption is 80%. In this case, 80% power at the central assumption may correspond to 50% power at a more realistic effect size, and the 80% figure is misleadingly reassuring.

Trials where the null is not the relevant comparison. In non-inferiority designs, the relevant comparison is not whether the treatment effect is zero but whether it is within the non-inferiority margin. The power in an NI design is the probability that the trial correctly concludes non-inferiority when the treatment is in fact non-inferior. The conventional 80% applies here as well, but the consequences of a false negative—failing to conclude non-inferiority for a treatment that is genuinely non-inferior—are different from failing to detect a superior treatment effect, and the appropriate power level should be considered in that context.

Trials in pediatric populations or rare diseases. When the patient population is small and trial replication is difficult, the conventional 80% power may not be achievable at an acceptable sample size. Regulators and ethics committees have recognized this: the FDA's guidance on trials in rare diseases and the EMA's guidance on pediatric trials both acknowledge that lower power may be acceptable when the patient population is inherently limited. In these contexts, the design team should document the actual power achievable, the rationale for accepting that power, and the implications for evidence strength of the anticipated result.

Allocating the risk budget

The power level is not the only design parameter that allocates the risk of a false negative. Several other design choices affect the probability of detecting a real effect, and they should be understood as part of the same risk budget.

Alpha allocation. For a fixed total alpha, a more stringent alpha for the primary test reduces power. Designing a trial with a pre-specified subgroup analysis that claims a share of the alpha—even a small share—reduces the power of the primary test. The primary test's power should be calculated at the adjusted alpha level, not at the unadjusted 5% (two-sided), if alpha sharing is planned. This interaction between alpha allocation (Chapter 6) and power (Chapter 3) is often not computed at design, leading to a primary test that is less powerful than intended.

Interim analyses. Pre-specified interim efficacy analyses reduce the power of the final analysis through alpha spending. If the interim analysis uses 1% of the nominal 2.5% one-sided alpha, the final analysis is conducted at the remaining 1.5%—and the power of the final analysis at that adjusted level is lower than the power at the unadjusted level. The power reported in the sample size report should reflect the adjusted levels after interim analysis alpha spending, not the unadjusted level. This is a standard computation, but it is not always performed or reported.

Stratified analyses. Stratified analyses—adjusting for pre-specified stratification factors in the primary analysis—typically increase power by reducing unexplained variability. If stratification factors are pre-specified in the randomization but not included in the primary

analysis, the power gain from stratification is not captured. The sample size calculation should reflect the analysis that will actually be performed.

Missing data. The assumed dropout rate affects the effective sample size. If the dropout assumption is wrong—as it frequently is—the effective sample size is smaller than planned, and the power is lower than calculated. The power reported in the sample size report is conditional on the assumed dropout rate; the power at the realized dropout rate is lower and not reported until the trial is over.

Each of these adjustments reduces the operating power below the headline figure. A trial with 80% power before accounting for interim analyses, alpha sharing, and realistic dropout may have an actual power of 70% or below. This is the risk budget that the design team is actually committing to, and it should be calculated and acknowledged—not obscured by reporting only the most favorable number.

The power report as a design document

The sample size and power report should be treated as a design document, not a technical appendix. It should include:

The assumed effect size, with sources, a plausible range, and the rationale for the chosen operating point. The power at the central assumption, at the lower end of the plausible effect size range, and at the minimum clinically important difference. The nuisance parameter assumptions—variance, dropout, event rate—with their sources and ranges. The power at joint pessimistic assumptions. The power at the adjusted alpha level accounting for interim analyses and alpha sharing. The rationale for the chosen power level—why this power, for this trial, for this patient population.

A power report that contains only "assuming a hazard ratio of 0.75 with 90% power at a one-sided alpha of 0.025, the required number of events is 312" has not done the job. It has produced a number. The design team needs a document that makes the risk budget explicit, assigns accountability for the key assumptions, and shows what happens when the assumptions are wrong.

That document is what a regulator reviewing the design rationale, a DSMB reviewing the monitoring plan, or a clinical team deciding whether to commit resources to the trial needs to see. It is not supplementary to the design. It is the design.

References: Cohen, Statistical Power Analysis for the Behavioral Sciences, 2nd ed. (1988); Lakens et al., "Justify Your Alpha," Nat Hum Behav 2018; Wittes, "Sample Size Calculations for Randomized Controlled Trials," Epidemiol Rev 2002; Altman and Bland, "Statistics Notes: Absence of Evidence Is Not Evidence of Absence," BMJ 1995.

3.5 Closing: The Commitment Behind the Number

What this chapter asked

Chapter 3 asked one question: what are we committing to?

It asked that question in four registers, each exposing a different layer of the commitment embedded in the sample size calculation.

Section 3.1 asked who owns the effect size assumption—the most consequential prediction in the entire design, and the one most subject to optimism bias, borrowed evidence, and unchallenged convention. The answer is the clinical team, not the statistician. When it is owned by the statistician, it is produced without the judgment that the prediction requires. When it is owned by the clinical team, it is examined with the care that a prediction of this consequence deserves.

Section 3.2 asked what happens when variance and dropout assumptions are wrong simultaneously—not just individually, which is bad enough, but in the same direction at the same time, which is how assumptions tend to fail in practice. The compound error of joint pessimistic realizations is the most common and least examined source of power loss in clinical trials. The correction is not more careful individual assumptions, though that helps; it is joint scenario analysis that explicitly evaluates the power at the boundary of what is plausible.

Section 3.3 asked why event rate uncertainty deserves its own section in a chapter about commitments. The answer is that in event-driven time-to-event trials, a wrong event rate assumption does not just affect power—it affects trial duration, enrollment requirements, DSMB operating characteristics, and the informational structure of the interim analysis. The secular trend problem—the systematic failure to account for declining background event rates—is not a subtle statistical concern. It is a predictable and preventable source of trial failure that has been documented repeatedly and still recurs.

Section 3.4 reframed power as a choice about risk, not a technical parameter. The 20% type II error rate that conventional 80% power accepts is a design decision about who bears the consequence of missing a real treatment effect. That decision should be explicit—connected to the consequences of a false negative in this indication, for this patient population, given the availability of alternatives—rather than defaulting to convention.

What this chapter decided

By the end of this chapter, six things must be documented.

The assumed effect size is specified in the units of the primary effect measure. It is connected to identified sources, assessed for optimism relative to those sources, and owned by a member of the clinical team who can defend it in a design review or regulatory meeting.

The power curve is reported—not just the power at the central assumption but the power at the lower end of the plausible effect size range and at the minimum clinically important difference. The curve shows the design's sensitivity to a wrong effect size assumption, and that sensitivity has been examined.

The variance, dropout, and event rate assumptions are documented with sources and plausible ranges. Each has been individually examined for the direction and magnitude of potential misspecification.

The joint pessimistic scenario has been analyzed. The power at simultaneous pessimistic realizations of all key nuisance parameters is documented. If it is unacceptable, the design has been adjusted—by a larger sample size, a pre-specified re-estimation rule, or an explicit acknowledgment of fragility—before enrollment.

The power level is justified. The rationale connects the chosen power to the consequences of a type II error in this indication: why this error rate is acceptable, who bears it, and whether alternatives—higher power, a different design, a different patient selection—would redistribute the risk more appropriately.

The power report is a design document, not a technical appendix. It contains the analysis that makes the risk budget explicit and the reasoning that justifies the operating point.

The characteristic mistakes of this chapter

Three mistakes recur in the territory Chapter 3 covers. They are not arcane failures. They are the normal product of design processes that move from estimand to sample size without pausing on the commitments in between.

The optimistic effect size that no one challenged. The assumed hazard ratio of 0.70 that was borrowed from a phase II trial in a more enriched population, with a smaller sample size and a shorter follow-up. The assumed mean difference of 3 points that came from the most favorable of three available prior studies. The assumed risk difference of 8% derived from a historical control arm event rate that reflects standard of care five years ago. Each of these assumptions is internally defensible—there is a source, a number, a rationale—but none of them reflects the question that should have been asked: is this assumption plausible for this trial, in this population, at this moment? When no one asks that question, the

calculation proceeds, the sample size is approved, and the underpowered trial follows two years later.

The dropout rate that no one examined. The assumed 10% dropout because the phase II had 10% dropout. The phase II was twelve weeks. The phase III is two years. The patient population in phase III includes patients with more advanced disease who are more likely to withdraw due to tolerability or lack of response. The 10% assumption was never examined against these differences because it was borrowed from a prior study and no one thought to challenge it. The result is a trial that completes enrollment with 25% actual dropout, effective power below 60%, and a primary result that is inconclusive for a treatment that might be real.

The power level that no one justified. Eighty percent because that is what we use. In a serious condition with no approved alternatives. In a trial that is the program's only confirmatory study. In a patient population where delay in approval means patients who will not live to benefit from a second trial. The 20% type II error rate was accepted by convention, not by reasoning—and the reasoning, if anyone had done it, might have produced a different answer.

What cannot be recovered

The commitments of Chapter 3—the effect size assumption, the nuisance parameters, the power level—cannot be revised after the trial is underway without invoking adaptation. And adaptation, in the form of sample size re-estimation, has costs and constraints that are addressed in Chapter 7. The relevant point here is that the fixed sample size design—the default design that most trials use—commits to these assumptions at the design stage, and the commitment cannot be unwound by analysis.

If the effect size assumption was too optimistic and the trial is underpowered, the result may be a non-significant finding for a treatment that is real. The options are: run an additional trial (time, resources, delay), pool with other trials in a meta-analysis (possible if the trials are compatible, but uncertain), accept the non-significant result and abandon the program, or report the result with an honest acknowledgment of the power limitation and accept a conditional conclusion. None of these options is as good as having powered the trial correctly from the beginning.

If the dropout rate was underestimated and the effective sample size is smaller than planned, the primary analysis may be the analysis of a trial that was not the trial that was designed. The registrations, the ethics approvals, the sample size justification, and the monitoring plan were all calibrated to a trial with different operating characteristics. This is not fraud; it is the predictable consequence of wrong assumptions materializing. But it is a trial whose result must be interpreted against the design's original commitments, and the interpretation is more complicated than it should be.

The lesson is not that commitments should be avoided. Trial design requires commitments, and the alternative—an indefinitely flexible design that adapts to every new piece of information—has its own costs and limitations, as Chapter 7 will show. The lesson is that commitments should be made consciously, by the right people, with examination of their consequences, and with acknowledgment of what happens when they are wrong.

The connection to Chapter 4

Chapter 4 asks when the trial might stop early—the question of interim analyses, stopping boundaries, and the governance of the decisions made at those boundaries. Everything in Chapter 4 depends on what Chapter 3 settled.

The information fraction at which interim analyses are planned is a fraction of the total information Chapter 3 specified. If the total information is wrong—if the event target was based on an optimistic event rate and will not be reached—the information fractions are wrong, and the stopping boundaries are wrong. The DSMB will be looking at analyses calibrated to a design that does not correspond to the trial it is actually monitoring.

The alpha spent at each interim analysis comes from the total alpha budget that was fixed at design. If the primary test alpha was not correctly specified—if the power calculation used the nominal alpha without accounting for interim analysis spending—the final analysis will be conducted at an alpha level that was not planned.

And the power that was committed to in Chapter 3 is the operating characteristic against which Chapter 4's stopping rules must be evaluated. An interim stopping rule that looks efficient in isolation—stopping early when the interim result is compelling—may look less efficient when its effect on the trial's operating power is computed. The integration of interim analysis planning and power calculation is not sequential. It is simultaneous, and it requires the commitments of Chapter 3 to be settled before the planning of Chapter 4 can proceed honestly.

Chapter 3 risk summary

The decision this chapter owns: how large must the trial be, under what assumptions, accepted by whom, to generate the evidence that Chapter 1 defined and Chapter 2 described?

The most common mistake: treating the sample size calculation as a formula whose inputs are technical parameters to be estimated by statisticians, rather than as a set of clinical predictions and risk allocations to be owned by the design team. The assumed effect

size is a clinical prediction. The dropout rate is a clinical prediction. The power level is a risk decision. When these are delegated to the statistician without clinical ownership, the calculation is correct and the commitments are unexamined.

The professional-level risk: the underpowered trial that cannot be recovered. Not underpowered because the assumptions were impossible to get right—they are always uncertain—but underpowered because they were not examined. The effect size was borrowed without calibration. The dropout rate was copied from a shorter trial. The power level was set by convention. The trial completes, the result is non-significant, and the question that no one asked at design—could we have done this differently?—has an answer. The answer is yes. But the time to give it was before enrollment began, not after the database locked.

Chapter 4: When Might We Stop Early?

The question Chapter 3 leaves open

Chapter 3 committed the trial to a sample size. It specified the assumed effect size, documented the nuisance parameters, justified the power level, and produced a target—a number of patients, or a number of events, at which the trial would complete its primary analysis.

What Chapter 3 did not address is what happens between enrollment and completion. Trials are not sealed experiments that run to their prespecified endpoint and then open. They run in real time, with real patients accumulating outcomes, and with the increasing discomfort that comes from suspecting the answer before it is officially known.

That suspicion creates pressure. If the treatment appears to be working dramatically, continuing to randomize patients to a control arm feels wrong. If the treatment appears to be harmful, continuing enrollment feels unconscionable. If the treatment appears to be doing nothing, continuing a long and expensive trial feels wasteful. These are real pressures, not hypothetical ones, and they have produced real consequences: trials stopped early for striking efficacy, trials stopped early for harm signals, trials continued past the point where futility was evident, and trials with interim results that contaminated the final analysis in ways that could not be corrected afterward.

The interim analysis plan—the pre-specified rules governing when the trial's accumulating data may be examined, by whom, under what conditions, and with what authority to stop—is the design's answer to these pressures. It is not a technical add-on to the sample size calculation. It is a governance document: a specification of who has authority over the trial's continuation, what information they may see, what decision rules constrain their authority, and how that authority is exercised when the interim data are surprising.

Why this chapter exists separately from the sample size

Interim analyses are sometimes presented as consequences of the sample size calculation—the trial needs 500 events, so we will look at the data at 250 events. This framing is backward. The interim analysis plan is a design choice that must be made alongside the sample size calculation, not after it, because interim analyses affect the operating characteristics of the primary test in ways that the sample size calculation must account for.

Specifically: every interim look at the accumulating data that uses a test of the primary hypothesis consumes alpha. Alpha is the probability of a false positive—the threshold the primary analysis must clear. If alpha is consumed at interim analyses, the final analysis must clear a higher threshold than the unadjusted alpha to maintain the overall type I error rate. The total alpha budget is fixed; the interim analyses and the final analysis share it. If the allocation is not pre-specified, the type I error rate is not controlled.

This is the statistical argument for pre-specifying interim analyses. But it understates the full purpose of the interim analysis plan, which is not just to control type I error but to govern a series of high-stakes decisions that will be made under conditions of uncertainty, time pressure, and competing interests. The plan must specify not just the statistical boundaries but the decision rules, the governance structure, the information access rights, and the documentation requirements that ensure the decisions are made correctly—which means made honestly, made by the right people, and made in a way that can be defended after the fact.

What this chapter covers

Section 4.1 — Why Sponsors Want Interim Analyses begins where the design conversation usually does not: with the interests that drive the request for interim analyses. Interim analyses are not value-neutral design features. They serve the sponsor's interest in early information, the regulatory interest in ongoing safety monitoring, the DSMB's interest in patient protection, and the clinical community's interest in not running trials past the point where the answer is evident. These interests are not always aligned, and the design of the interim analysis plan should acknowledge their divergence explicitly.

Section 4.2 — Alpha Spending examines the mechanism by which the type I error rate is controlled across multiple looks at the accumulating data. The alpha-spending framework—the O'Brien-Fleming, Pocock, and family of spending functions—is presented not as a statistical technique to be selected from a menu but as a design commitment: a pre-specified rule for how the total alpha budget is allocated across information fractions, with consequences for the shape of the stopping boundaries and the probability of stopping early under different scenarios.

Section 4.3 — Operating Characteristics addresses what the interim analysis plan actually commits the trial to in terms of probability statements. Under the null hypothesis, what is the probability of stopping early due to false efficacy? Under the alternative, what is the probability of stopping early due to correct efficacy detection? What is the expected sample size under each scenario? These operating characteristics are the trial's behavioral profile—the statistical analog of the power curve—and they should be computed and examined before the plan is finalized.

Section 4.4 — Futility, Efficacy, and Safety examines the three categories of interim stopping: stopping for overwhelming efficacy, stopping for probable futility, and stopping for safety. Each has a different decision logic, a different evidentiary standard, and a different governance implication. They are frequently conflated in design discussions and should not be.

Section 4.5 — Governance addresses the organizational structure within which interim decisions are made: the DSMB, its charter, its relationship to the sponsor, the information

flow at interim analyses, and the documentation requirements that ensure the interim analysis does not compromise the integrity of the final analysis.

Section 4.6 — Closing returns to the chapter's central theme: an interim analysis plan is a governance document as much as a statistical document. Its purpose is not only to control type I error but to structure the decisions that will be made when the trial's accumulating data are surprising—and to make those decisions defensible regardless of what they turn out to be.

The decision this chapter must make

Every trial must answer four questions before enrollment begins.

Will the trial have any interim analyses? If not, the answer is simple and honest: the trial will be analyzed once, at the end, and no interim decisions will be made based on the primary outcome data. If yes, the subsequent questions must be answered.

What will the interim analyses examine, and at what information fractions? The timing of interim analyses—expressed as fractions of the total planned information, not as calendar dates—must be pre-specified. Analyses at unspecified times, or analyses whose timing can be adjusted based on accumulating data, are not interim analyses in the controlled sense. They are unplanned looks.

Who will have access to the interim results, and in what form? Access to treatment-arm-specific interim data is the defining sensitivity of the interim analysis. Unrestricted access contaminates the final analysis. Restricted access—through an independent DSMB that acts on the interim data without revealing arm-specific results to the sponsor—preserves the integrity of the final analysis while enabling the decisions the interim plan requires.

What authority does the interim result carry? Recommendations, not decisions, is the appropriate framing for most DSMBs—the sponsor retains formal authority over trial continuation, but the DSMB's recommendation is the operative decision in practice. The charter must specify the conditions under which the sponsor may override a DSMB recommendation, and those conditions should be narrow.

These four questions have answers that must be written down before enrollment begins. If they are not written down—if the interim analysis plan is a verbal understanding rather than a document—the trial does not have an interim analysis plan. It has an informal arrangement that will be renegotiated under pressure, and pressure is precisely when the arrangement matters most.

What this chapter is not about

This chapter does not provide a guide to choosing between O'Brien-Fleming and Pocock stopping boundaries, or to selecting an alpha-spending function from the available literature. The statistical properties of these boundaries are well-documented, and the choice among them is a technical decision that follows from the design considerations this chapter discusses.

It also does not address Bayesian adaptive designs or response-adaptive randomization, which are covered in Chapter 7. The interim analysis plan discussed here is the classical frequentist plan for a trial with pre-specified information fractions and pre-specified alpha spending—the design that most regulatory agencies expect as the default and against which adaptive departures are evaluated.

What this chapter provides is the conceptual foundation: why interim analyses create risks as well as opportunities, what the interim analysis plan is actually governing, how the governance structure should be designed, and what the consequences are when the governance fails.

4.1 Why Sponsors Want Interim Analyses

The honest starting point

Interim analyses exist in clinical trial design because multiple parties want them, for reasons that are not identical and not always stated openly. Understanding those reasons—and their divergence—is the prerequisite for designing an interim analysis plan that serves the right purposes.

The statistician's framing of interim analyses is typically about efficiency: stopping a trial early when the evidence is overwhelming can save resources and accelerate access for patients. The DSMB's framing is about ethics: patients should not continue to be randomized to an arm that is clearly inferior, or to a trial that cannot answer its question. The regulatory agency's framing is about evidence quality: interim stopping must be controlled so that the final result, whatever it is, retains interpretive integrity.

The sponsor's framing is different from all three. Sponsors want interim analyses because they want early information about whether the development program is working. This is not an illegitimate interest. A sponsor running a trial for a treatment that is not working would benefit from knowing this early, before investing further resources in a program that will not succeed. A sponsor running a trial for a treatment that is working dramatically would benefit from knowing this early, to accelerate submission and access. These are genuine efficiency interests, and they are not opposed to patient welfare.

But the sponsor's interest in early information is also the source of the most significant risks that interim analysis plans must guard against. The risk is not that sponsors are dishonest; it is that the desire for early information creates incentives—some conscious, most not—that can distort the trial if the interim analysis plan does not constrain them carefully.

What early stopping for efficacy actually does

When a trial stops early because the interim efficacy result crosses a pre-specified boundary, the event looks like success. The treatment worked. The trial ended early because the evidence was overwhelming. Patients in the control arm were protected from continuing on an inferior treatment. The result is compelling.

What early stopping also does, systematically, is overestimate the treatment effect. This is not a statistical artifact of the analysis method; it is a mathematical property of the

process by which the decision was made. Trials that stop early for efficacy have, by definition, observed an interim result that exceeded a high threshold. Interim results that exceed high thresholds are, on average, larger than the true treatment effect—not because the researchers were wrong about the mechanism, but because extreme observations are more likely to be above the threshold when the true effect is large and when random variation has pushed the observation upward. The expected value of the observed effect in a trial that stops early for efficacy is larger than the true treatment effect, even when the stopping rule is pre-specified and valid.

The magnitude of this overestimation depends on the information fraction at stopping and the strength of the interim signal. Trials that stop at very early information fractions—at 25% or 30% of the planned information—can overestimate the treatment effect by 30% or more. This overestimate enters the literature as the point estimate from the trial. It enters meta-analyses. It enters label claims. It shapes clinical expectations for the treatment's benefit. And subsequent trials, often powered based on the overestimated effect from the early-stopping trial, tend to find smaller effects and fail at higher rates.

This is not a reason to prohibit early stopping. It is a reason to understand what early stopping does and to design for it—to plan early stopping at information fractions where the overestimation bias is acceptable, to report the result with appropriate uncertainty, and to communicate to regulators and the clinical community that the observed effect is likely an overestimate of the true long-term effect.

Stopping for futility: a different incentive structure

Stopping for futility is, in principle, the mirror image of stopping for efficacy: if the interim data suggest that the treatment is unlikely to achieve statistical significance at the planned final analysis, continuing the trial wastes resources and may expose patients to a treatment that is not working. Early stopping for futility is efficient and ethical.

In practice, the incentive structure for futility stopping is more complex than for efficacy stopping, and the complexity runs in both directions.

Sponsors are often reluctant to stop for futility even when the interim data support it. The investment in the development program, the belief in the mechanism, the hope that the remaining trial population will show a stronger signal—all of these create reluctance to declare that the program is not working when a chance remains that it might. A futility boundary that is binding—one that requires the trial to stop if the interim result falls below it—forces this decision regardless of reluctance. A non-binding futility boundary—one that is advisory rather than mandatory—is subject to override by a sponsor who is not ready to accept the interim conclusion.

The design choice between binding and non-binding futility boundaries is a governance decision as much as a statistical one. A binding futility boundary protects against the

waste of continuing a futile trial and the potential harm to patients enrolled in a trial that cannot succeed. It also removes the sponsor's discretion to continue based on information not available to the DSMB—commercial considerations, pipeline strategy, patient advocacy commitments—that may legitimately bear on the continuation decision. A non-binding boundary preserves that discretion but creates the risk of motivated continuation that biases the final result.

The appropriate design is case-specific. In indications where futile trials impose serious burdens on enrolled patients—through toxicity, through invasive procedures, through opportunity cost of participation—binding futility boundaries are appropriate and ethically required. In indications where continuation of a futile trial is relatively low-harm and where the sponsor has legitimate information about external factors that the DSMB does not, non-binding boundaries may be defensible. The design documentation should reflect this reasoning, not simply adopt a boundary type by convention.

Regulatory interest in interim analyses

Regulatory agencies have a specific and somewhat different interest in interim analyses than sponsors and DSMBs. The agency's primary concern is not early stopping per se—it is the integrity of the type I error rate and the interpretability of the final result.

From a regulatory perspective, the risk of interim analyses is not that they will produce incorrect stopping decisions. It is that they will contaminate the final analysis. Contamination can occur in several ways: the sponsor learns arm-specific interim results and makes changes to the trial—endpoint, eligibility criteria, treatment regimen—that are not fully pre-specified; the external community learns interim results through unblinded publications or conference presentations, affecting the behavior of sites and patients in the trial; or the statistical analysis is modified based on knowledge of the interim trend, even if the arm-specific data were not known.

The regulatory solution is governance: strict pre-specification of the interim analysis plan, independent conduct by a statistician who is not part of the trial team, DSMB review rather than sponsor review of arm-specific interim data, and documentation of what was seen, by whom, at each interim analysis. The FDA and EMA both have guidance on the conduct of interim analyses that reflects these concerns, and deviation from that guidance in the conduct—as distinct from the pre-specification—of the interim analysis is a significant finding in a regulatory review.

For the design team, the regulatory interest in interim analysis integrity has a practical implication: the interim analysis plan must be detailed enough to be monitored. A plan that specifies the statistical boundary but not the governance structure, or that specifies the governance but not the operational details of who receives which results, cannot be audited for compliance. A plan that can be audited for compliance must specify all of these elements before enrollment begins.

The DSMB's interest: protection, not optimization

The DSMB—Data Safety Monitoring Board—is the committee charged with reviewing interim data and making recommendations about trial continuation. Its interest in interim analyses is different from the sponsor's and from the regulator's: the DSMB's primary obligation is to the patients currently enrolled in the trial, and its secondary obligation is to future patients who would benefit from a clear and valid result.

This dual obligation creates a tension that the DSMB must navigate at every interim analysis. The immediate obligation to enrolled patients argues for stopping early when harm is evident or when futility means continued enrollment is purposeless burden. The obligation to future patients argues for continuing to a definitive result that will be interpretable and actionable—because a trial stopped early for inconclusive reasons may produce neither a clear positive nor a clear negative, failing both the current and future patient populations.

The DSMB charter—the document that specifies the DSMB's composition, authority, operating procedures, and decision criteria—is the design document that structures this tension. A well-written charter specifies what the DSMB is authorized to recommend, on what grounds, based on what information, and through what process. A poorly written charter leaves the DSMB with broad discretion and limited constraints, which creates risk when the interim data are ambiguous and the pressure to act is high.

Charter design is discussed in Section 4.5. The point here is that the DSMB's interest is not the same as the sponsor's and should not be conflated with it. The DSMB is not an agent of the sponsor. It is an independent body whose authority derives from the trust that enrolled patients and the scientific community place in the trial's integrity. When the DSMB acts as an agent of the sponsor—when its recommendations are heavily influenced by the sponsor's commercial interests, or when its independence is compromised by the information flow structure—the trial loses the integrity protection that the DSMB was constituted to provide.

Aligning the interests without conflating them

The design of an interim analysis plan that serves all four interests—sponsor efficiency, regulatory integrity, DSMB patient protection, and clinical community evidence quality—requires acknowledging that these interests are not identical and designing governance structures that allow each to be served without one overriding the others.

The sponsor's interest in early information is served by interim analyses that provide meaningful information at meaningful information fractions—not so early that the information

is noise, not so late that the trial cannot be stopped early even when the evidence is overwhelming.

The regulatory interest in integrity is served by strict pre-specification, independent conduct, governance documentation, and information access controls that prevent contamination of the final analysis.

The DSMB's interest in patient protection is served by stopping boundaries that are calibrated to detect harm and futility as well as efficacy, and by a charter that empowers the DSMB to act on these signals without requiring sponsor approval.

The clinical community's interest in evidence quality is served by operating characteristics that balance the efficiency of early stopping against the bias it introduces, and by transparency about the information fraction at stopping and the implications of stopping for the point estimate.

When these elements are designed together—as components of a single governance document rather than as separate technical choices—the interim analysis plan is capable of serving all four interests. When they are designed separately—or when the statistical boundaries are designed without the governance structure, or when the governance structure is designed without the statistical boundaries—the plan has gaps that will be filled under pressure, by whoever is present, in ways that favor the interest most immediately represented.

References: Pocock, "When to Stop a Clinical Trial," BMJ 1992; Montori et al., "Randomized Trials Stopped Early for Benefit," JAMA 2005; Bassler et al., "Stopping Randomized Trials Early for Benefit and Estimation of Treatment Effects," JAMA 2010; FDA Guidance for Industry on Adaptive Designs for Clinical Trials of Drugs and Biologics (2019).

4.2 Alpha Spending

What is being spent

The type I error rate—the probability of a false positive—is the trial's most fundamental statistical commitment. Set at 2.5% one-sided (the standard for regulatory submissions), it means that if the treatment has no effect, there is a 2.5% probability that the trial will produce a statistically significant result anyway. This probability is controlled by requiring that the test statistic exceed a threshold that corresponds to the chosen alpha level.

When a trial looks at its accumulating data more than once—at an interim analysis and at the final analysis—each look is an opportunity to reject the null hypothesis. If each look is conducted at the unadjusted alpha level of 2.5%, the overall probability of a false positive across all looks is higher than 2.5%. The exact inflation depends on the number of looks and the correlation structure of the sequential test statistics, but for two independent looks each at 2.5%, the probability of a false positive at either look exceeds 4%—a 60% inflation of the nominal type I error.

Alpha spending is the framework for controlling this inflation. Rather than conducting each look at the full alpha, the total alpha budget of 2.5% is allocated—spent—across the planned looks according to a pre-specified spending function. The spending function determines how much alpha is available at each information fraction: how stringent the stopping boundary is at the first interim, how stringent at the second, and how stringent at the final analysis. The total alpha spent across all looks, integrated over the spending function, equals 2.5%. The family-wise type I error rate is thereby controlled.

This is the statistical purpose of alpha spending. Its design purpose is different: it forces the design team to commit, before enrollment begins, to exactly how the trial will be analyzed at each interim time point. That commitment is not optional. A trial that performs interim analyses without a pre-specified spending plan has consumed alpha from an unaccounted budget, and the resulting family-wise type I error rate is unknown.

Spending functions and what they commit to

The spending function is a mathematical rule that maps the information fraction to the cumulative alpha spent up to that fraction. Three families of spending functions are

in common use, and they make different commitments about the shape of the stopping boundaries.

The **O'Brien-Fleming spending function** spends very little alpha at early information fractions and most of the alpha at or near the final analysis. The resulting boundaries are very stringent early—requiring an extremely large interim effect to stop for efficacy—and approach the unadjusted alpha at the final analysis. The final analysis is conducted at nearly the full nominal alpha level, with minimal penalty for the interim looks.

The practical implication is that the O'Brien-Fleming plan is conservative about early stopping: only a dramatic interim result—a treatment effect substantially larger than the assumed effect—will trigger an early stop. This conservatism is appropriate when early estimates are expected to be noisy and regression to the mean is a concern, and when the cost of an early stop—the overestimation bias discussed in Section 4.1—is judged to be worse than the cost of continuing. It is less appropriate when the treatment is expected to show an early, large effect and when early stopping carries lower overestimation cost.

The **Pocock spending function** spends alpha more evenly across information fractions, resulting in boundaries that are the same at each look. The effect is that early stopping is easier—a less extreme interim effect is needed to cross the boundary—but the final analysis is conducted at a more stringent alpha than the nominal level, with a meaningful penalty for the interim looks.

The practical implication is that Pocock boundaries are more sensitive to early efficacy signals and more protective of enrolled patients from continued randomization if the treatment is clearly working early. They are less appropriate when the scientific question requires a definitive final analysis with minimal alpha penalty, and when the clinical community expects the final result to be the evidentiary standard regardless of interim signals.

The **flexible spending functions** of Lan and DeMets allow the spending function to be specified in terms of alpha spent as a continuous function of the information fraction, with particular forms that approximate O'Brien-Fleming or Pocock shapes or occupy the space between them. The key advantage of the Lan-DeMets framework is that the number of interim analyses and their exact timing do not need to be specified in advance—only the spending function. If an interim analysis is added or its timing shifts, the boundaries can be recalculated from the spending function without violating the overall alpha control. This flexibility is valuable in practice, where interim analyses are often triggered by DSMB schedules that do not perfectly match the information fractions assumed at design.

The choice among these functions is not primarily statistical. It is a commitment about the relative weighting of early stopping sensitivity versus final analysis integrity, and it should be made based on the clinical characteristics of the trial—the expected shape and timing of the treatment effect, the cost of overestimation bias, the relative importance of early versus final evidence—not on convention or software defaults.

One-sided versus two-sided boundaries

The alpha-spending framework applies separately to the upper and lower tails of the test statistic distribution. The upper boundary—corresponding to the treatment appearing dramatically more effective than control—triggers stopping for efficacy. The lower boundary—corresponding to the treatment appearing dramatically worse than control—triggers stopping for harm or futility.

In a superiority trial, the primary test is typically one-sided: the hypothesis is that the treatment is better than control, and the trial is powered to detect improvement, not to exclude harm. The one-sided alpha of 2.5% is spent on the upper boundary. The lower boundary, if it exists, is a separate stopping rule for harm—often symmetric to the upper boundary in its statistical form but carrying different decision authority and different governance implications.

This asymmetry is important and frequently conflated. An interim result that crosses the upper boundary triggers a recommendation to stop for efficacy, subject to the governance constraints of the DSMB charter. An interim result that crosses the lower boundary triggers a recommendation to stop for harm—a recommendation with different clinical weight and different regulatory implications. The two boundaries are statistically symmetric but clinically different, and the interim analysis plan should treat them as different decision rules with different governance requirements.

For non-inferiority trials, the boundary structure is further complicated by the structure of the non-inferiority hypothesis. The efficacy test in an NI design is one-sided in the direction of the new treatment being no worse than the comparator by more than the margin. The stopping rules must reflect this structure, not the structure of a superiority test, and the spending function must be calibrated to the NI hypothesis.

The information fraction problem

The alpha-spending framework is specified in terms of information fractions—the proportion of the total planned information that has been observed at each interim look. For event-driven time-to-event trials, the information fraction is the fraction of the planned number of events observed. For continuous outcome trials, it is typically the fraction of the planned sample size enrolled and assessed.

The information fraction is the right currency for alpha spending because it reflects the statistical information content of the interim data, not its calendar timing. An interim analysis at 50% information fraction has half the statistical information of the final analysis, regardless of whether that corresponds to 18 months or 36 months of calendar time.

The problem arises when the planned total information—the target event count or the target sample size—is wrong. If the trial was designed for 500 events and will actually accumulate

only 400 due to a lower-than-assumed event rate, an interim analysis at 250 events is at the 50% information fraction of the actual trial but at the 50% information fraction of the planned trial only coincidentally. The spending function was calibrated to the planned 500-event design; applying it at the actual 400-event scale produces boundaries that are not quite right—not catastrophically wrong, but not operating as designed.

This is the interface between Chapter 3's event rate uncertainty and Chapter 4's alpha spending. When the total information target is misspecified, the spending function produces boundaries that are calibrated to a design that does not correspond to the trial being run. The correction requires either updating the spending function as the true total information becomes clearer—which requires pre-specification of the update rule—or accepting that the operating characteristics will deviate from the planned values, and documenting that deviation.

The practical implication is that interim analysis plans should specify not just the spending function but the rule for recalibrating the boundaries if the total information target changes. This rule must be pre-specified—applied by an independent statistician who does not see the treatment arm results—to prevent the recalibration from being influenced by knowledge of the interim trend.

What the spending function does not control

Alpha spending controls the family-wise type I error rate across the pre-specified interim analyses and the final analysis. It does not control the type I error rate for additional analyses—sensitivity analyses, subgroup analyses, secondary endpoints, exploratory analyses—that are conducted at the same data cut. It does not prevent information leakage from the interim results to parties who can use that information to influence the trial. And it does not protect against the trial being modified—endpoint changed, eligibility expanded, analysis plan revised—after knowledge of the interim trend is available to the design team.

These limitations are not defects of the alpha-spending framework. They are the boundary conditions within which the framework operates, and they define the governance requirements that must complement the statistical boundaries. A trial with a well-specified alpha-spending plan but a poorly specified governance structure—one that does not control who sees the interim results, does not prevent unblinded access from influencing protocol modifications, and does not document what was seen at each interim—is not a trial with controlled type I error in the full sense. It has controlled statistical type I error within the specified tests, and uncontrolled error everywhere else.

The full protection of the type I error rate requires the statistical spending plan and the governance structure to work together. The statistical plan specifies the boundaries. The governance structure ensures that the only decisions made based on interim data are the ones the statistical plan allows.

Reporting the spending plan

The alpha-spending plan must be reported in the protocol and in the statistical analysis plan before the first interim analysis. The report should specify the spending function by name and parameters, the planned information fractions at which interim analyses will be conducted, the stopping boundaries at each planned interim, the final analysis boundary after alpha spending, and the rule for recalibrating boundaries if the total information target changes.

A spending plan that specifies only the function name—"we will use an O'Brien-Fleming spending function"—without reporting the resulting boundaries is not complete. The boundaries must be computed and reported, so that the DSMB, the sponsor, and the regulatory agency can verify that the boundaries were applied correctly at each interim analysis. Discrepancies between the pre-specified boundaries and the boundaries applied at the interim analysis are a finding in a regulatory review, regardless of whether the primary result is positive or negative.

The spending plan is a commitment. It should be treated as one—documented with the same care as the estimand and the sample size justification, reviewed by the same people, and defended with the same rigor if it is questioned. A spending plan adopted because the statistical package defaults to it, without examination of what it commits the trial to, is not a plan. It is a setting.

References: Lan and DeMets, "Discrete Sequential Boundaries for Clinical Trials," Biometrika 1983; O'Brien and Fleming, "A Multiple Testing Procedure for Clinical Trials," Biometrics 1979; Pocock, "Group Sequential Methods in the Design and Analysis of Clinical Trials," Biometrika 1977; Jennison and Turnbull, Group Sequential Methods with Applications to Clinical Trials (2000).

4.3 Operating Characteristics

What the plan is actually committing to

A stopping boundary is a line on a graph. A spending function is a mathematical rule. Neither of these, on its own, tells the design team what the interim analysis plan will actually do—how often the trial will stop early, under what true treatment effects, at what information fractions, with what expected sample sizes, and with what probability of reaching an incorrect conclusion at each interim.

Operating characteristics are the answers to these questions. They are the behavioral profile of the interim analysis plan under the full range of scenarios the trial might encounter: the null hypothesis, the alternative hypothesis, and the intermediate hypotheses that span the gap between them. Just as the power curve in Chapter 3 showed the trial's probability of success as a function of the true effect size, the operating characteristics of the interim analysis plan show the trial's probability of each possible outcome—early stop for efficacy, early stop for futility, reaching the final analysis—as a function of the true effect size and the information fraction at which the trial happens to stop.

Computing and examining these operating characteristics before finalizing the interim analysis plan is not optional. It is the only way to know what the plan is committing the trial to.

Under the null hypothesis

The most important operating characteristic under the null hypothesis is the probability of stopping early for a false positive. This is the interim contribution to the family-wise type I error rate, and the alpha-spending framework guarantees it will be controlled at the nominal level if the plan is correctly specified and correctly implemented.

But the probability of stopping early for a false positive is not the only relevant null-hypothesis characteristic. The probability of stopping early for futility under the null—correctly concluding that the treatment is not working when it is not—is also important, because early futility stopping under the null means the trial was conducted efficiently: it stopped before consuming resources on a futile program. A futility boundary that correctly triggers early stopping under the null most of the time is a valuable feature of the design.

The expected sample size under the null hypothesis—the average number of patients enrolled across all possible null-hypothesis scenarios, weighted by the probability of stopping at each information fraction—is the primary measure of the interim analysis plan's efficiency. An interim analysis plan that stops early for futility with high probability under the null, and rarely continues to the final analysis when the treatment is not working, is an efficient plan: it saves the resources that would be consumed by a futile trial. This efficiency does not come for free—it requires a futility boundary that is powered to detect futility, which requires its own design investment.

Under the alternative hypothesis

Under the alternative hypothesis—when the treatment effect is as assumed—the critical operating characteristic is the probability of stopping early for efficacy. If this probability is high, the interim analysis plan is efficient in the beneficial direction: the trial will frequently stop early and deliver the evidence sooner than if it had run to the planned completion. If this probability is low, the interim analysis plan is providing minimal benefit over a design without interim analyses—it is adding governance complexity without adding efficiency.

The expected sample size under the alternative hypothesis quantifies this trade-off. A plan with high probability of early stopping under the alternative has a low expected sample size under the alternative—on average, the trial stops before reaching the planned maximum. A plan with low probability of early stopping has an expected sample size close to the maximum. The difference in expected sample size between a plan with and without interim analyses, under the alternative, is the efficiency gain from the interim analysis plan.

This efficiency gain comes at a cost that is less often quantified: the expected overestimation of the treatment effect at stopping. If the trial stops early for efficacy, the observed effect is expected to be larger than the true effect, as discussed in Section 4.1. The magnitude of this overestimation depends on the information fraction at stopping and the strength of the efficacy signal. If the trial stops at 50% information fraction with an effect twice as large as assumed, the overestimation bias is substantial. If it stops at 90% information fraction with an effect only modestly above the boundary, the bias is small.

The operating characteristics report should include, for each information fraction at which early stopping is possible, the expected overestimation of the treatment effect conditional on stopping at that fraction. This gives the design team and the regulatory agency a quantitative basis for evaluating whether the interim stopping rule produces a level of bias that is acceptable given the efficiency gain.

Under intermediate hypotheses

The operating characteristics under the null and alternative are the extremes. The intermediate hypotheses—true effects smaller or larger than the assumed effect—are where the interim analysis plan's behavior is most clinically informative and most often neglected.

The most important intermediate scenario is the pessimistic alternative: the true treatment effect is real but smaller than assumed. This is the scenario where the design team got the effect size prediction wrong in the optimistic direction—the most common failure mode of Chapter 3. Under this scenario, the trial will often not stop early for efficacy (because the interim effect is below the efficacy boundary) and may not stop early for futility either (because the interim effect, though below the assumed alternative, may still be above the futility boundary, if there is one). The trial continues to the final analysis—which, given the smaller-than-assumed effect, may or may not reach significance.

The probability of this scenario—continuing to the final analysis under the pessimistic alternative—and the probability of a significant result at the final analysis under the pessimistic alternative together characterize the trial's behavior in the most likely bad-case scenario. These probabilities should be reported in the operating characteristics, and the design team should examine them before finalizing the plan. If the probability of a significant final result under the pessimistic alternative is below an acceptable threshold, the interim analysis plan does not protect against the most common form of power loss.

Expected sample size and trial duration

The expected sample size under a given true treatment effect is the weighted average of the sample sizes at which the trial stops—early stopping at each information fraction, or reaching the maximum—with the weights being the probabilities of each outcome. This quantity summarizes the efficiency of the interim analysis plan under a given true effect in a single number.

For time-to-event trials, the expected number of events at stopping is the more relevant quantity—because the trial's power is determined by the number of events, not the number of patients. But the expected trial duration is also important: a trial that stops early at 50% information with a high probability under the alternative will have a shorter expected duration than a trial that rarely stops early. This duration reduction translates into earlier access to the evidence for patients and for the regulatory system.

Computing expected sample size and expected duration requires simulation, because the analytical approximations are accurate only under specific assumptions about the distribution of the test statistic. The simulation should be conducted across a grid of true treatment effects—at the null, at the assumed alternative, and at several intermediate values—to produce a complete picture of the plan's behavior. The result is a table or a plot that shows,

for each true effect size, the probability of each stopping outcome and the expected sample size or duration.

This simulation is the design team's most informative tool for evaluating the interim analysis plan. It is also the tool most often skipped in favor of reporting only the power at the primary alternative and the type I error rate under the null. The intermediate scenarios, which are the most likely scenarios in practice, are the scenarios whose characteristics should be known before the plan is locked.

The consistency requirement

The operating characteristics of the interim analysis plan must be consistent with the commitments of Chapter 3. Specifically:

The power at the final analysis—after alpha spending for interim analyses—must equal the power that the Chapter 3 sample size was designed to achieve. If the sample size was designed for 90% power at the unadjusted alpha, the power at the adjusted final analysis alpha (after interim spending) will be below 90%. Either the power calculation should have used the adjusted alpha, or the sample size should be inflated to restore the post-spending power to the target level.

This consistency check is performed more often in principle than in practice. The sample size calculation and the interim analysis plan are frequently produced by different members of the statistical team, at different points in the design process, without explicit reconciliation. The result is a trial whose actual power—accounting for alpha spending—is below the power that was presented to the clinical team and the regulatory agency. This is a silent error: the power is not wrong in the sense that the calculation was incorrect, but it is misleading in the sense that the reported power does not correspond to what the trial will actually achieve.

The consistency requirement is simple to state and straightforward to implement: the power for the sample size calculation should be computed at the final analysis alpha after spending, not at the nominal alpha. If this reduces the calculated power to below the target, the sample size should be increased, the power target should be revised, or the spending plan should be modified to leave more alpha for the final analysis.

Reporting the operating characteristics

The operating characteristics should be reported as part of the design documentation—in the protocol or in a separate statistical design document—before the first interim analysis is conducted. The report should include the probability of stopping at each interim analysis

under the null hypothesis, under the assumed alternative, and under at least two intermediate scenarios. The expected sample size or event count under each scenario. The expected overestimation of the treatment effect conditional on early stopping. The power at the final analysis at the adjusted alpha level.

This report is the operating manual for the DSMB. When the DSMB convenes at the first interim analysis and observes that the interim test statistic is at a specific value, the operating characteristics report tells them what that statistic means in terms of the probability of each future outcome: the probability that the trial will reach significance at the final analysis if it continues, the probability that it will not, and the expected sample size under each stopping decision. Without this information, the DSMB is making a high-stakes decision without the context that would make it well-informed.

The operating characteristics are not updated at the interim analysis unless the total information target has changed and the spending plan has been recalibrated. They are computed at design and applied throughout the trial as the reference document for all interim decisions. When they are not computed at design—when the DSMB convenes without a pre-computed operating characteristics report—the interim decision is being made in the dark. That is not a governed decision. It is an improvised one.

References: Jennison and Turnbull, Group Sequential Methods with Applications to Clinical Trials (2000); Proschan, Lan, and Wittes, Statistical Monitoring of Clinical Trials: A Unified Approach (2006); Emerson and Fleming, "Symmetric Group Sequential Test Designs," Biometrics 1989; DeMets and Lan, "Interim Analysis: The Alpha Spending Function Approach," Stat Med 1994.

4.4 Futility, Efficacy, and Safety

Three decisions that are not the same

Interim stopping comes in three varieties. A trial may stop early because the evidence of efficacy is overwhelming—the treatment is clearly working, and continuing to randomize to control is ethically unjustifiable. It may stop early because the trial has lost the realistic prospect of achieving its primary objective—the treatment is not working, or the accumulated evidence is insufficient to detect an effect even if one exists. Or it may stop early because of a safety signal—evidence that the treatment is causing harm that outweighs whatever benefit it provides.

These three stopping reasons are frequently described in the same design section, under the same DSMB charter, sometimes with the same statistical framework. This creates a misleading impression that they are variations on a common theme. They are not. They have different evidentiary standards, different decision authorities, different governance implications, and different consequences for the trial's scientific integrity.

The design of the interim analysis plan must treat them as three distinct decision types, with distinct rules, distinct authorities, and distinct documentation requirements. Conflating them—designing a single stopping boundary that handles all three—is a governance error that will not be apparent until the interim analysis convenes and the DSMB must make a decision that the plan has not adequately specified.

Efficacy stopping: what is being claimed

When a trial stops early for efficacy, it is making a claim: the evidence accumulated to this information fraction is sufficient to conclude that the treatment works. That claim has implications that extend beyond the trial. It enters the regulatory submission, the label negotiation, the clinical literature, and the clinical practice of prescribers who will treat patients not enrolled in the trial.

The evidentiary standard for this claim should therefore be the same standard that would be required for the claim if the trial had run to completion. The alpha-spending framework is designed to enforce this: the interim efficacy boundary is set so that the probability of a false positive at any interim look does not exceed the family-wise type I error. A trial that crosses the efficacy boundary at the interim analysis has cleared the same probabilistic

standard—over the full range of information fractions—as a trial that crosses the final analysis threshold.

But the probabilistic standard is not the only relevant standard. The clinical plausibility of the interim result is also relevant. An interim result that crosses the efficacy boundary at 25% information fraction, showing a hazard ratio of 0.45 in a disease where prior treatments have achieved hazard ratios of 0.75-0.85, should trigger scrutiny of whether the result reflects the true treatment effect or an extreme realization of sampling variation. The boundary was crossed, but crossing the boundary is a necessary condition for stopping, not a sufficient one.

The DSMB charter should specify the conditions under which the DSMB may recommend continuation despite crossing the efficacy boundary—when the result is implausibly large, when the duration of follow-up is insufficient to characterize the durability of the effect, when the sample size is so small that the confidence interval is wide despite significance. These conditions are judgment calls, and they should be specified in advance rather than improvised when the interim data are surprising.

Efficacy stopping: the information fraction question

The information fraction at which efficacy stopping is possible has two distinct consequences that pull in opposite directions.

Early stopping—at low information fractions—maximizes the efficiency gain: the trial ends sooner, resources are saved, and patients who are being randomized to control are protected from continued assignment to an inferior arm. It also maximizes the overestimation bias: the observed effect at early stopping is furthest from the true effect, and the confidence interval is widest. A result at 30% information fraction is dramatic but imprecise and potentially misleading.

Late stopping—at high information fractions—minimizes the overestimation bias and produces a result that is close to what the final analysis would have found. It also minimizes the efficiency gain: a trial that can only stop early at 80% information fraction is providing minimal benefit over a design without early stopping.

The resolution is not a universal formula but a judgment about the relative costs in the specific trial context. When early stopping efficiency is paramount—in seriously ill patients for whom delay in treatment access is measured in survival, when the infrastructure for a large trial is fragile, when the development timeline is constrained—the early-stopping benefits may outweigh the overestimation cost, and lower information fraction boundaries may be appropriate. When scientific integrity is paramount—when the treatment will become a cornerstone of clinical practice, when the result will be used as the evidence base for subsequent trials or for policy decisions that affect large populations—the overestimation cost

is higher, and boundaries should require higher information fractions before early stopping is possible.

This is a design decision that should be made explicitly, by the design team, before the plan is finalized—not selected from a menu of standard boundary types without examining what the selection commits to.

Futility stopping: binding versus conditional power

Futility stopping addresses a different question than efficacy stopping: not "is the evidence strong enough to claim the treatment works?" but "is there enough remaining potential for the trial to answer its question?" The decision is forward-looking, not backward-looking.

The most common statistical tool for futility assessment is conditional power: the probability of achieving a significant final result, given the interim data and the planned remaining sample. If the conditional power is below a pre-specified threshold—20% is common, though the threshold should be justified rather than conventional—the trial is unlikely to succeed even if it continues, and stopping may be appropriate.

Conditional power is computed under a specific assumption about the true treatment effect during the remaining trial period. The two natural assumptions are the originally assumed alternative (what is the probability of success if the true effect is what we assumed?) and the current estimate from interim data (what is the probability of success if the true effect is what the data currently suggest?). These two calculations produce different numbers and support different decisions.

Conditional power under the originally assumed alternative is the more optimistic calculation: it asks whether the trial can succeed if the treatment is doing what we hoped. Conditional power under the interim estimate is the more realistic calculation: it asks whether the trial can succeed if the treatment continues to behave as it has behaved so far. When the interim estimate is substantially below the assumed alternative—as it will be when the trial is heading toward a futility stop—the conditional power under the interim estimate will be much lower than under the assumed alternative.

The charter should specify which conditional power calculation is the basis for the futility recommendation. Using the more optimistic calculation produces later and less frequent futility stops—the trial continues longer in the hope that the effect will emerge. Using the more realistic calculation produces earlier and more frequent futility stops—the trial acknowledges sooner that the observed effect is below what was assumed.

The choice has ethical and efficiency implications. A futility standard based on the optimistic calculation is more permissive: it allows the trial to continue when the evidence is negative, on the grounds that the original hypothesis might still be true. A futility standard based on the realistic calculation is more stringent: it stops the trial when the current evidence does

not support continuation, on the grounds that continuing would not serve the patients being enrolled.

The non-binding futility problem

A non-binding futility boundary—one that the sponsor may choose to override—creates a specific and frequently realized governance problem. When the interim data suggest futility but the sponsor decides to continue, the trial proceeds based on a decision that was made with knowledge of the interim trend. This decision may be correct—the sponsor may have information about external developments, pending regulatory guidance, or complementary data that the DSMB does not have. But it may also be driven by reluctance to acknowledge a failing program.

When a non-binding futility boundary is crossed and the sponsor continues the trial, two things should happen. First, the sponsor's reasons for continuation should be documented in the DSMB charter response process, and those reasons should be available to regulatory reviewers who later examine the trial's conduct. Second, the operating characteristics of the trial conditional on continuation—the probability of achieving significance, the expected sample size, the expected information at the final analysis—should be recomputed and documented. These are not the operating characteristics originally planned; they are the operating characteristics of a trial that has been continued past its futility boundary, and the regulatory agency reviewing the final result will want to know what those characteristics were.

Failure to document the continuation decision and its basis creates a governance gap that undermines the interim analysis plan's integrity. If the trial subsequently fails at the final analysis, the continuation past the futility boundary may be interpreted as a decision motivated by something other than scientific judgment. If the trial subsequently succeeds, the concern is that the futility boundary was crossed and ignored, and the final result may reflect the influence of that decision on subsequent enrollment or conduct. Neither outcome is good, and both are avoidable by treating the non-binding futility decision as the high-stakes governance event it is.

Safety stopping: a different authority structure

Safety stopping is categorically different from efficacy and futility stopping, and it requires a different authority structure. The difference is in the nature of the evidence and the nature of the obligation.

Efficacy and futility decisions are about whether the trial is likely to answer its scientific question. They are governed by pre-specified statistical rules that balance type I and type II error. They are, ultimately, decisions about evidence quality and scientific efficiency.

Safety decisions are about whether enrolled patients are being harmed. They are governed not only by statistical thresholds but by the ethical obligation to protect research participants. They override the scientific objectives of the trial when necessary.

This means that safety stopping should not be constrained by the alpha-spending framework in the same way that efficacy stopping is. The alpha-spending framework controls the type I error for efficacy claims. A safety signal that falls below a pre-specified alpha-adjusted boundary is still a safety signal, and the DSMB must be empowered to act on it regardless of whether it crosses a formal statistical threshold. Requiring a pre-specified alpha level for safety stopping—as if safety and efficacy are the same kind of decision—is a category error that can result in patients being exposed to a harmful treatment because the safety signal was "not statistically significant."

The standard for safety stopping is not statistical significance. It is medical judgment: given the evidence of harm and the evidence of benefit, does it remain ethically justifiable to continue randomizing patients? This judgment may involve statistical evidence, but it is not reducible to it. The DSMB charter must empower the DSMB to make this judgment—to recommend stopping for safety on the basis of clinical judgment, independently of whether any pre-specified statistical threshold has been crossed.

The authority structure for safety stopping also differs from efficacy stopping. A recommendation to stop for safety by a fully constituted independent DSMB should be extremely difficult for a sponsor to override. The conditions under which a sponsor may continue a trial against a safety stopping recommendation from the DSMB should be narrow—limited to demonstrably incorrect data or clear procedural error—and the override should require regulatory notification. If a sponsor overrides a safety stopping recommendation without triggering regulatory review, the trial's governance has failed in a fundamental way.

When the three categories overlap

In practice, the three stopping categories are not always cleanly separated. A trial may observe a safety signal that is associated with biological efficacy—a finding that the treatment causes a side effect that is related to its mechanism of action. Continuing the trial is dangerous; stopping the trial means the efficacy question is unanswered. A trial may observe early futility on the primary endpoint alongside a signal of harm on a secondary endpoint—stopping for futility also removes the patients from harm, while continuing would resolve both questions more definitively. A trial may observe early efficacy in a subgroup alongside futility in the overall population—stopping for overall futility means the subgroup signal is never confirmed.

These overlaps require the interim analysis plan to specify not just the individual stopping rules but the decision hierarchy—which stopping category takes priority when multiple signals are present simultaneously. Safety always takes priority over efficacy and futility. Efficacy takes priority over futility, because a trial that is heading toward futility should not stop if there is also a compelling efficacy signal that requires more evidence. These hierarchies should be explicit in the charter, not resolved by the DSMB in real time under the pressure of an interim meeting.

References: Ellenberg, Fleming, and DeMets, Data Monitoring Committees in Clinical Trials (2002); Proschan et al., Statistical Monitoring of Clinical Trials (2006); Snapinn et al., "Monitoring a Clinical Trial Under an Adaptive Stopping Rule," Pharm Stat 2014; ICH E6(R2) Guideline for Good Clinical Practice.

4.5 Governance

The gap between the plan and the decision

A pre-specified alpha-spending function, a set of pre-computed stopping boundaries, and a table of operating characteristics are the statistical components of an interim analysis plan. They are necessary. They are not sufficient.

The gap between the statistical plan and the actual interim decisions is filled by governance: the organizational structure that determines who sees the interim data, in what form, with what authority, under what constraints, and with what documentation requirements. When governance is well-designed, the statistical plan is implemented faithfully, the interim decisions are made by the right people with the right information, and the final result retains interpretive integrity. When governance fails—when unblinded data reach parties who should not see them, when stopping decisions are made outside the pre-specified rules, when documentation is incomplete—the statistical plan is irrelevant. The trial has been compromised in ways that no post-hoc analysis can repair.

Governance is not the DSMB's charter alone, though the charter is its most visible document. It is the complete system of information access rights, decision authority, documentation requirements, and communication protocols that surrounds every interim analysis from the moment the data are extracted to the moment the decision is documented. Designing this system is as important as designing the statistical boundaries, and it requires as much deliberate attention.

The DSMB: composition and independence

The Data Safety Monitoring Board is the independent committee that reviews interim data and makes recommendations about trial continuation. Its composition and its independence are the first governance questions.

Composition requires expertise across the domains relevant to the interim decision: clinical expertise in the indication, statistical expertise in sequential trial monitoring, and—when safety signals are plausible—expertise in the specific safety domains at risk. A DSMB composed entirely of clinicians with no statistical monitoring expertise will have difficulty interpreting the alpha-spending context of an interim efficacy signal. A DSMB composed entirely of statisticians may lack the clinical judgment to evaluate whether a safety finding

meets the threshold for stopping. The composition should reflect the full range of decisions the DSMB may be asked to make.

Independence requires that DSMB members have no financial or professional relationship with the sponsor that would compromise their ability to make decisions contrary to the sponsor's interest. Advisory board membership, consulting relationships, equity positions, and institutional research funding all create potential conflicts that should be evaluated at the time of appointment. The standard is not the absence of any relationship—that standard would exclude many of the most experienced clinical trialists—but the absence of relationships significant enough to bias judgment in the direction of sponsor interests.

Independence also requires procedural protection: the DSMB's deliberations should be confidential, the interim data should be seen only by the DSMB and the independent statistician, and the DSMB's recommendation should be transmitted to the sponsor without disclosure of the arm-specific interim data that formed its basis. A recommendation that says "the DSMB recommends continuation" tells the sponsor what they need to know. A recommendation that reveals the interim hazard ratio, arm-specific event rates, or subgroup results tells the sponsor much more than they need to know for the continuation decision—and what they learn may influence their conduct of the trial in ways that were not planned.

The charter: what it must specify

The DSMB charter is the governance document that specifies, in advance of any interim data review, how the DSMB will operate. A charter that is specific enough to be useful is a charter that could be handed to a new DSMB member with no prior knowledge of the trial and would tell them everything they need to know to participate in an interim meeting.

The charter must specify the composition of the DSMB and the criteria for recusal. It must specify the information that will be presented at each interim analysis—what the DSMB will see, at what level of aggregation, with what annotation. It must specify the statistical boundaries that the DSMB will apply and the operating characteristics those boundaries produce. It must specify the decision rules—stopping for efficacy, stopping for futility, stopping for safety—with their respective evidentiary standards and decision authorities. It must specify the communication protocol between the DSMB and the sponsor: what the DSMB will communicate, in what form, on what timeline, and through what channel.

It must also specify what the DSMB is not authorized to do. This is as important as specifying what it is authorized to do. A DSMB that is not explicitly authorized to recommend eligibility changes, endpoint modifications, or analysis modifications should not make such recommendations—and if it does, the sponsor should not act on them without the full protocol amendment process that such changes require. The charter is the boundary of the DSMB's authority, and that boundary should be explicit.

Information flow: who sees what, when

The most consequential governance design decision is the structure of information flow at interim analyses. This decision determines what the sponsor knows about the interim results, and what they can do with that knowledge.

The cleanest governance structure is full firewall: the independent statistician prepares the interim analysis report for the DSMB, the DSMB reviews the arm-specific interim data and makes a recommendation, and the sponsor receives only the recommendation—not the data on which it was based. Under this structure, the sponsor cannot use knowledge of the interim trend to make decisions about the trial—commercial, regulatory, or operational—that are influenced by the interim result. The firewall protects the trial's scientific integrity by preventing the sponsor's response to interim data from becoming a source of bias.

This structure has costs. Sponsors have legitimate operational interests that may depend on interim trial characteristics: site management, supply chain planning, regulatory submission timelines, co-development partnerships. Knowing only "the DSMB recommends continuation" does not give the sponsor the information it needs to manage these operational decisions optimally. The cost of the firewall is operational inefficiency; the benefit is scientific integrity.

The intermediate structure—where the sponsor receives summary interim data without arm-specific comparisons—attempts to preserve operational utility while limiting the information that could bias the trial. This structure requires careful specification of what constitutes "arm-specific comparison" information. A report that shows overall event rates without arm attribution tells the sponsor something about the event rate assumption accuracy; a report that shows time trends in event accumulation may implicitly convey arm-specific information through pattern recognition. The intermediate structure requires active enforcement of the information boundary, which requires a statistician who understands what information is sensitive and what is not, and a process that systematically prevents sensitive information from crossing the firewall.

The weakest governance structure—where the sponsor has access to the full interim dataset, subject to a prohibition on using the information to modify the trial—relies on behavioral controls rather than structural ones. The prohibition is unenforceable in practice: the sponsor's decisions about the trial, from site management to communication strategy to regulatory interactions, will inevitably be influenced by knowledge of the interim trend, even if the influence is unconscious. This structure is appropriate only in situations where the sponsor's operational involvement in the trial requires real-time data access—certain adaptive designs, platform trials—and where compensating governance controls exist. For standard fixed designs with pre-specified interim analyses, it should not be the default.

The open session, closed session, and executive session

A well-designed interim meeting typically has three phases, and the governance implications of each are distinct.

The **open session** is attended by representatives of the sponsor, the clinical operations team, and the regulatory team, along with the DSMB. The open session presents data that do not reveal the treatment arm comparison: overall enrollment progress, site performance, protocol deviation rates, overall safety events without arm attribution, and any operational issues that require joint discussion. The DSMB can ask questions and the sponsor can present information, but the arm-specific interim efficacy and safety data are not discussed.

The **closed session** is attended only by DSMB members and the independent statistician. The arm-specific interim data—efficacy results, safety results, subgroup results—are presented and discussed in closed session. The DSMB deliberates, applies the pre-specified stopping rules, and formulates its recommendation.

The **executive session** is a DSMB-only session, without the independent statistician, in which the DSMB may discuss the recommendation before transmitting it to the sponsor. Not all interim analysis structures include this phase, but it is valuable when the DSMB's deliberations involve judgments about the charter's interpretation—what the stopping rules require—that the DSMB may wish to discuss without the constraint of external presence.

This three-phase structure is not bureaucratic excess. It is the mechanism by which the arm-specific interim data remain within the closed session while the non-arm-specific operational data are shared openly. When the session structure is not maintained—when arm-specific data are discussed in the open session, when sponsor representatives attend the closed session, when the DSMB's deliberations are communicated to the sponsor in more detail than the recommendation—the governance has failed, regardless of whether the stopping decision itself was correct.

Documentation: the governance record

Every interim analysis should produce a documentation record that specifies what was reviewed, who attended which sessions, what data were presented, what the DSMB deliberated on, what recommendation was made, and—critically—what the stopping rule status was at the time of the recommendation. This record is not primarily for regulatory inspection, though it will be examined during regulatory review. It is for the scientific record: the basis on which the interim decision was made should be preserved and interpretable independent of the memory of the participants.

The documentation should be specific enough to allow reconstruction of the interim analysis. It should identify the data snapshot date, the data completeness rate, the number of events in each arm (in the closed session record), the test statistic value relative to the boundary,

and the DSMB's reasoning if the recommendation involved judgment beyond mechanical application of the stopping rules.

When the stopping rules were not mechanically controlling—when the DSMB chose to continue despite approaching a futility boundary, or chose to continue despite crossing a non-binding efficacy boundary, or raised a safety concern without crossing a statistical threshold—the reasoning must be documented. An undocumented deviation from the pre-specified rules is a governance failure regardless of whether the deviation was scientifically justified. The regulatory reviewer who later asks "why did the DSMB continue after the interim test statistic was 1.89 against a boundary of 2.01?" must have a documented answer. "The DSMB exercised clinical judgment" is not a documented answer. The clinical judgment must be recorded.

When governance fails

Governance failures in interim analysis have a characteristic signature in the regulatory review: a timeline of events that is hard to reconcile with the premise that the trial was conducted without knowledge of interim results. Site activation patterns that change after the interim analysis. Eligibility criteria that are clarified in ways that affect enrollment composition after the interim. Communication strategy shifts that coincide with interim meetings. These patterns are not proof of misconduct—they may reflect legitimate operational decisions—but they raise questions that the governance documentation should answer.

When the governance documentation is incomplete—when the interim analysis records do not specify who attended which sessions, when the DSMB charter does not specify the information flow protocol, when the sponsor's response to the DSMB recommendation is not documented—the patterns cannot be explained, and the integrity of the trial is uncertain. This uncertainty does not necessarily prevent regulatory approval, but it complicates the review and can result in conditions or labeling language that the sponsor did not anticipate.

Governance design is not an afterthought. It is the mechanism by which the pre-specified interim analysis plan is actually implemented—faithfully, under conditions that may be very different from what the design team anticipated. A well-governed trial is one that can be explained from the inside: every decision at every interim analysis can be traced to a documented basis, and that basis is the pre-specified plan. A poorly governed trial is one that can only be explained from the outside: the decisions are defensible in retrospect, but the record that would establish their defensibility at the time they were made does not exist.

References: Ellenberg, Fleming, and DeMets, Data Monitoring Committees in Clinical Trials: A Practical Perspective (2002); ICH E6(R2) Good Clinical Practice (2016); FDA Guidance for Industry: Oversight of Clinical Investigations—A Risk-Based Approach to Monitoring (2013); Herson, Data and Safety Monitoring Committees in Clinical Trials (2009).

4.6 Closing: The Governed Decision

What this chapter asked

Chapter 4 asked one question: when might we stop early?

It asked that question in five registers, each addressing a different dimension of the same design challenge.

Section 4.1 asked why interim analyses are requested—and answered honestly: the interests driving that request are not identical across parties. Sponsors want early information about program viability. DSMBs want protection for enrolled patients. Regulators want evidence integrity. These interests are not opposed, but they diverge at the margins, and the divergence should be designed for explicitly rather than assumed away by the fiction that everyone wants the same thing.

Section 4.2 asked what the alpha-spending function actually commits to—not just the type I error calculation, but the shape of the boundaries across information fractions, and what that shape says about the relative weighting of early stopping sensitivity versus final analysis integrity. The spending function is a design commitment, not a technical parameter.

Section 4.3 asked what the interim analysis plan actually does across the full range of possible true effects—not just under the assumed alternative and the null, but under the intermediate scenarios that are most likely to be realized. The operating characteristics are the behavioral profile of the plan, and they must be examined before the plan is finalized.

Section 4.4 asked what it means to stop for efficacy, for futility, or for safety—and answered that these are three categorically different decisions with different evidentiary standards, different decision authorities, and different governance implications that should not be collapsed into a single stopping rule.

Section 4.5 asked who has authority over the interim decisions and how that authority is exercised—answering that the statistical boundaries are the scaffolding, and governance is the building. Without governance, the plan is a document. With governance, it is a decision system.

What this chapter decided

By the end of this chapter, five things must be documented and owned.

The interim analysis plan is pre-specified, in writing, before enrollment begins. The number of interim analyses, their information fractions, the spending function, the resulting boundaries, and the futility rules are all stated. If the plan is "no planned interim analyses," that too must be stated, as an explicit choice—not an omission.

The operating characteristics are computed and reviewed. The probability of early stopping under the null, under the assumed alternative, and under at least two intermediate scenarios is documented. The expected sample size under each scenario is reported. The expected overestimation conditional on early stopping is quantified. These are not optional sensitivity analyses; they are the design team's due diligence on what the plan actually commits the trial to.

The three stopping categories are explicitly separated. The efficacy boundaries are distinct from the futility boundaries, which are distinct from the safety monitoring rules. Each category has its own evidentiary standard, its own authority structure, and its own documentation requirement. The DSMB charter specifies the decision hierarchy when multiple stopping signals are present simultaneously.

The DSMB is constituted, and its charter is complete. The composition, the independence criteria, the information flow structure, the session format, the communication protocol, and the documentation requirements are all specified. The charter is a document that a DSMB member with no prior knowledge of the trial could read and know exactly what they are expected to do.

The governance system is in place before the first interim analysis. This means the independent statistician is identified, the data management team understands the firewall requirements, the documentation templates are prepared, and the communication chain from the DSMB to the sponsor is tested. The governance system does not begin at the first interim meeting. It begins at enrollment, because the information security that protects the interim analysis starts when the first patient is enrolled.

The characteristic mistakes of this chapter

Three failures recur in the territory Chapter 4 covers.

The plan designed for the boundary, not for the decision. The alpha-spending function is selected—O'Brien-Fleming because it is conservative, or because it is the software default—without examining the operating characteristics it produces or comparing them to alternative designs. The boundaries are computed and reported. The probability of early stopping under plausible scenarios is not computed. The DSMB convenes at the first interim and finds a test statistic below the efficacy boundary but also below the conditional power

threshold—and the charter does not specify what to do when both signals are present. The decision is improvised. The documentation of the improvisation is incomplete.

The governance that existed on paper but not in practice. The charter specified a full firewall. The independent statistician prepared the interim report for the DSMB. But the interim meeting included a sponsor representative in the "observer" role who was present during the presentation of arm-specific data. No one objected. The meeting minutes recorded "sponsor observer present during closed session" without noting the governance implication. Two years later, during the regulatory review, this entry in the meeting minutes raises questions about information access that the charter cannot answer because the charter's implementation was not monitored.

The DSMB recommendation that was not followed. The DSMB observed an interim result approaching the futility boundary and recommended continuation with heightened monitoring. The charter specified the futility boundary as non-binding, so the sponsor had the authority to continue. The sponsor continued. The basis for continuation—a pipeline argument that the program had companion treatments in development that needed the phase III result to proceed—was not documented in the continuation decision record. Eighteen months later, the trial failed at the final analysis. The regulatory reviewer noted the non-binding futility near-crossing and the undocumented continuation decision. The final result is credible but its context is not explained, and the explanation must be reconstructed from memory.

Each of these failures shares the same origin: the design team distinguished between the statistical plan and the governance structure, treating the first as the real design work and the second as administrative detail. The administrative detail is where the design lives during the trial. The statistical plan is what was committed to before enrollment; the governance is what ensures the commitment is honored after enrollment.

What cannot be recovered

Some governance failures can be managed after the fact. A documentation gap can be supplemented with declarations from the participants. An unexpected DSMB charter situation can be adjudicated by a chartered process. An interim analysis that deviated from the pre-specified boundaries by a small margin can be explained with statistical precision about the actual versus planned operating characteristics.

Some cannot. When arm-specific interim data have been seen by parties outside the DSMB—by sponsor representatives, by commercial teams, by investigators—the contamination of subsequent trial conduct cannot be quantified or corrected. It is possible that nothing changed. It is also possible that enrollment practices, patient counseling, investigator enthusiasm, or data collection procedures were influenced by knowledge that was not supposed to be known. The uncertainty is irreducible.

When an interim analysis is conducted outside the pre-specified plan—at an unplanned information fraction, with an unplanned scope, by an unplanned party—the type I error rate for the primary test is no longer controlled at the nominal level. This is not a regulatory technicality. It is the fundamental promise of the trial design: that the probability of a false positive, across all the analyses the trial will conduct, is no more than 2.5% one-sided. That promise cannot be retroactively restored.

When the DSMB charter is not specific enough to constrain the decisions made at the interim meeting—when the DSMB exercises discretion that the charter did not anticipate and did not limit—the interim decisions are governed by the judgment of the DSMB members at that moment, under those conditions, with those interests. That governance may be excellent. But it is not documented, it is not pre-specified, and it is not auditable. It is the decision that was made by whoever was in the room when the decision had to be made.

These are the failures that cannot be recovered because they are not mistakes in the implementation of the plan. They are failures of the plan itself—failures that became visible only after the plan was tested by real data, real pressures, and real decisions. The guard against them is not better implementation; it is a plan that is specific enough, governed well enough, and documented carefully enough that implementation failures are detectable before they become irreversible.

The connection to what follows

Chapter 5 addresses bias—the systematic distortion of the trial's primary comparison that arises when randomization, blinding, or allocation concealment fail. The interim analysis plan is a source of bias when it fails: information leakage from the interim to the sponsor creates bias in trial conduct; stopping decisions made outside the pre-specified rules create bias in the final result; documentation gaps create bias in the regulatory review. Chapter 5's tools for protecting against bias assume that the governance system of Chapter 4 has functioned correctly—that the blind has been maintained, that the interim data have not contaminated the conduct of the trial, and that the trial's operating characteristics correspond to what was pre-specified.

Chapters 4 and 5 are therefore not sequential in the operational sense. They are simultaneous requirements. The governance of the interim analysis is part of the bias protection system, not a separate design element that follows after bias protection is established. A trial that has good randomization and allocation concealment but poor interim governance is not a well-protected trial. It is a trial that has protected one dimension of its integrity and left another dimension open.

Chapter 4 risk summary

The decision this chapter owns: will the trial examine its accumulating data before completion, under what rules, with what stopping authority, governed by what structure?

The most common mistake: treating the interim analysis plan as a statistical document—a spending function and a set of boundaries—without designing the governance structure that ensures the plan is implemented faithfully. The boundaries control the type I error when they are correctly applied. The governance is what ensures they are correctly applied.

The professional-level risk: the interim analysis that was conducted—or that some party believes was conducted—outside the pre-specified plan. This risk is not primarily about fraud or misconduct. It is about the gap between what the plan specified and what actually happened at the interim meeting, in the information flow between the independent statistician and the DSMB, and in the documentation of the recommendation. When that gap is large enough, the trial's primary result cannot be fully defended—not because the result is wrong, but because the process that produced it is not fully transparent. That is the professional risk: a result that is probably correct but cannot be completely verified, because the governance record does not establish that the pre-specified plan was the plan that was actually followed.

Chapter 5: How Do We Protect Against Bias?

The question Chapter 4 leaves open

Chapter 4 governed the decisions made during the trial. It specified who could see the accumulating data, under what rules, with what authority, and with what documentation. When governance works, the trial's operating integrity is maintained across time—the blind holds, the interim data stay within the DSMB, and the final analysis reflects the design that was pre-specified.

What Chapter 4 assumed, without stating, is that the comparison between arms is valid. That the patients in the treatment arm and the control arm are exchangeable at baseline. That the difference observed at the primary endpoint reflects the treatment effect and not some systematic difference in who received which treatment, how they were assessed, or how their outcomes were handled.

That assumption is the subject of this chapter. Bias—the systematic distortion of the estimated treatment effect—is not one failure but many. It can enter through the assignment mechanism, through the assessment of outcomes, through the handling of protocol deviations, or through the differential loss of patients from each arm over time. Some forms of bias are obvious and dramatic; most are subtle and invisible, operating at a level below what any single person in the trial observes. Together, they can produce a trial that is statistically significant for a treatment that does not work, or that fails to detect a treatment that does.

The tools against bias are randomization, blinding, allocation concealment, and validation. They are not independent defenses; they are a system, and the system fails when any component fails. Chapter 5 examines each component, what it protects against, and what it cannot protect against.

What bias is, and what it is not

Bias in clinical trials is a systematic error in the estimated treatment effect—an error that does not diminish as the sample size increases. This distinguishes bias from random error, which diminishes with increasing sample size and is addressed by the power calculation. A biased trial with 10,000 patients is more precisely wrong than a biased trial with 1,000 patients. The additional patients add precision to the wrong estimate.

Bias operates at the level of the comparison, not the level of the individual measurement. A measurement that is systematically too high is a biased measurement, but in a randomized trial, if both arms are measured with the same systematic error, the bias in the measurement cancels in the comparison. The comparison is unbiased even if the measurements are not. Conversely, a measurement error that differs between arms—because assessors know which arm each patient is in and adjust their assessments accordingly—introduces bias into the comparison even if each arm's measurements are individually consistent.

This distinction matters for design. The protection against bias in clinical trials is not the accuracy of the individual measurements; it is the comparability of the conditions under which measurements are made across arms. Randomization ensures that the arms are comparable at baseline. Blinding ensures that post-randomization conditions—assessment, treatment, dropout—do not systematically differ between arms in ways that are not part of the treatment effect.

When either of these mechanisms fails, the trial's comparison is distorted. The distortion may be small or large, detectable or invisible, correctable or not. What it is, always, is systematic—favoring one arm over the other in a way that is not attributable to the treatment.

What this chapter covers

Section 5.1 — Randomization Logic examines what randomization actually does and what it does not do. Randomization guarantees that treatment assignment is probabilistically independent of patient characteristics—known and unknown—at the time of assignment. It does not guarantee balance; in small samples, randomization can produce imbalanced groups by chance. And it does not guarantee that post-randomization characteristics will be balanced; if the treatment affects which patients stay in the trial, which patients comply, or which patients are assessed, post-randomization imbalance is not a randomization failure—it is a feature of the treatment effect—but it will distort the primary comparison if it is not accounted for in the estimand and the analysis.

Section 5.2 — Stratification examines the specific design choice of whether to stratify randomization by pre-specified prognostic factors. Stratification is a tool for ensuring that the known predictors of the primary outcome are balanced between arms even in relatively small samples. It is not without cost: stratification factors must be specified before randomization begins, must be accurately ascertained at the time of randomization, and must be reflected in the primary analysis to capture their efficiency benefit. When stratification factors are misspecified, inaccurately measured, or not reflected in the analysis, stratification adds administrative complexity without the statistical benefit it was intended to provide.

Section 5.3 — Allocation Concealment examines the protection that must exist between the randomization sequence and the moment of treatment assignment. Allocation concealment prevents the person enrolling the patient from knowing the upcoming assignment before the patient is enrolled. Without allocation concealment, randomization can be subverted—enrollers who know the upcoming assignment will enroll or defer patients to ensure their preferred patients receive their preferred treatment. The result is a selection bias that compromises the validity of the randomized comparison as thoroughly as the absence of randomization would.

Section 5.4 — Validation examines the specific protections against outcome assessment bias—the systematic distortion of outcome measurements that occurs when the assessor

knows, or can infer, which treatment the patient received. Validation in this context refers to the procedures for ensuring that primary outcomes are assessed by people who do not know the treatment assignment, that assessment procedures are specified in enough detail to prevent interpretation bias, and that deviations from the assessment protocol are documented and handled according to pre-specified rules.

Section 5.5 — Open-Label Designs examines the special challenges of trials in which blinding is not feasible—where the treatment itself is visible, or where it is impractical to conceal the assignment from the patient or the clinician. Open-label designs sacrifice the blinding protection but can preserve other protections through endpoint choice, centralized assessment, and pre-specified analysis strategies. The section asks what is lost when blinding is not possible and what compensating designs can recover.

The system, not the components

The chapter's sections are organized around the specific tools of bias protection. But the chapter's argument is about the system.

Randomization without allocation concealment is not randomization in the sense that matters—it is a randomization sequence that has been revealed before it can prevent selection bias. Allocation concealment without randomization protects against selection bias at enrollment but cannot protect against the baseline differences that randomization eliminates. Blinding without randomization protects against assessment bias in a comparison that was not valid to begin with. And validation of outcome assessment does not compensate for a blinding failure that occurred six months earlier.

The system works when all components work together. The failure of any component creates a gap through which bias enters, and the remaining components cannot fully compensate. This is why the bias protection design is not a checklist—check randomization, check blinding, check allocation concealment—but a system design: an examination of how the components interact, where the gaps are, and what happens when each component is tested by the realities of trial conduct.

What this chapter is not about

This chapter is not about statistical adjustments for baseline imbalance. Post-hoc covariate adjustment for observed imbalance—however sophisticated—does not substitute for the design protections that prevent imbalance from occurring in the first place. Adjustment for observed imbalance corrects for known confounders; it cannot correct for unknown

confounders that were not measured, which randomization would have controlled for and post-hoc adjustment cannot.

It is also not about missing data in the sense of methods for handling outcomes that were not observed. Missing data handling—multiple imputation, mixed-effects models, pattern mixture models—is an analysis-stage tool that addresses a design-stage problem. The design-stage problem is protecting against the differential loss of patients from each arm in ways that are related to the outcome. If that differential loss is prevented by design—through follow-up procedures, through outcome collection from withdrawals, through endpoint selection that minimizes missing data—the missing data problem is smaller. If it is not prevented, the analysis tools address the symptom but not the cause.

The cause of most missing data problems in clinical trials is a bias protection failure: patients dropped out differentially between arms, or in ways related to their prognosis, because the design did not adequately protect against the forces that drive differential dropout. This chapter is about designing against those forces. Chapter 1, in its discussion of intercurrent event strategy, is about how the design commits to handling what dropout occurs despite those protections.

The question this chapter must answer

Every bias protection decision has a cost. Randomization requires a randomization system. Stratification requires pre-specification and accurate ascertainment of stratification factors. Allocation concealment requires central randomization or sealed envelopes. Blinding requires placebo formulations, sham procedures, or central assessment by blinded assessors. Each of these costs is real, and each involves trade-offs with recruitment feasibility, patient acceptability, trial complexity, and budget.

The question this chapter must answer is not whether to incur these costs—they should generally be incurred—but how to make the trade-offs explicit and defensible. When a design element that protects against bias is omitted, the omission should be acknowledged as a design decision with known consequences, not passed over as an operational necessity. The consequences of the omission—what kinds of bias become possible, how large they might be, and what compensating protections are available—should be documented.

That documentation is not primarily for the regulatory agency, though the agency will scrutinize it. It is for the design team: the acknowledgment of a known vulnerability is the prerequisite for designing a compensating protection. A team that does not acknowledge the vulnerability cannot protect against it.

5.1 Randomization Logic

What randomization guarantees

Randomization is the foundational protection of the randomized controlled trial. Its purpose is specific and often misunderstood: randomization ensures that treatment assignment is probabilistically independent of patient characteristics—measured and unmeasured, known and unknown—at the moment of assignment. This independence is what makes the randomized comparison valid. In its absence, the difference observed between arms may reflect differences in the patients who received each treatment rather than differences produced by the treatment itself.

The guarantee randomization provides is probabilistic, not deterministic. Randomization does not ensure that the treatment and control arms are balanced on any specific characteristic. In a small trial, randomization can produce arms that differ substantially on age, disease severity, or prognostic biomarkers by chance alone. The guarantee is that these imbalances, if they occur, are not systematic—they are attributable to chance, not to a selection mechanism that favors particular patients for particular arms. The probability of any specific pattern of imbalance is known and can be calculated; its occurrence is not evidence that randomization failed.

This distinction matters because it defines what randomization can and cannot protect against. Randomization protects against systematic selection bias: the assignment of sicker, healthier, more compliant, or more motivated patients to one arm based on characteristics that predict the outcome. It does not protect against chance imbalance in small samples, post-randomization imbalance that arises from differential dropout or compliance, or the confounding that enters through unmeasured characteristics after randomization begins.

Simple, blocked, and stratified randomization

The mechanism of randomization—simple, blocked, or stratified—determines how well the protection works in practice and what its vulnerabilities are.

Simple randomization assigns each patient independently to treatment or control with fixed probability. It is the purest form of the probabilistic guarantee: each assignment is an independent draw from the allocation distribution, entirely unpredictable from prior assignments. Its limitation is that in small trials, simple randomization can produce large

arm imbalances. A trial of 50 patients using simple 1:1 randomization has a non-trivial probability of ending with 30 patients in one arm and 20 in the other. This imbalance reduces statistical efficiency and may raise questions about whether the allocation was truly random.

Blocked randomization constrains the allocation within blocks of fixed size to maintain approximate balance at each step. Within each block of size four, for example, exactly two patients are assigned to each arm. The allocation is unpredictable at the patient level but predictable at the block level: if the first three patients in a block of four have all received the same treatment, the fourth must receive the other. This predictability is the vulnerability of blocked randomization: if an enroller knows the block size and can observe prior assignments within the block, they can predict the next assignment. The exploitation of this predictability is the most common mechanism by which allocation concealment fails in trials with unmasked block randomization, and it is addressed in Section 5.3.

Stratified randomization combines blocking with stratification: separate randomization lists are maintained for each stratum defined by the stratification factors, and blocked randomization is applied within each stratum. The effect is that the arms are balanced not only overall but within each stratum. The statistical benefit is an efficiency gain in the primary analysis—which, if it reflects the stratification, will have lower residual variance—and protection against chance imbalance in the stratification factors even in small samples. The additional complexity is that the stratification factors must be pre-specified, accurately ascertained at randomization, and consistent with the randomization system's operational implementation. Section 5.2 examines these requirements in detail.

What randomization does not protect against

The most important limitations of randomization are not its internal vulnerabilities—simple vs. blocked vs. stratified—but its boundaries: what it protects and what it leaves unprotected.

Randomization protects at the moment of assignment. Once a patient is assigned to an arm, the protection against selection bias is complete for that patient. What happens afterward—whether the patient complies with the assigned treatment, whether they drop out, whether they are assessed at the planned time points, whether they receive additional treatments outside the protocol—is not protected by randomization. If any of these post-randomization events are differentially associated with the treatment assignment—if patients in the treatment arm are more likely to drop out because of side effects, for example—the comparison of outcomes at the primary endpoint may be biased by the differential loss, even though the randomization was perfectly executed.

This is the source of the estimand framework's relevance to bias. The intercurrent event strategy—the decision about how to handle dropouts, rescue medication use, and protocol deviations in the primary analysis—is a decision about what to do with the post-randomization

imbalance that randomization could not prevent. Treatment policy strategies embrace the post-randomization imbalance as part of the treatment effect in context. Hypothetical strategies attempt to estimate what the comparison would have been without the post-randomization imbalance. Neither strategy removes the imbalance; they handle it differently in the analysis, with different implications for what the result means.

Randomization protects against baseline confounding. It does not protect against confounding that enters after randomization because of the treatment itself. If the treatment causes patients to seek additional care that the control arm does not receive, the additional care is a consequence of the treatment assignment—not a confounder in the classical sense—but it may affect the primary outcome in ways that are not attributable to the treatment directly. Whether this co-intervention effect is part of the treatment effect or a confound depends on the estimand: under a treatment policy strategy, it is part of the effect of being assigned to treatment in context; under a hypothetical strategy, it may be an unwanted component that the analysis attempts to remove.

Randomization does not protect against measurement bias. If assessors know which arm each patient is in and adjust their assessments accordingly—rating treated patients' outcomes more favorably, pursuing adverse events more aggressively in the treated arm, or applying outcome criteria differently across arms—the comparison will be biased by the differential assessment, even though the assignment was random. This is the source of the blinding requirement: randomization controls the assignment; blinding controls the assessment.

Covariate-adaptive randomization

Covariate-adaptive randomization—also called minimization—is an alternative to stratified randomization in which each assignment is made to minimize the current imbalance on a set of pre-specified covariates. Rather than maintaining separate randomization lists for each stratum, the algorithm assigns each new patient to the arm that would minimize the current imbalance across all covariates simultaneously.

The advantage of covariate-adaptive randomization is that it can balance on more covariates than stratified randomization can accommodate practically—stratified randomization with more than three or four factors creates too many strata for most trials to populate adequately—while maintaining marginal balance across each factor individually.

The disadvantage is that the statistical analysis must account for the adaptive nature of the assignment. Standard analyses that treat the assignment as if it were simple randomization may produce inflated or deflated test statistics when the assignment was adaptive. The correct analysis for a covariate-adaptive randomization scheme uses a test that conditions on the covariates used in the adaptive algorithm—which is typically an adjusted analysis, not a simple comparison of arm means or proportions.

This creates an administrative requirement: the covariates used in the adaptive algorithm must be pre-specified, documented, and reflected in the primary analysis plan. An adaptive randomization scheme that is not reflected in the primary analysis plan produces a primary result that is either conservative or anti-conservative, depending on the specific design—and in either case, not correctly calibrated to the stated alpha level.

Covariate-adaptive randomization is appropriate when many prognostic factors must be balanced and stratified randomization cannot accommodate them all. It is not appropriate as a convenience substitute for a simpler design that could achieve the necessary balance through fewer stratification factors.

Cluster randomization: when the unit of assignment is not the individual patient

Cluster-randomized trials assign groups of patients—clinics, hospitals, communities, households—rather than individual patients to treatment or control. The cluster assignment is random; the individual assignment is determined by membership in the cluster. This design is appropriate when the treatment is applied at the cluster level—a system-level intervention, a care process change, an environmental modification—or when contamination between individual patients assigned to different arms is unavoidable in the same clinical setting.

Cluster randomization introduces specific vulnerabilities that individual randomization does not. The effective sample size of a cluster-randomized trial is determined not by the number of individuals but by the number of clusters and the intraclass correlation coefficient—the degree to which individuals within the same cluster are more similar to each other than individuals in different clusters. When the intraclass correlation is high, the effective sample size can be substantially smaller than the number of enrolled individuals, and the trial may be severely underpowered if the clustering is not accounted for in the sample size calculation.

Selection bias in cluster-randomized trials operates at the cluster level, not the individual level. If the choice of which clusters are enrolled—and therefore which clusters are randomized to treatment or control—is not independent of the cluster's characteristics, the comparison between arms is confounded at the cluster level. This is the cluster-level analog of individual-level selection bias, and it is addressed by the same mechanism: random allocation of clusters to arms, concealed from the people who decide which clusters participate.

The statistical analysis of cluster-randomized trials must account for the clustering in the primary analysis—typically through mixed-effects models or generalized estimating equations that capture the within-cluster correlation. A standard analysis that treats cluster-randomized data as if it were individually randomized produces a variance estimate that is too small and a test statistic that is anti-conservative: the trial appears more significant than it actually is. This is not a subtle error; it can change the interpretation of the primary result from significant to non-significant.

What this section demands before proceeding

The randomization scheme must be specified before enrollment begins, with the randomization type chosen based on the trial's size, the number of prognostic factors that need to be balanced, and the feasibility of the randomization system. The choice of randomization type should be documented with its rationale and its implications for the primary analysis.

The boundaries of randomization's protection must be acknowledged. Post-randomization imbalance—differential dropout, differential compliance, differential additional treatment—must be addressed through the estimand and the analysis strategy, not assumed away by the randomization. When the estimand and analysis strategy have been settled (in Chapter 1 and Chapter 2), they should be examined for consistency with the expected post-randomization behavior: if differential dropout is expected and the estimand is a treatment policy estimate, the data collection plan must support outcome ascertainment from dropouts.

And the randomization scheme must be matched to the planned analysis. If covariate-adaptive randomization is used, the primary analysis must condition on the covariates. If cluster randomization is used, the primary analysis must account for the clustering. A randomization scheme that is not reflected in the primary analysis produces an analysis whose type I error rate is not controlled at the nominal level—an error that is invisible in the protocol but material in the regulatory review.

References: Schulz and Grimes, "Generation of Allocation Sequences in Randomised Trials: Chance, Not Choice," Lancet 2002; Lachin, "Statistical Properties of Randomization in Clinical Trials," Control Clin Trials 1988; Taves, "Minimization: A New Method of Assigning Patients to Treatment and Control Groups," Clin Pharmacol Ther 1974; Murray, "Design and Analysis of Community Trials," Am J Epidemiol 1998.

5.2 Stratification

What stratification is for

Stratification in clinical trial randomization serves two purposes, and they are often conflated.

The first purpose is balance: ensuring that known prognostic factors are distributed similarly across treatment arms, even in moderately small samples where chance imbalance is a real risk. Without stratification, a trial of 200 patients might randomize 65% of high-risk patients to the control arm by chance, producing a comparison that appears to favor the treatment not because the treatment works but because the treated patients were at lower baseline risk. Stratification prevents this by maintaining separate randomization lists within each stratum, so that each stratum contributes proportionally to both arms.

The second purpose is efficiency: when the primary analysis reflects the stratification—by including the stratification factors as covariates, or by using a stratified test—the residual variance in the comparison is reduced, increasing the power of the primary test. This efficiency gain is real and can be substantial when the stratification factors are strongly prognostic. It is also conditional: the efficiency gain is captured only when the primary analysis accounts for the stratification. A trial that uses stratified randomization but analyzes as if randomization were simple—ignoring the stratification in the analysis—has incurred the operational cost of stratification without capturing its statistical benefit.

Both purposes are served by stratification, but they make different demands on the design. Balance is served by having stratification factors in the randomization. Efficiency is served by reflecting those factors in the analysis. A design that stratifies without analyzing accordingly has achieved one purpose and missed the other.

Choosing stratification factors

The choice of stratification factors is a clinical judgment about which patient characteristics are the strongest predictors of the primary outcome, independent of treatment. The stronger the predictor, the larger the efficiency gain from stratification, and the more damaging a chance imbalance would be to the validity of the comparison.

The general principle is to stratify on factors that are known to be strongly prognostic, that can be ascertained accurately at the time of randomization, and whose distribution

across arms would materially affect the interpretation of the primary result if imbalanced by chance.

The specific candidates for stratification are indication-specific, but several categories recur: disease severity or stage at baseline, prior treatment history, key demographic variables that predict prognosis (age, performance status in oncology), and the presence or absence of a biomarker that modifies the treatment effect. For trials in which a subgroup analysis is pre-specified—an analysis by biomarker status, by disease severity, by region—the subgroup-defining variable should be a stratification factor, so that the subgroup analysis is not confounded by a chance imbalance in that factor between arms.

The constraints on stratification factor selection are practical. Each stratification factor multiplies the number of strata by the number of levels of that factor. A design with four stratification factors each at two levels has sixteen strata. In a trial of 400 patients, the average stratum has 25 patients—manageable. In a trial of 100 patients, the average stratum has approximately 6 patients, and many strata will have too few patients to maintain balance through blocking. When strata are small, stratified randomization degenerates toward simple randomization within strata, and the balance guarantee is weakened.

The practical limit on stratification factors is approximately three to four for most trial sizes. Beyond this, covariate-adaptive randomization—minimization—is a better tool for balance across multiple factors simultaneously, as discussed in Section 5.1.

Ascertainment at randomization

The stratification factor must be ascertained accurately at the moment of randomization. This requirement is more demanding than it appears.

For factors determined by laboratory results—a biomarker level, a genetic test, a disease stage defined by imaging—the test result must be available before the patient is randomized. In practice, this means the laboratory processing time must be shorter than the window between enrollment and randomization, or the patient must be enrolled before the test result is available but not randomized until it is—a distinction that requires careful operational specification. A patient who is enrolled (consented) before the test and randomized only when the result is available is handled differently from a patient who is enrolled and randomized simultaneously.

For factors that involve clinical judgment—disease severity ratings, performance status assessments, subtype classifications—the assessment must be made by a person who does not know which treatment the patient will receive, and the criteria for the assessment must be pre-specified with enough specificity to prevent interpretation variation across sites. An assessor who knows the patient is about to be randomized and who has an opinion about what treatment the patient should receive will, consciously or not, adjust their assessment to favor the arm they prefer for that patient. This is not a hypothetical concern; it is

a documented mechanism of bias in trials where the stratification assessment is made by personnel with knowledge of the upcoming assignment.

Misascertainment at randomization—assigning a patient to the wrong stratum—has two consequences. It reduces the balance guarantee, because the patient is counted toward the wrong stratum's allocation and may shift balance within the correct stratum. And it creates an analysis complication: the stratification factor as recorded at randomization and the stratification factor as it actually applies to the patient may differ, and the primary analysis must be pre-specified to handle this discordance. The standard approach is to analyze according to the stratum as randomized, regardless of the ascertained value—preserving the integrity of the randomization sequence at the cost of using a slightly incorrect stratification value. The pre-specification of this handling rule is not administrative; it is a substantive choice with implications for the validity of the stratified analysis.

Reflecting stratification in the analysis

The efficiency benefit of stratification is captured only when the primary analysis includes the stratification factors. For continuous outcomes, this typically means including the stratification factors as covariates in the primary regression model. For time-to-event outcomes, this means including them as covariates in the Cox model or using a stratified Cox model with separate baseline hazard functions within each stratum. For binary outcomes, this means including them in the logistic model or using a Cochran-Mantel-Haenszel test stratified by the stratification factors.

The alternative—analyzing the data without accounting for stratification—treats the stratified randomization as if it were simple randomization. This is not wrong in the sense that the estimate will be biased; the randomization still ensures that the comparison is valid in expectation. But it is inefficient: the analysis does not use the information encoded in the stratification, and the residual variance is larger than it would be if the stratification were reflected. The primary test is less powerful than it would be, and the confidence interval is wider.

This inefficiency is avoidable by pre-specification. The primary analysis plan must specify that stratification factors will be included in the primary model, and this specification must be consistent with the randomization scheme. A trial that stratifies on three factors and then runs a primary analysis without including those factors in the model has failed to capture the benefit of the stratification it paid the operational cost to implement.

The regulatory agencies are attentive to the consistency between the randomization scheme and the primary analysis. The FDA and EMA both expect the primary analysis to reflect the stratification—for continuous and binary outcomes by including the stratification factors, for time-to-event outcomes by stratification in the model or sensitivity analysis confirming that the unstratified result is consistent. A primary analysis that ignores the stratification

scheme is not incorrect, but it will receive questions, and the answers to those questions must explain why the pre-specified analysis was not used.

Stratification versus subgroup analysis

Stratification is often confused with pre-specified subgroup analysis, and the confusion creates design errors that are difficult to correct after enrollment.

Stratification ensures that the randomization is balanced on a factor and improves the efficiency of the overall treatment effect estimate. It does not, by itself, enable a confirmatory subgroup analysis within each stratum.

A pre-specified subgroup analysis within a stratum is a separate hypothesis test, requiring its own multiplicity control, its own power calculation, and its own estimand specification. The fact that the subgroup-defining variable is also a stratification factor does not give the subgroup analysis any special statistical status. The subgroup analysis is still subject to the same risks of false positives and underpowered true negatives as any other subgroup analysis, and it must be pre-specified with sufficient rigor to be treated as confirmatory rather than exploratory.

Stratifying on a variable and planning a subgroup analysis on the same variable serves both purposes simultaneously—it ensures the main analysis is efficient with respect to that variable and positions the subgroup analysis as pre-specified—but the two purposes require different documentation and different statistical controls. A trial that stratifies on biomarker status and intends to make a confirmatory claim in the biomarker-positive subgroup must: calculate sample size for the subgroup analysis separately, pre-specify the analysis in the protocol and SAP, and control the type I error across the overall analysis and the subgroup analysis according to a pre-specified hierarchy. Stratification alone does not accomplish any of these.

Stratification in multi-regional trials

In trials conducted across multiple geographic regions—global trials with sites in North America, Europe, and Asia—region is sometimes proposed as a stratification factor. The motivation is legitimate: regional differences in standard of care, patient population, measurement procedures, and regulatory requirements can all produce prognostic differences between regions that, if imbalanced between arms, could distort the primary comparison.

But region as a stratification factor requires careful consideration. If the number of regions is large and the number of patients per region is small, stratification by region creates many

strata with few patients each, reducing the effectiveness of the stratification. If the treatment effect is expected to be homogeneous across regions—as is typically assumed for a global trial—the efficiency benefit of regional stratification is small, because the regions are not strong independent predictors of outcome net of other factors.

The more important question is whether the primary analysis should include region as a covariate, independently of whether it is a stratification factor. In trials where regulatory submissions must be made in multiple regions, and where each regional regulatory agency will examine the trial's primary result, including region in the primary analysis ensures that the result is not driven by regional imbalance and that the estimate is interpretable as an average effect across regions rather than the effect in the largest or most influential region. This is a pre-specification decision that should be addressed in the protocol and SAP, whether or not region is formally a stratification factor.

What this section demands before proceeding

The stratification factors must be specified before enrollment begins, with the rationale for each factor documented: why this factor is prognostic, why it must be balanced across arms, and why the expected benefit of stratification justifies the operational complexity.

The ascertainment procedure for each factor must be specified at the same level of detail. Who assesses the factor, at what time relative to randomization, with what criteria, and what happens when the assessment is ambiguous or delayed. The handling of misascertainment must be pre-specified: analyze according to the stratum as randomized, not the stratum as corrected.

The primary analysis must be pre-specified to include the stratification factors, with the specific model form documented. And the relationship between stratification and any planned subgroup analysis must be explicitly addressed: if a stratification factor is also the basis for a pre-specified subgroup analysis, the subgroup analysis requires its own power calculation, multiplicity control, and estimand specification, independent of the stratification.

The stratification is a commitment. What it commits to—balanced arms on known prognostic factors, efficient primary analysis, pre-specified handling of misascertainment—must be documented before enrollment begins, not rationalized after the data are seen.

References: Kernan et al., "Stratified Randomization for Clinical Trials," J Clin Epidemiol 1999; Pocock and Simon, "Sequential Treatment Assignment with Balancing for Prognostic Factors in the Controlled Clinical Trial," Biometrics 1975; Kahan and Morris, "Reporting and Analysis of Trials Using Stratified Randomisation in Leading Medical Journals," BMJ 2012; EMA Guideline on Adjustment for Baseline Covariates in Clinical Trials (2023).

5.3 Allocation Concealment

The gap between randomization and assignment

Randomization produces a sequence. Allocation concealment protects that sequence from being known to the people who enroll patients until the moment of assignment—and not before.

This distinction—between the randomization sequence and the allocation—is where the bias protection system is most commonly breached. The randomization sequence may be perfectly generated: truly random, appropriately blocked, correctly stratified. But if the sequence can be determined—by the enroller, the investigator, or the site coordinator—before the patient is committed to trial participation, the randomization fails its purpose. The person making the enrollment decision can use knowledge of the upcoming assignment to determine whether to enroll the next eligible patient.

This is not a hypothetical mechanism. It is the most reliably documented source of selection bias in published clinical trials. Studies of trials with inadequate allocation concealment—where the sequence was accessible before enrollment or where the blocking made the next assignment predictable—consistently find inflated treatment effects compared to trials with adequate concealment. The effect estimate in a trial with inadequate allocation concealment is biased in the direction of the treatment, because investigators who believe in the treatment will preferentially enroll favorable patients when they know those patients will be assigned to the active arm.

The bias introduced by inadequate allocation concealment is not correctable after enrollment. It is embedded in the baseline characteristics of the enrolled population, invisible in the data, and impossible to distinguish from a true treatment effect using any analysis method. The only protection is preventing the breach from occurring.

What concealment requires in practice

Adequate allocation concealment requires that three conditions be met.

First, the assignment sequence must not be accessible to the enrolling person before the patient is committed to participation. “Committed to participation” means the patient has consented, the eligibility criteria have been verified, and the decision to enroll has been

made—independently of which arm the patient will be assigned to. Only after these steps should the assignment be revealed. This sequence—eligibility verification and consent before assignment revelation—is the operational core of allocation concealment.

Second, the sequence must be held by a party who is not the enroller and not the investigator. In practice, this means either central randomization through an interactive voice or web response system (IVR/IWR), which reveals the assignment only after the enrollment data are transmitted and logged, or sequentially numbered, sealed, opaque envelopes held by a person who is not part of the trial team at the enrolling site. Central randomization is the more robust mechanism, because the log of assignment requests provides an auditable record of when the assignment was revealed and to whom. Envelopes are prone to tampering—opening and resealing, or selecting between envelopes—and are increasingly regarded as inadequate for trials where the treatment effect claim requires high credibility.

Third, the assignment must not be inferable from prior assignments. This is where blocked randomization creates its characteristic vulnerability. If the block size is known and fixed, an enroller who has observed several assignments within a block can determine the remaining assignments by elimination. The standard mitigation is variable block size—blocks of different sizes, selected at random—so that the pattern of prior assignments does not determine the current assignment even to someone who knows the possible block sizes. The block sizes themselves should not be revealed in the protocol (though they should be specified in the statistical analysis plan, which is not publicly available during the trial). A protocol that specifies "randomization will be performed in blocks of six" provides the information needed to exploit the blocking, and should instead specify "randomization will be performed in blocks of variable size."

Inadequate concealment in common trial designs

The literature on allocation concealment distinguishes between adequate and inadequate mechanisms, and the distinction is not always the one that trial designers expect.

Alternating assignment—every other patient to treatment, every other to control—is not randomization and provides no concealment. The next assignment is always known. This is a historical control method that predates randomized trials, still occasionally used in situations where randomization is impractical, but providing none of the protection of true randomization.

Open allocation lists—a randomization list posted at the site, or accessible through the site's clinical trial management system—provide a sequence that is correct but not concealed. Any site staff member with system access can determine the upcoming assignments. The sequence is random; the assignment is predictable.

Sealed envelopes without opacity or sequential numbering—envelopes that can be held up to light, that are not numbered, or that can be opened and resealed—are inadequate because they provide insufficient protection against tampering. Opaque, sequentially

numbered, sealed envelopes are the minimum standard for envelope-based concealment, and even this minimum is subject to manipulation at sites with poor oversight.

Predictable block sizes without variable blocking—the documented vulnerability of fixed-block randomization to prediction at the end of blocks—is inadequate concealment even when a sealed central randomization system is used, because an enroller who has tracked prior assignments can predict the current assignment. Variable block sizes close this gap.

The regulatory agencies—FDA and EMA—both require description of the allocation concealment method in the protocol and in the clinical study report, and both assess the adequacy of the described method. A trial whose primary result is positive but whose allocation concealment was inadequate will face scrutiny about whether the treatment effect estimate is inflated by selection bias. The inadequacy cannot be corrected post-hoc, and the scrutiny may result in a requirement for additional trials or in labeling that reflects the uncertainty.

The relationship between blinding and concealment

Allocation concealment is frequently confused with blinding, and the distinction matters for design.

Allocation concealment protects the assignment sequence before randomization—it prevents prior knowledge of the assignment from influencing the enrollment decision. Blinding protects the comparison after randomization—it prevents knowledge of the assigned treatment from influencing the assessment of outcomes, the provision of care, or the dropout behavior of patients and clinicians.

A trial can have adequate allocation concealment and inadequate blinding: the assignment was not known before enrollment, but once enrolled, the patient and clinician both know which treatment was assigned, and this knowledge affects the subsequent course of the trial. An unblinded trial with adequate allocation concealment is valid in the sense that the enrolled population is not selectively biased; but it is vulnerable to assessment bias and care bias after enrollment.

A trial can have inadequate allocation concealment and adequate blinding: the assignment was predictable before enrollment—for example, through open allocation lists—but after enrollment, neither the patient nor the clinician knows which arm the patient is in, because the treatment is administered as identical capsules. The enrolled population is biased; the assessment is not.

In practice, the two mechanisms often fail together: trials that do not use central randomization—and therefore have inadequate concealment—also tend to have inadequate blinding, because the operational systems that support central randomization often also support blinded treatment dispensing. But they are distinct mechanisms protecting against distinct forms of bias, and their design requirements should be specified separately.

Concealment in platform and adaptive trials

Platform trials—multi-arm trials where new treatment arms can be added and closed as evidence accumulates—and adaptive trials—trials where the design parameters are modified based on accumulating data—introduce specific allocation concealment challenges that standard fixed-design trials do not face.

In a platform trial, the allocation ratios between arms may change over time as arms are added or closed. The changing ratios can, if visible to site personnel, reveal information about the current allocation sequence and potentially about the interim results that drove the change. Concealment in platform trials requires that the rationale for allocation ratio changes not be communicated in ways that reveal the interim efficacy data, and that the mechanisms for changing ratios be designed to minimize the inferential information they provide to site personnel.

In adaptive trials with sample size re-estimation, the decision to increase the sample size may itself convey information about the interim trend: a sample size increase signals that the interim effect was below expectations, and a trial that does not increase the sample size signals that the interim effect is on track or better. This inference cannot be fully prevented when the re-estimation decision is visible, but it can be mitigated by pre-specifying the re-estimation rules precisely enough that the decision conveys minimal information beyond what the rules would predict.

These are not reasons to avoid platform or adaptive designs. They are reasons to design the allocation concealment system for those trials with the same care that standard trials require—and with additional attention to the ways in which the adaptive features create new information flows that standard concealment mechanisms do not address.

What this section demands before proceeding

The allocation concealment mechanism must be specified before enrollment begins, with sufficient detail to be audited. The description should identify: who holds the randomization sequence, how assignment requests are made and logged, what information is revealed at the time of assignment and to whom, and what safeguards prevent the sequence from being inferred before the assignment is made.

The block structure of the randomization—whether fixed or variable, and if variable, what range of block sizes is used—must be documented in the SAP (not the protocol) and the variable structure must be confirmed before the first assignment. If central IVR/IWR randomization is used, the system must be validated before enrollment begins to confirm that assignments are not accessible before the enrollment request is submitted.

And the distinction between concealment and blinding must be clear in the design documentation. Both must be specified; neither substitutes for the other. The vulnerability that concealment protects against—selection bias at enrollment—and the vulnerability that blinding protects against—assessment and care bias after enrollment—are different mechanisms requiring different design solutions. Section 5.4 addresses the latter.

References: Schulz and Grimes, "Allocation Concealment in Randomised Trials: Defending Against Deciphering," Lancet 2002; Wood et al., "Empirical Evidence of Bias in Treatment Effect Estimates in Controlled Trials with Different Interventions and Outcomes," BMJ 2008; Higgins et al., Cochrane Handbook for Systematic Reviews of Interventions (2019), Section 8.3; Altman and Schulz, "Statistics Notes: Concealing Treatment Allocation in Randomised Trials," BMJ 2001.

5.4 Outcome Assessment and Validation

The assessor who knows too much

Randomization ensures that the groups are comparable at baseline. Allocation concealment ensures that knowledge of the upcoming assignment did not influence who was enrolled. These two protections together make the trial's starting comparison valid.

What happens between enrollment and the primary outcome measurement is where that validity can be lost.

If the person assessing the primary outcome knows which treatment the patient received—or can infer it from the patient's clinical course, side effect profile, or laboratory values—that knowledge will influence the assessment. Not through dishonesty; through the ordinary psychology of clinical judgment, which integrates all available information, including contextual information that should be irrelevant to the measurement. An assessor who knows a patient is in the treatment arm will, on average, be slightly more optimistic in rating improvement, slightly less aggressive in pursuing adverse events that could complicate the picture, and slightly more tolerant of ambiguity in the patient's favor. These adjustments are small individually, and they are invisible at the level of the individual assessment. Aggregated across hundreds of assessments, they produce a systematic bias in the treatment arm's direction.

This is assessment bias, and it is the mechanism that blinding protects against. Blinding—masking the treatment assignment from the assessor—removes the contextual information that would otherwise influence the assessment. When blinding is complete, the assessor's judgment is as close as human measurement allows to the truth about the patient's status. When blinding fails—partially, or for specific patients, or at specific time points—the assessment bias returns.

What must be blinded, and why

Three parties can be blinded in a clinical trial: the patient, the clinician providing care, and the assessor of the primary outcome. Blinding all three is the double-blind design; blinding only the patient and the assessor is sometimes called double-blind with a blinded assessor; blinding only the assessor is the single-blind assessor design. Open-label designs blind no one and are addressed in Section 5.5.

The patient must be blinded when the primary outcome is patient-reported. If the patient knows they are receiving the active treatment—and particularly if they expect to benefit from the active treatment—their self-assessment of symptoms, function, and health status will be influenced by that expectation. Placebo effects in patient-reported outcomes are large, well-documented, and not adequately controlled by any analysis method after the fact. They must be prevented by blinding.

The clinician providing care must be blinded when their clinical decisions—additional medications, referral patterns, the intensity of monitoring—will affect the primary outcome. If the clinician knows a patient is in the active arm and believes the treatment is effective, they may manage that patient differently from control arm patients in ways that produce differential outcomes independent of the direct treatment effect. This co-intervention bias is prevented by keeping the clinician unaware of the assignment.

The assessor of the primary outcome must be blinded when the assessment involves judgment. A binary outcome that is defined by a specific laboratory value above a specified threshold requires no assessor judgment; the measurement is objective and the assessor's knowledge of the assignment cannot distort it. But most primary outcomes in clinical trials—clinical ratings, imaging reads, endpoint adjudications, functional assessments, quality of life instruments—involve judgment at some stage, and judgment that integrates knowledge of the treatment assignment is biased judgment.

Blinding in practice: maintaining it under pressure

Blinding in clinical trials is not a binary state. It is a gradient that is established at randomization, maintained through trial conduct, and eroded by the clinical events that accumulate as the trial proceeds.

The most common sources of blinding erosion are the treatment's recognizable side effects. A drug with a characteristic side effect profile—a statin causing muscle aches, a beta-blocker causing bradycardia, an ACE inhibitor causing cough—reveals the assignment to observant patients and clinicians regardless of placebo control. When the side effect profile is distinctive and the blinding assessment shows that patients and clinicians correctly guess their assignment at rates substantially above chance, the blinding has been compromised in a way that the placebo controls cannot prevent.

The design responses to characteristic side effect profiles are limited. Active placebos—control treatments that mimic specific side effects of the active treatment—can restore blinding but add safety concerns and regulatory complexity of their own. Centralized outcome assessment by blinded assessors who do not see the clinical notes can remove the assessor from the information that erodes blinding, even when the treating clinician's blinding has been compromised. Independent endpoint adjudication by blinded committees is the standard approach for hard clinical endpoints.

Blinding assessment—formally asking patients and clinicians to guess their assignment at specified time points—provides a direct measure of the degree to which blinding has been maintained. The results of blinding assessments should be reported as part of the trial's transparency. When blinding assessment reveals substantial unblinding—more than 60-70% correct guesses, depending on the expected rate by chance—the trial's reliance on blinding as the primary protection against assessment bias is weakened, and the sensitivity analysis plan should include an assessment of how the primary result would change if the estimated assessment bias were removed.

Independent adjudication committees

For hard clinical endpoints—death, myocardial infarction, stroke, hospitalization for a specified cause—independent endpoint adjudication is the standard mechanism for ensuring that the primary outcome is assessed consistently and without knowledge of treatment assignment.

An endpoint adjudication committee (EAC) reviews the clinical documentation for each potential endpoint event—ECG tracings, hospital records, imaging reports, discharge summaries—without seeing the patient's treatment assignment, and classifies the event according to pre-specified criteria. The pre-specified criteria are the adjudication charter, which must be developed and finalized before any events are adjudicated.

The adjudication charter is a design document, not an analysis document. It specifies the case definition for each endpoint—what clinical and diagnostic criteria must be met for an event to be classified as the primary endpoint—with sufficient specificity that two adjudicators reviewing the same documentation would reach the same conclusion in the large majority of cases. When the charter is ambiguous—when a case definition is stated in terms that allow substantial interpretive variation—the adjudication is effectively an exercise in judgment that can be influenced by unblinding, by adjudicators' baseline beliefs about the treatment, or by subtle shifts in the standard applied across cases.

The adjudication charter must be finalized before the trial begins, or at minimum before any cases are adjudicated. A charter developed after cases have accumulated—after the adjudicators have been exposed to the clinical course of enrolled patients—is subject to the suspicion that its case definitions were shaped by the cases rather than the science. This is not necessarily true, but it cannot be excluded, and the uncertainty is damaging to the trial's credibility.

The independence of the EAC from the sponsor and from the site investigators is the same kind of independence required of the DSMB: no financial or professional relationships that would compromise the ability to adjudicate against the sponsor's interest. EAC members should not be investigators in the trial, should not have access to treatment assignments, and should not see unblinded interim results that might bias their adjudication of subsequent cases.

Outcome validation: the adjudication charter as evidence

The adjudication charter serves a second purpose beyond consistency: it defines what the primary endpoint means. A myocardial infarction in this trial is what the adjudication charter says it is—not what an individual investigator's clinical judgment says it is, not what a general cardiology textbook says it is, but what the pre-specified case definition specifies.

This definitional role is important for two reasons. First, it ensures that the primary endpoint is measured consistently across all sites and across the full duration of the trial. A primary endpoint whose definition varies across sites, across time, or across adjudicators is not measuring the same thing in each patient, and the comparison between arms reflects both the treatment effect and the measurement inconsistency. Second, it connects the trial's primary result to the estimand. The primary endpoint—as measured by the adjudication process—is the variable attribute of the estimand. If the adjudication charter does not produce a measurement that corresponds to the variable the estimand specifies, the primary analysis is estimating something different from what the estimand defines.

The adjudication charter must therefore be developed in coordination with the estimand specification. If the estimand defines the primary endpoint as a specific composite of events—each with specific clinical criteria—the adjudication charter must specify the case definitions for each component event in a way that operationalizes the estimand's variable attribute. This coordination is not always performed. Estimand specification and adjudication charter development often proceed on separate tracks, by different teams, without explicit reconciliation. The result is a disconnect between what the trial claims to be measuring and what the adjudication actually produces.

Blinding of the statistical analysis

A less commonly discussed form of blinding is the blinding of the statistician to the treatment assignment during the exploratory data analysis and sensitivity analysis development. In most trials, the statistician sees the unblinded data at the point of formal analysis. Before that point, some trials use a blinded review process—examining the distribution and quality of the data without the treatment assignment—to finalize the analysis plan before unblinding.

The blinded review is not a substitute for pre-specification; the primary analysis must be specified before the data are seen, regardless of whether the statistician sees unblinded data. But the blinded review serves a specific purpose: it allows the statistician to examine the data's adequacy—the distribution of the primary outcome, the extent of missing data, the characteristics of the enrolled population—and to address any issues that might affect

the primary analysis, without the opportunity to reverse-engineer the analysis to favor the observed treatment assignment.

A statistician who sees unblinded data before the analysis is finalized is not necessarily biased—professional integrity matters, and the pre-specified analysis constrains the interpretation—but the appearance of independence is compromised. For trials where the primary result will be contested—large trials in important indications, trials in contentious therapeutic areas—the blinded review process adds a layer of protection against the appearance of motivated analysis that is proportional to the importance of the result.

What this section demands before proceeding

The blinding design must specify who is blinded—patient, clinician, assessor—and through what mechanism. The mechanism must be sufficient to maintain blinding through the expected clinical course: if a characteristic side effect is likely to erode blinding, the design must address this explicitly, either through an active placebo, centralized assessment, or a documented acknowledgment of the expected erosion and its likely effect on the primary estimate.

The adjudication charter must be finalized before the first event is adjudicated. It must be specific enough to achieve high inter-rater agreement, and its case definitions must be consistent with the estimand's variable attribute. The independence of the adjudication committee must be established and documented.

The blinding assessment plan must be specified: when blinding will be formally assessed, by what method, and what will be done if the assessment reveals substantial unblinding. These are not administrative requirements. They are the measurement standards that make the primary outcome meaningful as an estimate of the estimand.

References: Boutron et al., "Methods and Processes of the CONSORT Group: Example of an Extension for Trials Assessing Nonpharmacological Treatments," Ann Intern Med 2008; Sackett, "Bias in Analytic Research," J Chron Dis 1979; Meinert, Clinical Trials: Design, Conduct, and Analysis (1986); Domanski et al., "Pitfalls in the Design of Cardiovascular Outcome Trials," Circulation 2003.

5.5 Open-Label Designs

When blinding is not available

Blinding is the design's protection against assessment bias. But blinding is not always achievable. Some treatments are inherently identifiable—a surgical procedure versus no surgery, a device implanted under imaging guidance versus a sham procedure, an intravenous infusion with visible characteristics versus an oral tablet, a behavioral intervention that requires patient engagement versus no intervention. Some patient populations cannot be masked to treatment without deception that ethics committees will not approve. Some comparators have side effect profiles so distinctive that placebo matching is impossible.

In these situations, the trial must be conducted open-label: the patient knows which treatment they are receiving, the clinician knows, and the assessor—if they receive clinical information about the patient—also knows. The blinding protection is absent, and the bias risks that blinding would prevent are real.

The appropriate response to this situation is not to abandon the trial. Open-label trials have produced clinically important evidence in many indications—surgical trials, device trials, behavioral intervention trials, and open-label extensions of blinded trials have all contributed to the evidence base for clinical practice. The appropriate response is to design the trial to compensate for the absent blinding protection, using the tools that remain available.

What open-label designs are at risk for

When the treatment assignment is known, three forms of bias become possible that blinded designs prevent.

Patient-reported outcome bias is the most serious risk in open-label trials where the primary endpoint is patient-reported. Patients who know they are receiving the active treatment expect to improve; their reports of symptoms, function, and quality of life are influenced by that expectation. The placebo response—the improvement that occurs because of the belief that one is receiving a beneficial treatment—is not a measurement error; it is a real biological and psychological phenomenon. But when the comparison between arms includes patients in the active arm who have benefited from both the treatment and the placebo response, and patients in the control arm who have not, the difference between arms reflects both effects. The open-label trial cannot separate them.

The design response is endpoint selection. If a patient-reported primary outcome can be replaced by an objective outcome—a laboratory measurement, an imaging finding, an event adjudicated by a blinded committee from clinical documentation—the placebo response is removed from the primary comparison. The objective outcome may be less sensitive to the treatment's effect than the patient-reported outcome; this is a cost. It is the cost of trading interpretive ambiguity for bias protection.

Assessment bias occurs when clinicians who know the treatment assignment make clinical decisions—ordering additional tests, adjusting concurrent medications, classifying clinical events—differently across arms. A clinician who believes the patient in the active arm is improving may be more likely to classify an ambiguous event as an adverse effect of the control treatment, or less likely to classify it as an adverse effect of the active treatment. These classification differences are subtle, largely unconscious, and difficult to prevent when the treatment assignment is known and the clinician has strong prior beliefs about the treatment's effect.

Dropout differential occurs when patients in the control arm, knowing they are not receiving what they believe to be the better treatment, withdraw from the trial at higher rates than patients in the active arm. When dropout is informative—when the patients who drop out are sicker, or less responsive, or more burdened—the control arm from which the dropouts have left is no longer representative of the originally enrolled population. The comparison at the primary endpoint is between the treated arm and a selected control arm survivor population.

Compensating designs for open-label trials

The tools available to compensate for absent blinding vary by the form of bias they address, and each involves a trade-off.

Blinded endpoint adjudication separates the assessment of the primary outcome from the clinical management of the patient. A central adjudication committee reviews clinical documentation—imaging, laboratory results, hospital records—without knowledge of the treatment assignment, and classifies events according to a pre-specified adjudication charter. This mechanism protects against assessment bias for adjudicable endpoints. It does not protect against patient-reported outcome bias, because the patient's self-report is generated with knowledge of the assignment. And it does not protect against dropout differential, because the dropout has already occurred before the adjudicator reviews the remaining patients' outcomes.

The effectiveness of blinded adjudication depends on the degree to which the clinical documentation itself reveals the treatment assignment. Imaging reports that include description of treatment-related changes, hospital notes that mention the patient's treatment, and laboratory values that are characteristic of the active treatment can unblind the adjudicator through the documentation they review. The adjudication process must be designed to

remove or redact this information from the documentation reviewed by the committee—a more operationally demanding requirement than is usually anticipated.

Objective endpoints replace subjective clinical assessments with measurements that do not require clinical judgment. A trial of a treatment for arrhythmia can use a device-recorded continuous ECG as the primary endpoint rather than a clinician's classification of arrhythmia severity; a trial of a treatment for bone density loss can use DEXA scan measurement rather than a clinician's assessment of fracture risk. Objective endpoints are not influenced by the assessor's knowledge of the treatment assignment, because the measurement is automated rather than judged.

The limitation is that objective endpoints may not capture the treatment effects that matter most to patients. A treatment that reduces arrhythmia burden on a continuous ECG but does not reduce symptoms may or may not improve quality of life; the ECG endpoint measures the mechanism, not the consequence. In open-label trials where symptoms are the primary concern, the conflict between bias protection and clinical relevance is not easily resolved by endpoint substitution.

Placebo-run-in and enrichment are sometimes used to reduce dropout differential by enrolling only patients who demonstrate the willingness and ability to adhere to the trial regimen before randomization. A run-in period in which all patients receive placebo or a standardized treatment, and only those who complete the run-in are randomized, reduces the dropout differential in the randomized phase at the cost of a more selected, less generalizable enrolled population. This is an enrichment strategy that changes the estimand—the trial is now estimating the treatment effect in the population that completed the run-in, not in the population initially eligible—and the change must be reflected in the design documentation.

The estimand for open-label trials

The estimand is particularly consequential for open-label designs because the choice of intercurrent event strategy determines what the trial is claiming to show about a treatment whose benefits and risks cannot be separated from the patient's knowledge of receiving it.

Under a treatment policy estimand, the open-label trial estimates the effect of receiving the treatment in a clinical context where patients and clinicians know the assignment—which is the natural context of clinical use. The result includes the placebo component of the benefit, which may be a legitimate part of the treatment's clinical value if it is present in practice. This is defensible but requires honesty: the treatment policy estimand from an open-label trial does not separate the biological effect from the knowledge effect, and the label should acknowledge this.

Under a hypothetical estimand—what would the treatment effect be if patients did not know which treatment they were receiving?—the open-label trial cannot produce a direct estimate. The hypothetical estimand requires a counterfactual condition that the open-label

trial did not implement. Estimating it requires modeling assumptions about the size of the knowledge-related component of the benefit, which cannot be validated from the trial's own data.

This is why endpoint choice is so important in open-label trials: objective endpoints, unlike subjective ones, produce outcomes that are less influenced by the knowledge of assignment, and therefore closer to the hypothetical estimand. If the trial's intended claim is about the biological effect of the treatment rather than its contextual effect including the patient's knowledge, the primary endpoint should be as objective as possible.

Open-label extensions

A specific and common form of open-label data arises in the open-label extension phase that follows a blinded controlled trial. In these extensions, patients who completed the blinded phase are offered continued treatment, usually open-label, to generate long-term safety and effectiveness data.

Open-label extension data have a specific vulnerability: the patients who enter the extension are not representative of the patients who were randomized. Patients who completed the blinded phase and chose to enter the extension are patients who tolerated the treatment well enough to complete the controlled phase and who were willing to continue. Patients who dropped out during the blinded phase—often for tolerability reasons or lack of response—are not in the extension. The extension population is therefore a selected survivor population, and estimates of long-term safety and effectiveness derived from it may be systematically favorable.

This selection bias is intrinsic to open-label extension designs and cannot be removed by analysis. The appropriate response is to acknowledge it in the design and in the reporting, and to be explicit about what the extension data can and cannot show. Extension data are appropriate evidence for characterizing the safety profile in long-term users, for examining whether the treatment effect is durable, and for informing dosing decisions in patients who tolerate the treatment well. They are not appropriate evidence for estimating the treatment effect in the full enrolled population or for making claims about what would happen if all patients who started the treatment continued it.

What this section demands before proceeding

The decision to use an open-label design—when blinding is not feasible—must be documented as a design choice with known consequences, not as an operational necessity that escaped scrutiny. The documentation should identify which forms of bias the open-label design

introduces, what compensating design elements are used to address each form, and what residual bias remains after the compensating elements are in place.

The compensating elements must be specified before enrollment begins. Blinded adjudication requires a finalized adjudication charter before the first event; central assessment by blinded assessors requires the assessors to be identified and trained before enrollment; objective endpoint selection requires the endpoint to be pre-specified in the estimand and the protocol.

And the estimand must reflect the open-label context. If the treatment policy estimand is used—estimating the effect of receiving the treatment in the context where the assignment is known—the result should be interpreted as inclusive of the knowledge component of the benefit. If the endpoint choice is designed to minimize the knowledge component, this should be stated as a design intent in the protocol.

An open-label trial is not a compromised trial. It is a trial that has confronted a specific set of bias risks honestly and designed to manage them explicitly. The management is imperfect—some bias remains—but the imperfection is known and documented, and the result is interpretable within the limits that the design acknowledges.

References: Wood et al., "Empirical Evidence of Bias in Treatment Effect Estimates in Controlled Trials with Different Interventions and Outcomes," BMJ 2008; Hrobjartsson et al., "Observer Bias in Randomised Clinical Trials with Binary Outcomes," BMJ 2012; Noseworthy et al., "The Impact of Blinding on the Results of a Randomized, Placebo-Controlled Multiple Sclerosis Clinical Trial," Neurology 1994; Boutron et al., "Reporting and Interpretation of Randomized Controlled Trials with Statistically Nonsignificant Results for Primary Outcomes," JAMA 2010.

5.6 Closing: The System That Holds the Comparison Valid

What this chapter asked

Chapter 5 asked one question: how do we protect against bias?

It asked that question in five sections, each addressing a different mechanism by which the trial's primary comparison can be distorted.

Section 5.1 established what randomization actually guarantees—probabilistic independence of assignment from patient characteristics at the moment of assignment—and what it does not protect against: post-randomization imbalance arising from differential dropout, compliance, or additional treatment. The boundaries of randomization's protection define the terrain that the other mechanisms must cover.

Section 5.2 examined stratification as a tool for balance and efficiency, and distinguished between the two purposes it serves. Balance is served by having the factors in the randomization; efficiency is captured only when those factors are reflected in the primary analysis. Stratification that is not reflected in the analysis has paid the operational cost without receiving the statistical benefit.

Section 5.3 isolated allocation concealment as the mechanism that prevents the randomization sequence from being known before the enrollment decision. Without this protection, randomization can be subverted by investigators who prefer specific assignments for specific patients—producing a selection bias that is invisible in the data and uncorrectable in the analysis.

Section 5.4 examined outcome assessment—the protections that must exist between the treatment assignment and the measurement of the primary outcome. Blinding, independent adjudication, and pre-specified assessment criteria are the tools; the target is preventing the assessor's knowledge of the assignment from influencing what they measure.

Section 5.5 addressed open-label designs—situations where blinding is not feasible and the design must compensate through endpoint choice, centralized adjudication, and honest acknowledgment of the bias that remains. An open-label trial with well-designed compensating protections is not a compromised trial. It is a trial that has confronted its limitations explicitly.

What this chapter decided

By the end of this chapter, four things must be documented and their consistency with each other confirmed.

The randomization scheme is specified—mechanism, block structure, stratification factors—and its implications for the primary analysis are documented. If covariate-adaptive randomization is used, the primary analysis conditions on the covariates. If cluster randomization is used, the primary analysis accounts for the clustering. If stratified randomization is used, the stratification factors are included in the primary analysis model. The randomization scheme and the analysis plan are consistent.

The allocation concealment mechanism is specified with sufficient detail to be audited. Who holds the randomization sequence, how assignments are revealed, what prevents prediction from prior assignments. If central IVR/IWR randomization is used, the system is validated. If envelopes are used, they are opaque, sequentially numbered, and held by a party independent of the trial team.

The outcome assessment plan is specified—who assesses the primary outcome, under what blinding conditions, according to what pre-specified criteria. The adjudication charter is finalized before the first event. The blinding assessment plan specifies when and how blinding will be formally evaluated, and what will be done if substantial unblinding is detected.

For open-label designs: the specific bias risks introduced by the absence of blinding are identified, the compensating design elements are specified, and the residual bias is acknowledged. The estimand reflects the open-label context. The endpoint choice is documented as a deliberate decision to minimize the knowledge-related component of the outcome, or to include it, with the rationale stated.

These four things—randomization, concealment, assessment, and open-label management—form the bias protection system. They are documented together, reviewed for consistency with each other and with the estimand, and owned by the design team as a system, not as independent checklist items.

The characteristic mistakes of this chapter

Three failures recur in the territory Chapter 5 covers.

The allocation concealment that was specified but not implemented. The protocol specified central randomization through an IVR system. The IVR system was not validated before enrollment; site coordinators were given the access credentials before the validation was complete; the system had a configuration error that allowed coordinators to query the assignment without committing the enrollment. The error was discovered after several patients were enrolled—and after the site coordinator at the highest-enrolling site had been querying the system before enrollment. The randomization sequence was compromised at

that site. The trial continued, the data from that site were included in the primary analysis, and the regulatory reviewer noted the IVR system validation timeline in the audit trail. The explanation required was possible; the explanation required was also evidence of the gap between specification and implementation.

The adjudication charter that was finalized after the events. The trial enrolled and began accumulating endpoint events before the adjudication charter was finalized. The adjudication committee reviewed interim events while the charter was still in draft. When the charter was finalized, several of the events already adjudicated were reclassified under the final case definitions—some from endpoints to non-endpoints, some from one endpoint type to another. The reclassification was disclosed in the clinical study report. It was also evidence that the adjudication process was influenced by exposure to the cases rather than preceded by the pre-specified standards. The regulatory reviewer's question—were the case definitions shaped by knowledge of the accumulated events?—could not be answered with complete certainty.

The open-label design whose bias was not acknowledged. The trial was open-label because the treatment—a structured exercise program—could not be concealed from patients. The primary endpoint was a patient-reported functional assessment. The design documentation noted that the open-label nature was unavoidable and did not specify any compensating protections. The result showed a large treatment effect on the functional assessment. The regulatory reviewer asked what proportion of the effect was attributable to the patient's knowledge of receiving the exercise intervention rather than to the physiological effects of the exercise itself. There was no design-stage answer to that question, because it was not asked at design. The trial was informative but its result was uninterpretable in the dimension that the regulatory question required.

What cannot be recovered

Selection bias from failed allocation concealment cannot be corrected by analysis. The enrolled population in the compromised trial is not a random sample from the eligible population; it is a selectively enrolled population that reflects the investigators' preferences. No covariate adjustment, no sensitivity analysis, and no post-hoc reweighting can restore the validity of the comparison, because the selection was made on characteristics that may not have been measured and cannot be identified after the fact.

Adjudication charter defects similarly cannot be corrected after the events are adjudicated. An adjudication that occurred under a draft charter, or that reclassified events after the charter was revised, is an adjudication whose case definitions were not fully pre-specified. The ambiguity about what the events were classified as—before the charter was revised, or after—means the primary endpoint count is uncertain, and the uncertainty is not reducible by further analysis.

Assessment bias from failed blinding leaves a residual uncertainty that the primary result must be interpreted in light of. If blinding assessments reveal that patients correctly guessed their assignment at a rate substantially above chance, the primary result is larger than the true biological effect by an unknown amount. The amount is not recoverable from the trial's own data. It requires a separate experiment—a trial of the same treatment with a different assessment method—which may not be feasible or funded.

Open-label bias from an unacknowledged and unaddressed patient-reported outcome in an unblinded trial is, in the most honest sense, the trial's own design decision made visible. The decision not to use compensating protections was a design decision. Its consequence—that the primary result cannot be attributed solely to the biological effect of the treatment—is the consequence of that decision. The result is real; it is the result of assigning patients to the treatment in the context where they know they are receiving it. Whether that context-inclusive result is the clinically relevant question is what the estimand should have specified.

The connection to Chapter 6

Chapter 6 asks what the trial is allowed to claim. The bias protection system of Chapter 5 determines the validity of the primary comparison that any claim must rest on.

A claim requires evidence. Evidence in a randomized trial is the comparison between arms that randomization, allocation concealment, and blinding together have made valid. When the comparison is valid—when bias has been adequately controlled—the evidence supports a claim in direct proportion to its statistical strength. When the comparison is compromised—when selection bias, assessment bias, or inadequate concealment have distorted the comparison—the evidence supports a weaker claim, regardless of the statistical significance.

Chapter 6 will introduce the concept of claim discipline: the principle that what can be asserted from a trial is bounded by what was pre-specified and what the evidence supports. The bias protection system of Chapter 5 defines the floor of that evidence: the minimum validity that the primary comparison must have to support any claim. A trial whose primary comparison is compromised by known bias has a lower floor, and its claims must be calibrated accordingly—claimed with narrower scope, with explicit acknowledgment of the bias risk, and with less confidence than an uncompromised trial would support.

This connection runs in both directions. Chapter 5 protects the comparison that Chapter 6 claims from. Chapter 6's discipline in claiming is what prevents a compromised comparison from being asserted with more confidence than the bias risk justifies. Together, they protect the integrity of the trial's result: the comparison is as valid as the design made it, and the claim is as confident as the validity supports.

Chapter 5 risk summary

The decision this chapter owns: what mechanisms protect the trial's primary comparison from systematic distortion, and are they specified, implemented, and documented with sufficient rigor to make the comparison valid?

The most common mistake: treating the components of bias protection—randomization, concealment, blinding, adjudication—as independent checklist items rather than as a system. A trial that has excellent randomization but inadequate allocation concealment has failed; a trial with adequate concealment and excellent randomization but no blinding assessment has an unknown bias risk at the primary endpoint. The components protect against different mechanisms of bias, and the absence of any one component leaves a gap that the others cannot close.

The professional-level risk: a primary result that is statistically significant but scientifically uninterpretable because the bias protection system was inadequate. Not fraudulent—the trial was conducted according to its protocol—but inadequate. The result cannot be attributed to the treatment because the comparison cannot be attributed to the randomization. The trial has produced a number. It has not produced evidence. The difference between those two things is the bias protection system, and the professional risk of Chapter 5 is discovering, after the database is locked, that the system was not in place.

Chapter 6: What Are We Allowed to Claim?

The question Chapter 5 leaves open

Chapter 5 protected the comparison. It established that the treatment and control arms are comparable at baseline, that the assignment was not predictable before enrollment, that the outcome was assessed without knowledge of the assignment. When all of this works, the primary comparison is valid: the observed difference between arms is attributable to the treatment, not to systematic distortion.

What Chapter 5 did not address is what that valid comparison allows the trial to assert.

A valid comparison and a defensible claim are not the same thing. A primary analysis that crosses the pre-specified significance threshold allows the trial to reject the null hypothesis that the treatment effect is zero. It does not automatically allow the trial to assert that the treatment works in a specific subgroup, that it works on a secondary endpoint that was not the basis for the power calculation, that it is superior to a comparator on a measure that was not the primary, or that its effect in a pre-specified subgroup is large enough to support a distinct label. Each of these additional assertions requires something beyond a valid primary comparison: it requires that the assertion was pre-specified, that the evidence supports it with appropriate error control, and that the framing of the assertion is honest about what the evidence can and cannot show.

This is claim discipline. It is not a statistical concept, though statistics enforces it. It is a scientific and ethical concept: the trial is allowed to claim what the evidence, correctly analyzed within the pre-specified plan, genuinely supports. Not more. The pressure to claim more—to extract every possible positive message from the data, to find the subgroup where the treatment appears most effective, to report the secondary endpoint that crossed the threshold when the primary did not—is real and constant. Claim discipline is the design structure that resists this pressure.

Why this chapter exists

The history of clinical trial reporting is substantially a history of claims that exceeded the evidence. Post-hoc subgroup findings reported as if confirmatory. Secondary endpoints elevated to primary when the primary failed. Composite endpoints disaggregated to highlight the component that moved. Surrogate results extrapolated to clinical benefit without establishing the surrogate-outcome relationship. These practices are not rare exceptions; they are common enough that the regulatory agencies, the clinical literature, and the evidence synthesis community have invested enormous effort in developing reporting standards—CONSORT, STROBE, COMET, SPIRIT—that attempt to make the gap between pre-specified and reported analysis transparent.

The standards help. They do not eliminate the problem, because the problem is not primarily one of disclosure. It is one of design. A trial that does not specify its claims before the data are seen—that does not define, before enrollment, what it is allowed to assert if it

succeeds—is a trial that will construct its claims from the observed data, rationalizing each assertion as a pre-specified analysis that was always intended. The reconstruction is usually invisible, because the protocol language is often ambiguous enough to support multiple interpretations.

Claim discipline begins at design, not at analysis. It requires that the claims the trial is capable of making—if it succeeds—are specified before the data are seen, connected to a pre-specified analysis with appropriate error control, and bounded by the power calculation that determined the sample size. A trial that is powered to detect the primary hypothesis is not automatically powered to detect secondary hypotheses, subgroup hypotheses, or multiplicative claims across co-primary endpoints. The design must specify not just what the trial is testing but what it is designed to claim if the tests succeed.

What this chapter covers

Section 6.1 — Claims, Not P-values establishes the foundational distinction between a statistical result and a scientific claim. A p-value below 0.05 is not a claim. It is a rejection of the null hypothesis within a specific test, conducted under specific assumptions, at a specific alpha level. The claim—what the trial asserts about the treatment's effect—is a translation of that statistical result into a statement about the treatment's benefit, its size, its durability, its generalizability. That translation requires pre-specification, honest framing, and acknowledgment of its limits. A trial that reports p-values as claims has done the calculation but not the science.

Section 6.2 — Co-Primary and Multiple Primary Endpoints examines the specific design challenge of trials that propose more than one primary endpoint. The motivations for multiple primary endpoints are legitimate—when the treatment is expected to affect multiple dimensions of patient outcome, and when demonstrating benefit on each is required for the claim—but the statistical and scientific implications are severe. Whether both endpoints must succeed for the trial to claim success, whether either one is sufficient, and how the type I error is controlled across the joint claim must be specified before enrollment. A trial with two primary endpoints and no pre-specified decision rule for their joint result does not have two primary endpoints. It has two opportunities to claim success with a total type I error rate that is not controlled.

Section 6.3 — Hierarchical Testing examines the framework that most trials use to manage multiple claims without inflating the type I error: a pre-specified hierarchy of hypotheses, tested sequentially, in which a hypothesis is tested only if all prior hypotheses in the hierarchy have been confirmed. The hierarchy is a discipline: it limits the claims the trial can make to the ones pre-specified and supported by the evidence, in the order pre-specified, without additional alpha for each test. But the hierarchy must be designed—its order must be chosen before the data are seen, its stopping rule must be pre-specified, and its connection to the clinical narrative must be explicit.

Section 6.4 — Subgroup Discipline addresses the most common and most consequential source of overclaiming: subgroup analyses. When the primary analysis succeeds, subgroup analyses identify the patients in whom the treatment appeared most effective. When the primary analysis fails, subgroup analyses identify the patients in whom it appeared effective despite the overall failure. In both cases, the subgroup finding is almost always interpreted more confidently than the evidence supports. The section examines what pre-specified subgroup analyses can and cannot claim, why post-hoc subgroup findings are hypothesis-generating rather than confirmatory, and what design features—including stratification by the subgroup-defining variable and adequate power for the subgroup comparison—are required to support a confirmatory subgroup claim.

The decision structure of this chapter

The four sections of this chapter address a single question—what can this trial claim?—from four different angles, each corresponding to a distinct design decision that must be made before enrollment.

The decision in Section 6.1 is about framing: every statistical result must be connected to a pre-specified claim before the trial begins, and the framing of that claim must be honest about what the evidence supports and what it does not.

The decision in Section 6.2 is about the decision rule for co-primary endpoints: whether success on both is required, whether success on either is sufficient, and how the type I error is controlled across the joint claim.

The decision in Section 6.3 is about the hierarchy: what the trial is allowed to claim if the primary succeeds, in what order, with what stopping rule if a hypothesis in the hierarchy fails.

The decision in Section 6.4 is about subgroup pre-specification: which subgroups are confirmatory, which are exploratory, what power is available for each, and what the claim is if the subgroup analysis succeeds.

These four decisions are not independent. The hierarchy governs the primary, the secondary, and the subgroup claims together. The co-primary decision rule determines whether the hierarchy even begins. The framing of each claim must be consistent with the analysis that generated it. A trial that makes these four decisions coherently—before enrollment, as components of an integrated analysis plan—has claim discipline. A trial that makes them piecemeal, or after the data are seen, does not.

What this chapter is not about

This chapter is not about the statistical methods for controlling multiplicity. The Bonferroni correction, the Holm procedure, the Hochberg method, the Dunnett test—these are tools for implementing a pre-specified multiplicity control plan, not the plan itself. Choosing among these tools is a consequence of the design decisions this chapter addresses, not a substitute for them.

It is also not about the reporting standards for clinical trial results—CONSORT, pre-registration requirements, trial registry disclosures. These standards enforce claim discipline after the trial; this chapter is about building claim discipline into the design before the trial. The reporting standards catch the failures of claim discipline; this chapter is about preventing them.

The connection to what surrounds this chapter

Chapter 6 is the last of the design chapters before the book turns to adaptive design (Chapter 7) and pre-specification requirements (Chapter 8). Its position is consequential.

The claim discipline of Chapter 6 inherits the validity of the primary comparison from Chapter 5. A compromised comparison supports a weakened claim. The stronger the bias protection, the stronger the claim that valid results can support—and the more damaging it is when the claim exceeds what the valid comparison actually shows.

Chapter 6 also sets the stage for Chapter 7, which asks what happens when the design adapts. Adaptive designs—sample size re-estimation, adaptive enrichment, seamless phase II/III designs—all create additional sources of multiplicity and claim complexity that require the same discipline as fixed designs, applied to a more complex situation. The foundation for managing multiplicity in adaptive designs is the claim discipline established for fixed designs in Chapter 6.

The question this chapter must answer, for each claim the trial might make, is: was this claim pre-specified, is the evidence sufficient to support it with the stated confidence, and is the framing honest about what the evidence shows and what it does not? If the answer to any of these three questions is no, the claim should not be made—or should be made with the explicit qualification that it answers a different question than the trial was designed to answer.

6.1 Claims, Not P-values

What a p-value is

A p-value is a probability. Specifically, it is the probability of observing a test statistic at least as extreme as the one observed, under the assumption that the null hypothesis is true. When this probability falls below the pre-specified alpha threshold, the null hypothesis is rejected.

This is a narrow, precisely defined statement. It refers to a specific test, conducted on a specific dataset, under specific assumptions, at a specific alpha level. It says nothing about the size of the treatment effect, its clinical relevance, its generalizability, its durability, or its applicability to patients who were not enrolled. It is a rejection of a null hypothesis. It is not a claim about the treatment.

The conflation of a p-value below 0.05 with a positive trial result—and of a positive trial result with a claim that the treatment works—is one of the most consequential confusions in clinical research. It allows a trial to cross the significance threshold on a test of a surrogate endpoint and claim clinical benefit. It allows a trial to achieve significance on the least important component of a composite endpoint and claim overall clinical efficacy. It allows a trial with a statistically significant result in a subgroup to assert that the treatment works in that subgroup. In each case, the p-value is real; the claim exceeds what the p-value supports.

What a claim requires

A claim—a statement about what the treatment does, for whom, by how much, under what conditions—requires more than a p-value. It requires four elements that must be established before the data are seen.

Pre-specification. The claim must correspond to a pre-specified analysis. This means the analysis was defined in the protocol or statistical analysis plan before unblinding, with sufficient detail that a second statistician, reading the plan, would conduct the same analysis and reach the same conclusion. A claim based on an analysis that was not pre-specified is a claim based on a post-hoc observation, and post-hoc observations are hypothesis-generating, not confirmatory.

Appropriate error control. The claim must be made at a pre-specified type I error rate that accounts for all the tests conducted to reach it. If the trial tested five hypotheses and claims success based on the second one, the claim is valid only if the error rate for the second test reflects the multiplicity of the five-test procedure. A claim made at the nominal alpha without accounting for multiplicity is a claim made at an inflated type I error rate—a claim that appears more certain than it is.

Adequate power. The claim must be supported by a trial that was adequately powered for the specific test generating the claim. A trial powered for the primary hypothesis is not necessarily powered for secondary hypotheses. A trial powered for the overall population is not powered for subgroup analyses. A claim based on a test that was not powered to detect the effect size required for the claim is a claim based on an underpowered test—which may be true but cannot be confirmed with the stated confidence.

Honest framing. The claim must not assert more than the evidence shows. A significant effect on a surrogate endpoint should not be framed as demonstrated clinical benefit without establishing the surrogate-outcome relationship. A significant effect in a pre-specified subgroup should not be framed as the primary claim without acknowledging that the trial was powered for the overall population. A secondary endpoint that crossed the threshold only because the primary succeeded under a hierarchical plan should not be framed with the same confidence as the primary.

These four requirements—pre-specification, error control, power, honest framing—are the constitutive elements of a claim. A statistical result that satisfies all four is a claim. A result that satisfies fewer is an observation, a hypothesis, or a finding—all of which are valuable, but none of which is a claim in the sense that a regulatory submission or a clinical practice change requires.

The gap between the result and the claim

The gap between a statistically significant result and a defensible claim is where most overclaiming occurs. The gap has three dimensions.

The scale gap. The result is expressed on the statistical scale—a test statistic, a p-value, a confidence interval—and the claim is expressed on the clinical scale—a reduction in mortality, an improvement in function, a decrease in hospitalizations. The translation from one scale to the other is the effect measure, discussed in Chapter 2. The claim must be expressed in the effect measure's units, at the effect size the trial observed, with the uncertainty reflected in the confidence interval. A claim that omits the confidence interval—that reports only the point estimate—is a claim that hides the uncertainty. A confidence interval that is wide enough to include clinically trivial effects is evidence that the trial is underpowered to support a strong claim.

The population gap. The result is generated from the enrolled population—defined by the eligibility criteria—and the claim may be intended for a broader or different population.

A trial that enrolled patients with moderate-to-severe disease cannot directly claim the effect in mild disease. A trial that enrolled primarily male patients cannot directly claim the effect in female patients. The claim must be bounded by the population in which it was generated, and extrapolations beyond that population require explicit justification, not implicit extension.

The time gap. The result is generated at the primary endpoint time point—the duration of follow-up specified in the protocol—and the claim may be intended to extend beyond that time point. A trial with twelve weeks of follow-up cannot directly claim that the effect persists at two years. The durability of the effect is a separate claim that requires separate evidence. When durability data are not available, the claim should be bounded to the duration of the trial.

Each dimension of the gap can be bridged—by a trial that was designed to generate the relevant evidence—or acknowledged—by a claim that is honest about its limits. What it should not be is ignored.

The p-value as a decision tool, not a truth statement

The p-value functions as a decision tool in the regulatory system: a pre-specified threshold, crossed or not crossed, determines whether the trial has met its primary objective. This function is appropriate—the binary decision rule provides clarity and prevents post-hoc adjustment of the decision criterion—but it is not the same as a truth statement about the treatment's efficacy.

A p-value of 0.049 and a p-value of 0.051 are nearly identical in their evidential content. The difference between them—crossing the threshold or not—is consequential for regulatory decisions but should not be consequential for scientific inference. A trial that fails by 0.002 is not a failed trial in the scientific sense; it is a trial whose evidence was insufficient to meet the pre-specified decision criterion. Conversely, a trial that succeeds with a p-value of 0.048 on a test of questionable relevance has met the decision criterion without necessarily supporting the clinical claim.

The regulatory system uses the p-value as a decision tool because it needs a decision rule that is objective, pre-specified, and not subject to post-hoc adjustment. This is the right tool for that purpose. It is not the right tool for evaluating what the trial has contributed to clinical knowledge. For that purpose, the effect size, the confidence interval, the consistency across subgroups, the clinical plausibility of the result, and the relationship between the surrogate and the clinical outcome are all more informative than the p-value alone.

The design implication is that the primary claim—the one that the regulatory decision rests on—should be specified as a decision rule: the trial succeeds if the primary endpoint test achieves significance at the pre-specified alpha level. The scientific claims—what the trial contributes to knowledge about the treatment—should be framed as evidence statements:

the trial showed an estimated effect of X with a 95% confidence interval of Y to Z, which is clinically meaningful by the standards established before the trial. These are different statements, and they should be distinguished.

When the primary fails: the secondary claim problem

When the primary endpoint fails to achieve significance, the statistical hierarchy—described in Section 6.3—closes: no secondary endpoint can be claimed as confirmatory, because the gate-keeping test has not been passed. This is the correct consequence of hierarchical testing. It is also the consequence most commonly violated in practice.

The violation takes several forms. The most common is elevating a secondary endpoint to clinical significance based on its numerical result, without acknowledging that the primary hierarchy has closed. A trial that fails on the primary endpoint but shows a statistically significant secondary endpoint can report the secondary result as an observation—but it should not be framed as evidence of efficacy in the same sense that a pre-specified primary endpoint success would be. The secondary result has been tested in the context of a failed primary, and its significance is not protected from the multiple testing that the hierarchical structure was designed to prevent.

A second form of violation is changing the framing of the primary failure. A trial whose primary endpoint failed but whose secondary results suggest a benefit may be presented as a trial that "showed positive trends" or "demonstrated signals of activity" or "achieved significance on multiple exploratory analyses." Each of these framings may be technically accurate. None of them acknowledges that the trial failed on its pre-specified primary objective and that the secondary results are hypothesis-generating rather than confirmatory.

The honest framing of a failed primary is: the trial did not demonstrate the pre-specified primary objective. The secondary results are as follows, and they generate the following hypotheses for future investigation. This framing does not prevent the secondary results from being informative. It does prevent them from being overclaimed.

What this section demands before proceeding

Before the multiplicity structure of Section 6.2 can be designed, the primary claim must be fully specified. This means: the statement the trial will make if the primary hypothesis is confirmed, expressed in terms of the effect measure (from Chapter 2), for the population defined by the estimand (from Chapter 1), at the time point and duration of the trial, with the confidence level established by the alpha and power of the design.

This statement is not a marketing message and not a regulatory summary. It is the scientific claim: what this trial is designed to show, specified in enough detail that after the trial is over, it is unambiguous whether the claim was supported. If the statement cannot be written before enrollment—if it depends on what the data show—it is not a pre-specified claim. It is a post-hoc rationalization waiting to be constructed.

References: Wasserstein and Lazar, "The ASA's Statement on p-Values," Am Stat 2016; Ioannidis, "The Importance of Predefined Rules and Prespecified Statistical Analyses," JAMA 2019; Gelman and Loken, "The Statistical Crisis in Science," Am Sci 2014; ICH E9(R1) Addendum on Estimands (2019).

6.2 Co-Primary and Multiple Primary Endpoints

When two primary endpoints appear

Two primary endpoints in a trial design are almost never a statistical decision. They are a signal—of unresolved scientific questions, of competing stakeholder interests, or of regulatory requirements that the single most clinically meaningful endpoint cannot satisfy alone.

The unresolved scientific question: the treatment may affect two distinct dimensions of the disease that are both clinically important and not reducible to a single measure. A treatment for heart failure might be expected to reduce mortality and to improve symptoms; neither one captures the full picture of benefit, and demonstrating benefit on only one might leave the treatment's value incomplete for clinical and regulatory purposes. A treatment for a progressive neurodegenerative disease might need to show both a slowdown in disease progression and an improvement in functional ability; one without the other would raise questions about whether the benefit is clinically meaningful.

The competing stakeholder interests: the clinical team wants the most clinically meaningful endpoint; the regulatory agency has established a precedent for a specific endpoint in this indication; the payer wants an economic endpoint; the patient advocacy community wants a patient-reported endpoint. When no single endpoint satisfies all stakeholders, the trial acquires multiple primaries—one for each constituency, each with the implicit expectation that it will succeed.

The regulatory requirement: in some indications, approval requires demonstration of benefit on both a disease-specific endpoint and a clinical outcome endpoint. The trial must show both to make the label claim the sponsor intends. Two primaries are not a design choice but a regulatory condition.

These motivations differ in their scientific legitimacy, and the design of the co-primary structure must reflect the motivation. When two primaries are scientifically necessary—because neither alone captures the full claim—the trial should require both to succeed. When two primaries reflect competing interests, the trial needs a pre-specified decision rule that resolves the competition without allowing post-hoc cherry-picking of the endpoint that happened to cross the threshold.

The conjunctive and disjunctive decision rules

A trial with two primary endpoints must specify, before enrollment, which of two decision rules applies.

Under a **conjunctive rule** ("and"), the trial claims success only if both primary endpoints achieve significance. The trial is designed to show that the treatment works on both dimensions simultaneously. The power calculation must account for the joint probability—not just the probability that each individual endpoint achieves significance, but the probability that both do. If the two endpoints are positively correlated—which they often are when they measure related aspects of the same disease—the joint power is higher than the product of the individual powers. If they are uncorrelated, the joint power is approximately the product. The sample size must be calculated for the joint criterion, not for each endpoint separately.

The conjunctive rule is the more demanding standard. It is appropriate when the scientific claim requires demonstrating both dimensions of benefit simultaneously—when the treatment's value cannot be established without showing it works on both endpoints. Its cost is that the trial fails if either endpoint fails, even if the other endpoint shows a large and clinically meaningful benefit.

Under a **disjunctive rule** ("or"), the trial claims success if either primary endpoint achieves significance. The power calculation is for the probability that at least one endpoint achieves significance, which requires specification of the correlation between endpoints. The type I error calculation must account for the probability of a false positive on either endpoint: if each is tested at alpha, the joint false positive probability exceeds alpha by an amount that depends on the correlation. The standard correction—testing each endpoint at a Bonferroni-adjusted alpha—is overly conservative when the endpoints are positively correlated. Better approaches use the correlation directly to compute the joint test.

The disjunctive rule is the less demanding standard. It is appropriate when demonstrating benefit on either dimension is clinically sufficient—when the treatment would be used if it demonstrates benefit on either endpoint, independently of the other. Its cost is that the type I error is inflated unless explicitly controlled, and its use is sometimes a rationalization for testing two endpoints in the hope that one will cross the threshold.

The regulatory agencies are attentive to which rule is in use and whether it is appropriate for the scientific claim. A conjunctive rule that is later proposed to be relaxed to disjunctive—after the data show that one endpoint succeeded and the other did not—is a post-hoc change to the decision rule, which the agency will reject as a violation of the pre-specification requirement. A disjunctive rule that was pre-specified is defensible; a disjunctive rule that was constructed from a conjunctive plan after unblinding is not.

The type I error challenge

The specific type I error challenge of co-primary endpoints is the joint family-wise error rate. If both endpoints are tested at alpha, the probability of a false positive on at least one is higher than alpha when the decision rule is disjunctive, and lower than alpha when the decision rule is conjunctive.

Under a conjunctive rule, the joint type I error rate is actually controlled at alpha without any correction, because both endpoints must achieve significance—and the probability that both are simultaneously false positives is lower than the probability that either individually is a false positive. The conjunctive rule is therefore liberal in the direction of type II error (lower power for the joint criterion) and conservative in the direction of type I error (lower probability of a joint false positive). No multiplicity correction is required for a conjunctive rule to control the type I error at alpha.

Under a disjunctive rule, the joint type I error rate exceeds alpha, and a correction is required. The Bonferroni correction—testing each endpoint at alpha/2—is the simplest and most conservative correction. More efficient corrections use the correlation between the endpoints: when the correlation is high, the joint false positive probability is close to alpha even without correction, and less correction is needed. The appropriate correction should be derived from the anticipated correlation between the two endpoints and documented in the statistical analysis plan.

The practical challenge is that the correlation between the two endpoints must be estimated before the trial, from prior data, to determine the appropriate correction. If the correlation is overestimated—if the design assumes high correlation but the actual correlation is lower—the correction will be insufficient and the type I error will be inflated. If it is underestimated, the correction will be excessive and the trial will be unnecessarily conservative. The sensitivity of the type I error to the correlation estimate should be examined in the design documentation.

Two primaries as a symptom of unresolved design

When two primary endpoints appear not because the science requires both but because the design process did not resolve which one should be primary, the two-endpoint structure creates problems that are not solved by any decision rule.

The most common symptom of this situation is a trial where one endpoint is clinically preferred but uncertain to achieve significance—because it is a hard clinical endpoint that accumulates slowly—and the other endpoint is more sensitive but less clinically compelling—because it is a surrogate or a composite with a high-frequency component. The trial tests both, hoping that the sensitive endpoint will cross the threshold if the hard endpoint does not. The implicit strategy is: if the hard endpoint succeeds, claim the clinically meaningful result; if it fails but the surrogate succeeds, claim the biological signal.

This strategy is not a co-primary design. It is a hedged design with a post-hoc decision rule: the claim will be framed around whichever endpoint succeeded. A genuinely pre-specified co-primary design specifies the decision rule—conjunctive or disjunctive—before enrollment and applies it regardless of which endpoint succeeds. A hedged design applies different framing to the results depending on what the data show.

The regulatory agencies distinguish between the two. A co-primary design with a pre-specified decision rule, applied faithfully, is a defensible design even if the result is mixed. A hedged design with post-hoc framing is a design that allowed the trial to claim different things depending on the data, which is exactly what the pre-specification requirement is designed to prevent.

The diagnostic question for any proposed co-primary design is: if the trial were to fail on endpoint A but succeed on endpoint B, what would the sponsor do? And: if it were to succeed on endpoint A but fail on endpoint B, what would the sponsor do? If the answer to both questions is "claim success," the design is disjunctive and should be explicitly designed as such. If the answer to the first question is "fail the trial" and the second is "claim success," the design is not co-primary—endpoint A is primary and endpoint B is secondary. Forcing it into a co-primary structure without this clarity is asking for post-hoc rationalization.

Multiple primary endpoints in regulatory context

The regulatory requirements for co-primary endpoint designs have become more specific in recent years, reflecting the agencies' experience with trials that used co-primary structures as vehicles for flexible post-hoc claiming.

The FDA expects co-primary designs to specify the decision rule, the type I error control for the joint test, the power for the joint criterion, and the scientific rationale for why both endpoints are necessary for the claim. A protocol that proposes two primary endpoints without addressing these four elements will receive a deficiency request before the trial begins, not a question after it is over.

The EMA's position is similar, with additional specificity about the minimum clinically important difference for each endpoint and the clinical interpretation of mixed results—when one endpoint succeeds and the other fails. Specifying the clinical interpretation of mixed results in advance is not standard practice in most trials, but it is the logical consequence of a conjunctive or disjunctive decision rule: if the rule is conjunctive and one endpoint fails, the trial has failed; the clinical interpretation of the failing endpoint's result should be stated. If the rule is disjunctive and one endpoint fails, the trial has succeeded; the clinical interpretation of the failing endpoint's result—which was not required for success—should also be stated, so that the claim is honest about what the trial showed on the endpoint that did not contribute to the success criterion.

Pre-specifying the interpretation of mixed results is uncomfortable—it forces the design team to confront scenarios they would prefer not to think about—but it is the minimum required

to claim that the co-primary structure was genuinely pre-specified rather than designed to allow post-hoc framing.

What this section demands before proceeding

Before the hierarchical testing framework of Section 6.3 can be designed, the co-primary structure must be resolved. If there are two primary endpoints, the decision rule—conjunctive or disjunctive—must be specified, the type I error control for the joint test must be derived from the anticipated correlation, and the power for the joint criterion must be calculated and reflected in the sample size. The interpretation of mixed results must be pre-specified, including what the trial will claim and what it will not claim if one endpoint succeeds and the other fails.

If the proposed co-primary structure is a symptom of unresolved design—if the two endpoints reflect competing interests rather than a genuine scientific requirement for both—the structure should be resolved before it is formalized. The resolution is a design decision: one endpoint is primary, the other is secondary, and the trial is powered accordingly. This resolution may be uncomfortable. It is less uncomfortable than a regulatory review that asks why the co-primary structure was designed to allow claiming success on either endpoint independently.

References: FDA Guidance for Industry, Multiple Endpoints in Clinical Trials (2017); EMA Guideline on Multiplicity Issues in Clinical Trials (2017); Sozu et al., "Sample Size for Clinical Trials with Co-Primary Composite Endpoints," Stat Med 2010; Dmitrienko and D'Agostino, "Multiplicity Considerations in Clinical Trials," N Engl J Med 2013.

6.3 Hierarchical Testing

The problem hierarchical testing solves

A clinical trial that tests only the primary hypothesis exhausts all of its alpha on a single claim. This is coherent but often insufficient: treatments that work on the primary endpoint frequently also work on secondary endpoints, and the evidence of that secondary benefit has clinical and regulatory value. A trial that cannot claim secondary benefits because it spent all its alpha on the primary is a trial that has left evidence on the table.

But a trial that tests ten secondary hypotheses, each at the nominal alpha level, has a family-wise type I error rate far above alpha. If the primary succeeds, and if each of the ten secondary tests is conducted at 5% one-sided, the probability that at least one secondary is a false positive—claimed as significant when the true effect is zero—is high. The accumulated false positives enter the label, the clinical guidelines, and the prescribing behavior of physicians who trust that each claim was supported by evidence controlled at the stated error rate.

Hierarchical testing—the sequential testing procedure that tests hypotheses in a pre-specified order, advancing to the next hypothesis only when the current one has been confirmed—solves this problem by borrowing the structure of the primary test to control the family-wise type I error across the entire hierarchy. When the primary succeeds, the full alpha is available for the first secondary. When the first secondary succeeds, the full alpha is available for the second secondary. At each step, the test is conducted at the full nominal alpha—not at a corrected alpha—because the sequential conditioning controls the error rate without individual corrections.

The cost is the stopping rule: when a hypothesis in the hierarchy fails to achieve significance, all subsequent hypotheses in the hierarchy are treated as exploratory, regardless of their individual p-values. The hierarchy is a one-way gate: once closed, it does not reopen.

Designing the hierarchy

The hierarchy is a sequence of clinical claims, ordered by scientific priority and clinical importance. Designing it requires answering two questions: what is in the hierarchy, and in what order?

What is in the hierarchy depends on what the trial is designed to claim if the primary succeeds. Secondary endpoints that are part of the label claim—that the sponsor intends to assert in the regulatory submission and that require controlled type I error—should be in the hierarchy. Secondary endpoints that are informative but not intended to support label claims can be analyzed without alpha allocation and reported as exploratory. Subgroup analyses that are pre-specified as confirmatory—for which the trial has adequate power and a specific claim is intended—may be included in the hierarchy. Safety endpoints are not typically in the efficacy hierarchy; they are monitored separately under a different statistical framework.

The order of the hierarchy is a clinical decision. The convention is to order hypotheses from the most to the least important—from the perspective of the clinical claim, not the statistical significance. The most important secondary endpoint should be first in the hierarchy because it is the one that, if confirmed after the primary succeeds, constitutes the second most important claim the trial can make. Ordering by expected statistical significance—putting the hypothesis most likely to succeed first, to protect the weaker hypotheses that follow—is a design choice that prioritizes claim efficiency over clinical importance, and it should be made consciously and documented, not adopted by default.

The specific ordering also has implications for what happens when a hypothesis in the middle of the hierarchy fails. If the hierarchy is ordered by clinical importance, a failure in the middle closes off less important claims below—which is the scientifically appropriate consequence. If the hierarchy is ordered by statistical likelihood, a failure in the middle closes off claims that might be clinically more important than the ones above the failed hypothesis—which is a consequence that the design team should examine before finalizing the order.

Fixed sequence testing versus graphical procedures

The simplest hierarchical testing procedure is fixed sequence testing: hypotheses are ordered before the trial begins, and testing proceeds sequentially until a failure occurs. The type I error rate is controlled without any alpha redistribution—when a hypothesis is confirmed, the full alpha is passed to the next hypothesis in the sequence.

Fixed sequence testing is straightforward and transparent, but it is brittle: a failure at any point in the hierarchy closes all subsequent hypotheses, regardless of their clinical importance. If the second hypothesis in the hierarchy is the one with the strongest clinical evidence and the most important label claim, but the first hypothesis—which must be tested first because of the pre-specified order—fails, the second hypothesis is lost.

Graphical procedures—also called graph-based multiple testing procedures or the Bretz-Maurer-Branson framework—address this brittleness by specifying not just the order of testing but the rules for redistributing alpha when a hypothesis is confirmed or when a hierarchy branch fails. In a graphical procedure, the hypotheses are represented as nodes in a

directed graph, and the edges of the graph specify how alpha flows between hypotheses when each is confirmed or rejected. When a hypothesis is confirmed, its alpha is redistributed to other hypotheses in the graph according to the pre-specified weights on the edges. The result is a more flexible testing procedure that can continue claiming secondary hypotheses even when some intermediate hypotheses fail, as long as the flow of alpha is pre-specified.

The flexibility of graphical procedures is their advantage and their vulnerability. A graphical procedure that is complex—with many nodes and many conditional alpha flows—can be designed to be highly resilient to failures at specific nodes, continuing to protect important hypotheses even when intermediate ones fail. But the same complexity makes the procedure difficult to verify, to explain to a regulator, and to audit for correct implementation. The design team must be able to show that the graphical procedure controls the family-wise type I error under all possible outcomes—all possible subsets of confirmed and rejected hypotheses—and this demonstration is not always straightforward for complex graphs.

The practical advice is to use graphical procedures when the added flexibility is necessary—when the clinical importance of the hypotheses does not align with the order in which fixed sequence testing would require them to be tested—and to keep the graph as simple as the clinical claims allow. A graphical procedure with two or three nodes and a small number of well-specified edges is defensible and auditable. A graphical procedure with eight nodes and conditional flows that depend on which combination of hypotheses have been confirmed is not.

The fallback and the fallback's failure modes

A common design element in hierarchical testing is the fallback: a lower-level hypothesis that is tested even if a higher-level hypothesis fails, because the lower-level hypothesis does not depend on the higher-level one for its clinical rationale. The fallback is not technically part of the main hierarchy—it is a parallel branch that is activated when the main branch closes.

For example, a trial might have a primary hypothesis about all-cause mortality, a secondary hypothesis about cardiovascular death, and a fallback hypothesis about cardiovascular death and hospitalization—a composite that is expected to be positive even if the mortality endpoint is not significant. The fallback is tested at a separately allocated alpha, independent of the main hierarchy, so that its significance does not depend on the mortality hypothesis having been confirmed.

The fallback is legitimate when the separate alpha allocation is pre-specified and the clinical rationale for the fallback endpoint is independent of the main hierarchy. It is illegitimate when the fallback is designed to guarantee a significant result even if the main hierarchy fails—when the fallback is the easy endpoint that the team expects to cross the threshold regardless of what happens to the primary.

The failure mode of the fallback is that it becomes the de facto primary endpoint—the endpoint that the trial is really powered for and really intended to claim—while the main hierarchy is maintained as a high-aspirational structure that is expected to fail. A trial with this structure is not a trial with a meaningful primary endpoint and a well-considered fallback. It is a trial with a primary endpoint that is not the primary and a fallback that is the real primary, and the design has obscured the actual scientific intent behind the appearance of a rigorous hierarchy.

What counts as pre-specified

The hierarchy is pre-specified only if it was fixed before the data were seen. This requires that the protocol or statistical analysis plan—finalized before unblinding—contains the complete hierarchy: every hypothesis, in the order it will be tested, with the decision rule for each.

Partial pre-specification—specifying the primary and first secondary but leaving the remaining hierarchy to be "finalized in the SAP"—does not constitute pre-specification if the SAP is finalized after interim data are available. The hierarchy must be final before any look at the unblinded outcome data. A hierarchy that was developed based on the observed direction of secondary endpoint results—even if developed in good faith—is a post-hoc hierarchy, and the claims it generates are post-hoc claims.

The distinction between pre-specified and post-hoc claims is the most consequential distinction in claim discipline. A pre-specified claim, confirmed at the stated alpha level, is a claim with a known and controlled type I error rate. A post-hoc claim, regardless of its p-value, is a claim with an unknown error rate—inflated by the unknown number of hypotheses that were considered but not selected, and by the influence of the observed data on the selection of hypotheses. The p-value of a post-hoc claim is a conditional probability, conditioned on the data having been seen, that is not the same as the unconditional type I error rate the nominal alpha implies.

Interaction between the hierarchy and interim analyses

Chapter 4's interim analysis plan and Chapter 6's hierarchical testing plan interact in a way that is not always addressed in the design documentation.

When the primary endpoint crosses the efficacy boundary at an interim analysis and the trial stops early, the secondary endpoints are analyzed at the same data cut. The hierarchical testing framework for the secondary endpoints must be pre-specified for this scenario: which secondary endpoints are tested at the interim if the primary succeeds, in what order, at

what information fraction of the secondary endpoint data, and with what alpha adjustment for the fact that the secondary data are also interim.

This interaction is frequently overlooked. The design documents the primary efficacy hierarchy for the final analysis and the primary stopping rule for the interim—but not the secondary hierarchy for the scenario in which the primary succeeds at the interim. When the trial stops early based on the primary, the analysis team must construct the secondary analysis at the interim, and in the absence of pre-specification, the construction will be post-hoc.

The pre-specification requirement for this scenario is: if the trial stops early based on the primary efficacy crossing the boundary, the secondary hypotheses will be tested in the following order at the interim data cut, with the following alpha allocation. This specification must be in the SAP before the first interim analysis, not constructed at the time the stopping decision is made.

What this section demands before proceeding

The hierarchical testing plan must be complete before enrollment begins. Complete means: every hypothesis in the hierarchy is named, the order is specified and justified on clinical grounds, the decision rule for each position in the hierarchy is stated, the alpha allocation across the hierarchy is defined, and the handling of the interim-stops scenario is pre-specified.

If graphical procedures are used, the graph must be documented with its edge weights and its alpha redistribution rules, and the team must be able to demonstrate that the procedure controls the family-wise type I error under all possible outcomes.

The hierarchy is the design's claim structure. What it contains—and what it does not contain—defines what the trial is and is not capable of claiming, regardless of what the data show. Designing the hierarchy is designing the claims. It must be done before enrollment, by the people who understand both the clinical importance of the hypotheses and the statistical consequences of the ordering.

References: Bretz et al., "Graphical Approaches for Multiple Comparison Procedures," Pharm Stat 2009; Maurer and Bretz, "Multiple Testing in Group Sequential Trials," Stat Biopharm Res 2013; Dmitrienko et al., "Key Multiplicity Issues in Clinical Drug Development," Stat Med 2013; FDA Guidance for Industry, Multiple Endpoints in Clinical Trials (2017).

6.4 Subgroup Discipline

The subgroup problem

No result in clinical trial reporting is more systematically misinterpreted than the subgroup analysis. When the primary analysis succeeds, subgroup analyses appear to identify the patients who benefited most—and clinicians adjust their prescribing toward those patients. When the primary analysis fails, subgroup analyses appear to rescue the trial by identifying a population in whom the treatment worked—and the sponsor reframes the program around that population. In both cases, the subgroup finding is typically interpreted with more confidence than the evidence supports.

The mechanism is mathematical and well-understood. When a trial with adequate power for the overall population is subdivided into subgroups, each subgroup is smaller than the overall trial and is therefore underpowered for the subgroup-specific effect. An underpowered analysis has low probability of detecting a real effect—and also a high probability of producing an apparently significant result by chance when tested against the background of all the subgroups that were examined. A trial that tests twenty subgroup hypotheses, each at the nominal 5% alpha level, expects one false positive by chance alone. When the primary analysis succeeds and the team then examines twenty subgroups to identify who benefited most, the subgroup that shows the largest apparent effect is likely the product of chance variation rather than a true differential effect.

This is not a theoretical concern. The empirical literature on subgroup analyses in clinical trials documents a systematic pattern: apparent subgroup effects are often not reproduced in subsequent trials powered for the subgroup hypothesis, are frequently not consistent with biological plausibility, and are asymmetrically reported—positive subgroup findings are more likely to be highlighted than negative ones.

The design response to this pattern is pre-specification, power, and discipline in claim framing.

Confirmatory versus exploratory subgroup analyses

The fundamental distinction in subgroup analysis is between confirmatory and exploratory.

A **confirmatory subgroup analysis** is one for which the trial was specifically designed to support a claim. It is pre-specified before enrollment in the protocol, has adequate power

based on a separate sample size calculation for the subgroup, is included in the hierarchical testing plan with appropriate error control, and is supported by a biological rationale that is plausible and documented. A confirmatory subgroup finding—when all of these conditions are met—supports a claim about the treatment's effect in that subgroup with the same logical force as the primary analysis supports the overall claim.

The conditions for a confirmatory subgroup analysis are demanding, and they are demanding for a reason: a confirmatory claim in a subgroup is a claim that the treatment works specifically in those patients, with an error rate that is controlled and stated. If the design conditions are not met—if the analysis was post-hoc, if the power was not calculated for the subgroup, if the error control did not account for the number of subgroups examined—the claim is not confirmatory regardless of how significant the p-value is.

An **exploratory subgroup analysis** is one that does not meet the conditions for a confirmatory claim. It is hypothesis-generating: it identifies patterns in the data that are potentially interesting and warrant further investigation. An exploratory finding may be consistent with a prior biological hypothesis; it may be the basis for a new trial designed to confirm the hypothesis in the subgroup population. What it is not is evidence of a differential treatment effect in the subgroup, at the claimed error rate, to a degree that would support a label claim or change clinical practice without further investigation.

The distinction between confirmatory and exploratory is not about the p-value. An exploratory subgroup finding with a p-value of 0.002 is not more confirmatory than an exploratory subgroup finding with a p-value of 0.02—because the p-value of an exploratory finding is conditioned on the data having been seen and the subgroup having been selected, which invalidates the error rate interpretation of the p-value.

What makes a subgroup analysis confirmatory

The four conditions for a confirmatory subgroup analysis are not all equally weight-bearing, and understanding which ones are most critical clarifies what the design must do.

Pre-specification is the minimum necessary condition. An analysis that was not specified before the data were seen cannot be confirmatory, regardless of any other feature. Pre-specification means the subgroup-defining variable, the cut-point defining the subgroup, and the analysis plan are documented in the protocol and the SAP before unblinding. A subgroup analysis that appears in the SAP but not the protocol is less well-established than one that appears in both. An analysis that appears in neither is post-hoc regardless of when it was first discussed internally.

Biological rationale is the prior probability element. A subgroup analysis that is pre-specified and biologically motivated—the treatment targets a pathway that is more active in the subgroup, or the subgroup is defined by a biomarker that the treatment mechanism directly engages—has a higher prior probability of being a true differential effect than a

subgroup analysis defined by a demographic variable with no mechanistic connection to the treatment. When the biological rationale is strong, a significant subgroup finding is more likely to be real. When the rationale is weak—when the subgroup is a demographic category that was examined because it was available—the significant finding is more likely to be a false positive.

Adequate power is the most operationally demanding condition. The subgroup analysis must have been powered to detect the subgroup-specific effect. This requires a separate sample size calculation for the subgroup, based on the expected subgroup-specific effect size and the expected size of the subgroup in the enrolled population. If the subgroup comprises 30% of the enrolled population, and the trial was powered for the overall effect, the subgroup analysis has at most 30% of the planned events or subjects, and its power for the subgroup-specific effect—which may differ from the overall effect—may be inadequate even for a large overall effect.

Error control is the multiplicity condition. A confirmatory subgroup analysis must be included in the hierarchical testing plan with alpha allocated to it, or its alpha must be separated from the primary and secondary hierarchy with an explicit Bonferroni or equivalent correction for the number of subgroup hypotheses being tested. A subgroup analysis that is tested at the nominal alpha, in addition to the primary and secondary hierarchy, has consumed alpha that was not allocated to it—inflating the family-wise type I error.

The biomarker subgroup: a special case

The biomarker-defined subgroup—patients who are positive or negative for a specific biomarker—has become the most clinically important and most carefully scrutinized form of subgroup analysis in modern clinical trial design. Companion diagnostics, targeted therapies, and precision medicine strategies all depend on biomarker-defined patient selection, and the claim that a treatment works specifically in biomarker-positive patients requires the same evidentiary standards as any other confirmatory claim.

The specific design challenge for biomarker subgroups is the interaction between the biomarker test performance and the trial's claim. If the biomarker test is imperfect—if some patients classified as biomarker-positive are actually negative, and some biomarker-negative patients are actually positive—the comparison between biomarker-positive and biomarker-negative patients is diluted by the misclassification. A trial that enrolls all patients regardless of biomarker status and then compares outcomes by biomarker status will observe a diluted biomarker effect, because the misclassified patients are in the wrong subgroup.

This dilution can be addressed by prospective biomarker enrichment—enrolling only biomarker-positive patients, or enriching the population toward biomarker-positive patients—rather than relying on retrospective subgroup analysis. Prospective enrichment transforms the biomarker subgroup from an exploratory analysis into the primary population of the trial, and the trial's claims are naturally bounded to the biomarker-positive population.

This is the design appropriate for a treatment whose mechanism specifically targets a biomarker-defined pathway and for which there is no clinical rationale for benefit in the biomarker-negative population.

When the biomarker's role is less certain—when there may be benefit in both biomarker-positive and biomarker-negative patients, or when the biomarker's clinical implementation is imperfect—an all-comers design with a pre-specified biomarker subgroup analysis is appropriate. The design must specify the primary analysis population (overall or biomarker-positive), the subgroup analysis for the other population, and the multiplicity control for the joint claim across both populations.

The FDA's guidance on companion diagnostics and enrichment strategies reflects these considerations, and the EMA's guidance on biomarker-defined subpopulations is similarly explicit. A biomarker subgroup claim—one that asserts the treatment works specifically in biomarker-positive patients—requires a companion diagnostic that is validated for the biomarker test, analytical validation of the assay, and clinical validation of the biomarker's predictive value. Meeting these requirements before the trial is designed, not after the subgroup analysis identifies an apparent differential, is the design standard.

Interaction tests and what they cannot show

The standard approach to assessing whether a treatment effect differs between subgroups is the statistical test of interaction: does the effect size in the subgroup differ from the effect size in the complement of the subgroup by more than would be expected by chance? A significant interaction test is taken as evidence of a differential treatment effect. A non-significant interaction test is taken as evidence that the treatment effect is homogeneous across the subgroups.

Both interpretations are problematic, and for the same reason: interaction tests are underpowered. A trial powered to detect the overall treatment effect at 80-90% power is typically powered for an interaction test at only 20-30% power, because the interaction test requires detecting a difference between effect sizes, and the effect sizes in each subgroup are estimated with more uncertainty than the overall effect size. An interaction test that fails to achieve significance does not demonstrate homogeneity; it demonstrates that the trial was not powered to detect heterogeneity.

The consequence is that the absence of a significant interaction does not validate subgroup-specific claims. A treatment that shows an apparently strong effect in men and an apparently weak effect in women is not validated as having a homogeneous effect across sex by a non-significant interaction p-value of 0.12—because the test was not powered to detect the apparent difference. The non-significant interaction is consistent with a true differential as large as the observed apparent difference.

Conversely, a significant interaction—a p-value below 0.05 for the test that the treatment effect differs between subgroups—does not confirm a clinically meaningful differential. The

interaction test is subject to multiple testing if many subgroup interactions are tested; a significant interaction from a large number of exploratory interaction tests is expected by chance. And even a pre-specified interaction test that achieves significance may reflect a difference in the nuisance parameters—the baseline event rates or the variance—across subgroups rather than a true differential treatment effect.

The honest framing of subgroup results is therefore one of consistent uncertainty: the observed effects in each subgroup, with their confidence intervals, as descriptive summaries of the data in those groups; the interaction test p-value as a measure of the evidence for heterogeneity, with explicit acknowledgment of the test's low power; and the clinical plausibility of the observed pattern as additional context. A confident claim about differential treatment effects across subgroups requires a trial that was designed to make that claim—with adequate power, pre-specified hypotheses, and controlled error—not a post-hoc analysis of a trial designed for the overall effect.

The forest plot and its misreading

The forest plot—the graphical display of subgroup-specific treatment effects, with confidence intervals, arrayed vertically by subgroup category—has become the standard visual presentation of subgroup results. It is also the source of one of the most systematic misreadings in clinical reporting.

The forest plot presents a visual impression of differential effects. Subgroups whose confidence intervals do not cross the null appear to "show benefit." Subgroups whose confidence intervals cross the null appear to "not show benefit." The visual impression is that the treatment works in some subgroups and not in others, based on the statistical significance of each subgroup's confidence interval.

This impression is produced by the underpowered nature of each subgroup's estimate, not by a genuine differential effect. When the treatment effect is homogeneous across subgroups, the subgroup estimates will vary randomly around the true effect, and some will cross the null by chance—not because those subgroups do not benefit, but because the subgroup samples are small and the estimates are imprecise. The forest plot, by displaying statistical significance visually, creates the impression that the variation reflects genuine differential effects.

The appropriate reading of a forest plot is consistency checking, not significance reading. The question the forest plot should answer is: are the subgroup estimates consistent with the overall treatment effect, or are there subgroups whose effects are implausibly discordant? Implausible discordance—an effect in one direction in a large subgroup and the opposite direction in another large subgroup, for a plausible mechanistic reason—is evidence of genuine heterogeneity. Random variation around the overall estimate, with some confidence intervals crossing the null and others not, is not.

The forest plot should therefore be accompanied by the overall treatment effect and its confidence interval—the reference against which each subgroup estimate is compared—and by

the interaction test results, with explicit acknowledgment of the test's power. Without these elements, the forest plot is a visual invitation to misread exploratory results as confirmatory claims.

What this section demands before proceeding

The subgroup analysis plan must be complete before enrollment begins. It must specify which subgroup analyses are confirmatory—meeting all four conditions for a confirmatory claim—and which are exploratory. The confirmatory subgroup analyses must be included in the hierarchical testing plan or assigned separately controlled alpha. The exploratory subgroup analyses must be labeled as hypothesis-generating in the protocol and the SAP, with explicit statements that they do not support confirmatory claims.

The biomarker-defined subgroups, if any, must be connected to a companion diagnostic with pre-specified analytical and clinical validation criteria, and the biomarker's role in the trial's primary population versus subgroup analysis must be specified.

And the forest plot interpretation guidelines must be pre-specified: what the trial will and will not conclude from a visual pattern of subgroup effects, what consistency checking criteria will be applied, and what the threshold for reporting a subgroup finding as hypothesis-generating rather than hypothesis-confirming is.

Subgroup discipline is not skepticism about subgroup analyses. It is precision about what they can and cannot show—and the precision must be established before the data are seen, not constructed from the data that were seen.

References: Rothwell, "Subgroup Analysis in Randomised Controlled Trials: Importance, Indications, and Interpretation," Lancet 2005; Assmann et al., "Subgroup Analysis and Other (Mis)Uses of Baseline Data in Clinical Trials," Lancet 2000; Brookes et al., "Subgroup Analyses in Randomized Trials: Risks of Subgroup-Specific Conclusions," Ann Intern Med 2004; FDA Guidance for Industry, Enrichment Strategies for Clinical Trials (2019).

6.5 Closing: The Boundary of What Can Be Said

What this chapter asked

Chapter 6 asked one question: what are we allowed to claim?

It asked that question in four registers, each addressing a different mechanism by which claims can exceed the evidence.

Section 6.1 established that a p-value is not a claim. It is a rejection of a null hypothesis under a specific test. The translation from a statistical result to a scientific claim requires pre-specification, appropriate error control, adequate power, and honest framing. Without all four, the claim asserts more certainty than the evidence supports.

Section 6.2 examined co-primary endpoints—and found that two primary endpoints without a pre-specified decision rule are not a design with two primaries. They are a design that allows post-hoc selection of the endpoint that happened to cross the threshold. The decision rule—conjunctive or disjunctive—must be specified before enrollment, with the type I error controlled for the joint test and the power calculated for the joint criterion.

Section 6.3 examined hierarchical testing as the framework by which the trial manages multiple claims without inflating the family-wise type I error. The hierarchy is a pre-specified sequence of claims, ordered by clinical importance, testable in order, with the stopping rule applied faithfully. A hierarchy that was designed post-hoc—ordered to protect the hypotheses that happened to succeed, or that omits hypotheses that failed—is not a hierarchy. It is a retrospective rationalization.

Section 6.4 examined subgroup analyses—the most common source of overclaiming in clinical trial reporting. A confirmatory subgroup claim requires pre-specification, biological rationale, adequate power for the subgroup, and error control. An exploratory subgroup analysis, regardless of its p-value, generates hypotheses. The distinction between confirmatory and exploratory is not about the magnitude of the finding; it is about the conditions under which the finding was made.

What this chapter decided

By the end of this chapter, four things must be documented and their consistency with each other confirmed.

The primary claim is fully specified—not as a statistical test but as a scientific assertion. The statement of what the trial will claim if the primary hypothesis is confirmed, expressed in terms of the effect measure for the population defined by the estimand, at the time horizon of the trial, at the confidence level established by the alpha and power of the design. This statement exists before enrollment begins and does not change based on what the data show.

The co-primary decision rule—if two primary endpoints are proposed—is specified as conjunctive or disjunctive, with the type I error controlled for the joint test and the power calculated for the joint criterion. The interpretation of mixed results is pre-specified: what the trial will claim if one endpoint succeeds and the other fails.

The hierarchical testing plan is complete and clinically ordered. Every hypothesis in the hierarchy is named, ordered by clinical importance, connected to the alpha allocation, and documented in the SAP before unblinding. The handling of the interim-stops scenario—if the primary crosses the efficacy boundary at an interim and the secondary endpoints are tested at the interim data cut—is pre-specified.

The subgroup analysis plan distinguishes confirmatory from exploratory, assigns alpha to confirmatory subgroup hypotheses, powers confirmatory subgroup analyses separately, and labels exploratory subgroup analyses as hypothesis-generating in the protocol and the SAP.

These four things together constitute the trial's claim structure—the complete specification of what the trial is designed to assert, and what it is not. The claim structure is as important as the estimand: the estimand defines what is being measured; the claim structure defines what the measurement is being used to say.

The characteristic mistakes of this chapter

Three failures recur with enough consistency to constitute the characteristic errors of claim discipline.

The hierarchy that was ordered after unblinding. The protocol specified a primary endpoint and four secondary endpoints, without ordering the secondary endpoints. The SAP was finalized after the database lock and before the primary analysis was conducted—technically before unblinding, but after the data quality review had given the analysis team a view of the overall data distribution that allowed the direction of each secondary to be inferred. The secondary endpoints were ordered in the SAP in the order they subsequently tested significant. The regulatory reviewer asked for the timestamps of the SAP finalization and the data quality review. The timestamps did not support the claim that the SAP

was finalized without knowledge of the secondary endpoint directions. The hierarchy was declared post-hoc and the secondary claims were reclassified as exploratory.

The subgroup that was confirmed without power. The trial pre-specified a biomarker subgroup analysis in the protocol. The biomarker-positive subgroup comprised 40% of the enrolled population. The trial was powered for the overall population. The subgroup analysis was included in the hierarchical testing plan. The biomarker-positive subgroup showed a significant result at the 5% level. The regulatory reviewer asked for the power of the subgroup analysis. The power for the biomarker-positive subgroup, at the observed subgroup-specific effect size, was 38%. The claim that the treatment works in biomarker-positive patients—based on a 38% powered analysis—was not supported as a confirmatory claim. The biomarker subgroup finding was reclassified as hypothesis-generating, and the sponsor was required to conduct a confirmatory trial in the biomarker-positive population.

The secondary that was elevated after the primary failed. The primary endpoint—all-cause mortality—was not significant. The first secondary endpoint—cardiovascular death—was significant at the pre-specified alpha level under the hierarchical plan. The sponsor proposed to submit the cardiovascular death result as the basis for approval, arguing that the hierarchical plan allowed testing of the secondary after the primary failed. The regulatory reviewer noted that the hierarchical plan requires the primary to succeed for the secondary to be tested within the controlled family-wise error rate. Cardiovascular death had been tested in the context of a failed primary, outside the protective structure of the hierarchy. The cardiovascular death p-value was not controlled at the stated alpha level. Approval was denied.

These three mistakes are not statistical errors. They are governance failures: the claim structure was specified ambiguously, implemented partially, or applied post-hoc. In each case, the design team had done the statistical work but not the claim discipline work—had designed the tests without fully designing the claims.

What cannot be recovered

A post-hoc hierarchy cannot be retroactively made confirmatory. The regulatory agencies have extensive experience with the sequence of events that indicates a post-hoc ordering: SAP finalization timestamps that follow data quality reviews that provided inferential information about the direction of secondary results; hierarchies that happen to be ordered in the order the endpoints subsequently test significant; SAP amendments that rearrange the hierarchy after an interim data look that revealed trends. When these patterns are present, the claim that the hierarchy was pre-specified cannot be established, and the secondary claims are treated as exploratory.

A subgroup analysis that was not adequately powered cannot be retroactively powered by post-hoc sample size calculations. The power of the subgroup analysis is determined by the number of patients in the subgroup at the time of the analysis. A claim that the subgroup

analysis was intended to be confirmatory but was inadvertently underpowered is a claim that the design was flawed. The remedy is a new trial designed for the subgroup population, not a reinterpretation of the subgroup finding from the underpowered analysis.

And a primary endpoint failure cannot be converted into a secondary endpoint success within the same trial's confirmatory claim structure. When the primary hypothesis fails, the hierarchical structure closes. The secondary endpoint that was next in the hierarchy can be analyzed and reported, but it cannot be claimed at the pre-specified alpha level within the trial's family-wise error control. The secondary finding, however significant, is an exploratory observation from a trial that did not achieve its pre-specified primary objective.

These irrecoverabilities are the practical consequences of the pre-specification requirement. The pre-specification requirement is not bureaucratic; it is the mechanism by which the error rate of the claims is established. Claims made outside the pre-specified structure have unknown error rates. A claim with an unknown error rate is not evidence in the sense that regulatory approval and clinical practice change require.

The connection to what follows

Chapter 7 asks what happens when the design itself adapts. Adaptive designs—where the sample size, the patient population, or the analysis strategy changes based on accumulating data—introduce additional sources of multiplicity and claim complexity that require the same discipline as fixed designs, applied to more complex structures.

The claim discipline of Chapter 6 is the foundation for managing multiplicity in adaptive designs. When a trial adapts its sample size based on interim data, the adaptation itself may be informative about the treatment's likely effect—and the information introduced by the adaptation must be accounted for in the final analysis. When a trial adapts its patient population—enriching for biomarker-positive patients after an interim—the claim about the biomarker-positive population is subject to the same pre-specification and power requirements as any other subgroup claim, plus the additional complexity of having been introduced adaptively.

Chapter 7 cannot be designed coherently without the foundations of Chapter 6. The claims an adaptive trial is allowed to make—at each stage of the adaptation and at the final analysis—are the adaptive extension of the claim discipline established for fixed designs. The adaptation does not relax the pre-specification requirement; it extends it to the adaptive rules themselves. The design team must specify, before the trial begins, not just what the final claims will be but what the claims will be at each possible adaptation point, under each possible interim result.

Chapter 6 risk summary

The decision this chapter owns: what is the trial allowed to assert if it succeeds on the primary, and what is the structure—hierarchical, co-primary, subgroup—within which secondary and tertiary claims are bounded?

The most common mistake: treating the statistical analysis plan as the claim document, when the claim document is the pre-specified hierarchy of hypotheses, ordered by clinical importance, each associated with an alpha allocation, each connected to a power calculation. The SAP specifies the analyses. The claim structure—which claims are confirmatory and in what order—must be in the protocol.

The professional-level risk: the secondary claim that was elevated after the primary failed, the subgroup claim that was made without adequate power, or the hierarchy that was ordered after the data provided inferential information about the direction of effects. Each of these is a claim that was made with an uncontrolled error rate—that was, technically, a false claim about the evidence even if the underlying treatment effect is real. The professional risk is not that the treatment does not work. It is that the evidence was not strong enough to support the claim at the stated confidence level, because the claim structure was not in place before the data were seen.

Chapter 7: What If the Design Itself Adapts?

The question Chapter 6 leaves open

Chapter 6 built the claim structure for a fixed design—a trial whose sample size, patient population, analysis plan, and hypothesis hierarchy are fully specified before enrollment and do not change in response to accumulating data. The claim structure was demanding: pre-specification, adequate power, error control, honest framing. Each requirement was defensible as a consequence of the pre-specification principle.

What happens when the design itself is not fixed?

Adaptive designs—trials in which one or more design features change during the trial based on accumulating data—are increasingly common in clinical development. They are used to address the uncertainty that exists at the time of design: uncertainty about the right dose, the right patient population, the right effect size assumption, the right endpoint. If those uncertainties can be resolved by data accumulated during the trial, the argument goes, the design can be improved mid-course rather than committed irreversibly to a design that may have been wrong.

This argument has genuine force. A trial designed with an effect size assumption that turns out to be optimistic will be underpowered; a pre-specified sample size re-estimation rule can detect the optimism early and correct the sample size before it is too late. A trial designed for an unselected population when a biomarker-defined subgroup shows substantially stronger benefit can enrich toward the responding subgroup rather than continue enrolling patients who are unlikely to benefit. These are real improvements that adaptive design can provide—at a cost that this chapter examines.

Why adaptation is a risk amplifier

The design of this chapter follows the writing discipline of this book: adaptive design is not a distinct technical specialty to be introduced on its own terms. It is a context in which the risks established in prior chapters are amplified.

Every prior chapter identified a decision with consequences. Chapter 1 defined the estimand and noted that intercurrent events distort it. Chapter 3 identified the effect size assumption and noted that optimism produces underpowered trials. Chapter 6 specified the hierarchy and noted that post-hoc ordering inflates the type I error. In each case, the risk was present in fixed designs and managed through pre-specification.

When the design adapts, the same risks are present—and they interact with each other and with the adaptation in ways that fixed designs do not experience. A sample size re-estimation changes the information fraction at which interim analyses are conducted, shifting the operating characteristics of the stopping boundaries established in Chapter 4. An adaptive enrichment toward a biomarker-positive subgroup changes the estimand—the population attribute shifts mid-trial—creating a connection to Chapter 1's requirement that

the estimand be settled before the design is finalized. A seamless phase II/III design that adapts the dose based on phase II data introduces a multiplicity structure that Chapter 6's claim discipline must address across both phases.

These interactions are not complications to be managed after the adaptation is designed. They are the reason that adaptive design requires the foundations of all prior chapters to be in place before the adaptation is layered on top. An adaptive design is not an escape from the commitments of the fixed design; it is an additional commitment—the commitment to pre-specified rules for how and when the design will change—built on top of the commitments that were already required.

What this chapter covers

Section 7.1 — Sample Size Re-estimation examines the most common and most straightforward form of adaptation: adjusting the planned sample size based on observed nuisance parameters from interim data. When the variance is higher than assumed, or the control arm event rate is lower than assumed, or the dropout rate is higher than assumed, a pre-specified re-estimation rule can correct the sample size to restore the planned power. The section examines what pre-specification of the re-estimation rule requires, what information the re-estimation can and cannot use without inflating the type I error, and what the operating characteristics of the adapted design look like relative to the fixed design.

Section 7.2 — Adaptive Enrichment examines the adaptation of the enrolled population—narrowing toward a subgroup that shows greater benefit, or widening to include patients initially excluded. Adaptive enrichment changes the estimand, because the population attribute of the estimand is determined by who is enrolled. The section examines what conditions must be met for adaptive enrichment to be scientifically defensible, what the estimand is in the adapted trial, and what the claim structure looks like across the pre- and post-adaptation populations.

Section 7.3 — Estimand Shift examines the ways in which adaptations that do not explicitly change the endpoint can nonetheless change what the trial is estimating. A sample size re-estimation based on the observed effect size—rather than on nuisance parameters alone—changes the implicit estimand by conditioning the sample size on an interim result. An adaptive enrichment changes the population attribute. A seamless phase II/III design with an adaptive dose selection changes the treatment attribute. Each of these estimand shifts must be identified, acknowledged, and addressed in the primary analysis and the claim structure.

Section 7.4 — Non-Inferiority Failure Modes examines the specific ways in which adaptive design interacts with non-inferiority trials—already the most conceptually complex design type—to amplify the risks that Chapter 2 identified. The NI margin depends on the constancy assumption; adaptations that change the enrolled population, the background therapy, or the trial duration affect whether the constancy assumption remains credible. The

section examines how these interactions can produce NI conclusions that are formally valid but scientifically misleading.

The regulatory context

Regulatory agencies have developed substantial guidance on adaptive designs, and their positions reflect hard-won experience with the ways in which adaptation can be used to improve designs and the ways in which it can be used to obscure pre-specification failures.

The FDA's 2019 guidance on adaptive designs distinguishes between well-understood adaptations—sample size re-estimation based on nuisance parameters, pre-specified enrichment—and less well-understood adaptations that require more extensive simulation and justification. The key regulatory principle is that the adaptation rule must be pre-specified in the protocol before enrollment begins, that the rule must be implemented independently of the treatment arm comparison by an independent statistician, and that the operating characteristics of the adapted design must be established by simulation before the trial begins.

The EMA's reflection paper on adaptive designs reflects similar principles, with additional emphasis on the estimand framework: adaptations that change any of the four estimand attributes require an explicit account of how the primary analysis will handle the population that was enrolled before and after the adaptation.

Both agencies are clear that adaptive designs do not relax the pre-specification requirement—they extend it. The fixed design pre-specifies the analysis. The adaptive design pre-specifies the analysis and the rules by which the design will change. The latter is more demanding, not less.

What this chapter is not about

This chapter does not address platform trials, master protocols, or basket and umbrella designs—multi-arm, multi-population designs that share infrastructure across multiple treatments or indications. These are important and growing parts of the clinical trial landscape, but they require separate treatment that goes beyond the scope of a chapter on adaptive modifications to a single trial's design.

It also does not address Bayesian adaptive designs in detail, beyond the principles that Bayesian and frequentist adaptive designs share in common. Bayesian designs use prior distributions and posterior probabilities rather than frequentist error rates, but the fundamental

governance requirements—pre-specification of the adaptation rules, independent implementation of the adaptation decisions, simulation-based characterization of the operating characteristics—apply equally.

The question this chapter must answer

By the end of this chapter, a trial team considering an adaptive design must be able to answer four questions.

What, exactly, is adapting? The specific design feature that changes must be identified—sample size, population, dose, endpoint—and distinguished from features that are fixed. Adaptations that are not precisely identified cannot be precisely pre-specified, and an adaptation that is not precisely pre-specified is a post-hoc modification.

What information triggers the adaptation? The adaptation rule must specify what data are used to make the adaptation decision—nuisance parameters only, or the primary endpoint trend, or a secondary endpoint result—and who has access to those data. The information used and the people who access it determine whether the adaptation inflates the type I error and whether it is consistent with the governance requirements of Chapter 4.

What does the adaptation do to the estimand? Every adaptation changes at least one attribute of the estimand. The question is whether the change is small enough to be handled within the pre-adaptation estimand or large enough to require a new estimand specification for the post-adaptation phase.

What are the operating characteristics of the adapted design? The probability of success under the null and under a range of alternative hypotheses—for the adapted design, simulated across the adaptation rule's input space—must be established before enrollment begins. An adapted design whose operating characteristics have not been established is not a design. It is a plan to make a decision whose consequences have not been examined.

If these four questions cannot be answered before enrollment begins, the adaptive design is not ready to run. The adaptation is being added to the design before the design implications of the adaptation are understood—which is the specific failure mode that regulators, DSMBs, and post-trial critics will identify, and that the pre-specification requirement exists to prevent.

7.1 Sample Size Re-estimation

The problem SSR solves

Chapter 3 documented the fragility of sample size assumptions. The effect size is predicted from prior studies with different populations. The variance is borrowed from contexts that may not transfer. The control arm event rate may be declining secularly. When these assumptions are wrong—as they frequently are—the trial is either underpowered and cannot detect a real effect, or overpowered and wastes resources and patient time.

Sample size re-estimation (SSR) is the adaptive response to this fragility. Rather than committing irrevocably to a sample size at design, the trial pre-specifies a rule by which the sample size can be adjusted based on data observed during the trial. If the interim data suggest the nuisance parameters are worse than assumed—variance higher, event rate lower, dropout higher—the sample size is increased to restore the planned power. The adaptation is prospective, rule-governed, and designed to address a well-identified source of design failure.

This is the correct motivation for SSR. Its statistical properties are well-established when implemented correctly. The challenge is in the implementation: what data can the re-estimation use without inflating the type I error, who performs the re-estimation, what are the constraints on the size adjustment, and how do the operating characteristics of the adapted design relate to those of the original design.

Nuisance-parameter SSR versus effect-size SSR

The most important distinction in SSR design is between re-estimation based on nuisance parameters and re-estimation based on the observed effect size.

Nuisance-parameter SSR uses blinded interim data—data that do not reveal the treatment arm comparison—to update the estimate of the variance, the control arm event rate, or the dropout rate. Because the data are blinded, the re-estimation does not depend on the interim treatment effect estimate and does not inflate the type I error in the way that unblinded data use would. A blinded SSR that adjusts the sample size based on the pooled variance, without knowing which arm the variance is coming from, is a pre-specified design adjustment that does not compromise the primary test.

This is the most defensible form of SSR. It addresses the most common source of power loss—wrong nuisance parameter assumptions—without introducing the type I error inflation that comes from conditioning the sample size on the interim treatment effect. The operating characteristics of the adapted design are close to those of the fixed design at the true nuisance parameters; the adaptation corrects the design toward what it would have been if the nuisance parameters had been known at the outset.

Effect-size SSR uses unblinded interim data—including the interim treatment effect estimate—to determine whether the sample size should be increased. The rationale is that if the interim effect is smaller than assumed, the trial is heading toward underpower; increasing the sample size can restore the probability of success. This form of SSR is more intuitive and more commercially attractive than nuisance-parameter SSR, because it directly addresses the scenario the sponsor cares most about: the trial that will miss significance by a small margin.

Its cost is significant. When the sample size depends on the interim effect estimate, the final test statistic is correlated with the interim estimate in a way that inflates the type I error if the standard fixed-sample critical value is used. The inflation can be substantial: without correction, an effect-size SSR can increase the type I error from 2.5% to 4-5% or more, depending on the specifics of the re-estimation rule. Controlling the type I error after effect-size SSR requires analytical methods—the combination tests of Lehmacher and Wassmer, the inverse normal method, or related approaches—that adjust the final test for the adaptive nature of the sample size selection.

These methods work—they control the type I error—but they come at a cost: reduced power relative to the pre-adaptation design, and reduced power relative to what a fixed design with the correct sample size would achieve. The efficiency of SSR—relative to a fixed design that happened to use the right sample size—depends on the magnitude of the nuisance parameter uncertainty, the design of the re-estimation rule, and the power of the correction method. For plausible scenarios, effect-size SSR with a correct correction method is more efficient than a fixed design with a wrong sample size assumption, but less efficient than a fixed design with the correct assumption.

What the re-estimation rule must specify

The SSR rule is a pre-specified function from interim data to a new sample size. It must specify, before enrollment begins, all of the following.

The interim data used. Is the re-estimation based on blinded data only, or on unblinded data? If blinded, what pooled statistics will be used—pooled variance, pooled event rate, pooled dropout rate? If unblinded, what information about the interim treatment effect will be used, and how?

The timing of the re-estimation. At what information fraction will the interim data be collected for the re-estimation? The timing affects the power of the correction method

and the operating characteristics of the adapted design. Re-estimation at early information fractions is more uncertain—the nuisance parameter estimates are less stable—but allows more time to enroll the additional patients if the sample size is increased.

The bounds on the adaptation. What is the minimum and maximum sample size after re-estimation? A re-estimation without bounds can produce a sample size that is impractically large or that so extensively extends the trial as to make the original design unrecognizable. Practical upper bounds—typically expressed as a maximum number of additional patients or a maximum multiple of the original sample size—are part of the pre-specification and part of the operating characteristic calculation.

Who performs the re-estimation. The re-estimation must be performed by an independent statistician who does not have access to information that would be available to the sponsor or the trial team beyond what the pre-specified rule allows. If the re-estimation is based on blinded data, the independent statistician receives the blinded pooled statistics and applies the rule. If it is based on unblinded data, the independent statistician must be isolated from the sponsor in the same way that the DSMB statistician is isolated—with the information firewall requirements of Chapter 4's governance framework applied to the re-estimation decision.

How the primary analysis accounts for the adaptation. For nuisance-parameter SSR, the primary analysis can typically proceed as if the trial were a fixed design with the re-estimated sample size, with minor corrections. For effect-size SSR, the primary analysis must use the adaptive test methodology—combination test or inverse normal method—and the critical value must be pre-specified for the combination test, not for the standard test.

Operating characteristics of the adapted design

The operating characteristics of the SSR design must be established by simulation before enrollment begins. The simulation must cover the full range of scenarios the trial might encounter: true effect sizes above, at, and below the assumed alternative; nuisance parameters at their assumed values and at their plausible extremes; various combinations of true effect size and nuisance parameter misspecification.

For nuisance-parameter SSR, the key operating characteristics are the power at the true effect size across the range of true nuisance parameters, and the probability that the re-estimation triggers a sample size increase across the range of nuisance parameter realizations. The simulation should show that the adapted design achieves the target power for a broader range of nuisance parameters than the fixed design, at the cost of a larger expected sample size when the nuisance parameters are at their pessimistic values.

For effect-size SSR, the key operating characteristics are more complex. They include the type I error under the combination test method, the power across the range of true effect sizes, the expected sample size under the null and under the alternative, and the conditional

power at the re-estimation time point—the probability of achieving significance at the final analysis, conditional on the interim result observed at the re-estimation.

The conditional power at the re-estimation time point is particularly important because it is the diagnostic that determines whether the re-estimation triggers an increase. If the conditional power is below a pre-specified threshold—50% is common, though the threshold should be justified—the sample size is increased. This threshold must be pre-specified; a threshold chosen after the interim conditional power is computed is a post-hoc decision rule, not a pre-specified one.

The governance requirements

SSR is a design adaptation, and it carries governance requirements that parallel those of interim analyses.

The blinded information used in the SSR must remain blinded—to the sponsor, to the analysis team, and to anyone whose behavior during the trial could be influenced by knowledge of the interim trend. The independent statistician who performs the re-estimation receives the blinded pooled statistics and returns a new sample size (or a determination that no adjustment is needed) without revealing the statistics to any other party. The process must be documented in the SSR charter—a document analogous to the DSMB charter that specifies the procedure, the data to be used, the decision rule, and the communication protocol.

When the re-estimation uses unblinded data—for effect-size SSR—the governance requirements are stricter. The unblinded data must be handled with the same firewall controls as DSMB interim analyses. The independent statistician sees the unblinded interim data, applies the re-estimation rule, and returns only the new sample size—not the interim treatment effect estimate—to the sponsor. The sponsor receives a sample size decision, not interim efficacy information.

This governance requirement is the source of a common operational failure: SSR that was designed correctly on paper but implemented in a way that allowed the treatment arm comparison to reach the sponsor. When this happens—when the sponsor learns the interim effect from the SSR process—the SSR has functioned as an unplanned interim analysis with unrestricted information access, contaminating the final analysis in exactly the way Chapter 4 described.

SSR in event-driven trials

Event-driven trials—where the primary analysis is triggered by accumulation of a target number of events rather than a target number of patients—introduce a specific SSR challenge. The relevant nuisance parameter is the control arm event rate, and the adaptation is a change in the target number of events or the target follow-up duration rather than the target sample size.

The mechanics are the same: blinded interim event rates are used to update the estimate of the rate, and the target is adjusted to restore the planned power. The governance requirements are the same. The additional complexity is that event-driven SSR affects the trial duration as well as the sample size—a lower-than-assumed event rate requires either more patients (to accumulate the same number of events faster) or longer follow-up (to allow events to accumulate in the existing cohort). These two responses have different cost and operational profiles, and the pre-specified rule must specify which response is used and under what conditions.

The interaction with the alpha-spending plan is also more complex in event-driven SSR, because the information fractions at which interim efficacy analyses are conducted are expressed as fractions of the target event count—and the target event count is now variable. The spending function must be applied at the information fractions defined relative to the new target, not the original target. This requires the pre-specification to address how the spending function is recalibrated when the target changes—which is the same problem identified in Chapter 4's discussion of event rate uncertainty.

What this section demands before proceeding

Before Section 7.2's discussion of adaptive enrichment, the SSR design must be complete in the following sense.

The re-estimation rule is fully specified: what data, at what time, by whom, with what bounds, corrected how in the primary analysis. The SSR charter is drafted—the document that governs the re-estimation process with the same specificity that the DSMB charter governs interim efficacy reviews. The operating characteristics are established by simulation across the relevant scenario space, and the adapted design is shown to achieve the target power for a broader range of nuisance parameters than the fixed design would, at an acceptable cost in expected sample size.

And the interaction between SSR and the efficacy interim analysis plan is addressed. If the efficacy interim analysis occurs at the same time as the SSR, the governance structures must be separated—the DSMB reviews efficacy data in closed session while the independent statistician performs the re-estimation based on blinded or separately unblinded data, with no information crossing between the two processes. This separation must be pre-specified and operationally verified before the trial begins.

References: Wittes and Brittain, "The Role of Internal Pilot Studies in Increasing the Efficiency of Clinical Trials," Stat Med 1990; Bauer and Köhne, "Evaluation of Experiments with Adaptive Interim Analyses," Biometrics 1994; Lehmacher and Wassmer, "Adaptive Sample Size Calculations in Group Sequential Trials," Biometrics 1999; Proschan, "Two-Stage Sample Size Re-Estimation Based on a Nuisance Parameter," J Biopharm Stat 2005.

7.2 Adaptive Enrichment

What adaptive enrichment does

Adaptive enrichment is the modification of the enrolled population—mid-trial, based on accumulating data—to concentrate enrollment in the patient group that shows the strongest signal of benefit. The most common form enriches toward a biomarker-positive subgroup: if the interim data show a stronger treatment effect in biomarker-positive patients than in the overall population, the trial narrows its eligibility to biomarker-positive patients for the remainder of enrollment.

The rationale is clinical and developmental. When a treatment's mechanism targets a specific biological pathway, biomarker-positive patients who express that pathway are more likely to benefit. If the all-comers design was chosen because the biomarker's predictive value was uncertain at the time of design, and the interim data clarify that the benefit is concentrated in the biomarker-positive subgroup, enriching toward that subgroup is a more efficient and more ethically defensible use of the remaining enrollment. Patients who are unlikely to benefit are not enrolled unnecessarily; patients who are likely to benefit are enrolled at a higher rate.

The cost is complexity—in the estimand, in the analysis, and in the claim structure. Adaptive enrichment does not merely change the trial's efficiency; it changes the trial's scientific question. The trial that was designed to answer the question "does this treatment work in the overall population?" becomes a trial that answers a different question: "does this treatment work in the overall population enrolled before the enrichment, and in the biomarker-positive population enrolled after?" These are not the same question, and the trial's primary result must address both—or must specify, before the enrichment, which population's result will be primary.

The estimand shift at enrichment

Chapter 1 required that the estimand be specified before enrollment begins. Adaptive enrichment appears to violate this requirement: if the enrolled population changes mid-trial, the population attribute of the estimand changes, and the estimand specified at the beginning of the trial no longer describes the trial that was actually run.

This is the central tension of adaptive enrichment, and it is resolved—not eliminated—by designing the estimand for the adapted design prospectively. Before enrollment begins, the trial must specify not just the estimand for the pre-adaptation phase but the estimand or estimands that will apply if the enrichment is triggered. The pre-adaptation estimand covers the overall population. The post-adaptation estimand covers the enriched population. The primary analysis must specify how these two estimands will be combined into a primary result.

Several frameworks exist for this combination. The simplest is to specify the overall population as the primary population and the biomarker-positive population as a secondary estimand—a pre-specified subgroup analysis that is powered by the enrichment. In this framework, the primary test uses all enrolled patients, and the enrichment improves the power of the biomarker-positive subgroup analysis without changing the primary claim. This is interpretively clean but may leave power on the table if the treatment has no effect in the overall population and the enrichment was scientifically correct.

A more ambitious framework—the stratified enrichment design of Simon and Simon, or the related approaches of Rosenblum and van der Laan—specifies the primary analysis as a combination of the overall and enriched populations, weighted by the information fraction in each phase. This framework allows a single primary claim that applies across the pre- and post-adaptation populations, at the cost of an analysis that is more complex and less familiar to regulatory reviewers.

The choice between these frameworks must be made before enrollment begins—before the enrichment decision is made—as part of the pre-specification of the adaptive rule. A framework chosen after the enrichment is triggered, to fit the data that triggered it, is post-hoc.

Pre-specifying the enrichment rule

The enrichment rule is a pre-specified function from interim data to an enrollment decision: continue enrolling all patients, enrich toward a subgroup, or close the all-comers design entirely and enroll only the subgroup.

The rule must specify what data trigger the enrichment—the interim treatment effect estimate in the subgroup, relative to the interim treatment effect in the overall population, relative to a pre-specified threshold—and how those data are handled from a governance perspective. Enrichment decisions based on the biomarker subgroup's interim treatment effect require access to unblinded interim data, and the governance requirements of Chapter 4 apply in full.

The most common governance structure for adaptive enrichment uses an independent DSMB or an independent data monitoring committee (IDMC) that reviews the unblinded interim data and makes the enrichment recommendation. The sponsor receives the enrichment

decision—enrich or do not enrich—without seeing the interim biomarker subgroup treatment effect that drove the recommendation. This is the same information firewall that Chapter 4 required for efficacy stopping decisions, applied to the enrichment decision.

The enrichment threshold must be pre-specified as a specific criterion rather than a general statement of intent. A rule that specifies "enrichment will be considered if the biomarker-positive subgroup shows a meaningfully stronger effect" is not a pre-specified rule; it is a statement that the IDMC will exercise judgment—which is a legitimate delegation of authority, but not a pre-specification. A rule that specifies "enrichment will be triggered if the interim hazard ratio in the biomarker-positive subgroup is at least 30% more favorable than the interim hazard ratio in the overall population, and the biomarker-positive conditional power under the null is above 50%" is a pre-specified rule that can be applied mechanically.

The specificity requirement is not pedantry. An enrichment rule that is vague enough to accommodate different decisions depending on the interim data is not a pre-specified rule—it is a rule that will be interpreted to support the decision that the IDMC or the sponsor prefers, post-hoc. The specificity requirement is what makes the rule genuinely adaptive rather than nominally adaptive.

The claim structure after enrichment

An adaptive enrichment design that is triggered—that switches from all-comers to biomarker-positive enrollment—faces a specific challenge at the primary analysis: what population does the primary result describe?

The population at the final analysis is a mixture: some patients who were enrolled before the enrichment and are in the overall population, some who were enrolled after the enrichment and are in the biomarker-positive population. If the primary analysis uses all enrolled patients in the denominator, the result describes a mixture population that does not correspond to any natural clinical population. If the primary analysis uses only the biomarker-positive patients, it excludes patients who were enrolled in the first phase under a different eligibility criterion.

The pre-specified analysis framework must address this explicitly. The most common approaches are:

A **two-stage combination test** that combines the test statistic from the first stage (all-comers) with the test statistic from the second stage (biomarker-positive), using the combination weights pre-specified in the design. The primary result is the combined test, and its operating characteristics—the type I error and power—are established by simulation across the space of possible first- and second-stage outcomes.

A **weighted analysis** that uses all enrolled patients but assigns weights based on the information fraction in each phase and the population in each phase. The weights must be pre-specified before any unblinded data are seen, not derived from the interim data.

A **hierarchical analysis** that tests the overall population first and the biomarker-positive population second, in the hierarchical framework of Chapter 6. This approach is interpretively clean—the claim for the overall population is tested first, and the biomarker-positive claim is secondary—but it is most powerful when the treatment has a real effect in the overall population, and may be underpowered for the biomarker-positive claim if the overall test fails.

None of these approaches is universally superior. Each is appropriate in some design contexts and inappropriate in others, and the choice must be made based on the scientific question—what population the primary claim is intended to address—before enrollment begins.

The biomarker: validated before the enrichment?

Adaptive enrichment based on a biomarker is defensible only if the biomarker can be measured reliably and consistently at the time of randomization, and if the enrichment rule specifies the biomarker measurement that will be used to make the enrichment decision.

If the biomarker assay changes between the first phase and the second phase—because a more sensitive assay becomes available, or because the threshold for positivity is revised—the patients enrolled in the two phases are not biomarker-defined by the same criterion. The enrichment has changed the population in a way that is more complex than the nominal enrichment rule describes, and the primary analysis must address the assay discordance explicitly.

This is not a hypothetical problem. In oncology, biomarker assays are frequently refined during the period of a trial. A trial that begins with a research-use-only assay and transitions to a companion diagnostic may classify some patients differently under the two assays. The pre-specification of the enrichment rule must include the specific assay that will be used to make the enrichment decision—not just the biomarker—and the handling of patients who might be classified differently under the two assays must be pre-specified.

The companion diagnostic validation is therefore not just a regulatory requirement for the commercial product; it is a design requirement for the adaptive enrichment rule. If the biomarker cannot be measured reliably enough to support the enrichment decision—if the assay has high false positive and false negative rates in the relevant patient population—the enrichment rule will be applied to misclassified patients, and the enriched population will not be the biomarker-positive population the design intended to enrich toward.

Adaptive enrichment in non-inferiority designs

Adaptive enrichment in a non-inferiority trial introduces the additional complexity that Chapter 2 identified for NI designs: the constancy assumption requires that the comparator's historical effect transfers to the enrolled population. When the enrolled population changes mid-trial—because of adaptive enrichment—the constancy assumption must be re-evaluated for the post-enrichment population.

If the comparator's historical effect in the biomarker-positive subgroup is different from its effect in the overall population—which it may be, because the biomarker-positive patients have a different disease biology—the NI margin derived from historical data on the overall population may not be the appropriate margin for the biomarker-positive population. A margin that was defensible before the enrichment may not be defensible after, because the population to which the constancy assumption applies has changed.

This interaction between adaptive enrichment and NI margin validity is rarely addressed in the design of adaptive NI trials. The pre-specification requirement must include an explicit evaluation of the constancy assumption for the post-enrichment population, and if the margin is not transferable, either the margin must be recalculated for the subgroup or the enrichment must be reconsidered.

What this section demands before proceeding

Before Section 7.3's discussion of estimand shift, the adaptive enrichment design must be complete. The enrichment rule is fully specified: the criterion for triggering enrichment, the biomarker assay that will be used, the governance structure for the enrichment decision, and the primary analysis framework that combines the pre- and post-adaptation populations.

The estimand for each phase is pre-specified, and the primary claim across phases is stated. The operating characteristics of the adapted design—the type I error for the combined test, the power across the range of true effects in the overall and subgroup populations, the probability of triggering enrichment under various scenarios—are established by simulation.

And the constancy assumption, for NI designs, is evaluated for the post-enrichment population. The margin's transferability to the enriched population is documented before the enrichment rule is pre-specified—not because the margin will certainly differ, but because the question must be asked and answered before the rule is locked.

References: Simon and Simon, "Adaptive Enrichment Designs for Clinical Trials," Stat Biopharm Res 2013; Rosenblum and van der Laan, "Simple, Efficient Estimators of Treatment Effects in Randomized Trials," Int J Biostat 2010; FDA Guidance for Industry, Enrichment

Strategies for Clinical Trials (2019); Bhatt and Bhatt, "Adaptive Designs for Clinical Trials," N Engl J Med 2016.

7.3 Estimand Shift

The adaptation that changes the question

Sections 7.1 and 7.2 examined specific forms of adaptation—sample size re-estimation and population enrichment—and noted, in passing, that each can shift the estimand. This section examines estimand shift directly: the ways in which adaptations that do not explicitly change the primary endpoint nonetheless change what the trial is estimating.

Estimand shift is the most conceptually underappreciated risk in adaptive design. A sponsor who adds patients through SSR, or who enriches toward a biomarker subgroup, understands that something has changed. What is less often recognized is that even when the nominal endpoint, the analysis model, and the enrolled population all remain unchanged, an adaptation can alter the conditional structure of the analysis in a way that changes the estimand implicitly.

The ICH E9(R1) framework requires that the estimand—population, variable, intercurrent event strategy, and population-level summary—be specified before the trial begins and remain stable throughout. When an adaptation changes any of these four attributes, the estimand has shifted, and the primary analysis must address the shift explicitly. When an adaptation changes the conditional structure of the analysis without changing any of the four attributes nominally, the estimand has shifted implicitly, and the shift is harder to detect and harder to correct.

Both forms of shift require design-stage attention—before enrollment, as part of the pre-specification of the adaptive rule—not analysis-stage correction.

Explicit estimand shifts

An explicit estimand shift occurs when the adaptation changes one of the four estimand attributes in a way that is visible in the design.

Population shift is the clearest case. Adaptive enrichment, as examined in Section 7.2, changes the population attribute: the patients enrolled after the enrichment are a different population than those enrolled before. The estimand that was specified for the pre-adaptation phase—a population-level treatment effect in all eligible patients—does not describe the post-adaptation phase's population. A new estimand, or a combined estimand, must be specified.

Treatment attribute shift occurs in seamless phase II/III designs that adapt the dose based on phase II results. The treatment attribute of the estimand is the specific dose that patients receive. When the dose is selected adaptively—based on phase II efficacy and safety data—the treatment attribute changes at the adaptation point: patients in phase III receive a different dose than some patients in phase II. If phase II patients who received the selected dose are included in the primary analysis—a common efficiency gain in seamless designs—the estimand for those patients is the same as for phase III patients. If phase II patients who received doses that were not selected are included, or if phase II patients are excluded entirely, the estimand for the pooled population must be specified to describe who is included and why.

Intercurrent event strategy shift can occur when the adaptation changes the background therapy or the trial's context in ways that affect how intercurrent events are expected to occur. An enrichment that shifts the population toward sicker patients may increase the rate of rescue medication use; a SSR that extends the trial duration may increase the rate of treatment discontinuation due to tolerability. When the intercurrent event rate or pattern changes as a consequence of the adaptation, the pre-adaptation intercurrent event strategy may not adequately describe the post-adaptation situation, and the combined analysis must address the change.

Implicit estimand shifts

Implicit estimand shifts are more dangerous because they are harder to recognize. They occur when the adaptation changes the conditional structure of the analysis without visibly changing any estimand attribute.

The most important example is effect-size SSR. When the sample size is adjusted based on the interim treatment effect estimate, the final analysis is conditional on the interim result—in the sense that the patients enrolled in the second stage are a sample selected from the population under the condition that the interim result was in the region that triggered an increase. The final test statistic is not an unconditional estimate of the population treatment effect; it is a conditional estimate, conditioned on the interim result falling in the region that triggered the adaptation.

The combination test methods used in effect-size SSR are designed to produce a valid test—one that controls the type I error—under this conditional structure. But the treatment effect estimate from the combination test may not be a valid estimate of the population treatment effect in the same sense that a fixed-design estimate would be. The conditional maximum likelihood estimate—the estimate obtained by conditioning on the adaptive decision—is approximately unbiased for the population treatment effect when the adaptation threshold is not too aggressive, but the approximation worsens as the threshold tightens.

This is an implicit estimand shift: the nominal estimand has not changed—same population, same endpoint, same intercurrent event strategy, same summary measure—but the analysis

is estimating a quantity that is conditioned on the interim result, which is not the same quantity as the unconditional population treatment effect.

Seamless phase II/III designs

Seamless designs—in which phase II and phase III are conducted in a single trial with a pre-specified adaptation at the phase boundary—are the most complex form of adaptive design because they introduce multiple potential estimand shifts simultaneously.

A typical seamless design enrolls patients in phase II across multiple doses, selects one or more doses for continuation based on phase II results, and continues enrollment in phase III at the selected dose or doses. The primary analysis combines phase II and phase III data for the selected dose, with the combination test method controlling the type I error for the adaptive dose selection.

The estimand for the combined analysis must specify who is included from each phase and why. If phase II patients at the selected dose are included, the estimand covers a population that was enrolled before and after the dose selection—which may have different characteristics if the dose selection affected site practices, patient counseling, or eligibility interpretation. If phase II patients at non-selected doses are excluded, the estimand covers only the patients at the selected dose, but the dose selection was based on all phase II data, creating a conditional structure that the combination test must account for.

The dose selection rule is itself a source of estimand shift. If the selected dose is the one with the highest efficacy signal in phase II—regardless of safety—the patients enrolled at that dose in phase III are a population selected under the condition that the efficacy signal was highest at that dose. The treatment effect estimate in phase III may be influenced by selection bias: the dose with the highest phase II signal is likely to have a higher phase II estimate than the true dose-specific effect, because of regression to the mean. The phase III estimate—which conditions on the dose having been selected—may partially correct for this, or may not, depending on the combination method.

Pre-specification of the dose selection rule—including the criteria for selection, the weighting of efficacy and safety signals, and the method for handling ties—is therefore not just a governance requirement. It is a scientific specification of what the primary estimand is, because the dose selection rule determines which patients are in the primary analysis and under what conditions they were enrolled.

The estimand for the adapted trial

Specifying the estimand for an adaptive trial requires working through the four attributes of the estimand for each phase of the trial and for the combined analysis.

Before the adaptation: the estimand is as specified at design. Population, variable, intercurrent event strategy, and population-level summary are all pre-specified and unchanged.

After the adaptation: the adaptation changes one or more attributes. The new values of the changed attributes must be specified as part of the pre-specification of the adaptive rule. If the population changes—because of enrichment—the post-adaptation population attribute is specified as part of the enrichment rule. If the treatment changes—because of dose selection—the post-adaptation treatment attribute is the selected dose, and the dose selection rule pre-specifies which dose or doses will be selected under which circumstances.

For the combined analysis: the combined estimand must cover the patients enrolled in both phases. The combined population, the combined variable, the combined intercurrent event strategy, and the combined summary measure must be specified. When the population changes at the adaptation, the combined estimand requires explicit specification of whether the pre- and post-adaptation patients are analyzed jointly or separately, and how the combination weights the two phases.

This four-part specification—before adaptation, after adaptation, combined—is the estimand framework for adaptive designs. It is more demanding than the estimand framework for fixed designs, and it must be completed before enrollment begins—which means before the adaptive rule can be finalized, because the estimand determines what the adaptive rule is adapting.

Documentation and the audit trail

The estimand shift at an adaptive decision point must be documented in real time, not reconstructed afterward. At the time of the adaptation, the following must be recorded.

What data triggered the adaptation. What the adaptation decision was and the basis for it—including the specific values of the criteria in the pre-specified rule that were met or not met. What the post-adaptation estimand is, based on the pre-specified framework. Who made the adaptation decision and under what governance structure. What information was made available to each party in the adaptation process.

This documentation is the audit trail for the adaptation. It is the evidence that the adaptation was made according to the pre-specified rule rather than in response to information beyond what the rule allowed. Without it, the adaptation cannot be verified as pre-specified, and the claim that the primary analysis is valid under the combination test method cannot be established.

The audit trail for adaptive designs is more demanding than for fixed designs, because the adaptive decisions are made in real time—under time pressure, with governance constraints, by multiple parties with different information access—and the record of those decisions must be complete and contemporaneous. A reconstruction of the adaptation decision based on meeting notes and email records, prepared for regulatory submission, is not an audit trail. It is a narrative, and narratives can be selective.

What this section demands before proceeding

Before Section 7.4's examination of non-inferiority failure modes in adaptive designs, the estimand shift analysis must be complete.

For each planned adaptation in the trial—SSR, enrichment, dose selection, or other—the estimand shift must be identified: which of the four attributes changes, how it changes, and what the post-adaptation estimand is. The combined estimand must be specified: how the pre- and post-adaptation patients are handled in the primary analysis, with the combination weights and combination method pre-specified. And the documentation requirements for the adaptation—the audit trail—must be specified before enrollment, so that the information to be recorded is known before the adaptation is triggered.

An adaptive design whose estimand shifts have not been identified and addressed is a design that will answer a different question than it was designed to answer—and the difference may not be recognized until the regulatory review asks what population the primary result describes.

References: ICH E9(R1) Addendum on Estimands and Sensitivity Analysis in Clinical Trials (2019); Bauer et al., "Adaptive Designs for Confirmatory Clinical Trials," Stat Med 2016; Wassmer and Brannath, Group Sequential and Confirmatory Adaptive Designs in Clinical Trials (2016); FDA Guidance for Industry, Adaptive Designs for Clinical Trials of Drugs and Biologics (2019).

7.4 Non-Inferiority Failure Modes in Adaptive Designs

Why NI and adaptive design compound each other

Chapter 2 introduced non-inferiority design as the context in which the effect measure's interpretation is most demanding and the margin's derivation is most consequential. The NI trial is not asking whether the treatment works; it is asking whether the treatment is not too much worse than a comparator whose historical effect provides the evidence base for the margin. The constancy assumption—that the comparator's historical effect transfers to the current trial—is the load-bearing assumption on which the entire NI argument rests.

Chapter 7 has been examining adaptive design as a context in which the risks of prior chapters are amplified. The amplification when NI design and adaptive design are combined is not additive—it is multiplicative. Every adaptation that changes the enrolled population, the trial duration, the background therapy context, or the analysis timing has the potential to undermine the constancy assumption on which the NI margin is based. When the constancy assumption fails in an adaptive NI design, the result is a trial that formally concludes non-inferiority—the pre-specified boundary was not crossed—while the scientific basis for that conclusion has been eroded by the adaptations that occurred during the trial.

This is the specific failure mode that this section examines: the adaptive NI trial that is formally valid and scientifically misleading.

The constancy assumption under population shift

Section 2.3 established that the NI margin requires two components: M1, the estimated effect of the comparator over placebo from historical trials, and M2, the clinically acceptable fraction of M1 to sacrifice. The constancy assumption is the claim that the comparator's true effect in the current trial is at least as large as M1—that the historical evidence about the comparator's effect transfers to the current population and context.

When an adaptive enrichment shifts the enrolled population toward a biomarker-positive subgroup, the constancy assumption must be re-evaluated for the enriched population. The comparator's historical effect in the overall population—which was the basis for M1—may not be the same as its effect in the biomarker-positive subgroup, because biomarker-positive

patients have a different underlying biology, a different disease trajectory, and potentially a different response to the comparator.

If the comparator's effect is smaller in the biomarker-positive subgroup than in the overall population, the margin M1 derived from historical overall-population data overstates the comparator's historical effect in the enriched population. The pre-specified NI margin is too permissive for the enriched population—it allows a larger inferiority than is scientifically justified. A trial that concludes non-inferiority against this permissive margin has shown that the new treatment is not worse than the comparator by more than an amount that was calculated for a different population.

This problem is not resolved by simply noting the enrichment in the trial report. It requires either: recalculating M1 for the biomarker-positive subgroup, based on historical data stratified by biomarker status; or restricting the NI claim to the pre-adaptation population, for which the original M1 is appropriate; or accepting that the NI conclusion applies to the enriched population with a margin that may be too permissive, and acknowledging this limitation explicitly.

None of these resolutions is costless, and the pre-specification requirement means that the chosen resolution must be specified before the enrichment is triggered—before the adaptation reveals whether the enriched population's constancy assumption is credible.

The constancy assumption under duration extension

SSR in an event-driven NI trial extends the trial's follow-up duration when the control arm event rate is lower than assumed. The extension may seem operationally straightforward—more time to accumulate the needed events—but it carries a specific constancy assumption problem.

The NI margin was derived from historical trials that had a specific follow-up duration. The comparator's effect estimated in those trials—the M1—is an effect over that specific duration. If the current trial extends its duration beyond the duration of the historical trials that generated M1, the constancy assumption is being applied to a longer follow-up period than the historical evidence covers.

In many indications, the treatment effect of both the comparator and the new treatment is not constant over time. Active treatments that prevent events in the short term may show waning efficacy over longer follow-up; treatments that require time to achieve their mechanism may show increasing efficacy over longer follow-up. When the follow-up is extended beyond the duration of the historical evidence, the M1 derived from that evidence may not apply, and the margin calculated from it may be too permissive or too conservative for the extended duration.

The standard response—that the constancy assumption is routinely violated and should be evaluated qualitatively rather than tested formally—does not resolve the problem when the

adaptation itself introduces the duration extension. In a fixed design, the trial duration is pre-specified, and the constancy assumption is evaluated for that pre-specified duration. In an adaptive NI design that extends the duration based on SSR, the constancy assumption must be evaluated for the extended duration, which was not known at the time of design.

Pre-specification of the constancy assumption evaluation for the extended duration scenario—including what historical data will be used and how the analysis will be modified if the constancy assumption is not credible at the extended duration—is part of the SSR rule specification for adaptive NI designs.

The assay sensitivity problem under adaptation

Chapter 2's discussion of the NI trial identified assay sensitivity—the trial's capacity to distinguish active treatments from inactive ones—as the property that makes the NI margin meaningful. A trial with adequate assay sensitivity, run under conditions where the comparator would show its historical effect, demonstrates that a treatment within the NI margin of the comparator is genuinely close to the comparator in efficacy.

Adaptive designs can undermine assay sensitivity in ways that are not always recognized.

An adaptive enrichment that shifts toward patients with better prognosis may reduce the trial's assay sensitivity. Better-prognosis patients have lower background event rates and a smaller absolute treatment effect, even if the relative effect is similar. A smaller absolute effect means the difference between the arms is smaller in absolute terms, making it harder to distinguish a genuinely effective treatment from a less effective one at the chosen margin. The NI margin—expressed in absolute terms, as a risk difference, or in relative terms, as a hazard ratio—may not have the same discriminatory power in the enriched population as in the original population.

An SSR that extends the trial beyond the original duration may similarly affect assay sensitivity. If the comparator's effect diminishes over longer follow-up—because of habituation, resistance, or natural history changes—the trial at the extended duration may have lower assay sensitivity than at the original duration, because the comparator is less effective at distinguishing its own effect from the null than it was at the original duration.

These are not categorical objections to adaptive design in NI trials. They are the specific questions that must be asked—and answered—before the adaptive rule is finalized. If the enrichment or the SSR can be shown, by simulation and by review of relevant historical data, to not materially affect assay sensitivity, the concern is addressed. If it cannot be shown, the adaptation must be modified or the NI conclusion must be bounded to the conditions under which assay sensitivity was maintained.

The switching strategy: from NI to superiority

A common adaptive strategy in NI trials is to pre-specify a switch from an NI primary objective to a superiority objective if the interim data show the new treatment is substantially better than the comparator. The rationale is efficiency: if the new treatment is not merely non-inferior but actually superior, the trial should be capable of claiming the stronger result.

This strategy is statistically valid when pre-specified correctly. A trial that tests superiority after failing to show non-inferiority—or that tests NI after the superiority test fails—is testing two hypotheses and must control the family-wise type I error across both. The pre-specified switching strategy must specify the conditions under which the primary test switches, the alpha allocation between the NI and superiority tests, and the decision rule for each possible outcome.

The switching strategy also has an estimand implication. The NI test and the superiority test are estimating the same effect—the treatment versus comparator difference—but interpreting it against different thresholds and for different conclusions. This is not a new estimand; it is the same estimand with two evaluation criteria. The pre-specification must be clear that the estimand does not change at the switch—the population, the variable, the intercurrent event strategy, and the summary measure remain the same—and that only the conclusion rule changes.

What the switching strategy cannot do is allow the choice of NI versus superiority objective to be made based on the interim treatment effect without controlling the type I error for the adaptive selection. A trial that tests NI if the interim effect is weak and superiority if the interim effect is strong—and that claims whichever test succeeds without adjusting for the adaptive selection—has an uncontrolled type I error rate. The switching strategy must pre-specify both tests, both thresholds, and the combined family-wise error rate.

Pre-specifying the constancy re-evaluation

The constancy assumption cannot be tested within the trial. It is a claim about the comparator's behavior in the current trial relative to its behavior in historical trials, and the historical trials are not randomized to the current or historical context. The constancy assumption is evaluated qualitatively—by examining whether the current trial's population, background therapy, and context are similar enough to the historical trials that the M1 is transferable.

In an adaptive NI trial, the constancy re-evaluation must occur at the adaptation point. When the adaptation changes the population—enrichment—or the duration—SSR—the design team must re-examine whether the constancy assumption remains credible for the post-adaptation trial. This re-evaluation must use the same criteria that were used to justify

the constancy assumption at design: a structured comparison of the pre-adaptation and post-adaptation trial contexts against the historical trials that generated M1.

This re-evaluation cannot be conducted by parties who have access to the interim treatment effect data. It must be conducted by an independent group—which may be the same independent statistician or IDMC that makes the adaptation decision, but acting on the constancy question separately from the adaptation decision—using only the protocol-defined population characteristics, the background therapy information, and the historical evidence base.

Pre-specifying the constancy re-evaluation means specifying, before enrollment, the criteria that will be applied to evaluate whether the constancy assumption remains credible after the adaptation. If the criteria are not pre-specified, the re-evaluation will be conducted post-hoc, using criteria that may be chosen to support the NI conclusion the sponsor wants to make.

What this section demands before proceeding

Adaptive NI design is the intersection of two demanding design frameworks, each of which amplifies the other's risks. Before this section's concerns can be addressed in the design, the following must be specified.

The constancy assumption for the original trial population is documented with the specific historical evidence—the trials, the M1 calculation, the credibility evaluation for the current population. The constancy re-evaluation criteria are pre-specified: what will be examined if the population shifts through enrichment or if the duration extends through SSR, by whom, at what point in the adaptation process, and with what consequence for the NI margin or the primary analysis if the constancy is no longer credible.

The assay sensitivity analysis for the adapted trial is documented: what historical data support the conclusion that the adapted population and duration will have adequate assay sensitivity to distinguish effective from ineffective treatments at the pre-specified margin. If the adaptation changes the assay sensitivity, the consequence for the NI margin is pre-specified—either the margin is adjusted for the adapted population, or the claim is bounded to the conditions under which assay sensitivity was maintained.

And the switching strategy, if used, pre-specifies both the NI and superiority tests, the conditions for switching, the alpha allocation, and the combined family-wise error rate.

These requirements are demanding because the combination of adaptive design and NI design is demanding—not because the combination is inherently invalid, but because each framework requires careful pre-specification of assumptions that the other framework can undermine. The demands are proportionate to the risk.

References: D'Agostino et al., "Non-Inferiority Trials: Design Concepts and Issues," Stat Med 2003; Hung et al., "Adaptive Statistical Analysis Following Sample Size Modification in Clinical Trials," J Biopharm Stat 2006; EMA Reflection Paper on Methodological Issues in Confirmatory Clinical Trials Planned with an Adaptive Design (2007); ICH E10, Choice of Control Group and Related Issues in Clinical Trials (2000).

7.5 Closing: The Pre-Specified Adaptation

What this chapter asked

Chapter 7 asked one question: what if the design itself adapts?

It asked that question not as an introduction to a technical specialty, but as an application of the book's central theme. The risks established in every prior chapter—optimistic effect size assumptions, fragile nuisance parameters, estimand ambiguity, interim governance failures, claim structure inflation—do not disappear when a design adapts. They are amplified, because the adaptation introduces new decision points, new information flows, and new interactions between design features that fixed designs manage separately.

Section 7.1 examined sample size re-estimation as the tool for correcting the nuisance parameter misspecifications that Chapter 3 identified. Nuisance-parameter SSR—based on blinded data—corrects the most common power failure without inflating the type I error. Effect-size SSR—based on unblinded data—corrects a broader range of failures but requires combination test methods to control the type I error and introduces governance requirements that parallel the Chapter 4 interim analysis framework.

Section 7.2 examined adaptive enrichment as a design that changes the enrolled population mid-trial based on the interim biomarker subgroup result. Enrichment changes the population attribute of the estimand and requires pre-specification of the combined estimand across pre- and post-adaptation phases, the enrichment rule including the biomarker assay specification, and the governance structure for the enrichment decision.

Section 7.3 identified estimand shift as the most conceptually underappreciated risk in adaptive design: the ways in which adaptations change what the trial is estimating even when the nominal endpoint and analysis model remain unchanged. Explicit shifts—population, treatment attribute, intercurrent event strategy—are visible. Implicit shifts—the conditional structure introduced by effect-size SSR, the dose selection conditioning in seamless designs—are harder to recognize and equally consequential.

Section 7.4 examined adaptive NI design as the intersection of two demanding frameworks, each of which amplifies the other's risks. The constancy assumption—the load-bearing scientific justification for the NI margin—must be re-evaluated at each adaptation point, using pre-specified criteria, by parties independent of the unblinded interim data.

What this chapter decided

By the end of this chapter, a trial team proposing an adaptive design must have answered four questions with documented, pre-specified answers.

What, exactly, is adapting? The specific feature—sample size, population, dose, analysis—is named. The pre-adaptation value and the post-adaptation value under each adaptation scenario are specified. The boundary between what adapts and what does not is drawn explicitly, so that a reviewer can verify that features claimed to be fixed were not modified in response to the interim data.

What information triggers the adaptation, and who handles it? For nuisance-parameter SSR: the blinded pooled statistics and the independent statistician who applies the rule. For effect-size SSR or adaptive enrichment: the unblinded interim data, the IDMC that reviews it, the firewall that prevents the unblinded result from reaching the sponsor, and the communication protocol that conveys the adaptation decision without conveying the interim treatment effect. For seamless designs: the phase II efficacy and safety data, the selection rule, the parties involved in the selection, and the information that each party receives.

What does the adaptation do to the estimand? The explicit estimand shifts are identified: which of the four attributes changes at the adaptation, and what the post-adaptation values are. The implicit estimand shifts are identified: how the combination test method handles the conditional structure introduced by the adaptation, and what the primary analysis is estimating as a result. The combined estimand is specified for the pooled population.

What are the operating characteristics of the adapted design? The simulation covers the full scenario space: true effect sizes above, at, and below the assumed alternative; nuisance parameters at assumed and pessimistic values; various adaptation outcomes under each scenario. The type I error under the combination test method is verified at the nominal level. The power under the target alternative is established. The expected sample size—or expected event count—under the null and alternative is documented.

The characteristic mistakes of this chapter

Three failures define the characteristic risk profile of adaptive designs.

The adaptation triggered by the wrong information. The SSR was designed to use blinded nuisance parameters. At the re-estimation time point, the independent statistician had access not only to the blinded pooled variance but to the interim clinical data report—a document prepared for the DSMB that included arm-specific event summaries. The re-estimation was completed by the independent statistician before the DSMB meeting, and

the new sample size was communicated to the sponsor. The sponsor did not receive the arm-specific event data, but the independent statistician who performed the re-estimation did—and the re-estimation was no longer based purely on nuisance parameters. The governance firewall had a gap that was not identified until the regulatory inspection examined the independent statistician's data access log.

The enrichment whose estimand was not pre-specified. The trial pre-specified an enrichment rule—enrich toward biomarker-positive patients if the interim biomarker subgroup shows substantially stronger benefit. The enrichment was triggered at the first interim. The primary analysis framework for the combined pre- and post-adaptation population had not been pre-specified in the SAP; the SAP addressed the unadapted design only. After the enrichment, the analysis team proposed a weighted combination test using the pre- and post-adaptation patient data. The regulatory reviewer asked when the combination weights and the combined estimand were specified. The answer—after the enrichment decision—was the answer that made the primary analysis post-hoc.

The NI adaptive trial whose constancy assumption was not re-evaluated. The trial was an NI design with a pre-specified SSR. The SSR triggered an extension of the follow-up duration from 24 to 36 months, because the control arm event rate was lower than assumed. The NI margin had been derived from historical trials with 18-24 months of follow-up. The primary analysis used the pre-specified margin, without re-evaluating whether the constancy assumption remained credible at 36 months. The trial concluded non-inferiority. The regulatory reviewer noted that the margin had been derived from trials with substantially shorter follow-up than the adapted trial's final duration, and that the constancy assumption had not been evaluated for the extended duration. The NI conclusion was challenged, and the sponsor was required to provide a post-hoc analysis of whether the constancy assumption was credible at 36 months. The analysis was inconclusive. The label was approved with a note about the limitation.

Each of these failures has the same structure: a pre-specification that was incomplete in a specific way, and an adaptation that operated in the gap left by the incompleteness. The pre-specification was not fraudulent—the incomplete element was not a deliberate omission. It was a gap that was not recognized as a gap until the adaptation was triggered and the gap became consequential.

What cannot be recovered

A governance breach in an SSR—unblinded data reaching parties who should not have seen it—cannot be undone. The information, once accessed, cannot be erased from the knowledge of the parties who accessed it. The adaptation decision that followed may have been made on the same basis as the pre-specified rule would have required, but the process by which it was made is contaminated, and the contamination is not correctable by analysis.

A combined estimand that was not pre-specified—that was constructed after the adaptation to fit the adapted population—cannot be retroactively pre-specified. The combination weights, the analysis method, and the decision about which patients to include in the primary analysis were all made with knowledge of which patients were in the pre-adaptation cohort and which were in the post-adaptation cohort. The analysis is post-hoc even if the analysis plan was written before the final database lock.

An NI margin that was not validated for the adapted population cannot be retroactively validated. The constancy assumption for the adapted population requires historical data that may not exist—because the historical trials did not enroll the enriched population or run to the extended duration. A post-hoc analysis of the constancy assumption is an analysis that was not required at design and is now being conducted to support a conclusion that was already made. The analysis may be informative, but it is not the evaluation that the pre-specification requirement demands.

These irrecoverabilities define the professional risk of adaptive design. The risk is not that adaptive designs are invalid—they are valid when pre-specified correctly. The risk is that the pre-specification of an adaptive design is more demanding than the pre-specification of a fixed design, and the gaps in the pre-specification are not always visible until the adaptation is triggered and the gap becomes a problem.

The connection to Chapter 8

Chapter 8 asks what must be locked, and when. It is the chapter that closes the book's design sequence by examining what the pre-specification requirement means in practice—what documents must be finalized before what events, by whom, with what review and approval process.

The connection to Chapter 7 is direct. Adaptive designs require more to be locked, earlier, and with more specificity than fixed designs. The SSR rule must be in the SAP before the re-estimation time point. The enrichment rule—including the biomarker assay, the enrichment threshold, and the combined estimand framework—must be in the protocol before the first patient is enrolled. The constancy re-evaluation criteria for adaptive NI designs must be in the protocol before enrollment. The combination test method and the critical values for the adaptive primary analysis must be in the SAP before the first interim.

Each of these lock requirements is the adaptive extension of a fixed design requirement. The fixed design locks the sample size at design; the adaptive design locks the re-estimation rule at design. The fixed design locks the estimand at design; the adaptive design locks the pre- and post-adaptation estimands, and the combined estimand, at design. Chapter 8 will establish the lock requirements for the full design—fixed and adaptive—as a governance discipline that makes the pre-specification requirement operational.

Chapter 7 risk summary

The decision this chapter owns: is the planned adaptation pre-specified in sufficient detail to be implemented without discretion, governed with sufficient rigor to prevent information from crossing the boundaries the rule requires, and documented in sufficient specificity to be audited against the pre-specification?

The most common mistake: treating the adaptation as a design improvement without treating the pre-specification of the adaptation as a design requirement. The adaptation is specified in the protocol—SSR will be conducted, enrichment is planned—but the details that determine whether the adaptation controls the type I error, maintains the estimand, and is governable without contamination are left to the SAP or to the IDMC charter, which are completed later and with less scrutiny than the protocol.

The professional-level risk: the adaptive design that was pre-specified at the level of intent and implemented at the level of discretion. The SSR was triggered when the independent statistician judged the nuisance parameters to be sufficiently pessimistic—not when a specific threshold was crossed. The enrichment was recommended by the IDMC based on their clinical judgment about the biomarker subgroup result—not based on a pre-specified threshold. The combined estimand was constructed by the analysis team to be as favorable as possible to the adapted population—not selected from pre-specified options. Each individual decision may have been made in good faith. The cumulative effect is a trial whose adaptations were not pre-specified in the sense the regulatory agencies require—and whose primary result, however valid statistically, cannot be verified as the product of the pre-specified design rather than of the data that were seen.

Chapter 8: What Must Be Locked, and When?

The question the entire book leaves open

Every prior chapter asked a design question and demanded an answer before enrollment begins. Chapter 1 required the estimand. Chapter 2 required the effect measure. Chapter 3 required the sample size with owned assumptions. Chapter 4 required the interim analysis plan. Chapter 5 required the bias protection system. Chapter 6 required the claim structure. Chapter 7 required the adaptive rules.

Each of these requirements was framed as pre-specification. Pre-specify the estimand. Pre-specify the nuisance parameters. Pre-specify the hierarchy. Pre-specify the enrichment rule.

What none of those chapters addressed is what pre-specification means in practice. Pre-specification is not a state of mind—a sincere intention to specify before looking. It is a documentary condition: a document exists, with sufficient specificity to be audited, finalized at a time that can be verified, by people whose authority to finalize it is established. Without the documentary condition, the sincere intention is indistinguishable from a post-hoc rationalization, and the regulatory agency reviewing the trial has no way to verify which it was.

Chapter 8 addresses the documentary condition. It is the governance chapter—not in the sense of the DSMB governance addressed in Chapter 4, but in the deeper sense: the governance of the design process itself. Who finalizes what document, when, under what review, with what consequence for the trial if the document is not finalized on time. These are the questions that determine whether the book's prior demands were actually met or merely stated.

Why governance of the design process matters

The pre-specification principle is widely understood in clinical trial methodology. It is widely violated in clinical trial practice. The gap between understanding and practice is not primarily a gap in statistical knowledge; it is a governance gap.

The governance gap has a specific anatomy. The protocol is finalized before enrollment—this is a regulatory requirement—but the statistical analysis plan is sometimes finalized after interim data are available, sometimes after the database is locked, occasionally after the primary analysis has been run. The hierarchical testing order is specified in the protocol in general terms and detailed in the SAP, and if the SAP is finalized after data trends are visible, the hierarchy detail may be shaped by what the data show. The subgroup analyses listed in the SAP may be more numerous than those in the protocol, and the additions may reflect questions generated by the data rather than questions the design was built to answer.

None of this requires dishonesty. It requires only that the design process does not enforce documentary discipline—that the team does not treat the finalization of each design document as a governance event with a hard deadline, independent review, and consequences for the trial's credibility if the deadline is missed. When the process does not enforce discipline, the documents slide toward the data, and what looks like pre-specification from the outside is shaped by the results from the inside.

Chapter 8 establishes the documentary discipline that the pre-specification principle requires. It does so not as a compliance exercise—not as a list of things that must be in the file—but as a design requirement: the documents that govern the trial must be finalized before the events that could compromise their independence, and the finalization must be verifiable.

What this chapter covers

Section 8.1 — What Must Be Pre-specified establishes the complete list of design elements that must be documented before specific trial events, with the rationale for each deadline. The list is not the same for every trial; a fixed-design superiority trial has a different pre-specification requirement than an adaptive NI trial with a seamless phase II/III structure. But the principle is constant: every element that, if finalized after the triggering event, could be shaped by knowledge of the interim data or the final result, must be finalized before that event.

Section 8.2 — Adaptive Rules and Their Lock Requirements addresses the specific pre-specification demands of adaptive designs—the designs whose rules must be finalized before enrollment begins, not before the adaptation is triggered, because the adaptation will be governed by parties who must not exercise discretion beyond what the pre-specified rule allows.

Section 8.3 — The Decision Log examines the documentation of decisions made during the trial—not the pre-specified decisions, but the governance decisions that occur in response to unexpected events, DSMB recommendations, protocol deviations, and regulatory interactions. The decision log is the audit trail for the trial's conduct, and its completeness determines whether the trial's integrity can be verified from the outside.

Section 8.4 — Design Review and the Independent Challenge addresses the role of independent review in the design process—the prospective scrutiny of the design by parties who were not part of its construction and who can identify gaps, inconsistencies, and vulnerabilities that the design team, close to their own work, may not see. Independent review is not a quality control step at the end of the design process; it is a governance requirement at each stage of the process.

The four lock events

The documentary discipline of Chapter 8 organizes around four events in the trial's timeline at which specific documents must be finalized and locked.

Lock 1: Protocol finalization before first patient enrolled. The protocol contains the estimand, the primary endpoint, the effect measure, the eligibility criteria, the randomization scheme, the blinding design, the interim analysis plan (at the level of information fractions and stopping criteria categories), the co-primary decision rule if applicable, and the pre-specified subgroup analysis list with their confirmatory or exploratory status. The protocol is the public commitment to the trial's design, registered in a clinical trial registry before enrollment begins.

Lock 2: Statistical analysis plan finalization before any unblinded data access. The SAP contains the complete hierarchical testing plan in order, the specific analysis models for each endpoint, the pre-specified sensitivity analyses, the detailed definitions of all endpoints and their adjudication criteria, the handling of missing data under each intercurrent event strategy, the adaptive rule details (SSR rule with bounds, enrichment threshold and criteria), and the combination test method for adaptive trials. The SAP must be finalized before any member of the analysis team or the sponsor has access to unblinded data from the trial—including interim DSMB reports that contain arm-specific results.

Lock 3: DSMB charter and SSR charter finalization before first interim analysis. The DSMB charter—as described in Chapter 4—is finalized before the first interim. The SSR charter, for adaptive designs, is finalized on the same timeline. Both documents are governance instruments for events that will occur during the trial; they must be in place before the events they govern.

Lock 4: Decision log completeness before database lock. The decision log documents every governance decision made during the trial—every protocol amendment, every deviation from the pre-specified plan, every DSMB recommendation and the sponsor's response, every interaction with the regulatory agency that affected the trial's conduct. The decision log must be complete before the database is locked, so that the analysis team and the regulatory reviewer have a complete record of the decisions that shaped the trial's conduct and the data it produced.

What this chapter is not about

This chapter is not about regulatory submission formatting, clinical study report structure, or the specific content requirements of ICH E3 for clinical study reports. Those are post-trial documentation requirements; this chapter is about pre-trial and during-trial governance.

It is also not about data management—the systems and processes for collecting, cleaning, and locking the clinical database. Data management is the operational infrastructure for the

trial; this chapter is about the governance of the design decisions that determine what data are collected and how they will be analyzed.

The book's final question

Chapter 8 is the last of the eight design chapters. Its subject—the governance of the design process—is last not because it is least important, but because it can only be understood after all the prior decisions have been made. The governance requirements of Chapter 8 are the governance of the decisions established in Chapters 1 through 7. Without those decisions, there is nothing to govern. With them, the governance question is: how do we ensure that the decisions that were made become the decisions that were implemented?

That question is the book's final question. It is also, in a specific sense, the most consequential one—because a design that was correctly conceived but incorrectly documented is, from the outside, indistinguishable from a design that was post-hoc rationalized. The trial's scientific integrity and its regulatory defensibility both depend on the answer being verifiable, not just sincere.

8.1 What Must Be Pre-specified

The pre-specification principle, made operational

The pre-specification principle has appeared throughout this book as a design requirement: specify before the data, not from the data. Every chapter demanded pre-specification of a specific design element—the estimand, the effect measure, the hierarchy, the adaptive rule. Chapter 8 asks what pre-specification means as a documentary condition, not a state of mind.

The documentary condition has two components: specificity and timing. Specificity means the document contains enough detail that an independent statistician, reading it without access to the data, could implement the pre-specified analysis and arrive at the same result as the analysis team. Timing means the document was finalized before the event that could compromise its independence—before the data that might shape its content were available to the people who wrote it.

Both components are verifiable. Specificity can be assessed by a reviewer who examines whether the document contains the decisions it claims to contain, or whether it defers decisions to later judgment—using language like "will be determined at analysis" or "to be specified based on observed data." Timing can be assessed by examining the document's finalization timestamp, the trial's enrollment and unblinding timeline, and the data access records of the people who wrote the document.

When both components are satisfied, pre-specification is established. When either is missing—when the document is vague, or when it was finalized after the data were available—pre-specification is not established, regardless of the sincerity of the intent.

The protocol: what it must contain

The protocol is the first pre-specification document and the one with the hardest deadline: it must be finalized and registered before the first patient is enrolled. Its content requirements for design integrity are distinct from its regulatory content requirements, and both must be met.

From a design integrity perspective, the protocol must contain the following elements with enough specificity to be binding.

The estimand. The four attributes of the estimand—population, variable, intercurrent event strategy, and population-level summary—must be stated in terms that correspond to the scientific question and that can be operationalized in the statistical analysis. The estimand is not a narrative description of the intended treatment benefit; it is a formal specification of what is being estimated. If the protocol's estimand section cannot be translated directly into the SAP's primary analysis without additional decisions, it is not specific enough.

The primary endpoint definition. The operational definition of the primary endpoint—the specific criteria by which the outcome is determined for each patient—must be in the protocol. For composite endpoints, the definition of each component must be in the protocol. For patient-reported outcomes, the instrument, the scoring algorithm, and the threshold for response must be in the protocol. A primary endpoint defined as "will be operationalized in the adjudication charter" is not pre-specified in the protocol; it defers a binding decision to a later document.

The intercurrent event strategy. The pre-specified strategy for handling the primary intercurrent events—discontinuation of study treatment, initiation of rescue medication, switching to alternative therapy, death before the primary endpoint—must be in the protocol. For each intercurrent event, the strategy (treatment policy, hypothetical, composite, while-on-treatment, principal stratum) must be named and briefly justified.

The interim analysis plan. The number of planned interim analyses, the information fractions at which they will be conducted, the category of stopping rule at each (efficacy, futility, safety), and the spending function must be in the protocol. The specific boundaries need not be in the protocol if they will be computed from the spending function in the SAP, but the spending function itself—identified by name and parameters—must be in the protocol.

The co-primary decision rule. If two primary endpoints are proposed, the conjunctive or disjunctive decision rule must be in the protocol. "Success will be defined as achievement of significance on both endpoints" or "success will be defined as achievement of significance on either endpoint" must appear in the protocol before enrollment. The protocol may not defer this decision to the SAP.

The pre-specified subgroup analyses. The list of pre-specified subgroup analyses—with each subgroup's defining variable and the confirmatory or exploratory status of the analysis—must be in the protocol. Subgroup analyses added in the SAP but not in the protocol are pre-specified at the SAP level, which is a weaker pre-specification than the protocol level.

The randomization scheme. The randomization type, the stratification factors, and the block structure (variable block size, noted without specifying the specific sizes) must be in the protocol. The actual randomization sequence need not be in the protocol; it is maintained by the randomization system.

The adaptive rule, at the level of type and criteria. For adaptive designs, the type of adaptation (SSR, enrichment, dose selection) and the general criteria that will trigger it must be in the protocol. The specific numerical thresholds—the conditional power threshold

for SSR, the hazard ratio comparison threshold for enrichment—may be in the SAP, but the existence and nature of the adaptation must be disclosed in the protocol.

The SAP: what it must contain and when it must be locked

The statistical analysis plan is the second pre-specification document. Its deadline is softer than the protocol's—it need not be finalized before enrollment, because it can be developed and refined as the trial progresses—but it must be finalized before any unblinded data from the trial are accessible to the analysis team or the sponsor.

"Accessible" means not just formal access to the analysis database, but any channel through which unblinded information could reach the people who write the SAP. A DSMB interim report shared with the sponsor through a governance gap, a summary of interim event rates prepared for operational planning, a conversation in which a DSMB member mentioned the direction of the interim trend to a sponsor employee—any of these constitutes access to unblinded information that could shape the SAP.

From a design integrity perspective, the SAP must contain the following.

The complete hierarchical testing order. Every hypothesis in the hierarchy, in the order it will be tested, with the decision rule for each position—including what happens if the hypothesis at that position fails. The hierarchy is not complete if it specifies the primary and first secondary and defers the rest; it is complete only when every hypothesis the trial will test confirmatorily is named and ordered.

The primary analysis model. The specific regression model, the covariates included, the method for handling missing data under each intercurrent event strategy, and the test statistic that will be reported. The model must be specific enough that two statisticians, running the same software on the same data, would produce the same result.

The sensitivity analyses. The pre-specified sensitivity analyses for the primary endpoint—the analyses that will be conducted to assess the robustness of the primary result to its assumptions. Each sensitivity analysis must specify the assumption being varied, the direction of the variation, and the method.

The detailed missing data plan. The specific imputation method or model-based approach for each category of missing data, the assumptions under each method, and the criteria for when the sensitivity analysis plan for missing data will be triggered.

The adaptive rule details. For adaptive designs: the specific numerical thresholds for the SSR (conditional power threshold, sample size bounds), the specific criteria for the enrichment decision (the comparison threshold, the biomarker assay specification), and the combination test method with the pre-specified critical value. These must be in the SAP before the re-estimation time point or the enrichment interim analysis, not before the final analysis.

The exploratory analysis list. The list of analyses that are exploratory—not part of the confirmatory hierarchy, not controlled for multiplicity—with explicit labeling of each as exploratory. The exploratory list should be exhaustive in identifying which analyses will be reported, so that the distinction between pre-specified exploratory and post-hoc analysis is clear.

The timing of SAP finalization: what compromises it

The most common source of SAP timing compromise is the blinded data review—the review of the pooled, unblinded data that is sometimes conducted before finalization of the SAP to assess data quality. A blinded review that examines only pooled statistics—overall event rates, overall missing data rates, overall protocol deviation rates—without revealing the treatment arm comparison does not compromise the SAP. A blinded review that reveals the distribution of outcomes in ways that allow the direction of the treatment effect to be inferred does compromise it.

The boundary between acceptable and compromising information from a blinded review is not always clear, and the SAP finalization process must specify in advance what information will and will not be examined in the blinded review. Typical safe elements: overall sample size, overall missing data rate, overall event rate (pooled), protocol deviation types and rates. Potentially compromising elements: time-to-event distributions that reveal the treatment arm comparison, outcome distributions that allow inference about which arm is performing better, subgroup sizes that have changed in ways suggesting differential dropout.

When the blinded review is conducted by the analysis team—the same team that will finalize the SAP—the review creates a channel for implicit unblinding even when formal blinding is maintained. The analysis team may not see the treatment arm labels, but if the review reveals that the outcome distribution is concentrated in a specific range, and if the team knows which direction of the primary endpoint would be favorable, the SAP that follows the review may be shaped by this implicit knowledge.

The cleanest solution is to finalize the SAP before the blinded review. When this is not operationally feasible—because the SAP development requires information about data quality that the blinded review provides—the review and the SAP finalization must be separated in time, with the review conducted by a party who is not part of the SAP development team and who communicates only non-inferential data quality information to the SAP team.

When the deadline is missed

The SAP is sometimes finalized after the database is locked, or after the primary analysis has been run in draft. This is not always disclosed clearly; the SAP document may bear a finalization date that is technically before the final analysis report, without disclosing that the SAP was finalized after interim data were examined or after the primary analysis was run in a blind-broken version.

When this occurs—when the SAP finalization was not independent of the data—the consequence for the trial's credibility is specific. The primary analysis is not wrong in the sense that it was incorrectly executed. It may be wrong in the sense that the analysis choices—the model specification, the sensitivity analysis list, the hierarchical order—were shaped by the data rather than by the scientific question. The type I error rate for the primary test is nominally controlled by the pre-specified alpha level, but the actual error rate is inflated by the adaptive choice of analysis method—an inflation that is not reflected in the p-value.

This is the professional consequence of a missed SAP deadline. Not a finding of misconduct, but a finding of insufficient documentary evidence that the analysis was pre-specified. The trial's result is credible only to the extent that the analysis choices can be shown to have been made before the data, and when the SAP finalization timestamp does not support that showing, the credibility is weakened.

The remedy for a missed deadline is not a corrective action after the fact. It is a governance system that prevents the miss—that treats the SAP finalization as a binding event on the project timeline, with the same urgency and the same escalation path as regulatory submission deadlines.

What this section demands before proceeding

Before the adaptive rule lock requirements of Section 8.2 can be addressed, the protocol and SAP content requirements must be mapped to the specific trial being designed.

For each required element—estimand, endpoint definition, ICE strategy, interim plan, co-primary rule, subgroup list, randomization scheme, adaptive rule at protocol level—the responsible party and the finalization deadline must be identified. For the SAP—the hierarchy, the primary analysis model, the sensitivity analyses, the missing data plan, the adaptive rule details—the finalization deadline relative to the data access timeline must be established, and the blinded review boundary must be specified.

This mapping is not bureaucratic planning. It is the design team's recognition that the quality of the trial's evidence depends as much on the governance of these documents as on the scientific quality of the decisions they contain. A scientifically correct estimand in a protocol that was finalized after three patients were enrolled is a scientifically correct estimand that the regulatory agency cannot verify was not modified after enrollment revealed

something about the population. A hierarchical testing order that reflects genuine clinical priorities but was finalized after the SAP development team had seen blinded outcome distributions is a hierarchy that cannot be verified as independent of the data.

The document and the timestamp together establish the pre-specification. Neither alone is sufficient.

References: ICH E9 Statistical Principles for Clinical Trials (1998); ICH E9(R1) Addendum on Estimands (2019); FDA Guidance for Industry, Adaptive Designs for Clinical Trials (2019); EMA Guideline on the Investigation of Subgroups in Confirmatory Clinical Trials (2019); CONSORT 2010 Statement.

8.2 Adaptive Rules and Their Lock Requirements

Why adaptive rules require earlier locks

Section 8.1 established that the SAP must be finalized before unblinded data access. For fixed designs, this means the SAP is finalized before the database lock—a deadline that is challenging in practice but clear in principle.

For adaptive designs, the relevant deadline is earlier and more demanding. The adaptive rule must be finalized before the event it governs—before the re-estimation time point, before the enrichment interim analysis, before the dose selection. And the rule must be finalized in the sense that every party who will be involved in implementing the adaptation knows, before the adaptation event, exactly what data they will see, what rule they will apply, and what decision authority they have. Ambiguity in the rule—resolved at the adaptation event by judgment rather than by pre-specified criteria—is post-hoc discretion, not pre-specified adaptation.

This earlier, more demanding lock requirement is the governance consequence of Chapter 7's insistence that the adaptation rule be pre-specified in sufficient detail to be implemented without discretion. If the rule requires discretion, it is not pre-specified. If it is not pre-specified, the type I error control that the adaptive methodology provides—the combination test, the information fraction recalibration, the operating characteristics established by simulation—is not valid. The method controls the type I error for the pre-specified rule. It does not control the type I error for the rule plus undocumented discretion.

The SSR charter: content and lock timing

The SSR charter is the governance document for the sample size re-estimation. Its analog in fixed designs is the statistical analysis plan; its analog in the DSMB framework is the DSMB charter. Like both, it must be finalized before the event it governs—before the re-estimation time point.

The SSR charter must specify the following with operational precision.

The re-estimation trigger. The specific information fraction or calendar time at which the pooled data will be collected for the re-estimation. Not "approximately 50% of planned

events" but "when the independent statistician receives the data extract dated no earlier than [date criterion] and containing at least [specific number or fraction] of the planned events."

The data to be used. For nuisance-parameter SSR: the specific pooled statistics—pooled variance, pooled event rate, pooled dropout rate—that will be calculated from the blinded data. The data extract specification must exclude any information that would allow arm-specific comparisons to be inferred: no subgroup summaries that reflect the randomization, no time-to-event distributions that reveal arm-specific accumulation patterns.

The re-estimation formula. The specific formula by which the new sample size is calculated from the interim nuisance parameter estimates. The formula is not an algorithm; it is a specific function with specific parameters, pre-specified before the interim data are collected.

The bounds. The minimum and maximum sample size after re-estimation. The minimum prevents the re-estimation from reducing the sample size below the original plan (or below a conservative threshold). The maximum prevents the re-estimation from producing an impractically large trial. Both bounds must be specific numbers, not vague limits like "reasonable" or "operationally feasible."

The independent statistician. The specific individual or team who will perform the re-estimation, their data access rights (limited to the blinded pooled statistics specified above), and their communication protocol (they communicate only the new sample size, not the interim nuisance parameter estimates, to the sponsor or the operations team).

The primary analysis adjustment. For nuisance-parameter SSR: whether the primary analysis will use the standard test or a combination test, and if the standard test, whether any adjustment is needed for the re-estimation. For effect-size SSR: the combination test method, the combination weights (pre-specified as a function of the information fraction at re-estimation), and the critical value for the combination test.

The SSR charter must be finalized before the re-estimation time point, with the same formal review and approval process as the protocol. A charter that exists in draft at the re-estimation time point, or that is being finalized concurrently with the re-estimation, is not a finalized charter—and the re-estimation conducted under a draft charter is a re-estimation conducted without a pre-specified rule.

The enrichment charter: content and lock timing

The enrichment charter—the governance document for the adaptive enrichment decision—must be finalized before enrollment begins, not before the enrichment interim analysis. This earlier deadline reflects the fact that the enrichment rule determines what data will be collected throughout the trial, not just at the interim. The biomarker assay that defines the enriched population must be validated before the first patient is enrolled; the data collection

plan for the biomarker must be in the trial's operations before the first patient's biomarker is assessed.

The enrichment charter must specify the following.

The biomarker assay. The specific assay, the specific threshold for positivity, the specific laboratory that will perform the assay, and the handling of ambiguous results (results near the threshold, results that change between initial and confirmatory testing). The assay specification must be the same assay that will be used in the commercial companion diagnostic, or a documented bridging study must establish the comparability between the trial assay and the commercial assay.

The enrichment interim timing. The specific information fraction or calendar time at which the interim data will be reviewed for the enrichment decision. The timing must be specified in the enrichment charter; a charter that says "enrichment will be considered when the IDMC deems it appropriate" is not a pre-specified rule.

The enrichment threshold. The specific criterion by which the interim biomarker subgroup result is compared to the interim overall result to determine whether enrichment is triggered. The criterion must be expressed as a specific comparison—"the hazard ratio in the biomarker-positive subgroup is at least [specific value] more favorable than the hazard ratio in the overall population, as determined by [specific statistical comparison]"—not as a general statement of intent.

The primary analysis framework. The pre-specified framework for combining the pre- and post-enrichment patient populations in the primary analysis: the combination test weights, the combined estimand specification, and the critical value for the combination test. This must be pre-specified before enrollment because it determines what data collection is needed from the pre-enrichment phase.

The IDMC composition and authority. The members of the independent data monitoring committee that will review the interim data and make the enrichment recommendation. Their data access rights—they see the unblinded interim biomarker subgroup comparison—and the communication protocol: they communicate the enrichment decision (enrich or do not enrich) to the sponsor without revealing the interim treatment effect estimates that drove the decision.

Protocol amendments during the trial

Not all design changes during a trial are adaptive adaptations. Some are responses to unexpected operational challenges, new safety information, or regulatory requests. Protocol amendments—changes to the protocol during enrollment—have their own governance requirements, distinct from the adaptive rule governance.

A protocol amendment that changes a design feature that was pre-specified requires disclosure to the regulatory agency and justification for the change. The critical question is whether

the amendment was driven by operational necessity or by knowledge of the interim data. An amendment that changes the eligibility criteria because an operational barrier was not anticipated at design is different from an amendment that changes the eligibility criteria after an interim data look revealed that enrolled patients were not responding as expected.

The amendment itself does not establish whether it was data-driven. The timing—relative to the interim data access—and the documented rationale establish this. An amendment that occurs immediately after a DSMB interim report has been reviewed by the sponsor, without a documented rationale that is independent of the interim data, will be examined for the possibility that the amendment was driven by knowledge of the interim result. The decision log—discussed in Section 8.3—must record the rationale for every amendment, the timeline of the amendment relative to interim data access, and the basis on which the design team concluded that the amendment was not driven by interim results.

Protocol amendments that are genuinely data-driven—that respond to information generated by the trial itself—are a form of adaptation. If they are pre-specified as possible responses to specific interim findings, they are pre-specified adaptations governed by the adaptive rule framework of this section. If they are not pre-specified—if they are responses to unexpected findings that were not anticipated in the adaptive rule—they are unplanned adaptations, and the regulatory agency will treat them as such: potentially valid, but requiring demonstration that the type I error is still controlled and that the estimand has not shifted in a way that was not acknowledged.

The verification requirement

Pre-specification of the adaptive rule is necessary but not sufficient for the adaptive methodology's type I error control to apply. The rule must also be implemented as specified. Implementation as specified means that the data used were the data specified, the formula applied was the formula specified, the parties involved were the parties specified, and the information that crossed the governance boundary was the information specified.

Verification of implementation requires documentation: the data extract used for the re-estimation, timestamped and content-verified; the calculation log showing the formula applied and the result; the communication record showing what was transmitted to the sponsor; and attestations from the independent statistician and the IDMC members confirming their data access and the independence of their work.

This documentation is not prepared for regulatory inspection as an afterthought. It is prepared concurrently with the adaptation, because the parties involved know at the time of the adaptation what records are required and produce them in real time. A reconstruction of the adaptation documentation from memory, from email records, and from meeting notes is not a verification record. It is evidence that the verification system was not in place.

The verification requirement is the final governance requirement for adaptive designs. Without it, the pre-specified rule and the governance structure are commitments on paper whose

implementation cannot be confirmed. With it, the adaptive design's claim to type I error control under the combination test methodology is supported by the evidence that the rule was applied as specified.

What this section demands before proceeding

Before the decision log of Section 8.3 can be designed, the adaptive rule lock requirements must be mapped to the specific trial's adaptive features.

For each adaptive feature—SSR, enrichment, dose selection, other—the governance document (SSR charter, enrichment charter, dose selection charter) must be identified, its content requirements must be specified, and its finalization deadline must be established relative to the trial timeline. The independent statistician or IDMC that will implement each adaptation must be identified before the trial begins, not recruited when the adaptation event approaches.

The verification records for each adaptation must be specified before the adaptation occurs: what documents will be produced, by whom, at what time, and where they will be stored. The verification system is part of the adaptive design's pre-specification, not a post-adaptation administrative task.

References: FDA Guidance for Industry, Adaptive Designs for Clinical Trials of Drugs and Biologics (2019); EMA Reflection Paper on Methodological Issues in Confirmatory Clinical Trials Planned with an Adaptive Design (2007); Wassmer and Brannath, Group Sequential and Confirmatory Adaptive Designs in Clinical Trials (2016); Proschan et al., Statistical Monitoring of Clinical Trials: A Unified Approach (2006).

8.3 The Decision Log

What the decision log is for

The pre-specification documents—protocol, SAP, DSMB charter, SSR charter—record the decisions that were made before the trial began. The decision log records the decisions made during the trial: the unexpected events that required responses, the governance decisions about protocol deviations, the DSMB recommendations and the sponsor's actions in response, the regulatory interactions that affected the trial's conduct, the protocol amendments and their rationales.

The decision log is the audit trail for the trial's conduct. It answers the question that every regulatory reviewer, every DSMB member, and every independent statistician will eventually ask: when a decision was made, what information was available, who made it, and why? Without the decision log, these questions can only be answered from memory, from email chains, and from meeting records that may be incomplete or selective. With it, the decisions are contemporaneously documented in a single record that is designed to answer exactly these questions.

The decision log is not a medical record. It is not a regulatory submission document. It is a governance record—the management record of the decisions that shaped the trial's conduct between the moment the first patient was enrolled and the moment the database was locked. Its audience is anyone who later needs to understand whether the trial was conducted as designed and whether deviations from the design were handled according to pre-specified governance rules.

What the decision log must contain

The decision log has four categories of entry, each corresponding to a different type of decision made during the trial.

Protocol amendment entries. Every protocol amendment—every change to the pre-specified design—must be entered in the decision log at the time the amendment is implemented. The entry must record: what was changed, what the rationale for the change was, whether the change was driven by operational necessity or by information generated by the trial (and if the latter, what information), whether the change was reviewed by the regulatory agency before implementation, and whether the change required re-consent of

enrolled patients. The entry must be specific enough that a reviewer can determine whether the amendment was pre-planned or responsive to trial data.

DSMB interaction entries. After every DSMB interim meeting—open session, closed session, or executive session—the decision log must record: what the DSMB reviewed (without revealing the arm-specific interim data), what recommendation the DSMB made, what information the recommendation was communicated to the sponsor, and what action the sponsor took in response. When the sponsor's action differs from the DSMB's recommendation—when the sponsor continues a trial against a non-binding futility recommendation, for example—the entry must record the sponsor's documented rationale for the divergence.

Regulatory interaction entries. Every interaction with the regulatory agency that affects the trial's conduct—a Type B or Type C meeting response, a clinical hold and its resolution, a complete response letter's requests—must be entered in the decision log. The entry must record the agency's specific request or guidance, the sponsor's response, and the timeline of implementation. Regulatory interactions that are not recorded in the decision log cannot be traced when the regulatory reviewer asks, during the submission review, when a specific change to the trial's conduct was made and why.

Operational deviation entries. Protocol deviations—departures from the pre-specified protocol that were not amendments—must be entered in the decision log when they are identified. The entry must classify the deviation (major or minor, by the pre-specified classification criteria), record the action taken in response, and note whether the deviation affects the primary analysis. Major deviations—those that affect the primary endpoint, the eligibility of the patient, or the blinding—require a more detailed entry that includes the assessment of the deviation's potential impact on the primary analysis and the pre-specified handling of deviating patients.

When entries must be made

The timing of decision log entries is as important as their content. An entry that is made days after the decision, from reconstruction, is not the same as an entry made contemporaneously. The contemporaneous entry reflects what was known at the time of the decision; the reconstructed entry reflects what was known afterward, including the decision's consequences.

The discipline for decision log entries is: the entry is made within a specified time window after the decision—24 to 72 hours for most decisions, immediately for DSMB recommendations and regulatory interactions. The time window must be specified in the decision log's operating procedures, which are part of the trial's governance documents. A decision log with no operating procedures—no specification of who enters what, by when, in what format—is a document that will be completed retrospectively, and its retrospective nature will be apparent to the reviewer who examines the entry timestamps.

For DSMB interactions, the entry timing is particularly critical. The DSMB closed session recommendation—the arm-specific recommendation that the DSMB makes after reviewing the unblinded interim data—must be recorded in the decision log immediately after the DSMB meeting, before the sponsor has time to consult internal advisors about the recommendation and respond. A recommendation entry made days after the meeting, after the sponsor has formulated its response, is an entry that reflects the sponsor's considered response rather than the DSMB's immediate recommendation.

Who maintains the decision log

The decision log must be maintained by a person or team that is independent of the sponsor's analysis function—the team that will eventually analyze the primary endpoint data. This independence requirement mirrors the independence requirements established throughout this book for interim data access: the people who maintain the decision log must not be in a position to use the log to shape the primary analysis.

In practice, the decision log is most commonly maintained by the clinical operations team, with entries made by the study manager or the project manager for each category of decision. The clinical operations team typically does not have access to unblinded interim efficacy data—they see overall enrollment and safety data, but not the arm-specific primary endpoint comparison—so their independence from the primary analysis is maintained.

The analysis team's role in the decision log is limited to reviewing entries that affect the primary analysis—major protocol deviations, endpoint adjudication clarifications, adaptive rule implementations—and confirming that the entries accurately reflect the decisions made. The analysis team does not write the entries; they review them for completeness and accuracy in matters that bear on the analysis.

The DSMB's interaction with the decision log is also limited. The DSMB meeting minutes—which may contain arm-specific interim data—are not part of the public decision log. The entry in the public decision log records only the DSMB's recommendation and the sponsor's response, not the data on which the recommendation was based. The DSMB's closed session minutes are maintained separately, available to the regulatory agency under confidential review but not part of the standard decision log.

The decision log and the clinical study report

At the end of the trial, the decision log becomes the primary source document for the clinical study report's account of protocol deviations, amendments, and governance events. The clinical study report—the regulatory submission document that describes the trial's

conduct—must account for every major deviation, every amendment, and every DSMB recommendation. Without the decision log, this account is reconstructed from disparate sources; with it, the account is extracted from a single contemporaneous record.

The decision log also provides the evidence that the primary analysis was conducted according to the pre-specified plan. When the clinical study report states that the primary analysis was pre-specified in the SAP and conducted without modification, the decision log supports this statement by showing that no entries record a change to the analysis plan after the SAP was finalized. When the clinical study report states that the sponsor followed the DSMB's recommendations at each interim, the decision log supports this by showing the recommendation and the sponsor's response for each interim.

A clinical study report that cannot be supported by a contemporaneous decision log is a report that rests on reconstruction. The regulatory reviewer who examines the submission will ask for the source documents supporting the report's account of the trial's conduct. If the source documents are the decision log, the support is direct. If the source documents are email chains, meeting minutes from various committees, and declarations from the study team, the support is indirect and the reviewer will need to reconcile disparate sources to assess whether the trial was conducted as described.

The decision log as a governance discipline

The discipline of maintaining the decision log creates a specific kind of accountability that is otherwise absent from the trial governance system. When every decision is entered within 72 hours, by a specified party, using a specified format, the people making the decisions know that their decisions will be recorded—promptly, in the terms defined in the operating procedures, without the opportunity to frame the decision in a more favorable light after its consequences are known.

This accountability structure changes the decisions made during the trial. Decisions that would be embarrassing in the decision log—decisions that reflect commercial pressure on the DSMB, decisions to continue a trial past a futility boundary without a documented rationale, decisions to change a protocol element in response to an interim trend that was not supposed to reach the sponsor—are decisions that the decision log's accountability structure discourages. The decision log does not prevent these decisions, but it makes them more visible and more consequential.

The goal of this visibility is not punishment. It is honesty: the trial's conduct should be defensible from the inside—every decision traceable to a documented basis, every deviation handled according to pre-specified rules—not just explainable from the outside after the fact. The decision log is the mechanism by which the inside and the outside of the trial's conduct are made the same.

References: ICH E6(R2) Good Clinical Practice (2016); FDA Guidance for Industry, Oversight of Clinical Investigations—A Risk-Based Approach to Monitoring (2013); Herson, Data and Safety Monitoring Committees in Clinical Trials (2009); Ellenberg, Fleming, and DeMets, Data Monitoring Committees in Clinical Trials: A Practical Perspective (2002).

8.4 Design Review and the Independent Challenge

Why independent review is a governance requirement

The design decisions documented in Chapters 1 through 7 are made by people who are close to the trial—the clinical scientists who understand the disease, the statisticians who understand the analysis, the clinical operations team who understand the logistics. These are the right people to make the decisions. They are also people who have invested in the design, who believe in the treatment, and who are subject to the optimism biases and the organizational pressures that Chapter 3 identified as sources of effect size overestimation and Chapter 4 identified as sources of governance failure.

Independent review is not a challenge to these people's competence. It is a structural protection against the specific vulnerabilities that competent, motivated people exhibit when they are close to their own work. The same human tendencies that make an expert design team effective—deep familiarity with the science, strong commitment to the development hypothesis, pattern recognition that integrates experience into judgment—make them poor reviewers of their own design. They know what the design is supposed to do; they are less able to see what it actually does when their assumptions are wrong.

An independent reviewer brings the complementary virtue: distance. They do not know what the design is supposed to do; they know only what it says. This distance allows them to ask questions that the design team stopped asking because they seemed obvious: Does the estimand actually correspond to the endpoint definition? Does the power calculation use the adjusted alpha after interim analyses? Does the hierarchy order reflect clinical importance or statistical convenience? Is the DSMB charter specific enough to govern the decisions the DSMB will be asked to make?

These questions may have good answers. The independent reviewer's job is not to find problems but to verify that the answers are defensible—that the design as documented can withstand the scrutiny it will face from regulatory reviewers, from DSMB members who did not participate in the design, and from clinical audiences who will interpret the trial's result.

What independent review must examine

The independent review of a clinical trial design is not a single event. It is a staged process that mirrors the document lock sequence of Section 8.1: a review of the protocol before finalization, a review of the SAP before finalization, a review of the DSMB and SSR charters before the first interim, and a review of the decision log before the database lock.

At each stage, the reviewer examines the document against the design principles established in the prior chapters.

Protocol review. The reviewer examines whether the estimand is specified with sufficient precision to be operationalized in the SAP without additional decisions. Whether the endpoint definition is specific enough to be applied consistently across sites and assessors. Whether the intercurrent event strategy is consistent with the estimand's scientific intent. Whether the co-primary decision rule, if present, specifies the conjunctive or disjunctive criterion clearly. Whether the interim analysis plan specifies the spending function. Whether the pre-specified subgroup list correctly labels each analysis as confirmatory or exploratory.

SAP review. The reviewer examines whether the hierarchical testing order reflects clinical importance rather than statistical convenience. Whether the primary analysis model is fully specified—no deferred decisions about covariate selection or missing data handling. Whether the sensitivity analyses address the assumptions most likely to be challenged. Whether the adaptive rule details—the SSR formula, the enrichment threshold, the combination test method—are specific enough to be implemented without discretion.

Charter review. The reviewer examines whether the DSMB charter specifies the decision hierarchy for scenarios when multiple stopping signals are present simultaneously. Whether the SSR charter specifies the data to be used, the formula, and the communication protocol with sufficient precision. Whether both charters specify the documentation requirements that will enable verification of implementation.

Decision log review. The reviewer examines whether the decision log is complete—whether every major protocol amendment, every DSMB recommendation, every regulatory interaction is recorded. Whether the entries were made within the specified time windows—whether the timestamps are consistent with contemporaneous recording rather than retrospective reconstruction. Whether the entries for DSMB interactions record the recommendation and the sponsor's response, without revealing the arm-specific data that drove the recommendation.

Who should conduct independent review

The independence requirement has two dimensions: organizational and technical.

Organizational independence means the reviewer has no reporting relationship to the people who made the design decisions, no financial stake in the trial's outcome, and no involvement in the development of the documents being reviewed. A statistician from a different division

of the same company, reviewing a design produced by the clinical development team, is organizationally independent of the specific design team but not independent of the company's interests. A statistician from an academic institution with no current consulting relationship to the sponsor is more independent. A statistician from a competing company is fully independent but typically not available for confidentiality reasons.

The practical standard for most trials is a senior statistician from a different function within the same organization—a regulatory statistics team reviewing a clinical development design, or an external consultant with no current engagement on this program—for routine design review, and an external academic statistician for trials whose design complexity or importance warrants a higher standard of independence.

Technical competence means the reviewer has sufficient expertise in the specific design elements being reviewed. A reviewer who is expert in frequentist hypothesis testing but unfamiliar with adaptive design methodology cannot meaningfully review the combination test specification for an adaptive enrichment design. A reviewer who is expert in oncology trial design but unfamiliar with the estimand framework cannot meaningfully review the intercurrent event strategy for a cardiovascular outcome trial. The reviewer's expertise must be matched to the design's complexity.

For complex designs—adaptive designs, NI designs with multi-regional populations, seamless phase II/III designs—the independent review may require a team of reviewers rather than a single reviewer: a statistician for the analysis plan, a clinical methodologist for the estimand and endpoint specification, and a governance expert for the DSMB and SSR charters.

The design review as a discipline, not an event

The independent review of a clinical trial design is most valuable when it is structured as a discipline—a regular, staged process that is embedded in the design timeline—rather than as a one-time event at the end of the design process.

When review is a one-time event, it occurs when the design is largely complete, the team has significant investment in the current approach, and the cost of substantive changes is high. The reviewer identifies problems; the team resists changing them because the cost of change is real and the benefit of the change—avoiding a regulatory challenge or a post-trial credibility problem—is hypothetical. The review becomes a negotiation rather than an examination.

When review is a discipline—when there are staged review points at the estimand specification stage, at the power calculation stage, at the interim analysis plan stage, and at the final document finalization stage—the design is reviewed when the cost of change is still manageable. The estimand is reviewed before the power calculation is built on it. The power calculation is reviewed before the interim analysis plan is built on it. The interim analysis

plan is reviewed before the governance charters are built on it. Problems are identified and addressed before they compound.

The staged review structure requires the design team to articulate the design decisions at each stage—to make the decisions visible before they are embedded in subsequent decisions—and this articulation itself has value. A team that cannot explain, at the estimand review stage, why the chosen intercurrent event strategy corresponds to the scientific question has not fully thought through the estimand. The review forces the articulation, and the articulation forces the thinking.

After the design review: the design summary

At the completion of the design process—after the protocol has been finalized, the SAP has been locked, and the charters have been approved—a design summary document should be produced. The design summary is not a regulatory submission document; it is an internal governance record.

The design summary records, in one place, the key decisions made at each design stage: the estimand and its scientific justification, the effect measure and the rationale for its selection, the assumed effect size and the clinical team member who owns it, the power level and the justification for accepting the implied type II error, the interim analysis plan and its operating characteristics, the bias protection design and any acknowledged limitations, the claim structure and the hierarchical order with its clinical rationale, and for adaptive designs, the adaptive rules and their governance structure.

The design summary is the record of the design review's conclusions—the documented basis for the independent reviewer's determination that the design is defensible. It serves two purposes. First, it provides a reference document for the DSMB, the clinical operations team, and the regulatory agency that can orient each party to the trial's design without requiring them to read the full protocol and SAP. Second, it provides the basis for the clinical study report's account of the design rationale—the post-trial document that explains why the trial was designed as it was and how the primary analysis reflects the pre-specified design.

A trial that produces a design summary before enrollment begins is a trial whose design process was governed with sufficient discipline that the summary can be written. A trial that cannot produce a design summary—because the design decisions were made piecemeal, without documentation, by parties who are no longer available or who disagree about what was decided—is a trial whose governance system failed before the first patient was enrolled.

References: ICH E8(R1) General Considerations for Clinical Studies (2021); FDA Guidance for Industry, Establishing Effectiveness and Safety of Drugs Intended to Treat Serious Conditions (2023); Chow and Chang, Adaptive Design Methods in Clinical Trials (2011); Pocock, Clinical Trials: A Practical Approach (1983).

8.5 Closing: The Defensible Design

What this chapter asked

Chapter 8 asked one question: what must be locked, and when?

It asked that question in four sections, each addressing a different component of the governance system that makes pre-specification operational.

Section 8.1 established the content and timing requirements for the protocol and the SAP—the two primary pre-specification documents. The protocol is finalized before the first patient is enrolled. The SAP is finalized before unblinded data access. Each document must contain specific elements with sufficient detail to be binding. The absence of either specificity or timing independence from the data is a governance failure that weakens the trial's credibility without necessarily invalidating the primary result.

Section 8.2 addressed the specific lock requirements for adaptive designs—the earlier and more demanding pre-specification requirements that come from the need to govern adaptation events with pre-specified rules, not post-hoc discretion. The SSR charter and the enrichment charter must be finalized before the events they govern, with content requirements that mirror the DSMB charter's operational specificity.

Section 8.3 examined the decision log as the audit trail for the trial's conduct during enrollment. The decision log records every governance decision—amendment, DSMB interaction, regulatory interaction, operational deviation—contemporaneously, by specified parties, within specified time windows. Its completeness and its timing determine whether the trial's conduct can be verified from the outside.

Section 8.4 examined independent review as the structural governance requirement that protects against the optimism and insider blindness of design teams. Staged review—at the estimand stage, the power stage, the interim plan stage, and the document finalization stage—is more valuable than a single terminal review, because it identifies problems before they compound into subsequent design decisions.

The characteristic mistakes of this chapter

Three failures define the governance breakdowns that Chapter 8 is designed to prevent.

The SAP finalized after the trend was visible. The protocol was finalized on time. The SAP development began eighteen months before the planned primary analysis, with good intentions. The blinded data review conducted six months before the database lock showed overall event rates that, given the known randomization ratio and the trial's expected event rates, allowed the analysis team to infer the approximate direction of the primary endpoint comparison. The SAP was finalized two weeks after the blinded data review. The hierarchical testing order in the SAP was different from the order that had been discussed at the design stage—one secondary endpoint had moved up in the hierarchy, and it was the secondary endpoint that the blinded data review's implied direction suggested would be most likely to succeed. The finalization timestamp was before the database lock. The regulatory reviewer's question—was the SAP finalized before or after the analysis team had access to inferential information about the treatment arm comparison?—did not have a clean answer.

The decision log completed retrospectively. The trial completed enrollment, completed follow-up, and locked the database. At the time of clinical study report preparation, the medical writing team requested the decision log. No decision log had been maintained during the trial. The study manager and the project director spent three weeks reconstructing the trial's major decisions from email archives, DSMB meeting summaries, and regulatory correspondence. The reconstructed log was comprehensive by the standards of memory and available records, but it was a reconstruction—not a contemporaneous record. The regulatory inspection that followed the submission asked for the source documents for each decision log entry. The response—"the entries were reconstructed from available records"—was accurate and damaging.

The design review that was a formality. The trial's design was reviewed by a senior statistician from the regulatory affairs group before the SAP was finalized. The review lasted four hours. The reviewer had been briefed by the design team on the trial's rationale and had received the documents three days before the review. The review meeting produced a set of minor comments on the sensitivity analysis specifications and confirmed that the major design elements were satisfactory. The DSMB charter was not reviewed because it had not been finalized in time for the review meeting; it was finalized two weeks later without review. The combination test method for the adaptive SSR was not reviewed because the reviewer's expertise was in fixed designs. The trial proceeded with a combination test specification that had a parameter error—the combination weights were specified for a balanced two-stage design but the SSR rule could produce an unbalanced allocation between the two stages. The error was discovered during the primary analysis. The correction required a post-hoc analytical justification that the regulatory agency accepted but flagged.

Each of these failures has the same origin: the governance requirement was understood as a compliance obligation—something that must be done to satisfy the protocol, the regulatory guidance, or the inspection checklist—rather than as a design requirement whose purpose is

to make the evidence credible. When governance is compliance, the minimum is achieved on paper. When governance is design discipline, the minimum is not the target.

What cannot be recovered

A SAP finalized after inferential information from the data was available cannot be verified as independent of the data. The analysis team's sincere belief that the SAP choices were not influenced by the data is not verifiable—not by the team, and not by the regulatory reviewer. The uncertainty is irreducible.

A decision log that was not maintained during the trial cannot be reconstructed with the fidelity of a contemporaneous record. The reconstruction is an approximation of what was decided, colored by the knowledge of what subsequently happened and by the selective availability of the records from which it is reconstructed. The reconstruction may be accurate. It is not verifiable as accurate with the same confidence as a contemporaneous record.

An independent review that did not examine the adaptive rule details—because the reviewer lacked the expertise, or because the charter was not available—cannot be supplemented by a post-hoc review after the adaptive design has been implemented and the data have been analyzed. The post-hoc review is valuable for planning future trials, but it does not establish that the adaptive design was correct before the data were examined.

These irrecoverabilities define the stakes of Chapter 8. The design decisions of Chapters 1 through 7 can sometimes be corrected after the trial begins—with protocol amendments, with DSMB guidance, with regulatory consultation. The governance failures of Chapter 8 cannot. When the SAP was finalized after the data were seen, when the decision log was not maintained, when the independent review did not cover the critical design elements—the evidence basis for the trial's claims is permanently weakened, regardless of how strong the underlying science is.

The four questions, answered

This book began with four questions that it promised to ask about every design decision. The governance framework of Chapter 8 closes by applying those questions to the governance process itself.

What evidence are we defining? The design documents—the protocol, the SAP, the charters—define the evidence by specifying what will be measured, in whom, under what conditions, with what analysis. The governance system ensures that this definition was made before the data, not from the data.

What risk are we committing to? The governance system commits the trial to the risk of having made wrong decisions before the data were seen—and of not being able to correct those decisions by revising the documents after the data reveal the error. This is the pre-specification premium: the cost of committing before knowing, paid in the form of decisions that cannot be modified once the data are observed.

How do we prevent failure? The governance system prevents the specific failure of post-hoc rationalization—the construction of design documents that appear pre-specified but were shaped by the data. Staged review, document lock timing, and the decision log together make post-hoc rationalization more difficult and more detectable.

What can we finally say? The trial can say what its pre-specified analysis shows, with the confidence level that the pre-specified alpha controls, for the population defined by the pre-specified estimand, according to the claim structure defined by the pre-specified hierarchy. These bounds on what can be said are not limitations; they are the definition of what the evidence supports. A claim bounded by pre-specification is a claim with a known and controlled error rate. A claim that exceeds pre-specification has an unknown error rate and is, in the full sense of the word, speculation.

The defensible design

A defensible design is not a perfect design. No trial is designed with perfect information; the assumptions about effect size, nuisance parameters, event rates, and patient behavior are all predictions, and predictions are wrong at rates that should be humbling. The defensible design acknowledges the uncertainty in its assumptions, specifies the consequences of those assumptions being wrong, and commits to a governance system that makes the design decisions verifiable before the data reveal whether the decisions were correct.

Defensibility is not primarily about the regulatory agency. It is about the clinical community, the patients who enrolled in the trial and those who were not enrolled, and the scientific record that will outlast any individual trial team or development program. A trial that demonstrates a treatment works—if it is defensible—contributes an increment of knowledge that is reliable, replicable, and interpretable by others without access to the trial team's memory. A trial that demonstrates a treatment works—if it is not defensible—contributes a result that cannot be fully evaluated and that may mislead the clinical community in ways that persist long after the trial team has moved on.

Defensibility is built into the design before the first patient is enrolled. It is maintained through the governance system during the trial. And it is verified through the decision log and the design summary at the end. The book has been about building that defensibility—not as a compliance exercise, but as the condition under which clinical evidence can be trusted.

Chapter 8 risk summary

The decision this chapter owns: are the design documents that govern this trial finalized at the right time, by the right parties, with sufficient specificity, and independently reviewed—so that the trial's primary result can be verified as the product of the pre-specified design?

The most common mistake: treating the SAP as an analysis document rather than a pre-specification document—finalizing it under the pressure of the submission timeline rather than under the governance requirement of independence from the data. The SAP is not a description of how the analysis was done. It is a commitment to how the analysis will be done, made before the data make it possible to choose the analysis that favors the result.

The professional-level risk: a primary result that is statistically significant, clinically plausible, and reproducible in biological terms—but that cannot be fully defended as the product of a pre-specified design because the governance record does not establish that the key documents were finalized before the data were seen. This is not a result that will be rejected. It is a result that will be accepted with reservations—labeled with qualifications, discussed in regulatory review meetings, questioned in subsequent trials that are powered to test the result under tighter governance constraints. The professional risk is not the failure of the program. It is the permanent uncertainty that attaches to a result that could have been unambiguous, if the governance system had been in place.

Appendix

Design Review Hearing

Questions That Must Be Answered Before Enrollment

A clinical trial does not begin when the first subject signs consent. It begins when the organization declares that its design can withstand scrutiny.

Before enrollment, the following questions must be answered — not casually, not implicitly, but in a form that could be defended before regulators, investors, and patients.

I. Definition of the Claim

What precisely is the treatment effect this trial intends to estimate?

In which population does that effect apply? Relative to what comparator? Over what time horizon? Under what handling of discontinuation, switching, rescue therapy, or death?

If this quantity cannot be stated without referencing the statistical model, it has not yet been defined.

If different members of the design team describe the objective differently, the design is not yet stable.

II. Ownership of Assumptions

What effect size is being assumed — and who takes responsibility for that assumption?

What event rate or variance assumption anchors the sample size? What happens if those assumptions are materially wrong?

Is dropout being treated as a logistical inconvenience to inflate for, or as a change in the scientific question?

In a non-inferiority trial, who determined the margin? On what historical and clinical basis? Is assay sensitivity defensible under current practice? Does the constancy assumption plausibly hold?

If these answers depend on optimism rather than justification, the risk has not been acknowledged.

III. Allocation of Error and Early Stopping

Is an interim analysis planned? For what purpose?

How is the probability of false positive redistributed across analyses? What are the operating characteristics under plausible true scenarios?

When someone states that "the stopping rate is too high," which probability are they referring to — stopping under null, stopping under benefit, or stopping under harm?

Who will see interim data? Who has authority to recommend stopping? Is that authority documented?

If governance is unclear, the design is unstable.

IV. Protection Against Bias

How does the randomization structure protect against selection bias?

Is allocation concealment operationally secured?

Does the design rely on balance in ways that cannot be guaranteed?

Has the randomization plan been independently validated to a level commensurate with its risk?

If the trial is open-label, what bias pathways have been anticipated, and how are they mitigated?

If these protections are assumed rather than demonstrated, interpretation will later fracture.

V. Discipline of Claims

What constitutes trial success?

Is success defined by a single estimand, or multiple?

If multiple objectives exist, what logical rule governs success?

How is multiplicity controlled across endpoints, timepoints, and subgroups?

Which findings will support labeling claims? Which are exploratory?

If a subgroup contradicts the overall result, what principle governs interpretation?

If these boundaries are not defined in advance, interpretation will drift.

VI. Adaptation and Structural Change

Will any aspect of the design adapt after initiation?

Could adaptation alter the estimand?

How is type I error preserved under that adaptation?

Does enrichment redefine the population midstream?

If the answer to these questions is deferred until after interim data are seen, adaptation becomes revision rather than design.

VII. Pre-Specification and Accountability

Which elements of the design are fully locked prior to enrollment?

Which may adapt under rule-based procedures?

Is there a documented record of design decisions and their justifications?

Could an external reviewer reconstruct the logic of the design without oral explanation?

If not, the design exists in conversation, not in structure.

Final Condition

A trial design is ready for enrollment only when:

The treatment effect is precisely defined.

The assumptions are owned.

The risks are acknowledged.

The governance is explicit.

The claims are bounded.

Enrollment converts assumptions into exposure. After that point, ambiguity is no longer theoretical.

If these questions cannot be answered clearly and consistently, the trial is not yet ready to begin.

Design Scenarios

Where the Principles Meet the Room

The eight chapters of this book have addressed design decisions in terms of their consequences: who owns them, what happens when they are wrong, and what governance structures make them defensible. This appendix addresses the same decisions from a different angle — the angle of the statistician in the room when the decision is being made.

Each scenario describes a situation that recurs in clinical trial design. The situation is not hypothetical; it is the kind of meeting, the kind of request, the kind of disagreement that happens before a trial begins. The scenario does not tell you what to say. It maps the request to the design question underneath it — the question that, if answered correctly, makes the design decision defensible regardless of who is in the room or what they want.

The scenarios are organized by the point in the design process at which they arise. They are not exhaustive. They are representative of the places where the principles of this book are most often tested by practice.

Scenario 1: "We Want an Interim Analysis"

The request arrives at the first design meeting, usually from the sponsor's development team or the clinical lead. The justification is often stated simply: "We want to be able to stop early if it works." Sometimes the word "budget" is in the room. Sometimes it is not, but it is present anyway.

What is actually being requested

The request for an interim analysis is rarely one request. It is usually three overlapping requests that the requester has not distinguished:

The first is **efficacy protection**: if the treatment is clearly working, stop enrolling patients to the control arm and accelerate submission. This is the request most often stated.

The second is **resource protection**: if the treatment is clearly not working, stop spending money on a futile program. This request is present in almost every development context but

is sometimes not stated because acknowledging it requires acknowledging the possibility of failure.

The third is **information access**: someone in the organization wants to know how the trial is going before it is over, for reasons that may be scientific, commercial, or political. This request is the most dangerous to the trial's integrity and is almost never stated as such.

Each of these requests leads to a different design. Efficacy protection requires an upper stopping boundary calibrated to the expected effect and the information fraction at which the treatment's benefit would be large enough to be clinically unambiguous. Resource protection requires a futility boundary calibrated to the conditional power under pessimistic scenarios. Information access — if it is the real request — requires a governance discussion about who sees what, not a stopping boundary discussion.

The design conversation cannot proceed until the request is separated into its components. The statistician's first task is not to design the interim analysis. It is to ask, and get a documented answer to: what decision are we trying to make with this interim, and who will make it?

The budget conversation

When the interim analysis request is budget-driven — when the real question is "can we stop the program before it consumes the full development budget if the treatment is not working" — the design implications are specific.

A budget-protective interim analysis is a futility analysis. Its statistical logic is conditional power: given what we have seen so far, what is the probability of achieving the primary objective if we continue to the end? If the conditional power is below a pre-specified threshold, stopping is warranted.

The budget conversation has a design consequence that is not always made explicit: the threshold for stopping must be set before the trial begins, not recalibrated when the interim data reveal a disappointing trend. A conditional power threshold of 20% means the trial will stop if the probability of success, given the interim data, is below 20%. A threshold set at 30% or 40% stops the trial sooner and saves more resources, but stops some trials that would have succeeded with continued enrollment. The threshold is a risk allocation decision — between the patients who might have benefited from a continued trial and the resources that will not be spent if it stops early.

This decision belongs to the sponsor's clinical and development leadership, not to the statistician. The statistician's role is to make the decision visible — to show what the stopping probability is under various true effect scenarios for each candidate threshold — and to ensure that the chosen threshold is documented before the trial begins. Chapter 4 addresses the operating characteristics that make this calculation possible. The budget conversation is the context in which those operating characteristics matter most directly.

The question the sponsor is not asking

When a sponsor requests an interim analysis, they are usually not asking: what does an interim analysis do to the final result?

Two consequences are reliably underappreciated. First, stopping early for efficacy produces an overestimate of the treatment effect. The observed effect at early stopping is expected to be larger than the true effect, because trials that stop early have, by definition, observed an interim result above a high threshold, and extreme observations are on average above the true value. The overestimate enters the label. It shapes clinical expectations. Subsequent trials powered on the overestimated effect tend to fail at higher rates.

Second, every interim efficacy analysis consumes alpha. After the interim, the final analysis is conducted at a more stringent threshold than the nominal alpha. The power reported in the sample size report — which is usually calculated at the nominal alpha — is not the power of the trial as designed. The power of the trial as designed is lower, and the sample size should have been calculated at the adjusted threshold. When this is not done, the trial is less powerful than the team believes it to be.

Both consequences should be in the room when the interim analysis request is made. They are rarely there unless the statistician puts them there.

Scenario 2: "The Stopping Rate Is Too High"

This objection arises most often from the clinical lead or the medical monitor, after the statistician has presented the interim analysis design. The stopping rate — the probability of stopping early — looks alarming. "We'll stop half the time before we even get to full enrollment. That can't be right."

Three probabilities, one phrase

When a clinician says "the stopping rate is too high," they are almost always referring to one of three distinct probabilities, and they do not know which one. The conversation cannot be resolved until the statistician identifies which probability is causing the concern.

The probability of stopping under the null. This is the probability of stopping for efficacy when the treatment has no effect — the false positive contribution from the interim analysis. For a well-designed O'Brien-Fleming boundary, this probability is very small at early information fractions: stopping at 25% information fraction requires a test statistic corresponding to a p-value well below 0.001. If the clinician has been told that the trial might stop at the first interim, and they are imagining a false positive, the actual probability is much smaller than their intuition suggests.

The probability of stopping under the alternative. This is the probability of stopping early when the treatment works as assumed — the efficiency gain of the interim analysis. For a compelling treatment effect, this probability can be substantial: a trial designed with 80% power at the final analysis, using an O'Brien-Fleming boundary, might have a 30-40% probability of stopping at the first interim if the treatment effect is exactly as assumed. This is the probability the clinician is most likely objecting to. Their concern: if we stop at 30-40% of planned events, is the result reliable? This concern is legitimate and connects to the overestimation discussion in Scenario 1.

The probability of stopping under futility. This is the probability that the futility boundary is crossed when the treatment is not working. For a well-designed futility rule, this probability should be high under the null — the whole point of the futility analysis is to stop efficiently when the treatment is failing. A clinician who objects that "the futility stopping rate is too high" is objecting to the trial's efficiency, which means they are not willing to accept early evidence of failure. This objection is not statistical. It is a belief that the treatment will work despite an early negative signal, and it should be addressed as a clinical judgment, not a statistical problem.

How to have this conversation

The most efficient resolution is a table. Not a formula, not a p-value, not a description of the alpha-spending function — a table that shows, for each information fraction at which an interim is planned, the probability of stopping under the null hypothesis, the probability of stopping when the treatment effect is exactly as assumed, and the probability of stopping when the treatment effect is half of what was assumed.

This table makes all three probabilities visible simultaneously. The clinician can look at the row for the first interim and see: the probability of stopping for a false positive here is 0.3%, the probability of stopping because the treatment works is 28%, and the probability of stopping because the treatment is not working is 55%. These numbers, together, tell the clinician what the interim analysis actually does — not as an abstraction but as a set of probabilities under scenarios they can evaluate clinically.

When the clinician sees all three probabilities, their objection usually becomes more specific. If they are concerned about the false positive probability, the design is already conservative and the concern dissolves. If they are concerned about the probability of stopping when the treatment works, the conversation turns to the overestimation bias and whether early stopping is worth the tradeoff. If they are concerned about the futility stopping probability, the conversation turns to how much evidence of failure they need before they would accept stopping.

Each of these is a real design decision. None of them can be resolved by adjusting "the stopping rate" as an undifferentiated quantity.

Scenario 3: Continuous, Binary, or Time-to-Event?

The endpoint discussion is usually scheduled early in the design process and rarely resolved in the first meeting. The clinical team comes with a preferred outcome — often the one that has been used in prior trials in the indication, or the one that is most familiar. The statistician has a view about efficiency, regulatory precedent, and what the analysis will require. The conversation needs to produce a decision that is both scientifically defensible and operationally achievable.

What the statistician should not do

The statistician should not arrive at this meeting with a recommendation and defend it. The choice of endpoint type is not a statistical decision. It is a clinical decision that has statistical consequences, and the statistician's role is to make those consequences visible, not to make the choice.

Arriving with a recommendation — "we should use a time-to-event endpoint because it's more efficient" — puts the statistician in the position of defending a position rather than facilitating a decision. The clinical team will push back, the conversation will become adversarial, and the decision will be made on the basis of who argued more persuasively rather than on the basis of what the trial is designed to show.

The three questions the statistician must answer

Before the meeting, the statistician should be able to answer three questions for each candidate endpoint type, and those answers should be the material the meeting uses to make the decision.

What does this endpoint actually measure, and does it correspond to the estimand? A continuous endpoint measures the average magnitude of the treatment's effect. A binary endpoint measures the proportion of patients who cross a clinically meaningful threshold. A time-to-event endpoint measures the speed at which the outcome occurs, under the proportional hazards assumption. These are different quantities, and they answer different scientific questions. The first task is to identify which quantity the estimand specifies — not which endpoint type is most familiar or most efficient.

What happens to power if the assumptions are wrong? For continuous endpoints, the critical assumption is the variance. For binary endpoints, the critical assumption is the baseline event rate. For time-to-event endpoints, the critical assumptions are the event rate, the proportional hazards assumption, and the follow-up duration. Each of these assumptions has a characteristic failure mode. The statistician should show the power curve across the plausible range of assumption violations for each candidate endpoint type, so the clinical team can see which endpoint is most fragile to the most plausible errors.

What does the regulatory agency expect? In most indications, there is a precedent for the primary endpoint type, and departing from that precedent requires a justification that

will be scrutinized. The statistician should know the regulatory precedent and present it as a constraint, not a recommendation. If the clinical team wants to depart from precedent, the departure needs a documented rationale and ideally a pre-submission regulatory interaction.

The dropout complication

The choice between continuous and binary endpoints has a specific interaction with dropout that is not always addressed in the endpoint discussion.

For continuous endpoints assessed at a fixed time point, dropout before the assessment time produces missing data. The missing data must be handled according to the intercurrent event strategy — under a treatment policy estimand, outcome data from dropouts should be collected; under a hypothetical estimand, the analysis must model what would have happened if the patient had not dropped out. The handling is complex, and the complexity is carried by the analysis.

For time-to-event endpoints, dropout produces censoring. Censoring is handled analytically by the survival model, without the complexity of imputation. In indications where dropout is expected to be high and informative — oncology trials where disease progression leads to treatment switches, for example — the time-to-event endpoint's censoring framework may be more defensible than the continuous endpoint's missing data framework, not because it is statistically simpler but because the censoring assumptions are more transparent.

This is not a recommendation for time-to-event endpoints. It is an example of the kind of consequence that should be in the room when the endpoint discussion is happening, and that the statistician is responsible for putting there.

Scenario 4: "We Need Five Co-Primary Endpoints"

This request arrives in various forms. Sometimes it is stated directly: "All five of these outcomes are important and we want to claim success on all of them." Sometimes it is stated as a list of "primary" endpoints that turns out to contain five items when the statistician counts them. Sometimes it arrives as a regulatory requirement that the sponsor has interpreted more broadly than the regulation requires.

What five co-primary endpoints actually means

Five co-primary endpoints, each tested at the nominal alpha, produce a family-wise type I error rate that is not 5%. Under the Bonferroni approximation, testing five independent hypotheses each at 5% produces an overall false positive probability of approximately 23%. If the endpoints are positively correlated — which they usually are, because they are measuring related aspects of the same disease — the inflation is smaller but still substantial.

This is not a statistical objection to the request. It is a description of what the request entails. The sponsor is proposing to run a trial in which the probability of claiming success when nothing works is substantially above the stated type I error. Whether that is acceptable depends on the clinical context and the regulatory framework, but it should be an explicit decision, not an implicit consequence of a design choice that was not examined.

The conversation that needs to happen

The statistician's role in this conversation is to translate five co-primary endpoints into a decision question that the clinical team can actually answer.

The decision question is: which of the following are you prepared to accept as the trial's result?

Option A: The trial requires all five endpoints to succeed. This is the conjunctive rule. The family-wise type I error is controlled below the nominal alpha without correction, but the power for the joint result is substantially lower than the power for any individual endpoint. If four endpoints succeed and one fails, the trial has failed.

Option B: The trial requires any one of the five endpoints to succeed. This is the disjunctive rule. The family-wise type I error is inflated and requires explicit correction. If the correction is applied, the threshold for each individual endpoint is more stringent than the nominal alpha, and the trial is less sensitive to each individual effect.

Option C: One of the five endpoints is actually primary, and the others are secondary. The trial is powered for the primary, the others are in a pre-specified hierarchy, and the label claim is structured around the primary with supportive evidence from the hierarchy.

When the clinical team is asked to choose among these three options, the five co-primary structure usually resolves quickly. The conjunctive rule is typically unacceptable — the team does not want the trial to fail because one of five endpoints missed significance. The disjunctive rule is typically unacceptable when the correction is explained — the team does not want each individual test to be harder to pass. Option C — one primary, others secondary — is almost always the right structure, and the conversation about which one is truly primary is the conversation that should have happened at the beginning.

The statistician's job is not to impose Option C. It is to make Options A and B concrete enough that the team chooses Option C on their own. A clinical team that understands what a conjunctive five-endpoint design actually commits them to, and what a Bonferroni-corrected disjunctive design actually requires, will usually arrive at the correct design decision without being told.

When the regulatory requirement is real

Sometimes the request for multiple primary endpoints is driven by a genuine regulatory requirement — an indication where the agency has established that both a functional

endpoint and a disease-specific endpoint must demonstrate benefit for approval. In this case, the conjunctive rule is not an option to be evaluated; it is the regulatory condition.

When the conjunctive rule is required, the power calculation must address the joint criterion, not the individual endpoints. The sample size is powered for the probability that both endpoints achieve significance simultaneously, accounting for the correlation between them. This joint power is typically lower than the power for either individual endpoint, and the sample size required to achieve 80% joint power is larger than the sample size for either individual endpoint alone. The design team needs to understand this before the sample size is presented to the development committee as the trial's enrollment target.

Scenario 5: Dropout — Sample Size Problem or Estimand Problem?

The dropout discussion usually arises in the context of sample size calculation: the statistician proposes inflating the sample size by the expected dropout rate, and someone asks what dropout rate to use. This question, if answered correctly, leads to a much larger discussion. If answered incorrectly — by picking a number and moving on — the discussion will recur at analysis in a more damaging form.

The question under the question

The question "what dropout rate should we use?" is actually two different questions that require different answers.

The first question is: **what proportion of enrolled patients will not complete the primary endpoint assessment?** This is a logistical prediction about patient behavior — how many patients will withdraw consent, be lost to follow-up, die before the assessment time point, or be administratively censored. This is the dropout rate that inflates the sample size. It is a nuisance parameter, and it should be estimated from prior studies with similar populations, similar trial durations, and similar treatment burdens.

The second question is: **what will happen to patients who discontinue study treatment before completing the trial?** This is not a logistical question. It is a scientific question about what the trial is measuring. A patient who discontinues treatment because of an adverse event, and then switches to an alternative therapy, has an outcome that is informative about the treatment's tolerability and its real-world effect. How that outcome is handled in the analysis depends on the estimand — and the estimand must be settled before the dropout rate question can be answered correctly.

Under a **treatment policy estimand**, the patient's outcome after discontinuation is part of the primary analysis. The patient is not dropped from the analysis; they are followed and their outcome is included regardless of whether they completed the assigned treatment. In

this case, the "dropout" that inflates the sample size is only the patients who are truly lost — who cannot be followed for any reason — not the patients who discontinued treatment. The inflation factor is smaller, but the data collection requirements are larger.

Under a **hypothetical estimand**, the patient's outcome after discontinuation is not the quantity of interest. The analysis attempts to estimate what would have happened if the patient had remained on treatment. In this case, the patient's post-discontinuation data are not used in the primary analysis, and the patient is effectively missing for the purposes of the primary comparison. The dropout rate that inflates the sample size is the discontinuation rate, which is typically higher than the administrative loss-to-follow-up rate.

The same event — a patient discontinuing treatment — has different implications for the sample size calculation depending on the estimand. If the estimand has not been settled when the dropout rate is being discussed, the sample size will be calculated for the wrong quantity.

The conversation that clarifies this

The most efficient way to surface the estimand question in the dropout discussion is to ask: if a patient discontinues treatment at week 12 in a 52-week trial, what do we do with them at the primary analysis?

The answers that reveal a treatment policy estimand: "We follow them and collect their week 52 outcome regardless of what treatment they are on." "We include their actual week 52 data in the primary analysis." "We want to know what happens to patients who receive this treatment in the real world, including the ones who stop early."

The answers that reveal a hypothetical estimand: "We want to know what would have happened if they had stayed on treatment." "We exclude or impute their data from the primary analysis." "We want to estimate the biological effect of the treatment, not the effect of trying to take it."

Once the estimand is identified, the dropout rate discussion can proceed correctly. Under treatment policy, the question is how many patients will be unreachable at 52 weeks for any reason — the administrative loss rate, which the statistician can estimate from logistics. Under hypothetical, the question is how many patients will discontinue treatment for any reason before 52 weeks — the discontinuation rate, which the clinical team must estimate from prior disease-area experience.

Both estimates carry uncertainty. The uncertainty should be examined in the joint pessimistic scenario — as Chapter 3 requires — not resolved by picking a round number and proceeding.

Scenario 6: The Statistician's Boundary

This scenario is different from the others. It is not about a specific design feature. It is about the range of questions the statistician can and cannot answer — and the damage that occurs when the boundary is crossed in either direction.

Questions the statistician can answer

The statistician can answer questions about probability, about operating characteristics, about the consequences of design choices under specified assumptions, and about the consistency between the design's components.

What is the probability of stopping early under each scenario? The statistician can answer this, given the interim analysis design and the assumed effect sizes.

What happens to power if the event rate is lower than assumed? The statistician can answer this, given the power calculation and the plausible range of event rate values.

Is this hierarchy consistent with controlling the family-wise type I error? The statistician can answer this, given the pre-specified order and the alpha allocation.

Does the primary analysis model correspond to the estimand? The statistician can answer this, given both the estimand and the analysis plan.

These are questions with technical answers, and the statistician is responsible for providing them clearly, in terms that the clinical team can use to make decisions.

Questions the statistician cannot answer

The statistician cannot answer questions about what the treatment will do in patients, what magnitude of effect is clinically meaningful, what risk of a false negative is acceptable to the patients who might benefit, or what the right comparator is for the indication.

What effect size should we assume for the power calculation? The statistician can present the range of plausible values and their sources, and can translate each into a required sample size. But the choice of which value to commit to is a clinical prediction about what the treatment will do, and it belongs to the clinical team.

Is 80% power acceptable for this trial? The statistician can show what 80% power means — a 20% probability of missing a real treatment effect — and can show how the required sample size changes at 85% or 90%. But the decision about what probability of failure is acceptable is a risk allocation decision that belongs to the sponsor's clinical and development leadership.

Is this margin clinically acceptable for a non-inferiority trial? The statistician can calculate M1 from historical data and can show the range of M2 values that the literature discusses. But the judgment about what fraction of the comparator's effect can be sacrificed without clinical harm is a clinical judgment, not a statistical one.

The damage of crossed boundaries

When the statistician answers questions that belong to the clinical team — when they propose the effect size, set the power level, and determine the NI margin without clinical input — two things happen. First, the decisions are made without the clinical judgment that is required to make them correctly. The effect size assumption reflects what the statistician thinks is reasonable rather than what the clinical team believes the treatment will do, and these are not the same.

Second, when the trial fails — when the effect size assumption turns out to be optimistic, or when the power level turns out to be insufficient — the accountability is assigned to the statistician rather than to the clinical team. The clinical team did not own the assumption; they accepted it without engagement. The statistician made the prediction. The prediction was wrong. In the debrief, the statistician is the one who cannot defend the number.

The correct boundary is not the statistician saying less. It is the statistician being more precise about what kind of input is needed from the clinical team, and more insistent on getting it. "I can calculate the sample size once you tell me the effect size you believe the treatment will produce" is not an abdication. It is the correct allocation of responsibility.

The damage of not crossing boundaries that should be crossed

The opposite failure is also real. When the statistician defers on questions that are genuinely statistical — when they say "it depends" or "there are different approaches" in response to questions that have clear technical answers — the design proceeds without the statistical input it needs.

Does this analysis control the type I error? This has a clear answer. The statistician should give it.

Is this hierarchy pre-specified in a way that would survive regulatory scrutiny? This has a clear answer. The statistician should give it.

Is this sample size consistent with the stated power, given the adaptive rule? This has a clear answer. The statistician should give it.

The statistician's value to the design team is not in managing uncertainty — it is in reducing the uncertainty that can be reduced, and being precise about the uncertainty that cannot. A statistician who treats every question as having multiple valid answers provides less value than one who distinguishes the questions with technical answers from the questions that require clinical judgment.

Scenario 7: Randomization and Double Programming

The randomization plan discussion is often brief — the statistician proposes a stratified, blocked randomization with central IVR/IWR, the team agrees, and the meeting moves on. The question of whether the randomization plan will be independently programmed — double programmed — is often not raised at all, or is treated as a QC formality rather than a design decision.

What double programming protects against

The randomization sequence is the foundation of the trial's validity. A randomization sequence that is incorrectly generated — because of a programming error in the randomization algorithm, an incorrect stratification structure, an error in the block structure, or a system configuration error — produces a trial in which the assignment is not what the protocol specified. The trial may still produce a valid result if the error is non-informative and symmetric, but it may not, and the error may not be discovered until the clinical study report is being prepared and the randomization sequence is audited.

Independent programming — implementing the randomization algorithm in a second programming environment, by a programmer who did not write the first version, and comparing the two outputs — is the standard protection against programming errors in the randomization sequence. When the two implementations agree, the probability of a systematic error in the generation is low. When they disagree, there is a programming error in one of them, and the error can be identified and corrected before enrollment begins.

When double programming is required

Double programming of the randomization is not always required, and treating it as a universal requirement adds cost and time without proportionate benefit in every case. The decision should be based on the complexity of the randomization structure and the consequences of a randomization error.

Simple randomization — equal allocation, no stratification, no blocking — is generated by a well-understood algorithm with a well-characterized output. The probability of a systematic error that would affect the trial's validity is low, and the output can be verified by inspection. Double programming adds limited value.

Stratified, blocked randomization with multiple stratification factors — particularly when the stratification factors interact, or when the allocation ratio differs across strata, or when the block structure is adaptive — is complex enough that programming errors are both more likely and harder to detect by inspection. Double programming is appropriate.

Covariate-adaptive randomization — minimization or related methods — uses an algorithm that updates with each enrollment and cannot be fully verified by examining a static sequence. Double programming is required, and the verification must be conducted prospectively —

by simulating enrollments and confirming that both implementations produce the same assignments — not retrospectively.

Central IVR/IWR systems that implement the randomization algorithm in a validated system require system validation, not double programming in the statistical sense. The IVR/IWR system's randomization module is validated as part of the system validation process, which is a different kind of independent verification. A trial that uses a validated IVR/IWR system has not skipped double programming; it has implemented the equivalent protection through a different mechanism.

What cannot be double programmed

The randomization sequence can be double programmed. The blinding cannot.

Blinding is maintained operationally — through identical appearance of treatment and placebo, through sealed treatment packs, through central dispensing systems. It cannot be verified by independent programming because it is not a program; it is an operational practice. The protection against blinding failures is the governance structure described in Chapter 5, not a technical verification step.

This distinction matters because it defines the boundary of what the randomization verification can assure. A successfully double-programmed randomization confirms that the sequence was generated correctly. It does not confirm that the sequence was concealed correctly, that the treatment packs were correctly labeled, or that the IVR/IWR system correctly assigned the treatment pack to the patient. Those protections require operational validation, not independent programming.

The design team should understand both what double programming provides and what it does not. A trial whose randomization was correctly generated but whose allocation concealment was compromised is a trial with selection bias, regardless of the randomization's technical correctness.

Scenario 8: "The Primary Endpoint Failed, But Look at This Subgroup"

This conversation happens after unblinding. The primary analysis did not achieve significance. Someone — often the clinical lead, sometimes the sponsor's development head — points to a subgroup result: a specific patient population where the treatment effect looks compelling. "Can we build a submission around this?" The pressure in the room is real. The program has been running for years. There is a treatment that may work in some patients. The statistician needs to say something.

What has just happened, statistically

When a primary endpoint fails and a subgroup appears positive, two things are simultaneously true. The subgroup result may reflect a genuine differential treatment effect — the treatment may really work better in this population. And the subgroup result is almost certainly inflated, because it was identified from the data rather than before the data were seen.

These two things are not contradictory. A result can be both potentially real and statistically unreliable. The question is not whether to believe the subgroup finding. The question is what the finding justifies doing next.

The statistician's role is to explain, precisely, why the subgroup finding — however compelling it looks — cannot substitute for the confirmatory evidence the failed primary was designed to provide. The explanation has three components.

First, the multiple testing problem. When the primary endpoint fails and the team examines subgroups to find a positive result, the number of comparisons being made — across all the subgroups examined, even informally, even in a single meeting — inflates the probability of finding a false positive. A single subgroup result with a p-value of 0.03, identified after examining twenty subgroups in a failed trial, does not have a 3% false positive probability. Its false positive probability is substantially higher, and the nominal p-value does not reflect the search that produced it.

Second, the regression to the mean problem. Subgroups identified from positive results in one trial tend to show smaller effects in subsequent trials, because the first trial's observation included both the true effect and the chance variation that pushed the result into significance. The effect in the next trial — powered for the subgroup, with the subgroup pre-specified — is expected to be smaller than what was observed in the subgroup analysis of the failed trial.

Third, the estimand problem. The failed primary trial was not designed for this subgroup. The eligibility criteria, the stratification, the sample size, and the analysis plan were all designed for the overall population. The subgroup comparison is conducted in a population that was not prospectively enriched, with a sample size that was not calculated for the subgroup effect, and without the stratification that would have been used if the subgroup had been the primary population. The subgroup result is not a failed primary that happened to succeed in a smaller population. It is a different analysis, in a different population, under different conditions.

What the finding does justify

The subgroup finding from a failed primary trial is hypothesis-generating. This is not a dismissal — hypothesis generation is valuable. It defines the next trial.

The next trial for this subgroup has a specific design requirement: the subgroup must be the primary population, prospectively defined, with eligibility criteria that operationalize the subgroup definition, with a sample size calculated for the subgroup-specific effect, and

with the biomarker or characteristic that defined the subgroup validated as a companion diagnostic or pre-specified selection criterion.

The statistician can help the team design that trial. What they cannot do is help the team construct a submission based on the subgroup result from the failed trial without acknowledging that the result is exploratory, that the effect is likely overestimated, and that the regulatory agency will evaluate the submission with exactly these concerns.

The specific risk of this conversation

The pressure to claim the subgroup finding as confirmatory is greatest when the finding is clinically plausible — when there is a biological rationale for why the treatment would work better in this population, when the subgroup effect is large, and when the clinical team believes the treatment is genuinely beneficial for these patients.

Clinical plausibility does not convert a post-hoc finding into a confirmatory result. It increases the prior probability that the finding is real. It does not change the statistical properties of the analysis. A finding that is both post-hoc and biologically plausible is a compelling hypothesis for a confirmatory trial. It is not a confirmatory result from an exploratory analysis.

The statistician who allows this conflation — who does not clearly distinguish between "this finding is worth pursuing" and "this finding supports a label claim" — has not served the clinical team or the patients who will receive the treatment based on an inflated efficacy estimate. The clinical team can believe the subgroup effect is real and still understand that the evidence does not yet support a claim. These are not contradictory positions.

Scenario 9: The NI Margin Discussion

The non-inferiority margin discussion is scheduled as a single meeting and almost never resolved in one. It begins with the statistician presenting the historical data on the active comparator's effect, proceeds to a discussion of what fraction of that effect should be preserved, and stalls when the clinical team realizes they are being asked to specify how much worse the new treatment is allowed to be. Nobody wants to say a number out loud.

Why the conversation stalls

The NI margin requires two quantities. M1 is the comparator's estimated effect relative to placebo, derived from historical trials. M2 is the fraction of M1 the new treatment must preserve to be considered clinically acceptable.

M1 is a statistical estimation problem. The statistician presents the historical evidence, accounts for the uncertainty in the estimate, and proposes a conservative estimate of M1 —

typically the lower bound of the confidence interval around the historical estimate, not the point estimate itself. This is technical work and the statistician should lead it.

M2 is a clinical judgment. It requires answering the question: if the new treatment preserves X% of the comparator's effect, is it still an acceptable treatment for patients? The answer depends on the treatment's other properties — its side effect profile, its route of administration, its cost, its convenience — relative to the comparator. If the new treatment is substantially more convenient or substantially less toxic, a lower M2 may be acceptable. If it offers no advantages except the ones being tested, M2 should be close to 1.

The conversation stalls because M2 requires the clinical team to commit to a clinical judgment in public, in writing, before the trial begins. The judgment will later be scrutinized — by the regulatory agency, by the DSMB, by clinical audiences reading the publication. If the judgment turns out to be too generous — if the new treatment is approved on the basis of a margin that later seems clinically unacceptable — the team members who specified M2 are accountable.

How to move the conversation forward

The statistician can move the conversation forward by reframing the question. Instead of asking "what fraction of M1 is acceptable," ask: "What would a clinician recommend for a patient who was told the new treatment might be up to X% less effective than the comparator, but had the following advantages?"

This reframe makes the clinical judgment concrete. The clinical team is not being asked to specify a statistical parameter. They are being asked to describe the clinical context in which the new treatment is acceptable. That context — the balance of benefit, risk, and practical considerations — is what M2 is supposed to capture, and specifying it in clinical terms is easier than specifying it as a fraction.

Once the clinical context is specified, the statistician translates it into a numerical M2. The translation should be documented — the clinical description that led to the number — so that the margin can be defended not just as a statistical choice but as a clinical judgment with a traceable rationale.

The constancy assumption and assay sensitivity

The margin discussion has two additional components that are often skipped because the meeting has already been long by the time M1 and M2 are resolved.

The constancy assumption asks whether the comparator's historical effect — the M1 estimated from past trials — transfers to the current trial. The question is whether the comparator performs as well in the current trial's population, under current standard of care, with current concomitant medications, as it did in the historical trials. If the answer is no — if the comparator's effectiveness has diminished because background therapy has improved — then M1 is overestimated and the derived margin is too permissive.

The assay sensitivity question asks whether the trial, as designed, is capable of detecting a difference between active treatment and inactive treatment if one exists. A trial with poor assay sensitivity will not reliably detect inferiority even when it is present — which means a non-inferiority conclusion from such a trial is not informative.

Both questions require engagement from the clinical team, not just the statistician. The statistician can present the historical data that bear on the constancy assumption. The clinical team must judge whether the historical context is sufficiently similar to the current context. The statistician can design the trial to maximize assay sensitivity. The clinical team must confirm that the enrolled population and the trial conditions are appropriate for detecting the comparator's effect.

If these discussions do not happen before the margin is finalized, the margin will be finalized on an incomplete basis — and the regulatory agency will ask the questions that the design team did not.

Scenario 10: "Can We Add an Endpoint?"

The request arrives during enrollment. Sometimes it is mid-way through, sometimes near the end, sometimes just before the primary analysis. The rationale is usually one of three: new external evidence has emerged that suggests an additional endpoint is important, a competitor's trial result has changed the landscape, or — and this is the one the statistician must be alert to — someone has seen something in the accumulating data that makes a particular endpoint look promising.

The three sources and why they matter

The source of the request determines everything about how it should be handled.

New external evidence — a published trial in a related indication, a regulatory guidance update, a biomarker discovery in a parallel program — can legitimately motivate adding an endpoint during enrollment, provided the motivation is documented and the timing is consistent with the evidence being genuinely new. A request driven by a paper published last week, documented with the citation, reviewed by the DSMB before implementation, and processed as a formal protocol amendment with regulatory notification is a defensible mid-trial modification. Its implications for the analysis — the endpoint is exploratory, not confirmatory, unless a multiplicity correction is applied — must be clearly stated.

Competitive intelligence — a competitor's trial succeeded on an endpoint this trial is not measuring — is a more complicated trigger. The sponsor's legitimate interest in understanding whether their treatment has the same properties as the competitor's creates pressure to add the endpoint. But the timing is informative to a regulatory reviewer: an endpoint added after a competitor's positive result looks like an attempt to claim the same

benefit without the same evidence. The request should be reviewed carefully, the rationale should be documented fully, and the statistical consequences — exploratory status, no multiplicity protection — should be clearly stated.

Internal data signal — the most dangerous trigger. When the request to add an endpoint arrives shortly after an interim DSMB review, or when the requester has any plausible channel to information about the trial's interim trends, the statistician must ask — directly, in writing — whether the request was motivated by knowledge of the accumulating data. This is not an accusation. It is a governance question. If the answer cannot be given with confidence — if the timeline is ambiguous, or if the requester had access to the DSMB's unblinded report through any channel — the amendment should not be treated as externally motivated.

An amendment driven by internal data signal is an unplanned adaptation. It is not necessarily invalid — unplanned adaptations can sometimes be accommodated with appropriate analytical corrections — but it cannot be treated as a pre-specified addition, and the new endpoint cannot be claimed as confirmatory without addressing the type I error implications of the data-driven selection.

The analytical consequence

Any endpoint added during enrollment is exploratory unless a specific provision was made for it. This is not a punitive rule. It is the consequence of the pre-specification requirement: the type I error rate for confirmatory claims is controlled by the pre-specified analysis. Endpoints added after the trial begins were not part of the pre-specified analysis and are not protected by its error control.

The clinical team must understand this before the amendment is filed. An endpoint added to the protocol is not automatically elevated to confirmatory status because it appears in the protocol. Its confirmatory status depends on whether it was in the original pre-specified hierarchy, with alpha allocated to it, before the trial began.

If the team wants the new endpoint to be confirmatory, the alpha allocation for the entire testing hierarchy must be revised. The revision requires adjusting the planned sample size if the new endpoint is the primary, or adjusting the hierarchical plan if it is secondary. Either adjustment must be made before any unblinded data bearing on the new endpoint are seen. If the adjustment is made after such data are available, the revision is post-hoc and the endpoint remains exploratory regardless of how the protocol is written.

Scenario 11: The Alpha Allocation Meeting

The hierarchical testing plan is being finalized. The clinical team has five secondary endpoints they consider important, and they want all of them in the confirmatory hierarchy. The

statistician has explained that the hierarchy must be ordered, and that testing stops when a hypothesis fails. The meeting has been going for two hours. Three different people have proposed three different orderings, and each ordering reflects a different view of which endpoint matters most. Nobody wants to say that their preferred endpoint is less important than someone else's.

What is actually being decided

The argument about ordering is not really about statistics. The statistician can implement any ordering the team chooses; no ordering is technically wrong. The argument is about which finding the organization will lead with, which subpopulation will be featured in the label, and which clinical hypothesis the team believes in most strongly.

These are clinical and strategic decisions, and the statistician should name them as such. The ordering of the hierarchy is the organization's public statement of its clinical priorities, expressed in a form that will be tested by data and scrutinized by regulators. If the team cannot agree on the ordering, it is because they cannot agree on those priorities — and that disagreement is the real problem, not the statistical plan.

The statistician's role is to make the decision's consequences concrete. For each proposed ordering, the statistician shows what happens to each endpoint's claim under each possible outcome sequence. If Endpoint A is first and fails, Endpoint B is not testable at the controlled error rate regardless of its p-value. If Endpoint B is first and fails, the ordering that followed was irrelevant. The team sees, for each ordering, what they are committing to — which claims they will lose if specific endpoints fail.

The efficiency argument versus the importance argument

There is a temptation to order the hierarchy by the probability of success — to put the endpoint most likely to achieve significance first, to protect the later endpoints by ensuring the hierarchy stays open as long as possible.

This approach has a statistical logic but a clinical cost. An ordering based on statistical likelihood rather than clinical importance sends a specific message: the organization is optimizing for the probability of making some claim, rather than testing its most important hypothesis first. A regulatory reviewer who examines the ordering and finds that it corresponds to the ranking of expected p-values rather than the ranking of clinical importance will ask why the most clinically important endpoint is not first.

The correct ordering is by clinical importance, and the clinical team must specify that ordering explicitly. The statistician can point out if an ordering appears to be driven by statistical convenience rather than clinical priority. But the judgment of what is most clinically important belongs to the clinicians, not to the statistician.

When the hierarchy genuinely cannot be ordered

Sometimes the secondary endpoints are genuinely of equal clinical importance — three different functional assessments in a rehabilitation trial, two different safety endpoints in an oncology trial with a complex risk profile. When clinical importance does not resolve the ordering, the statistician has two options.

The first is a graphical testing procedure — a procedure that allows alpha to flow between hypotheses in multiple directions, so that a failure in one endpoint does not necessarily close the path to testing others. Graphical procedures require more upfront design work and are more complex to explain to the DSMB and the regulatory agency, but they handle genuine ties in importance without forcing an artificial ordering.

The second is to reconsider whether all five endpoints should be in the confirmatory hierarchy at all. Three of the five may be genuinely confirmatory; the other two may be supportive evidence that does not need to be in the primary claim structure. Reducing the hierarchy to the endpoints that are truly confirmatory often dissolves the ordering disagreement, because the remaining endpoints have a clearer clinical priority ranking.

Scenario 12: "We Need to Show Results in All Regions"

The trial is global — sites in North America, Europe, East Asia, and sometimes additional regions. At the design stage, or at the analysis planning stage, the request arrives: we need to show that the treatment works in each region separately, because each regional regulatory agency will require evidence in their population. The statistician is asked to power the trial for regional subgroup analyses.

What regional consistency actually requires

The regulatory agencies in different regions — FDA, EMA, PMDA, NMPA — do not generally require a separate confirmatory analysis in each region. What they require is evidence that the overall trial result is not driven entirely by one region, and that the result in their region is consistent with the overall result in a way that makes it plausible the treatment works in their population.

This is a consistency requirement, not a confirmatory requirement. The regional subgroup analysis is evaluated as a consistency check: does the treatment effect in this region have the same direction and approximate magnitude as the overall effect? Is the confidence interval for the regional effect consistent with the overall estimate? Is there any evidence of substantial regional heterogeneity that would call the overall result into question for this region's population?

The statistician should explain this distinction clearly. Powering the trial for a confirmatory regional subgroup analysis would require a sample size that is a multiple of what the overall primary analysis requires — and would likely make the trial operationally unfeasible. What is required is a sample size that provides reasonable precision for the regional estimates, so that the consistency check can be conducted informatively.

The minimum regional sample size question

The practical question is how many patients per region are needed for the regional estimate to be informative for the consistency check. There is no single answer — it depends on the expected magnitude of the overall treatment effect, the variance of the regional estimate, and the threshold for consistency that the regional regulatory agency uses.

The PMDA has published specific guidance on what it considers an adequate regional contribution — a framework based on the proportion of the overall sample size contributed by the Japanese population and the consistency of the Japanese subgroup estimate with the overall result. Other regional agencies have less formal frameworks but similar underlying logic.

The statistician can present the precision of the regional estimate as a function of the regional sample size, under the assumed overall treatment effect. This allows the team to see, concretely, what a regional sample of 100 patients versus 200 patients versus 300 patients provides in terms of the width of the regional confidence interval. The team can then decide whether the regional sample is adequate for the consistency check, informed by the expected requirements of each regional agency.

The heterogeneity problem

When regional results are heterogeneous — when one region shows a strong positive result and another shows no effect — the trial's primary result becomes difficult to interpret for at least one region. This heterogeneity may reflect genuine biological or practice variation, or it may reflect chance variation in smaller regional samples.

The statistician should address the heterogeneity question prospectively in the design: what will the trial conclude about regional consistency if the regional estimates differ substantially? The pre-specified consistency criterion — the statistical test or the clinical threshold that will be used to evaluate whether the regional results are consistent enough to support the overall claim — should be in the SAP before unblinding. A consistency criterion constructed after the regional results are seen is a post-hoc criterion, and the regional agency reviewing the submission will treat it as such.

Scenario 13: Surrogate Endpoints

The discussion usually begins with a practical problem: the clinically meaningful endpoint — overall survival, major cardiovascular events, definitive disease cure — requires a trial too long and too large to be feasible in the current development context. Someone proposes a surrogate: progression-free survival instead of overall survival, a biomarker instead of a clinical event. "The field uses this surrogate. Regulators have accepted it before. Can we build the primary trial around it?"

What the statistician can and cannot say

The statistician can characterize the empirical relationship between the surrogate and the clinical outcome — the correlation in historical datasets, the proportion of the treatment effect on the clinical outcome that is mediated through the surrogate in meta-analyses, the consistency of the surrogate-outcome relationship across different treatments and populations. This is quantitative work, and the statistician should do it before the meeting, not during it.

What the statistician cannot say is whether the surrogate is valid for this trial. Surrogate validity is a scientific and regulatory judgment, not a statistical calculation. A surrogate that is highly correlated with the clinical outcome at the patient level may not be a valid surrogate at the trial level — the treatment may affect the surrogate without affecting the clinical outcome through the mechanism the surrogate is supposed to represent. The distinction between patient-level correlation and trial-level validity is technical, but the judgment of whether a specific surrogate is valid for a specific treatment mechanism belongs to the clinical and regulatory domain.

The statistician's role is to present the quantitative evidence that bears on the surrogate's validity and to identify the gaps in that evidence — without filling the gaps with statistical confidence the evidence does not support.

The accelerated approval context

In the United States, the FDA's accelerated approval pathway permits approval based on a surrogate endpoint that is "reasonably likely to predict" clinical benefit, conditioned on a post-approval confirmatory trial that demonstrates actual clinical benefit. The EMA's conditional approval has a similar structure.

The design implication is specific: the trial built on the surrogate endpoint is not the full evidentiary package. It is the first stage of a two-stage evidentiary development. The surrogate endpoint trial must be designed to support the accelerated approval claim, and the confirmatory trial must be designed — and ideally initiated — before the surrogate endpoint trial is complete.

The statistician should ensure the team understands both stages of the design. The surrogate endpoint trial's sample size, endpoint definition, and claim structure are designed for the

accelerated approval. The confirmatory trial's design — which will be scrutinized by the agency after the accelerated approval is granted — should be pre-specified in the accelerated approval submission. A development program that uses a surrogate endpoint for initial approval without a credible confirmatory trial plan is a program that may face post-approval pressure it was not designed to handle.

When the surrogate fails to predict

Historical precedent in clinical development includes cases where treatments improved surrogate endpoints substantially and failed to improve — or worsened — clinical outcomes. Antiarrhythmic drugs that reduced ventricular ectopy but increased mortality. Cancer treatments that improved tumor response rates but did not extend survival. These cases are not anomalies; they reflect the mechanism-dependence of surrogate validity.

The statistician should present these precedents in the surrogate endpoint discussion, not as arguments against using the surrogate but as evidence for why the surrogate-outcome relationship must be examined specifically for the treatment's mechanism. A surrogate that has predicted benefit for one drug class may not predict benefit for a drug with a different mechanism targeting the same surrogate. The validity of the surrogate must be evaluated for this mechanism, in this population — not assumed from the general literature on the surrogate's performance across treatments.

Scenario 14: Sample Size in Rare Disease and Small Populations

The indication has 5,000 patients in the United States. A realistic enrollment target for a confirmatory trial is 200 patients, perhaps 300 with global sites. The statistician presents a power calculation showing that at 80% power, the trial can detect a hazard ratio of 0.60 — a 40% reduction in the event rate. The clinical team asks: what if the true effect is 0.70? The answer is that the trial would have 45% power to detect that effect. The room goes quiet.

What 80% power means in this context

In a large cardiovascular trial enrolling 10,000 patients, a power of 80% is a design choice that accepts a 20% probability of missing a real treatment effect in exchange for a feasible sample size. The choice is defensible because there may be additional trials, because meta-analyses will eventually incorporate the result, and because the missed effect can be detected in subsequent development.

In a rare disease trial enrolling 200 patients, the same 80% power has a different meaning. There may be no subsequent trial — the population is too small, the development economics will not support another attempt, and the 200 patients enrolled in this trial may represent a substantial fraction of the identifiable patient population. A trial that fails to detect a

real treatment effect in this context does not lead to a second trial. It leads to patients not having access to a treatment that works.

This asymmetry — the higher cost of a false negative in a rare disease population with no alternative development path — is the argument for accepting a lower power level or for accepting a larger type I error. Both adjustments require documentation and justification, but both are defensible in the context of rare disease development.

The statistician should present this argument explicitly. Not as a recommendation to lower standards, but as a description of the decision the team is making and the consequences on both sides. The conventional 80% power was calibrated for a development context — multiple trials, large populations, post-market studies — that does not apply to rare disease. The team should make a power decision appropriate to their context, not default to the convention.

The regulatory framework for rare disease

The FDA's Rare Pediatric Disease designation, Orphan Drug designation, and Breakthrough Therapy designation — and the EMA's PRIME designation and orphan medicinal product designation — all reflect the regulatory system's acknowledgment that rare disease development requires different standards in some respects.

The key regulatory implication for trial design is that the FDA and EMA are generally willing to engage on trial design questions through the pre-submission process earlier and more extensively for rare disease programs than for large-indication programs. A statistician working on a rare disease trial should recommend that the sponsor use this access — that the trial design, including the power level and the effect size assumption, be discussed with the regulatory agency before enrollment begins, not after the trial is complete.

A power level of 70% or even 60%, discussed with and accepted by the regulatory agency before enrollment, is a defensible design choice for a rare disease with no alternatives. The same power level, imposed without regulatory discussion and revealed in the clinical study report, will face questions the design team should have anticipated and addressed in advance.

The single-arm trial question

In the most extreme cases — ultra-rare diseases, conditions where a randomized controlled trial cannot be conducted because the patient population is too small to support a control arm — the question arises of whether a single-arm trial with an external control or a natural history comparison can serve as the evidentiary basis for approval.

This is not primarily a statistical question. The regulatory standard for approval is controlled evidence of benefit and safety. Departing from the randomized controlled trial design requires a scientific justification for why the controlled design is not feasible and why the proposed alternative provides sufficient evidence to support approval in the absence of a concurrent control.

The statistician's contribution to this discussion is characterizing the uncertainty introduced by the absence of randomization — the confounding that cannot be controlled, the secular trend that cannot be accounted for, the selection bias in the natural history comparator — and quantifying its likely magnitude relative to the observed treatment effect. If the observed effect is an order of magnitude larger than the plausible confounding, the single-arm trial may provide adequate evidence despite the design limitations. If the observed effect is of similar magnitude to the plausible confounding, it does not.

Scenario 15: When the Statistician Disagrees With the Design

The design has been finalized. The protocol is about to be submitted for IRB approval. Enrollment is scheduled to begin in six weeks. And the statistician believes the trial is fundamentally flawed — not flawed in a correctable way, but flawed in a way that means the trial will produce evidence that is either misleading or uninterpretable. The effect size assumption is not just optimistic; it is implausible. The primary endpoint does not correspond to the estimand. The NI margin was set by the sponsor's commercial team without clinical input and cannot be defended. What does the statistician do?

The professional obligation

The statistician has a professional obligation that exists independently of their employment relationship with the sponsor. That obligation is to the integrity of the evidence the trial will produce, and through that evidence, to the patients who will ultimately be treated based on what the trial shows.

This obligation is not abstract. It is specified in the statistical professional codes of conduct — the ASA's Ethical Guidelines for Statistical Practice, the PSI's Code of Conduct — which are explicit that statisticians have a responsibility to ensure that their work is used appropriately, to point out when conclusions appear to be unsupported by the data, and to avoid participating in work that misrepresents evidence.

When the statistician believes the trial is fundamentally flawed, the obligation is to say so — clearly, in writing, with specific identification of the design elements that are problematic and the specific consequences they will produce. The saying must be documented. An oral concern expressed in a meeting and not followed up in writing has not been discharged.

The escalation path

The first step is to say it to the person who can change it. The statistician should communicate the concern to the clinical lead, the project statistician's manager, and the regulatory affairs lead — not simultaneously, but in sequence, beginning with the person closest to the design decision. The communication should be specific: this margin cannot be defended because

the constancy assumption fails under current standard of care; this effect size assumption implies a treatment effect that is larger than any effect observed in this drug class; this endpoint does not correspond to the estimand as written.

If the first step does not produce engagement — if the concern is acknowledged but not addressed — the second step is escalation. In most organizations, this means escalating to the medical director, the head of biostatistics, or the chief medical officer. The escalation should be documented: the concern was raised on this date, to this person, with this specific content, and the response was this.

If escalation within the organization does not produce engagement, the statistician is in a position where they must decide whether to continue their involvement with the trial. This is a personal decision with professional and economic consequences. The decision is easier to make when the concern has been clearly documented — when the statistician can demonstrate that they raised the issue, that they escalated appropriately, and that the response did not address the substantive concern.

What documenting disagreement achieves

Documented disagreement serves two purposes.

The first is organizational accountability. When a trial fails — when the effect size assumption proves to be implausible, when the NI conclusion is challenged because the margin was not defensible, when the primary endpoint result is challenged because it does not correspond to the estimand — the design decisions that produced the failure are examined. A statistician who documented their concern before enrollment has a clear record of having identified the problem. A statistician who did not document the concern is in the same position as everyone else: responsible for the design they contributed to.

The second is systemic learning. The clinical trial enterprise improves when design failures are understood. Design failures are understood when the concerns raised before the trial began are documented and can be examined after the trial ends. A culture in which statisticians raise concerns but do not document them — because documentation creates friction, because it signals distrust, because the organization does not reward it — is a culture that does not learn from its mistakes.

The specific risk of not saying it

The most common failure is not the statistician who disagrees and is overruled. It is the statistician who has reservations but does not raise them — who assumes that someone else has already evaluated the concern, who does not want to delay enrollment, who is not certain enough of their position to put it in writing.

The reservation that was not raised becomes the design flaw that was not corrected. The trial that proceeds with the unexamined flaw produces the result that cannot be fully defended. The statistician who did not raise the concern is, in the analysis that follows, part of the team that produced the design — not a party who identified the problem and was overruled.

Saying it clearly, in writing, at the right time, is the professional act. What happens afterward is partly outside the statistician's control. What the statistician says — and documents — is not.

On These Scenarios

These scenarios do not replace the chapters that precede them. They are anchored to those chapters: each scenario is a translation of a design principle into a meeting-room situation where that principle is tested. They do not cover every situation a statistician will encounter in clinical trial design. They cover the situations where the distance between good design and defensible design is most often created — not by technical errors, but by conversations that did not go far enough, decisions that were not owned clearly enough, and concerns that were not raised clearly enough.

The statistician's role in each scenario is not to advocate for a position — not to argue for O'Brien-Fleming over Pocock, or for time-to-event over binary endpoints, or for 90% power over 80%. It is to make the decision's consequences visible before the decision is made, and to ensure that the person who owns the decision has the information they need to make it defensibly. It is also to be the person who asks the question that the design cannot proceed without answering, and who insists on a written answer before enrollment begins.

That insistence is uncomfortable. It creates friction. It slows meetings. It occasionally makes the statistician unpopular. It also produces the only kind of trial design that can be defended when it matters — when the result is contested, when the regulatory agency asks how the margin was determined, when the DSMB asks what authority they have to stop the trial, when a post-hoc subgroup finding is being positioned as confirmatory evidence.

A design that withstands scrutiny is not one where the statistician won every argument. It is one where every decision was made by the right person, with the right information, at the right time, and documented in a form that can be verified. The design that withstands scrutiny was built by a team that had the uncomfortable conversations before enrollment. This appendix is about what those conversations look like.